Yang–Mills on a finite torus along Bałaban's route: a verified reduction

Claude

October 2026

Abstract

Bałaban's renormalization-group construction of four-dimensional SU(2) lattice Yang–Mills theory rests on the control of the running coupling, his Theorem 2, whose proof was never published. We prove that a weak form of Theorem 2 (two-sided bounds on $1/g_k^2$ with an additive constant), which is enough for Bałaban's Theorem 1, follows from one uniform inequality R-025 (H2) on a two-step minimizer map together with named fixed-level and history inputs; the shooting lemmas, the arithmetic of L-009 (Theorem K), and the bridge lemmas of the certifier are formalized in Lean 4. We prove that, on a fixed torus, a uniform moment bound H-116 (H-U) gives compactness of the lattice Schwinger functions of the centred plaquette density, and that every limit point satisfies the torus forms of the Osterwalder–Schrader axioms OS0–OS3. The bound H-116 (H-U) is reduced, not proved: it rests on sourced versions of Bałaban's bounds, on a hypothesis R-043 (FX) that is now reduced to a holomorphy bound R-040/R-041 (H-hol) for Bałaban's increments together with history bounds and three named inputs of the large-field step, and on a bridge from the source form to H-116 (H-U), and one dependency table records the status of each block. At $L = 13$, inside Bałaban's range $L > 11$, R-025 (H2) is a certified theorem: an explicit majorant and an outward-rounded interval branch-and-bound give $\rho_1(j, K)^2 \leq 10 < 13$ for every level j (including $j = \infty$) and every momentum K, so $\rho_2 \leq 10 < 13$, given our reading of Bałaban's averaging operator; the result is not a solution of the Clay Millennium problem.

Contents

1 Introduction **4**
1.1 The problem . . . 4
1.2 Bałaban's programme and the missing Theorem 2 . . . 4
1.3 The two targets and the main results . . . 5
1.4 Status conventions . . . 6
1.5 Relation to other work . . . 7
1.6 Organization . . . 9

2 Setup and notation **9**
2.1 Lattice, measure, and scales . . . 9
2.2 Range of L . . . 10
2.3 The coupling sequence . . . 10
2.4 One-loop notation . . . 10
2.5 Observables and Schwinger functions . . . 10
2.6 Names . . . 11

3 The coupling flow and Theorem 2 **11**
3.1 What Bałaban's Theorem 1 actually needs . . . 11
3.2 The weak form proved conditionally . . . 12
3.3 Literal (0.31) . . . 13
3.4 Shooting lemmas . . . 13
3.4.1 L-004 (Lemma D′) (shooting without early positivity) [PROVED+VERIFIED; LEAN] . . . 13
3.4.2 L-005 (Lemma D″) (tail-sum form; no index J) [PROVED+VERIFIED; LEAN] . 14
3.4.3 L-005 (Lemma D″) meets Balaban's proof: the cross-scale sweep [PROVED, UNVERIFIED (SWEEP); 4 OF 19 TABLE ROWS VERIFIED] . . . 15
3.4.4 L-006 (Lemma D) and L-007 (Lemma E) [PROVED+VERIFIED; LEAN] . . . 15
3.4.5 History dependence (L5) [PROVED RELATIVE TO LISTED INPUTS; VERIFIED (ROUND 7)] . . . 16
3.5 The exact structure of the one-loop sums . . . 16
3.5.1 Telescoping [PROVED+VERIFIED, CORRECTED SCOPE] . . . 16
3.5.2 Anchor-free identity [PROVED+VERIFIED] . . . 17
3.5.3 The constant c [PROVED+VERIFIED] . . . 17
3.5.4 The decomposition $S_k = ck \log L + \kappa_W^F - \kappa_k$ . . . 17
3.5.5 Level-uniform symbol facts used below [PROVED+VERIFIED] . . . 18
3.6 Bounded window sums: two formally distinct conditional routes . . . 18
3.6.1 Route K: L-009 (Theorem K) [PROVED+VERIFIED, CONDITIONAL ON (B2) AND THE FIXED-LEVEL INPUTS] . . . 19
3.6.2 Route B: Theorem $\mathrm{B_{PW}}$ [PROVED+VERIFIED, CONDITIONAL (LOGIC OF THEOREM $\mathrm{B_{PW}}$)] . . . 20
3.6.3 Route B inputs after `BETA_FP_U` [PROVED, UNVERIFIED; NO VERIFIER FILE] . 21
3.6.4 Comparison . . . 21
3.7 The hypothesis (B2) of Route K, and the single uniform inequality R-025 (H2) . . . 22
3.7.1 Proved . . . 22
3.7.2 Refuted, and the failed one-step bound . . . 23
3.7.3 R-025 (H2) and the two-step theorems [PROVED+VERIFIED, CONDITIONAL ON R-025 (H2)] . . . 23
3.7.4 One-step sufficiency [ARITHMETIC PROVED; INPUT CERTIFIED AT $L = 13$] . . 24
3.7.5 The precise remaining inequality . . . 24
3.8 Conclusion of the reduction . . . 24
3.9 The convergence route: what it adds, and its status . . . 25
3.9.1 Components and status . . . 25
3.9.2 Early steps and j_* . . . 26

4 The finite-torus existence theorem **27**
4.1 What "exists" means here . . . 27
4.2 Hypothesis delivered by the blocks . . . 27
4.3 Theorem (finite-torus existence, compactness form) . . . 27
4.4 Theorem (nontriviality), conditional on H-081 (B-NT) . . . 28
4.5 Proof modulo blocks . . . 29
4.5.1 H-084 (B-T2) Theorem 2: the running coupling . . . 29
4.5.2 R-035 (B-S) Sourced bounds E14-a..g . . . 29
4.5.3 R-045 (B-N1) Bridge: source-form bounds ⇒ H-116 (H-U) [OPEN] . . . 29
4.5.4 H-074 (B-C) Compactness [PROVED HERE, GIVEN H-116 (H-U)] . . . 30

4.5.5 H-082 (B-REG) Regularity (OS0) [PROVED HERE, GIVEN H-116 (H-U)] . . . 31
4.5.6 H-085 (B-TI) Translation and hypercubic invariance (OS1) [PROVED HERE, GIVEN H-116 (H-U)] . . . 31
4.5.7 H-083 (B-RP) Reflection positivity (OS2) . . . 31
4.5.8 (OS3) Symmetry and reality [PROVED HERE] . . . 32
4.5.9 H-081 (B-NT) Nontriviality: what is needed . . . 32
4.6 Dependency table . . . 33
4.7 What the theorem does not give . . . 45
4.8 Corrections after verification . . . 46

5 The sources block **46**
5.1 Target 1: inputs of L-001 (Theorem 2w) . . . 47
5.2 Target 2, step 2: the sourced Theorem 1 (ledger R-035) . . . 49
5.3 Step 3: the R-045 (S) clauses and R-043 (FX) (ledgers R-045, R-043) . . . 53
5.4 Steps 4–5: N3, R-040/R-041 (H-hol), and the large-field inputs (ledgers R-042, R-040, R-041) . . . 55
5.5 Step 6: thresholds and second history derivatives (ledgers R-039, R-038) . . . 59
5.6 Step 7: history bounds and the gain (ledger R-037) . . . 61
5.7 Step 8: Bałaban-level propagator inputs (ledgers R-001–R-005) . . . 63
5.8 Step 8′: the three write-outs of the propagator expansions . . . 68
5.9 Step 8″: round 5, the remaining Group A rows and L-174 (Lemma G3) . . . 70
5.10 Step 8‴: round 6, locality, the boundary, the 2D gap, and the order of constants . . . 74
5.11 Step 8⁗: round 7, the implicit Group A rows, (M_j), L5, and the sweep . . . 77
5.12 Step 9: round 8, the sub-items of R-039, R-044, R-045, the condition H-057 (C), and H-137 (L-LL) . . . 81
5.13 Step 10: round 9, a proof of R-027 (F), H-168 (S′), the s-rule, the p. 254 transfer, and CMP 99 Theorems 3.7 and 3.9 . . . 85
5.14 Step 11: the alternative (E_U) route . . . 88

6 Numerical and certified results **89**
6.1 First increment $\beta_1(0)$ . . . 90
6.2 Later increments ($L = 2$ only) . . . 90
6.3 Quantities of the Route K proof . . . 91
6.4 Values of R-025 (H2) [NUMERICAL; R-025 (H2) AT $L = 13$ IS CERTIFIED IN SECTION 6.7] . . . 92
6.5 One-step values . . . 93
6.6 A rigorous point enclosure at $L = 3$ [INTERVAL ENCLOSURE, VERIFIED] . . . 93
6.7 R-025 (H2) at $L = 13$ [CERTIFIED (INTERVAL ARITHMETIC, OUTWARD ROUNDING; VERIFIED)] . . . 94
6.8 Round 7: more certified L, the gap and neutral-sector checks, and an open tension . . . 95
6.9 Round 8: the β_1 tension rewritten, the cell cross-check, H-137 (L-LL), and H-057 (C) 96
6.10 Round 9: the seam test of H-057 (C), H-137 (L-LL) at smaller φ, and the β_1 run . . . 98
6.11 Certification plan and j_0 [DESIGNED; SUPERSEDED AT $L = 13$ BY SECTION 6.7] . . . 99

7 What remains **100**
7.1 Certification . . . 100
7.2 Fixed-level and structural inputs of L-001 (Theorem 2w) . . . 100
7.3 Items of the sources block . . . 103

7.4 Verification debt . . . 106
7.5 Open for stronger statements . . . 106
7.6 Why the mass gap is not on this route . . . 106
7.7 Trust ledger . . . 107

A Lean formalization **122**

B Index of results by status **124**

C Label map **129**

D Revision history **132**

1 Introduction

1.1 The problem

The Clay problem on quantum Yang–Mills theory asks for a four-dimensional quantum gauge theory with compact simple gauge group that satisfies axioms at least as strong as those of Osterwalder and Schrader [19, 20], and that has a mass gap [15]. One constructive approach starts from Wilson's lattice action on a torus with lattice spacing ε and takes $\varepsilon \to 0$. The bare coupling $g_0(\varepsilon)$ must then go to zero at the rate that asymptotic freedom prescribes, so that the effective coupling at a fixed physical scale stays fixed. This paper treats the first limit only: lattice spacing to zero on a fixed torus. It does not treat the infinite-volume limit or the mass gap (Section 4.7).

1.2 Bałaban's programme and the missing Theorem 2

Between 1984 and 1989 Bałaban built a renormalization-group (RG) analysis of lattice gauge theories [1, 2, 3, 5, 4, 6, 7, 8, 10, 9, 11, 12]. His main output for $d = 4$ is Theorem 1 of CMP 122 II: uniform bounds on the effective densities at every RG scale, under the hypothesis that the sequence of effective couplings stays small. The hypothesis was to be discharged by his Theorem 2, which controls the running coupling. The two statements read as follows.

Theorem 2 (CMP 109 (1987), p. 259): "Let $d = 4$, $G = SU(2)$, and let γ be a sufficiently small positive constant, then for a sufficiently small positive g there exists a bare coupling constant $g_0 = g_0(\varepsilon, g)$ such that the sequence of the effective coupling constants g_k is contained in the interval $]0, \gamma]$, and $g_K = g$. Moreover, there exist constants β, β', $0 < \beta \le \beta'$, such that g_k satisfy the inequality $1/g^2 + \beta \log(L^k\varepsilon)^{-1} \le 1/g_k^2 \le 1/g^2 + \beta' \log(L^k\varepsilon)^{-1}$. (0.31)" Balaban adds that the proof "will be given in a separate paper". It was never published (CMP 122 II p. 355, verbatim: "has not been published yet, so we have Theorem 1 with the assumption").

Theorem 1 (CMP 122 II p. 355): "If the sequence of the effective coupling constants is contained in an interval $]0, \gamma]$ with a sufficiently small positive γ, then the effective densities ρ_k have the form, and satisfy all the conditions and bounds, described in Sect. 2 [III]." The bound (0.1) there reads $\chi_k \exp[-(1/g_k^2)A(U_k) - E_-|T_\eta|] \le \rho_k \le \exp E_+|T_\eta|$ (corrected reading, `BALABAN_EXCERPTS_CHECK`[1] item 3).

Without Theorem 2, the hypothesis of Theorem 1 is an assumption on the unknown coupling flow. Every continuum statement built on Bałaban's bounds inherits this assumption.

[1] Project files named in this paper live in the `notes/` directory of the `yang-mills` repository [14].

1.3 The two targets and the main results

The paper has two targets. The first target is a proof of enough of Theorem 2 to remove the coupling hypothesis from Theorem 1. The second target is a finite-torus existence theorem for the continuum limit in compactness form, with the Osterwalder–Schrader properties proved on the limit. Both results are reductions: each holds under named hypotheses, and the paper records the status of every hypothesis.

Theorem A (weak Theorem 2; proved in Section 3)**.** *Let $G = SU(2)$, $d = 4$, and L odd, $L > 11$ (or L odd, $L \geq 3$, with Bałaban's inductive assumption H-202 (H0) and his construction of $\beta_j(g)$ at this L as explicit hypotheses).*

(i) Assume the uniform inequality R-025 (H2) (Section 3.7.3) [CERTIFIED AT $L = 13$ (THEOREM 6.1); CERTIFIED AT $L = 13$, AND IN ROUND 7 AT $L = 15, 17, 21$ (FULL RERUNS, VERIFIED, SECTION 6.8); A HYPOTHESIS AT OTHER L]*, the fixed-level inputs R-026 (PI), R-027 (F), $R - 028(M_j)$, V1, R-030 (G2) (Section 3.6.1), the history hypothesis L5 (Section 3.4.5; round 7: proved relative to listed inputs, Theorem 5.15), and the bound 7′, $|\beta_{k+1}(h, g) - \beta_{k+1}(0)| \leq C_\beta g$ on every (non-monotone) past (item 7′ of Section 7)* [PROVED+VERIFIED, CONDITIONAL ON H-202 (H0) (LEDGER R-006) OR ON THE H-111 (G-GAIN) INPUTS (LEDGER R-037)]*. Then the conclusions (b), (c′), K-018 (S) of Theorem 3.1 (L-001 (Theorem 2w)) hold. Status of the implication:* [PROVED+VERIFIED, CONDITIONAL]*, by Route K (Theorem 3.8 and Lemma 3.4).*

(ii) Given the conclusions of L-001 (Theorem 2w), Bałaban's Theorem 1 holds for the bare coupling $g_0(K, g)$, for every K and every $g \in]0, g_1]$, without his coupling assumption. Status: [PROVED; THE SWEEP VERIFIED IN ROUND 7 (LEDGER R-034), CONDITIONAL ON ROUTE K OR ROUTE B INPUTS, ON L5, ON THE FINITE-VOLUME IDENTIFICATION OF THE STEP (LEDGER R-032, REPLACED BY A PROVED FORM RELATIVE TO LISTED INPUTS AND THE VOLUME HYPOTHESIS K-032 (VOL) OF SECTION 2.1), AND ON BALABAN'S PUBLISHED PROOFS] *(Section 3.8).*

A second, formally distinct route (Route B, Theorem B_{PW} in Section 3.6.2) also gives (i), under its own input set (items B1–B7 of Section 7.2). Both routes rest on R-025 (H2).

At $L = 13$ the hypothesis R-025 (H2) is discharged by Theorem 6.1, so the weak form of Theorem 2 rests only on the fixed-level inputs, L5, item 7′, and the Bałaban-level items of the trust ledger (with the residue of Theorem 6.1, Section 6.7).

Theorem B (finite-torus existence; proved in Section 4)**.** *Let $G = SU(2)$, $d = 4$, L odd and large, and $\ell \in 2\mathbb{N}$ fixed. Let $g_0 = g_0(\varepsilon, g)$ be a shooting solution whose effective couplings satisfy $g_k \in]0, \gamma]$ for $0 \leq k \leq K$ and $g_K = g$. Assume the uniform bound H-116 (H-U) of Definition 4.1. When H-116 (H-U) is taken from the blocks rather than assumed, the chain also uses the $\kappa \to \kappa/2$ decay recovery of the sourced small-field step (ledger R-046), a* [VERIFIED REDUCTION] *by $L/2$-rescaling that rests on the unprinted large-field T-step rate loss (ledger R-020), and the volume hypothesis K-032 (Vol), $\ell > 22M$ (Section 2.1). Then the lattice Schwinger functions of the centred plaquette density are relatively compact as $\varepsilon = L^{-K} \to 0$, and every limit family satisfies the torus forms of (OS0)–(OS3). This is Theorem 4.2,* [PROVED MODULO NAMED BLOCKS]*. Nontriviality is the separate Theorem 4.3, conditional on (NT).*

The sourced step also uses the unit-scale condition K-024 (U), $\rho_1 \leq c_* r_0 / M^a$ at $\lambda_K = 1$ (K-024): this is a choice of the source-gradient constant, made once after r_0 and before K, not a hypothesis

on test functions, and the bridge B-N1 meets it by rescaling at a C^n cost (this K-024 (U) is not the Route B hypothesis (U)).

The trust boundary of the two theorems is recorded in the trust ledger (Section 7.7): Theorem A rests on 4 Bałaban-level statements and one external result, and Theorem B rests on 20 Bałaban-level statements and four external results. Part (i) of Theorem A, L-001 (Theorem 2w), rests on one of them, R-006 (which includes Bałaban's assertion that the whole-lattice limit of CMP 109 (1.22) exists), and its use of R-006 is a verified reduction. For L-001 (Theorem 2w) no genuine hypothesis remains; the chain rests on Bałaban's printed material, verified project work, conventions, readings, scope restrictions, and constant conditions. One input of L-001 (Theorem 2w), R-027, is a verified reduction whose full proof is pending verification (Theorem 5.26), and R-027 carries an open numerical tension (Section 6.9). Of the 22 Bałaban-level statements in the ledger, 15 are printed by Bałaban without a proof (3 of the 4 behind Theorem A and 13 of the 20 behind Theorem B), and one more (R-021) is a set of printed CMP 99 statements, partly with sketch proofs. After round 7 each of the cited-but-unproved rows of rounds 1–7 and each implicit row has a verified proof relative to listed inputs or a verified reduction (Sections 5.9, 5.10, 5.11, and 7.7). Round 9 adds three rows that have neither: R-021 (printed inputs of the condition H-057 (C)), and R-022, R-023 (CMP 109 pp. 253, 275, write-outs pending). In return the condition H-057 (C) is now a verified reduction (Proposition 5.23), and the torus-side hypothesis is discharged by K-032 (Vol). No Group B row of Theorem B is without a proof or a verified reduction: H-137 (L-LL) is proved and verified (Theorem 5.25), and H-168 (S′) is proved and verified (Proposition 5.27). Round 7 also proves the fixed-level input (M_j) and the history hypothesis L5 relative to listed inputs (Theorems 5.14 and 5.15). The propagator and covariance bounds R-001–R-005 are verified reductions to published theorems of CMP 99 and CMP 102; their joint residue is Bałaban's walk expansions of the final propagators (through R-002, with CMP 99 Thms 3.7 and 3.9 now written out, Theorems 5.30 and 5.29), the complex walk bound (W^c) in the exponential gauge and the decay of $C^*\Delta_k C$ at real U (R-008) under R-003, the regular-class identification R-018, the printed inputs R-021–R-023, and the conventions K-044 and K-046.

The hypothesis H-116 (H-U) is the output of three blocks: Theorem 2 (B-T2, that is, Theorem A), sourced versions of Bałaban's bounds R-035 (B-S), and a bridge from the source form to H-116 (H-U) R-045 (B-N1). The dependency table, Section 4.6, gives the status of each block and is the single status table of the paper. Section 7 lists what remains.

What the paper does not give: literal (0.31) at every K (Section 3.3), the limit of $\beta_j(0)$ and a rate (Section 3.9), uniqueness of the limit, infinite volume, rotation invariance, and a mass gap (Section 4.7). Theorem B is a compactness ("weak-existence") result. Under Jaffe–Witten footnote 2 it is not an admissible solution of the Millennium problem (Section 4.1).

1.4 Status conventions

Every result carries a status tag in small capitals. The tags mean the following.

- [PROVED+VERIFIED]: proved in a project file and confirmed by an independent verifier file.
- [PROVED+VERIFIED, CONDITIONAL ON X]: the implication is confirmed; X is open or assumed.
- [PROVED, UNVERIFIED]: proved in a project file; no verifier has graded it yet.
- [LEAN]: formalized in Lean 4 + Mathlib, no `sorry`, axioms propext, Classical.choice, Quot.sound only.

- [VERIFIED QUOTE]: read on the page image of Balaban's paper (`reports/BALABAN_EXCERPTS_CHECK.md` or a verifier file).
- [NUMERICAL, REPRODUCED]: floating-point result reproduced by a verifier. [NUMERICAL, RUN BY THE VERIFIER]: run once, by a verifier. [NUMERICAL]: not reproduced. None is certified.
- [ARGUED]: a plausibility argument, not a proof. [OPEN]: no proof. [REFUTED]: shown false.
- [BALABAN]: an input taken from Balaban's published papers, used as stated there.
- [NOT WRITTEN]: a proof plan; no project file contains the proof.
- [CITED]: a standard result; its text was not re-read. [CITED, SOURCE CHECKED]: the source was read, the transfer to the lattice is standard and not proved here.
- [PROVED HERE]: a short proof given in full in this paper (Section 4). [HYPOTHESIS]: assumed, and owned by a named block. [FORMAL]: a formal step whose analytic justification is not written. [DERIVED SKETCH]: a route with steps listed, not a proof.
- [INTERVAL ENCLOSURE, VERIFIED]: a rigorous interval-arithmetic enclosure at a point, rerun by a verifier.
- [CERTIFIED (INTERVAL ARITHMETIC, OUTWARD ROUNDING; VERIFIED)]: a statement over a continuum of parameters (all K, all j), proved by an analytic majorant plus a branch-and-bound in interval arithmetic with outward rounding, and confirmed by an independent verifier file. It differs from [INTERVAL ENCLOSURE, VERIFIED] (one point) and from [NUMERICAL] (floating point, no rounding control).
- [VERIFIED REDUCTION]: a reduction confirmed by a verifier file; the inputs stay open.
- [DRAFT PROOF, VERIFICATION PENDING], [PROVED, UNVERIFIED; VERIFIER RUNNING]: [PROVED, UNVERIFIED], with an independent verifier started and not yet reported.
- [PROVED MODULO NAMED BLOCKS]: proved given the hypotheses owned by the named blocks.

A compound tag such as [PROVED+VERIFIED, CONDITIONAL ON R-025 (H2)] combines these meanings. Tags refer to the project files cited next to them. In [VERIFIED QUOTE], a quote of a source other than Bałaban's papers was checked verbatim against a copy of that source in the project. Project files are cited as `NAME.md`; a section sign after a file name (for example `KAPPA_BOUNDED_VERIFY` §4) refers to a section of that file, not of this paper. "Grade n" and "item n" after a verifier file name refer to the numbered verdicts of that file. The paper as a whole is not verified as one document.

1.5 Relation to other work

Two recent works overlap with this paper. The record of how their content reached this project is given below; it is kept in the form fixed when the overlap was found.

Related work 1. E. Sussman and L. Alpöge (with Claude; README: "written largely by LLM ... Novel mathematical ideas should be attributed to Claude (under the guidance of LA)"), "The Yang–Mills continuum limit along Bałaban's renormalization group: a complete account of Theorem 0" (LaTeX, files `theorem_0_complete_account_raw_{master,I,II,III}.tex`, 424k

lines), and "SU(2) Yang–Mills theory in four dimensions via Bałaban's renormalisation group: construction, axioms and non-triviality" (`ym4_manuscript_zeroth_draft.pdf`, dated 6 October 2026) [26, 25]. Repository: `https://github.com/ethanwsussman/Yang--Mills-LLM-experiment`, commit `1b92bd0347e3b8763cd8ecc1f96dc3c2a92dd310` (2026-10-27T05:45:28-05:00), accessed 2026-10-27. Repository created 2026-10-06. Disclaimer in README: "This repository contains work-in-progress, written largely by LLM. Its content has not been satisfactorily vetted. All claims should be understood as tentative, pending verification." Suggested keys: SA26-T0 (account), SA26-YM4 (draft).

Related work 2. OpenAI, "The canonical massive continuum limit of the two-dimensional O(3) model" (October 4, 2026), `github.com/openai/math`, family 215 (preprints folder; read locally from the PDF, `reports/ONLINE_SWEEP.md` §1.2; text in `reports/o3_continuum.txt`) [18]. Suggested key: OAI26-O3.

How content from these works reached this project. The record (file times, `PLAN.md`, read lists; `THEOREM2_PROOF_v2_VERIFY` §I) shows the following.

1. Section 3 was written without reading the Sussman–Alpöge repository. Only the citation block above (`reports/SUSSMAN_ALPOGE_MAP.md` §5.1) was read.

2. First channel. On the night of 2026-10-27, `reports/ONLINE_SWEEP.md` §2 (about 20:20) read the Sussman–Alpöge repository directly. It reported that they prove a weak two-sided flow with an additive constant, never need per-step positivity, use a first-passage cutoff, and invoke only containment $g_k \le \gamma$. This is the first project file with Sussman–Alpöge content.

3. Second channel. `reports/SUSSMAN_ALPOGE_MAP.md` (about 20:28) was written by a separate agent under a firewall. Through it, the bounded-κ idea entered the project: the weak form of Theorem 2 needs only κ_k bounded, not convergence (their Lemma 4.19 structure; `PLAN.md` line 146). `KAPPA_BOUNDED.md` (about 20:39) received this idea through its prompt. Its read list excludes the map and ONLINE_SWEEP.

4. L-004 (Lemma D′) and the cross-scale sweep predate any Sussman–Alpöge content in project files. The shooting-without-positivity lemma and the band with additive constant appear in `THEOREM2_PROOF_VERIFY` Point 5(b) (about 20:05) and `LEMMA_DPRIME.md` (about 20:10). `LEMMA_DPRIME_SWEEP.md` (about 20:16–20:23) overlaps ONLINE_SWEEP in time, but its read list is LEMMA_DPRIME and its verifier only; it did not read ONLINE_SWEEP. L-005 (Lemma D″) is in `KAPPA_BOUNDED.md`, written after the map.

5. L-009 (Theorem K). Its result form, $|S_k - ck \log L| \le \kappa_*$, matches Sussman–Alpöge Lemma 4.2(b), $|\sum_l \beta_l^0 - kc_0| \le c_2$ (as reported in `PLAN.md` line 146; cited there, not proved). The proof in this project (Prop. 1.1, Lemma 2.1, Prop. 2.2 of `KAPPA_BOUNDED`) was written by firewalled agents from Balaban's papers and project files.

6. O(3) paper. Route B's doubling comparison follows the template of OAI26-O3 Lemma 6.1 (doubling difference; `BETA_FP_PW` uses the "O(3) Lemma 6.1 form"). The idea of a coupling band in place of per-step positivity also appears there (ONLINE_SWEEP §1.2(a)). `BETA_FP_PW` read ONLINE_SWEEP §1.2, §5, §8 and o3_continuum Lemma 6.1, and did not read ONLINE_SWEEP §2.

7. Route B (other than the O(3) template), the numerics, and the symbol results of Sections 3.5.5 and 3.7 were derived by firewalled agents from Balaban's papers and project files.

In short: two agents read the Sussman–Alpöge repository, the web sweep (item 2) and the mapping agent (item 3), and neither wrote a derivation. The derivation agents were firewalled; their read lists exclude the repository, the map, and ONLINE_SWEEP §2. One idea crossed the firewall: the bounded-κ route reached `KAPPA_BOUNDED.md` through the map (item 3). Their public, timestamped account predates this one. Any write-up must cite SA26-T0, SA26-YM4, and OAI26-O3, and state these overlaps.

1.6 Organization

Section 2 fixes the notation for both parts of the paper. Section 3 proves the reduction of Theorem 2 (Theorem A). Section 4 proves the finite-torus existence theorem modulo named blocks (Theorem B) and contains the dependency table. Section 5 gives one paragraph for each verified source file of the sources block. Section 6 collects the numerical and certified results. Section 7 lists what remains. Infinite volume, uniqueness, clustering, and the mass gap are outside the scope of this paper. Appendix A lists the Lean formalization, Appendix B indexes the results by status, Appendix C maps the numbering of this paper to the names used in the project files, and Appendix D records the revision history.

2 Setup and notation

This section fixes the notation for the whole paper. Sections 2.1–2.4 serve the coupling flow of Section 3. Section 2.5 serves the existence theorem of Section 4. Section 2.6 lists the names that would otherwise collide.

2.1 Lattice, measure, and scales

Notation. L is odd, $L > 11$, fixed (Balaban, CMP 109 p. 256 [8]). $K \in \mathbb{N}$, $\varepsilon = L^{-K}$. The physical side $\ell \in \mathbb{N}$ is in units of the unit lattice. $T_\ell^4 = (\mathbb{R}/\ell\mathbb{Z})^4$ and $T_\varepsilon = (\varepsilon\mathbb{Z}/\ell\mathbb{Z})^4$, so $N_\varepsilon = \ell L^K$ sites per direction. The Wilson action is $A(U) = \sum_p P_p(U)$, $P_p(U) = 1 - \frac{1}{2}\operatorname{Re}\operatorname{tr} U(\partial p) \in [0,2]$, over unoriented plaquettes p. The measure is $\mu_\varepsilon = Z_\varepsilon^{-1} \exp(-g_0^{-2} A(U)) \prod_b dU_b$ (Haar). x_p is the centre of p.

The bare coupling $g_0 = g_0(\varepsilon, g)$ is any solution of the shooting problem of Lemma 3.4 (L-005 (Lemma D″)) with terminal coupling $g_K = g$. L-005 (Lemma D″) gives existence, not uniqueness. Any choice works below.

Volume hypothesis K-032 (Vol) (ledger K-032). The torus side obeys $\ell > 22M$, where $M = L^{m_M}$ is the localization cube of CMP 109 p. 257 (the block size of ledger K-025, not the M of Δ_M in Section 2.4). Justification: at every step $k \le K$ the local set $\tilde{\square}^5$, of side $11M$ level-k cells, must lift to $\mathbb{Z}^4$. Then the whole-lattice β subtraction leaves only Bałaban's irrelevant class. This needs $N_k > 22M$ level-k cells per side, and $N_k = \ell L^{K-k}$ is smallest at $k = K$ (Proposition 5.11). K-032 (Vol) does not depend on K. The proof of L-001 (Theorem 2w) does not use it; L-001 (Theorem 2w) $\Rightarrow$ Theorem 1 (through ledger R-032) and Theorem B use it. Even form, for Theorem B: $\ell = 2L^{m_T}$ with $m_T \ge m_M + 1$, which keeps ℓ even and gives $\ell \ge 2LM > 22M$, because $L > 11$. The power form of `THEOREM2_PROOF_v4.md` §1.1, $\ell = L^{N-K}$ with $N \ge K + m_M + 2$, also suffices, but it makes ℓ odd. CMP 109 prints both exponents m_T and m_M as m; here they are fixed independently.

2.2 Range of L

Balaban assumes L odd, $L > 11$, as a standing assumption for the whole of CMP 109 (pp. 251–252; OCR line 124, verbatim confirmed in `THEOREM2_PROOF_VERIFY` Step 6) [8]. His construction of $\beta_j(g)$ for $g > 0$ and his inductive assumption H-202 (H0) are therefore available only at odd $L > 11$. Every statement below that uses H-202 (H0) or $\beta_j(g)$ at $g > 0$ is stated for odd $L > 11$, or carries "H-202 (H0) and Balaban's construction at this L" as an explicit hypothesis. Statements about the one-loop quantities $\beta_j(0)$, S_k, and the symbols σ_j hold for odd $L \geq 3$.

2.3 The coupling sequence

$\varepsilon = L^{-K}$, so $\log(L^k\varepsilon)^{-1} = (K-k)\log L$. $u_k := 1/g_k^2$. Recursion (CMP 109 (0.18)/(0.20)): $u_{k+1} = u_k - \beta_{k+1}(g_k)$. The increment β_{k+1} depends on the whole history $g_0, \ldots, g_k$ (CMP 109 p. 298, [VERIFIED QUOTE]). $\beta_j(0)$ is the one-loop (Gaussian) increment; it is history-independent by definition.

Shooting notation: $u_k = 1/g_k^2$, $t = 1/g^2$, $T(k,K) := \sum_{i=k+1}^{K} \beta_i(g_{i-1})$. Clamp $\varphi(u) := \max(u, 1/\gamma^2)^{-1/2}$.

Here γ is the upper bound for the couplings in Bałaban's Theorem 1. Bałaban's small constant β_0 (1/7 in CMP 122 II p. 389) is not a coupling (Section 3.1). In Lemma 3.6 (L-007 (Lemma E)) the symbols β_0, β_0' instead denote lower and upper bounds on $\beta_j(0)$, as in `LEMMAS.md`.

2.4 One-loop notation

The letter K has two uses. As an integer it is the number of RG steps, $\varepsilon = L^{-K}$. As a vector $K \in \mathrm{BZ} = [-\pi, \pi]^4$ it is a momentum in the Brillouin zone of the coarse unit lattice; then $k_l = (K + 2\pi l)/L$ are the fine momenta over K. The meaning is always clear from context. The other one-loop objects are defined where they are first used:

- S_k, the $p \to 0$ F^2-coefficient of the composed k-step one-loop integral, and its telescoping into $\beta_j(0)$: Section 3.5.1;
- $c = 11N^2/(12\pi^2)$ and $b := c\log L$: Section 3.5.3;
- the f^m Taylor symbols $\sigma_j^{(m)}(K)$ of the coarse kernel at level j, and d_j, κ_j, κ_W^F: Section 3.5.4;
- Δ_M (Gaussian RG kernel at block size $M = L^j$), Δ_∞, and the bordered matrix $\mathfrak{B}_j(K)$: Sections 3.5.5 and 3.9;
- the one-step minimizer map $H_j(K)$, ρ_1, ρ_2, and the inequality R-025 (H2): Section 3.7;
- the plane-wave quantities $\beta_{j+1}(p)$, Γ^{ex}, θ_* of Route B: Section 3.6.2.

2.5 Observables and Schwinger functions

Observable (class C1 of `SOURCES.md` §1.2). $\varphi_p := P_p - c_\varepsilon$, $c_\varepsilon := \langle P_p \rangle_{\mu_\varepsilon}$. By hypercubic invariance (Section 4.5.6) c_ε does not depend on p.

Test space. $\mathcal{D}_n := \{F \in C^\infty(T^{4n}) :$ every derivative of F vanishes on every diagonal $\{x_i = x_j\}$, $i \neq j\}$. The norm is $\|F\|_m := \max_{|\alpha|\leq m} \sup |\partial^\alpha F|$. On the compact torus this is the Osterwalder–Schrader (OS) Schwartz seminorm without weights.

Lattice Schwinger functions (`SOURCES.md` §1.3). $S_{n,\varepsilon}(F) := \sum_{p_1,\ldots,p_n} F(x_{p_1}, \ldots, x_{p_n}) \langle \varphi_{p_1} \cdots \varphi_{p_n} \rangle_{\mu_\varepsilon}$, $F \in \mathcal{D}_n$. These are moments, not cumulants

(SOURCES_VERIFY item 5(a)). No field renormalization factor is inserted: SOURCES_VERIFY item 8 gives $B_K = \prod c_j$ bounded above and below ($c_j = 1 + O(g_{j-1}^3)$, $\sum g_j^3 < \infty$), so a bounded factor B_K^n changes only the normalization of the limit.

Scaling check (SOURCES_VERIFY item 4, derived there). $\langle \varphi_p \varphi_{p'} \rangle \sim \varepsilon^8 g^4 |x_p - x_{p'}|^{-8}$ at tree level. So $S_{2,\varepsilon}(F) = O(1)$ for $F \in \mathcal{D}_2$ with no rescaling, while $\sum_p f(x_p)\varphi_p$ has variance $\sim \varepsilon^{-4}$. The limit object is a family of distributions off the diagonals, not a random field.

Extra hypothesis found while writing the source. ℓ is even (sufficient for the reflection argument; see Section 4.8, item 1). For a reflection of the periodic lattice to exchange two halves with both seams being link planes, N_ε must be even. With L odd, $N_\varepsilon = \ell L^K$ is even iff ℓ is even.

2.6 Names

Hypotheses are named by short labels in parentheses, such as R-025 (H2), R-026 (PI), or R-043 (FX); the dependency table (Section 4.6) lists each one with its status. Three names of the project files collide in one document and are changed here; Appendix C records the old names.

- The plane-wave theorem of BETA_FP_PW (its Theorem 3.2, called "Theorem B" there) is called Theorem $\mathrm{B_{PW}}$ (Theorem 3.9), so that "Theorem B" means the main Theorem B.
- The three global conditions of the dependency table, called (G1)–(G3) in ASSEMBLY.md, are called H-200 (GC1)–H-201 (GC3). The labels (G1) (constraint conditions of Proposition 3.7) and R-030 (G2) (per-cell infinite-volume densities) keep their meaning from the Theorem 2 files.
- The final sourced bound of SOURCES_D4D5.md, called (S) there, is called ($\mathrm{S_{src}}$). The label K-018 (S) means the smallness condition of Section 3.1.

3 The coupling flow and Theorem 2

This section proves Theorem A. Section 3.1 lists what Bałaban's proof of Theorem 1 uses of the coupling sequence; the answer is weaker than his (0.31). Section 3.2 states the weak form, L-001 (Theorem 2w), that supplies exactly this. Sections 3.4–3.7 prove L-001 (Theorem 2w) from bounded window sums of the one-loop increments, give two routes to these window sums, and reduce the uniform part of both routes to the single inequality R-025 (H2). Section 3.8 assembles the result. Its output, the trajectory conditions (b), (c′), K-018 (S), is the block H-084 (B-T2) of Section 4.

3.1 What Bałaban's Theorem 1 actually needs

Source: LEMMA_DPRIME_SWEEP §2, §7. The statements of Theorems 1 and 2 are quoted in Section 1.2.

The quoted hypothesis of Theorem 1 is only $g_k \in]0, \gamma]$. The proof uses more. CMP 119 Theorem 1 (p. 262) assumes "(I.0.33)", a misnumbered reference to (0.31) of CMP 109. Its working form is CMP 119 p. 255–256 (2.6)–(2.9), for scales $m < n$:

- (2.6) $g_n \le (1 + g_n^2 \beta'(n-m))^{1/2} g_m$ and $g_m \le (1 + \beta_0) g_n$;
- (2.7) $(\log g_n^{-2})^p \le (1 + \beta_0)(\log g_m^{-2})^p$ and $(\log g_m^{-2})^p \le (1 + g_n^2 \beta'(n-m))^{\beta_0} (\log g_n^{-2})^p$;
- (2.8), (2.9) the same for $\varepsilon_j = g_j p_0(g_j)$ and R_j.

Here β_0 is Balaban's small constant (1/7 in CMP 122 II p. 389), not a coupling. [VERIFIED QUOTE] for (2.6)–(2.9): page images, `LEMMA_DPRIME_SWEEP` §1; the sweep itself is single-author, see Section 3.4.3.

In u-variables these follow from two pairwise bounds:

- H-156 (P-up) $u_m \le u_n + B'(n-m)$, with no additive constant;
- H-155 (P-low) $u_m \ge u_n + \beta(n-m) - D'$, with $\beta > 0$, $D' \ge 0$ fixed,

plus the smallness condition

- K-018 (S) $\gamma^{-2} \ge c_{sw} D'$, $c_{sw} = \max(2, 1/(1-(1+\beta_0)^{-2})) = 64/15$ at $\beta_0 = 1/7$, and $\gamma \le \gamma_*(\beta_0, \beta, B', p_0, p_1, r, \nu, \kappa_0)$ (the conditions that literal (0.31) also needs). γ_* is free of the D'-ratio conditions, but it depends on R through B' (it contains $\gamma^2 B' \le (1+\beta_0)^2 - 1$ with $B' = 5b/4 + 2R$). The order of constants in Section 3.8 (R before γ) handles this.

β_0 must be fixed before γ.

3.2 The weak form proved conditionally

Theorem 3.1 (L-001 (Theorem 2w), weak form)**.** *Let $G = SU(2)$, $d = 4$, L odd, $L > 11$ (equivalently: L odd ≥ 3 with H-202 (H0) and Balaban's construction of $\beta_j(g)$ at this L as an explicit hypothesis). There are $\gamma > 0$, $g_1 \in]0, \gamma]$, $\beta > 0$, $B' \ge \beta$, $D' \ge 0$ such that for every $K \ge 1$ and every $g \in]0, g_1]$ there is a bare coupling $g_0 = g_0(K, g)$ with*

- *(b) $g_k \in]0, \gamma]$ for $0 \le k \le K$, and $g_K = g$;*
- *(c′) for all $0 \le m \le n \le K$: $u_n + \beta(n-m) - D' \le u_m \le u_n + B'(n-m)$;*
- *K-018 (S) holds for γ, D'.*

Consequence: Balaban's (0.31) holds up to the additive constant D' on the lower side, with constants $\beta/\log L$ and $B'/\log L$. The two-sided bound fixes $1/g_k^2$ up to constants, as required.

Status: [PROVED+VERIFIED, CONDITIONAL ON (B2) AND THE FIXED-LEVEL INPUTS OF SECTION 3.6] by Route K (L-009 (Theorem K) + L-005 (Lemma D″)). (B2) is reduced to the single inequality R-025 (H2) of Section 3.7 [PROVED+VERIFIED REDUCTION: `B2_TWOSTEP_VERIFY` GRADE 3]. Second route: Route B (`BETA_FP_PW` Theorem 3.2 + L-005 (Lemma D″)). The logic of Theorem $\mathrm{B_{PW}}$ is [PROVED+VERIFIED, CONDITIONAL] (`BETA_FP_PW_VERIFY` §6 items 1–4). Its inputs (U), (DF), (T1) from `BETA_FP_U` are [PROVED, UNVERIFIED]; the route is conditional on (U), ($B\theta$), and the plane-wave defects. `BETA_FP_U` reduces (U) to (T1) + R-025 (H2) [PROVED, UNVERIFIED]. Both routes also need H-202 (H0) at this L and L5 (Section 3.4.5). See Section 3.6.

Setting. The torus obeys the volume hypothesis K-032 (Vol) of Section 2.1: side $\ell > 22M$ final-scale cells, or the even form $\ell = 2L^{m_T}$, $m_T \ge m_M + 1$. The proof of L-001 (Theorem 2w) does not use K-032 (Vol), because Lemmas D′ and D″ use the whole-lattice β_j. K-032 (Vol) enters where L-001 (Theorem 2w) is applied on the torus: Theorem 1 through ledger R-032, and Theorem B.

Trust boundary (round 7; Section 7.7). By Route K, L-001 (Theorem 2w) rests on one Bałaban-level item, R-006, on the Group B items R-025–R-031, and on the external result X-005. R-015 is not in this set: L-001 (Theorem 2w) uses only the definition of β_j (Proposition 5.8 and the remark after it). The use of R-006, item 7′, is a [VERIFIED REDUCTION] (L-183 (Proposition H0-red) of `SOURCES_T2_H0.md`) with residue N1–N9; its item N2, L-174 (Lemma G3), is proved relative to listed inputs (Theorem 5.9). Round 6 removes the last bare assertion of R-006: the existence of the

whole-lattice limit of CMP 109 (1.22) at $g > 0$ now follows from Proposition 5.8 and Theorem 5.10. R-006 stays in the set, because the rest of the H0-red residue is not proved. If one takes R-006 through this reduction instead of Bałaban's assertion, L-001 (Theorem 2w) also rests on R-008, R-009 (Theorems 5.4, 5.5), R-017, the inputs of Bałaban's sup induction, K-009, the conditions K-026, K-027, and the locality conventions K-033–K-037. Round 7: of R-025–R-031, only R-027 lacks a proof or a verified reduction; R-028 and R-031 are proved relative to listed inputs (Theorems 5.14 and 5.15).

3.3 Literal (0.31)

- (L1) Literal (0.31) for every $K \geq K_0$ and every $g \in]0, \gamma]$. Needs eventual uniform positivity: there are J and $\beta_* > 0$ with $\beta_j(0) \geq \beta_*$ for all $j > J$, plus $\sup_j |\beta_j(0)| < \infty$ (Lemma 3.2 (d)). [PROVED+VERIFIED; LEAN] as an implication. L-009 (Theorem K) and Theorem B_{PW} do NOT give eventual positivity: they bound window sums, not single increments. Eventual positivity follows from the convergence route (Section 3.9), which is open.
- (L2) Literal (0.31) for every $K \geq 1$. Needs $\inf_j \beta_j(0) > 0$ and $\sup_j \beta_j(0) < \infty$ (Lemmas D + E). If $\beta_j(0) \to \beta_\infty > 0$, then "every $\beta_j(0) > 0$" is equivalent to $\inf_j \beta_j(0) > 0$ (L-004 (Lemma D′) Step 8). [PROVED+VERIFIED; LEAN] as an implication. Inputs open.
- Literal (0.31) is not needed for Theorem 1: Section 3.1 shows the proof uses (c′) + K-018 (S). Literal (0.31) is used directly in one place only, the alternative choice of ε_0 in CMP 119 p. 248, which Balaban does not adopt.
- Necessity. Literal (0.31) at every $K \geq 1$ forces $\beta_K(0) \geq \beta \log L > 0$ for every K (window $k = K - 1$, let $g \to 0$ using H-202 (H0); so L odd > 11). [PROVED, UNVERIFIED]; elementary; `THEOREM2_PROOF_VERIFY` Point 5(a) agrees. So certified early steps are needed only for (L2), not for Theorem 1.

3.4 Shooting lemmas

The shooting lemmas turn bounds on the increments β_j into a trajectory with prescribed terminal coupling. Lemma 3.2 needs eventual positivity of $\beta_j(0)$; Lemma 3.4 needs only bounded window sums, and is the one used for L-001 (Theorem 2w). The notation is that of Section 2.3. Lemmas D′, D″, D, E are abstract statements about a sequence of functions β_j. Applied to Balaban's $\beta_j(g)$, they need L odd > 11 (or H-202 (H0) at this L) and L5.

3.4.1 L-004 (Lemma D′) (shooting without early positivity) [PROVED+VERIFIED; LEAN]

Sources: `LEMMA_DPRIME.md` §1–§2; `LEMMA_DPRIME_VERIFY.md` grades 1, 2, 5 (CONFIRMED). Step 7 of that file also holds grades 3, 4, 6, which are CORRECTED (Remark 1 overclaim, misreads in §5, ledger); this chapter uses none of them except through the sweep (Section 3.4.3). Lean: `lean/YmLean/LemmaDPrime.lean` (`YmLean.lemma_DPrime`, 405 lines), statement check `lean/VERIFY_LemmaDPrime.md`.

Lemma 3.2 (L-004 (Lemma D′))**.** Hypotheses. *(i)* β_{k+1} *continuous on* $[0, \gamma_0]$ *and* $|\beta_{k+1}(g) - \beta_{k+1}(0)| \leq C_1 g$, C_1 *independent of* k, K. *(ii)* $\sup_j |\beta_j(0)| \leq B$. *(iii′) there is* J *with* $\beta_j(0) \geq \beta_\infty/2$ *for all* $j > J$ *(this is all the proof uses of "*$\beta_j(0) \to \beta_\infty > 0$*").*

Constants. $\gamma := \min(\gamma_0, \gamma_{Bal}, \beta_\infty/(4C_1))$; $B' := B + \beta_\infty/4$; $D := J \cdot B'$; $D' := J(B + \beta_\infty/2)$; $g_1 := (1/\gamma^2 + D)^{-1/2}$; $K_0 := J + \lceil 8D'/\beta_\infty \rceil$.

Conclusions.

- *(a) $T(k,K) \geq -D$ for all $k < K$ along any couplings in $]0,\gamma]$.*
- *(b) For every $K \geq 1$ and $g \in]0,g_1]$ there is g_0 with all $g_k \in]0,\gamma]$ and $g_K = g$.*
- *(c) $u_K + (\beta_\infty/4)(K-k) - D' \leq u_k \leq u_K + B'(K-k)$. The upper side has no constant.*
- *(d) For $K \geq K_0$: literal (0.31) with lower constant $\beta_\infty/8$, and shooting for every $g \in]0,\gamma]$. If every $\beta_j(0) > 0$: literal (0.31) for all K (Step 8 = Lemmas D + E).*

Proof idea. Clamped flow $\tilde{u}_{k+1} = \tilde{u}_k - \beta_{k+1}(\varphi(\tilde{u}_k))$ is continuous in u_0. Its tail sums lie in $[-D, B'(K-k)]$ for every u_0, because clamped couplings lie in $]0,\gamma]$. IVT on $[t-D, t+B'K]$ gives $\tilde{u}_K = t$. Then $\tilde{u}_k \geq t - D \geq 1/\gamma^2$ for $g \leq g_1$, so the clamp is never active. The bound is derived for the clamped flow first; there is no circularity. For history-dependent β this proof needs β_{k+1} defined and continuous on the whole cube $]0,\gamma]^{k+1}$; see L5 in Section 3.4.5.

Lean. The Lean hypotheses are weaker (ContinuousOn instead of C^1; (iii′) instead of convergence). Step 8 is `lemma_DPrime_step8` (`lean/YmLean/LemmaDE.lean`). Lean covers one-argument $\beta_j(g)$ only, not the history form.

Remark 3.3 (Remark 4 of `LEMMA_DPRIME.md`). Without $g \leq g_1$, (b) can fail at small K ($\beta_1 = \beta_2 = \beta_3 = -0.8$, $K = 3$, $g = \gamma$). [PROVED+VERIFIED]

3.4.2 L-005 (Lemma D″) (tail-sum form; no index J) [PROVED+VERIFIED; LEAN]

Source: `KAPPA_BOUNDED.md` §5.3; `KAPPA_BOUNDED_VERIFY` §6 (CONFIRMED: "No J needed"); hand rederivation in `THEOREM2_PROOF_v2_VERIFY` §A.

Lemma 3.4 (L-005 (Lemma D″))**.** Hypotheses. *(i) as in 2.1, and the window bound*

$$\text{(5.1)} \quad T_0(k,K) := \sum_{i=k+1}^{K} \beta_i(0) \in [\, b(K-k) - 2R,\ b(K-k) + 2R\,] \quad \text{for all } 0 \leq k < K,$$

with $b := c \log L > 0$ and some $R \geq 0$. (Route K gives $R = \kappa_*$; Route B gives $R = C_B$.)

Constants. $\gamma := \min(\gamma_0, \gamma_{Bal}, b/(4C_1))$; $\beta := 3b/4$; $B' := 5b/4 + 2R$; $D = D' := 2R$; $g_1 := (1/\gamma^2 + D)^{-1/2}$.

Conclusions. *(b) of 2.1, and the pairwise bound*

$$\text{(c′)} \quad u_n + \beta(n-m) - D' \leq u_m \leq u_n + B'(n-m), \quad 0 \leq m \leq n \leq K,$$

upper side without constant.

Proof. $|T - T_0| \leq C_1\gamma(K-k) \leq (b/4)(K-k)$. Add (5.1). Steps 2–5 of L-004 (Lemma D′) use only continuity and the bracket $\tilde{T} \in [-D, B'(K-k)]$; this bracket now comes from (5.1). The pairwise form is the same window argument with $K \to n$. The proof is the clamped-flow proof of 2.1; for history-dependent β the L5 caveat of Section 3.4.5 applies.

Lean: `lean/YmLean/LemmaDDoublePrime.lean` (`YmLean.lemma_DDoublePrime`, 146 lines, no `sorry`, axioms propext, Classical.choice, Quot.sound; `lean/README.md`). It proves (a″), (b) with the clamp inactive on the solution, (c′) with $\beta = 3b/4$, $B' = 5b/4 + 2R$, $D' = 2R$, and (d) below. The Lean hypotheses are weaker than the note: continuity instead of C^1; any γ with $0 < \gamma \leq \gamma_0$ and $C_1\gamma \leq b/4$; $R \geq 0$ derived; $K = 0$ allowed; (c′) and (d) hold for every trajectory with couplings in $]0,\gamma]$. One-argument β only (L5 not formalized).

(d) [LEAN]: for $n - m \geq 8R/b$, $u_n + (b/2)(n-m) \leq u_m \leq u_n + B'(n-m)$. So literal (0.31) from (5.1) alone holds, with lower constant $b/2$, only for windows with $n - m \geq 8R/b$; short windows near K are not controlled (see Section 3.3).

3.4.3 L-005 (Lemma D″) meets Balaban's proof: the cross-scale sweep [PROVED, UNVERIFIED (SWEEP); 4 OF 19 TABLE ROWS VERIFIED]

Source: `LEMMA_DPRIME_SWEEP.md` (all 12 Balaban files; OCR grep plus page image for every non-single-scale hit). `LEMMA_DPRIME_VERIFY` Steps 4–5 checked comparisons A–E on page images (CMP 122 I p. 192, p. 198 (1.94); II p. 381 ×2, p. 386) and corrected two misreadings of A (factor $(1+\beta_0)^2(k-j)^{\beta_0}$; $\delta'_j = g_j A_1 p_1(g_j)$). The five comparisons map to four table rows: A = row 11, E = row 12, C and D = row 16, B = row 18.

Result. 19 cross-scale uses of the coupling sequence (table in SWEEP §7), all in CMP 119 and CMP 122 I–II; none in CMP 95, 96, 98, 99, 102, 109, 116. Each is covered by (b) + (c′) + K-018 (S) of Section 3.1:

- items 1, 7, 10, 12, 17 need H-156 (P-up) with no constant (e.g. (2.6) line 1 is exactly H-156 (P-up));
- items 2, 3, 11, 13, 15, 16, 18 need H-155 (P-low) with fixed multiplicative slack $(1+\beta_0)$ or $(1+\beta_0)^2$; item 2 ((2.6) line 2) gives the constant 64/15; item 11 needs only $2.42D'$;
- items 4 ((2.8)–(2.9), ε and R ratios), 6 (CMP 119 (3.7), (3.8), (3.12), (3.13)), and 9 (CMP 122 I (1.52)), the last two of the form $\varepsilon_{k+1} \le (1+\beta_0)\varepsilon_k$, use both sides; they follow as items 1 and 3 do (item 4 from items 1–3);
- item 14 (CMP 122 I p. 200, R_{k-N_0+1} vs R_k) uses the lower side; it follows as item 3 does;
- item 5 (CMP 119 (2.31), (2.46): $\sum_j g_j^{\kappa_0} \le g_n^{\kappa_0-6}$) needs linear growth H-155 (P-low) and $\gamma^{-2} \ge 2D'$;
- item 0 is CMP 119 Theorem 1 assuming (0.31); its proof works with (2.6)–(2.9);
- item 8 (CMP 119 p. 284) is in a sketch, not load-bearing.

Caveats [OPEN]: (1) the sweep is a grep plus page images, not a line-by-line read; (2) "Similar inequalities hold for other constants, which will be introduced later" (CMP 119 p. 256) was not located; (3) Balaban says β_0 "can be chosen arbitrarily small, if g is sufficiently small"; under H-155 (P-low) the floor is $\beta_0 \gtrsim D\gamma^2/2$, so β_0 must be fixed first; no step was found that needs $\beta_0 \to 0$; (4) the sweep has no independent verifier beyond rows 11, 12, 16, 18; (5) `BALABAN_EXCERPTS_CHECK` item 11 lists readings not checked on page images: Π (4.37)/(5.1), CMP 119 (2.42), the (2.12)/(2.23) ambiguity, the "L odd" page, the Sect. 3 estimates, and the 1989 bound lists.

3.4.4 L-006 (Lemma D) and L-007 (Lemma E) [PROVED+VERIFIED; LEAN]

Sources: `LEMMAS.md` (corrected), `LEMMAS_DE_VERIFY.md` §1.4, §1.5, §2.4; `lean/YmLean/LemmaDE.lean` (`lemma_D`, `lemma_E`, `lemma_DE`).

Lemma 3.5 (L-006 (Lemma D)). *If $b \le \beta_j(g) \le b'$ on $]0,\gamma]$ with $b > 0$, then for every K and $g \in]0,\gamma]$ shooting succeeds and $t + b(K-k) \le u_k \le t + b'(K-k)$.*

This is literal (0.31). An earlier draft used a wrong first IVT interval, which the verifier refuted; the corrected interval is $[t+bK,\ t+b'K]$. The nested-interval variant for β defined only on admissible histories (`LEMMAS_DE_VERIFY` §1.5) is [PROVED+VERIFIED], not in Lean.

Lemma 3.6 (L-007 (Lemma E)). *If $\beta_0 \le \beta_j(0) \le \beta'_0$ and $|\beta_j(g) - \beta_j(0)| \le Cg^2$, then for $\gamma \le \sqrt{\beta_0/(2C)}$: $\beta_0/2 \le \beta_j(g) \le \beta'_0 + \beta_0/2$.*

The verifier checked this Cg^2 form; the C^1 form ($C_1 g$, $\gamma \le \beta_0/(2C_1)$) is the same two-line proof [PROVED, TRIVIAL VARIANT]. Lean has the Cg^2 form.

3.4.5 History dependence (L5) [PROVED RELATIVE TO LISTED INPUTS; VERIFIED (ROUND 7)]

Round 7 closes this item by the variant of the third bullet below: nested intervals on true trajectories, with no clamp and with the constants of Lemma 3.4 (Theorem 5.15, `SOURCES_T2_B7_CLAMP.md` Route 2 + VERIFY). The bullets record the gap as it stood before.

Balaban's β_{k+1} depends on $(g_0, \dots, g_k)$. L-004 (Lemma D′) Remark 3 (`LEMMA_DPRIME.md`) states the cube-continuity version (same proof). The gap is as follows (`THEOREM2_PROOF_v2_VERIFY` §F).

- *Domain.* Balaban defines β_{k+1} only on histories for which his Theorem 1 induction has run to step k (`LEMMAS_DE_VERIFY` §2 item 1). By the sweep (§2.3), that induction uses (2.6)–(2.9), that is (c′) + K-018 (S), on the history, not only $g_i \in]0, \gamma]$.

- *The new gap.* The clamped shooting of Lemmas 3.2 and 3.4 evaluates β on clamped histories $\tilde{g}_i = \varphi(\tilde{u}_i)$. A clamped history can violate the growth side of (c′): from $\max(a,c) - \max(b,c) \ge \min(a-b, 0)$ one gets only $u_m \ge u_n - D'$, not $u_m \ge u_n + \beta(n-m) - D'$. So the clamped shooting evaluates β on histories outside Balaban's domain. The clamp proof is valid only under the cube hypothesis, which Balaban does not supply.

- *A variant that avoids it* [NOT WRITTEN]. Use a nested-interval proof in the style of `LEMMAS_DE_VERIFY` §1.5, with domain $D_k = \{u_0, \dots, u_k \ge \gamma^{-2}\}$ of true (unclamped) trajectories. On such trajectories, (c′) up to step k follows from the tail-sum bound (a″) of L-005 (Lemma D″), and K-018 (S) holds by the choice of γ. So every evaluation of β stays inside Balaban's domain. At each step k, keep the set of u_0 whose trajectory stays in D_k; show by IVT, step by step, that this set contains a closed interval on which u_k sweeps a range that contains the next target interval. For L-006 (Lemma D) (all increments positive) this is written. For L-005 (Lemma D″) the increments can be negative, so the nested intervals need new envelopes, built from the window bounds $[b(n-m) - 2R - (b/4)(n-m), \dots]$. The adaptation is plausible; it is not written and not verified.

- Lean is one-argument only.

3.5 The exact structure of the one-loop sums

The window sums of the shooting lemma are sums of one-loop increments $\beta_j(0)$. This section expresses these sums as level-k quantities of a coarse Gaussian theory and isolates the remainder κ_k whose boundedness is the content of L-009 (Theorem K). It also lists the symbol facts, uniform in the level, that Section 3.6 uses.

3.5.1 Telescoping [PROVED+VERIFIED, CORRECTED SCOPE]

$S_k := p \to 0$ F^2-coefficient of the composed k-step gauge-fixed one-loop integral (U1 (1.2)). Then $S_k = \sum_{j=1}^{k} \beta_j(0)$, $S_0 = 0$ (`BETA_FP` Prop. 1.1; `BETA_FP_VERIFY` item 1). Scope: exact for the $p \to 0$ coefficient at fixed k, not for $\log Z_k$ at fixed p (the plane wave is off-shell).

3.5.2 Anchor-free identity [PROVED+VERIFIED]

Source: `BETA_FP_LEMMA43` Prop. 2.1; `BETA_FP_LEMMA43_VERIFY` items 1, 3–4 (hand rederivation; lattice check with the real CMP 109 averaging at random SU(2) backgrounds, relative difference 10^{-14}).

Proposition 3.7 (anchor-free identity; Prop. 2.1 of `BETA_FP_LEMMA43`). *Under H-194 (G1) =* $\{\Delta D = 0,\ \tilde{Q}D = D_c \iota_a^T\}$*:*

$$\log Z_\Lambda(V) = \big[\log\det X - \tfrac{1}{2}\log\det K\big] - \big[\log\det(D_c^T D_c) - \tfrac{1}{2}\log\det(\Delta_\Lambda(V) + D_c D_c^T)\big].$$

In words: constrained one-loop = fine Wilson one-loop in background Feynman gauge minus the one-loop of the coarse Gaussian theory with kernel $\Delta_\Lambda(V)$. All point-anchor objects cancel; the constant is exactly 0.

- $\tilde{Q}D = D_c \iota_a^T$ is exact at any background (lattice 10^{-15}). $\Delta D = 0$ is load-bearing (raw Hessian gives $O(1)$ errors).
- At the constant abelian field V_f the background is critical, so $\Delta D = 0$ is exact and Prop. 2.1 holds exactly (`BETA_FP_H_VERIFY` §2, CONFIRMED). At a plane wave it holds up to the defect $\mathfrak{d}_Y$ (`BETA_FP_PW` Lemma 2.3) [PROVED, UNVERIFIED].
- Per-cell infinite-volume densities only: on a torus the constant zero modes must be removed. This is input R-030 (G2) [VERIFIED REDUCTION AT FIXED j] (`SOURCES_T2_G2.md` + VERIFY).

3.5.3 The constant c [PROVED+VERIFIED]

$c = 11N^2/(12\pi^2) = 2N \cdot b_0/(8\pi^2)$, $b_0 = 11N/3$; SU(2): $c = 0.371511$. Normalization $1/g_B^2 = 2N/g^2$ (`R2_VERIFY`; `BETA_FP_LEMMA43_VERIFY` §5). The log coefficient of the fine piece at constant field is $c_2 = c/2$ per $\ln(1/\varphi)$, checked against Savvidy, arXiv:1910.00654 eq. (1.4) [23] (`BETA_FP_H_VERIFY` §4) = input R-027 (F) [CITED, SOURCE CHECKED; LATTICE = CONTINUUM LOG COEFFICIENT IS STANDARD, NOT PROVED IN PROJECT FILES].

$b := c \log L$: 0.25751 ($L = 2$), 0.40815 (3), 0.51502 (4), 0.59793 (5), 0.72293 (7).

3.5.4 The decomposition $S_k = ck \log L + \kappa_W^F - \kappa_k$

- As bookkeeping the split is empty: any c' gives an exact identity if κ_k is defined as the remainder (`U1_VERIFY`). Boundedness of a remainder-defined κ_k is a restatement, not progress.
- Content comes from defining κ_k by the coarse theory alone. `KAPPA_BOUNDED` Prop. 1.1 [PROVED+VERIFIED, CONDITIONAL ON R-026 (PI), R-027 (F), (M_j), V1, R-030 (G2)]: with the Wilson level $j = 0$ as reference,

$$S_j = cj \log L - d_j, \qquad \kappa_j = \kappa_W^F + d_j,$$

$$d_j := -\frac{1}{2n_2}\int_0^1 dt \int_{\mathrm{BZ}} \frac{dK}{(2\pi)^4}\ \mathrm{tr}\big[b_{t,2}\,\delta\sigma_j^{(0)} + b_{t,1}\,\delta\sigma_j^{(1)} + b_{t,0}\,\delta\sigma_j^{(2)}\big],$$

where $\delta\sigma_j^{(m)} = \sigma_j^{(m)} - \sigma_0^{(m)}$ are differences of the f^m Taylor symbols of the coarse kernel at level j and level 0, and $b_{t,m}$ are the Moyal inverse coefficients of the interpolated coarse operator. No limit symbols σ_∞ are used.

- Checks [NUMERICAL, REPRODUCED]: $\rho(1) = \kappa_W^F$ (Lüscher–Weisz (5.7) [16]); $L = 2$: $d_1 = -0.372$, $\kappa_1 = 1.114$; $L = 3$: $d_1 = c \log 3 - \beta_1 = -0.256 \pm 0.01$ (`KAPPA_BOUNDED_VERIFY` §4: fits give $-0.254 \cdots - 0.258$). This checks the definition of d_j through S_j; it does not evaluate (1.1) from the symbols independently.

- Cor. 3.1 of `BETA_FP_LEMMA43` ($\mathcal{C}(\Lambda) = c \log \Lambda + \kappa_W^F - \rho(\Lambda)$) is exact as an identity. "The whole $\log \Lambda$ sits in the fine piece" is equivalent to $\rho(\Lambda) = o(\log \Lambda)$. It is not a separate fact (`BETA_FP_LEMMA43_VERIFY` item 2; an earlier overclaim is removed).

- Given convergence $\beta_j(0) \to \beta_\infty$, Cesàro gives $\beta_\infty = c \log L - \lim \kappa_k / k$. So the hypothesis (H′) "$\kappa_k = o(k)$" of the first version of these notes is EQUIVALENT to $\beta_\infty = c \log L$, not weaker (`THEOREM2_PROOF_VERIFY` Point 4). L-009 (Theorem K) (Theorem 3.8) gives κ_k bounded, hence $\kappa_k = o(k)$ trivially.

3.5.5 Level-uniform symbol facts used below [PROVED+VERIFIED]

Source: `U3_AW.md`, `U3_AW_VERIFY.md` (grades 1–4 CONFIRMED); `U3_REPAIRS_VERIFY`.

- O5/A0: Δ_M real-analytic on a fixed strip, $|\partial^\alpha(\Delta_M - \Delta_\infty)| \le CM^{-2}$, uniform to $K = 0$ (odd L, centered blocks). Prop. B: $c_0|\hat{K}|^{-2} P_T' \le P_T' S_M P_T' \le C_1 |\hat{K}|^{-2} P_T'$, $c_0 = (2/\pi)^{12}$, $C_1 \le 1363$. Existential constants.

- Prop. E (evenness): $\sigma_j(f; -K) = \sigma_j(f; K)$ in the midpoint trivialization, all f, j. Tested in $d = 2$ only.

- L-308 (Theorem W): $\sigma_j^{(1)}(0) = \tau_c = (3ie/2)(E_{10} - E_{01})$ for every $j \ge 0$, from the exact Ward identity $\Delta_j(f) D_f = 0$. Numerically 0.750000, 0.75015, 0.75051 at $j = 0, 1, 2$ vs 0.75 ($d = 2$ proxy).

- L-071: $|\sigma_j^{(1)}(K) - \tau_c| \le C_A |K|^2$, real K, uniform in j (C^0 only).

- Prop. R-003: $\sigma_j^{(m)}$, $m \le 2$, analytic on the same strip at every fixed j (not uniform in j for $m = 1$ derivatives, $m = 2$).

- L-219 (Lemma M): Maxwell coefficient exactly 1 at every level.

3.6 Bounded window sums: two formally distinct conditional routes

This section gives the two routes to the window bound (5.1) that Lemma 3.4 needs.

Both routes give the window bound (5.1) of Lemma 3.4, $|S_K - S_k - b(K - k)| \le 2R$, with $b = c \log L$. L-005 (Lemma D″) then gives L-001 (Theorem 2w) (at odd $L > 11$, or with H-202 (H0) at this L as a hypothesis, and with L5). Neither route uses convergence of $\beta_j(0)$, a rate, (N), or certified early steps. The two routes have formally distinct named hypotheses. Their mathematical content is not independent: both reduce their uniform-in-level content to the same inequality R-025 (H2) on the two-step minimizer map (Section 3.7), plus A3-type facts (`BETA_FP_U` §3.5, §4.3 [PROVED, UNVERIFIED]; Route K side: `B2_TWOSTEP_VERIFY` grade 3 [PROVED+VERIFIED]).

3.6.1 Route K: L-009 (Theorem K) [PROVED+VERIFIED, CONDITIONAL ON (B2) AND THE FIXED-LEVEL INPUTS]

Source: `KAPPA_BOUNDED.md`; `KAPPA_BOUNDED_VERIFY.md` (items (i)–(v) and (vii) CONFIRMED; (vi) CORRECTED in reasoning, conclusion unchanged).

Theorem 3.8 (L-009 (Theorem K))**.** *L odd ≥ 3. Assume (B2) (Section 3.7; (B2) $\Leftarrow$ R-025 (H2) [PROVED+VERIFIED]) and the fixed-level inputs below. Then $\sup_k |d_k| \leq \kappa_* < \infty$, so $\sup_k |\kappa_k| \leq |\kappa_W^F| + \kappa_*$ and $|S_k - ck \log L| \leq \kappa_*$ for all k. Window sums satisfy (5.1) with $R = \kappa_*$. (L-009 (Theorem K) is a statement about one-loop quantities. Its use in L-005 (Lemma D″) for Balaban's $\beta_j(g)$ needs L odd > 11 or H-202 (H0) at this L.)*

Proof structure.

1. Prop. 1.1 (Section 3.5.4): $S_j = cj \log L - d_j$. [PROVED+VERIFIED, CONDITIONAL ON R-026 (PI), R-027 (F), (M_j), V1, R-030 (G2)]
2. Lemma 2.1: the only K-derivative of $\sigma^{(1)}$ in d_j enters linearly with f^0 coefficients. Integration by parts moves it onto f^0 data; boundary terms vanish (periodic up to a common constant conjugation; inner sphere $O(r^2)$). So $\sigma^{(1)}$ is needed only in C^0, which L-071 gives. [PROVED+VERIFIED]
3. Prop. 2.2: with (E0) f^0 bounds (Prop. B, O5/A0, L-219 (Lemma M)), (E1) L-071 + L-308 (Theorem W), and (B2), every term of the d_j integrand is $\leq C|K|^{-2}$, uniformly in j and t. In $d = 4$ this has two powers of infrared margin. $|d_j| \leq \kappa_* := (C(L)/n_2)(1 + Y_1^2 + C_A(1 + Y_1) + Y_2) \cdot I_2$. [PROVED+VERIFIED, CONDITIONAL ON (B2)]
4. Infrared: the degree -4 part of the coarse f^2 integrand is the same at every level (L-219 (Lemma M) + L-308 (Theorem W)), so the $\tau_c \log$ cancels exactly in level differences. [PROVED+VERIFIED]

Constants are existential; κ_* is not evaluated.

Fixed-level inputs of Route K (needed at each j separately; no uniformity in j):

- R-026 (PI) probe independence: plane-wave p^2 coefficient = constant-field f^2 coefficient [ARGUED; PER-CELL LIMIT FOR THE LINEAR POTENTIAL NOT WRITTEN].
- R-027 (F) fine log coefficient c_2 at constant field [CITED, SOURCE CHECKED], Section 3.5.3.
- (M_j) the formal f^2 coefficient d_j equals the true one at fixed j. The resolvent step of the first argument was false (Nielsen–Olesen zero at $k_\perp^2 = |ef|$; `BETA_FP_H_VERIFY` §5). `BETA_FP_M` repairs it with a direct $\log|\det|$ bound given spectral bounds H-244 (S1)–H-247 (S3) [OPEN ON THE LATTICE]; Prop. 4.4 there is [ARGUED]; `BETA_FP_M` has no verifier file. The t-subdivision of KAPPA §1.3 removes the small-ε requirement [ARGUED, CONFIRMED AS PLAUSIBLE]. Status: [ARGUED].
- V1: the Maxwell coefficient of the per-step determinant equals $\beta_{j+1}(0)$, i.e. order-g^0 completeness of CMP 109 (1.3)–(1.5) [PROVED+VERIFIED AT FIXED LEVEL, CONDITIONAL ON R-030 (G2), BAŁABAN'S INDUCTIVE BOUNDS, AND THE AVERAGING (0.4)] (`SOURCES_T2_V1.md` + VERIFY; Section 5.1).
- R-030 (G2) per-cell infinite-volume densities [VERIFIED REDUCTION AT FIXED j] (`SOURCES_T2_G2.md` + VERIFY; was [FORMAL]); the j-uniform version is not proved.

- H-202 (H0) Balaban's inductive assumption: β_{j+1} smooth on $[0,\gamma]$, "uniformly bounded on this interval together with all derivatives" (CMP 109 pp. 263–264, (1.22)) [BALABAN; AVAILABLE AT ODD $L > 11$ ONLY]. CMP 116 has no theorem statement and no C^m-in-g bound in words (`BALABAN_EXCERPTS_CHECK` item 10); the smoothness in g is CMP 109's inductive assumption, asserted by CMP 109 Theorem 3, whose proof CMP 116 completes.

- L5 history-dependent β, its domain, and its continuity (Section 3.4.5) [PROVED RELATIVE TO LISTED INPUTS; VERIFIED (ROUND 7)].

3.6.2 Route B: Theorem $\mathrm{B_{PW}}$ [PROVED+VERIFIED, CONDITIONAL (LOGIC OF THEOREM $\mathrm{B_{PW}}$)]

Source: `BETA_FP_PW.md` §1–§5 (Theorem 3.2 = Theorem $\mathrm{B_{PW}}$, Theorem 5.1). `BETA_FP_PW_VERIFY.md` §6 grades it: items 1–4 CONFIRMED (doubling identity and Lemma 1.3; Prop. 2.2 / Lemma 2.3; Theorems A, B as conditional logic; Lemma 4.1, conditional), items 5 (comparison) and 6 (numerics) CORRECTED. The tag follows the decision recorded in `PLAN.md` (2026-10-28): the logic of Theorem $\mathrm{B_{PW}}$ is [PROVED+VERIFIED, CONDITIONAL] on its hypotheses, citing `BETA_FP_PW_VERIFY` §6 items 1–4. Its inputs (U), (DF), (T1) from `BETA_FP_U` stay [PROVED, UNVERIFIED]. The file rederives the Lemma 1.3 arithmetic and the Lemma 2.3 / Prop. 2.2 algebra correctly, confirms the Prop. 2.2 determinant identity on a 4^4 lattice (4e−13), and notes that identity 3.0 is a tautology ($\Gamma^{ex} := \hat{\Pi} - S$), so all content sits in $(B\theta)/(C\theta)$.

Theorem 3.9 (Theorem $\mathrm{B_{PW}}$; Theorem 3.2 of `BETA_FP_PW`)**.** *L odd ≥ 3. Assume*

- *(U) there are $p_* > 0$, C_U with $|\beta_{j+1}(p) - \beta_{j+1}(0)| \leq C_U p^2$ for all $j \geq 0$, $0 < p \leq p_*$* [HYPOTHESIS; ARGUED FROM UNIFORM ANALYTICITY IN p OF THE ONE-STEP PLANE-WAVE DATA PLUS ONE-STEP O6]*;*

- *($B\theta$) there is $\theta_* \in (0, \min(p_*, 1))$ with $B_* := \sup_k |\Gamma^{ex}(\theta_*; L^k)| < \infty$, where Γ^{ex} is the coarse remainder at fixed coarse momentum θ_** [HYPOTHESIS]*;*

- *Lüscher–Weisz (5.7) [16]* [CITED].

Then $|S_k - ck \log L| \leq C_B := B_* + c_2 \ln(1/\theta_*^2) + |\kappa_W^F| + \sup_{p \leq \theta_*} |\eta(p)| + C_U \theta_*^2/(L^2 - 1)$, and window sums satisfy (5.1) with $R = C_B$. $\theta_* < 1$ is required so that $c_2 \ln(1/\theta_*^2) \geq 0$; otherwise the term must be replaced by its absolute value (`BETA_FP_PW_VERIFY` §3).

Key step. Lemma 1.3: $|S(\theta; L^k) - S_k| \leq C_U \theta^2/(L^2 - 1)$ from (U). The massless coarse side is then needed only at a fixed $\theta_* > 0$, where there is no infrared problem. No (W), R-026 (PI), (M), IR weights, or spectral theory.

What ($B\theta$) reduces to (`BETA_FP_PW` §4) [ARGUED]: Lemma 4.1(i) (trivial-background convergence, from O5, odd L) [PROVED]; vertex bounds at fixed θ_* (the uniform bounds cited by U3 Theorem 7.1 to Balaban; the fixed-θ data hypothesis ($D\theta$) is needed only in its "bounded" form); and boundedness in k of the defect $\mathfrak{d} = \mathfrak{d}_Y + \mathfrak{d}_{gf}$.

- $\mathfrak{d}_Y$ (Prop. 2.1 defect at the non-stationary plane wave): $O(a\Lambda^{-2})$ relative at fixed θ, from zero capacity of point anchors in $d = 4$ [ARGUED; REGION $|K| \lesssim \Lambda^{-1}$ NOT WRITTEN].

- $\mathfrak{d}_{gf}$ (composite gauge vs D_k^T gauge at the plane wave): $O(\theta^2)$ relative; uniformity in k not shown [ARGUED]. Data: 0.953 (direct) vs 0.9595 (recursion) at $L = 2$, $p_0 = 0.2$; difference below the tabulation error, so unresolved.

Identification. Route B's $\beta_j(0)$ is the per-level plane-wave increment (the quantity the code computes). It matches Balaban's second-moment β up to the expansion point (fine wave vs constrained minimizer), argued to change S by $O(\theta^2)$ relative (`SOURCES_D1_VERIFY` item 8) [ARGUED]. This identification and the $\mathfrak{d}_{gf}$ bound need expansion-point bounds uniform in k. So the R-026 (PI)-type content of Route K is not removed in Route B; it moves into a uniform-in-k input (`BETA_FP_PW_VERIFY` §3).

Three points from `BETA_FP_PW_VERIFY` §3 (item 5 of its §6 grades them CORRECTED):

1. $\theta_* < 1$ is required (above).
2. The β identification (R-028) is a uniform-in-k input, not a fixed-level input.
3. (U) is a uniform-in-j analyticity of the level data. It plausibly needs the plane-wave analogue of a uniform C^1 bound on $\sigma^{(1)}$: uniform p-smoothness of the a^1 vertex. That is the same uniform first-order vertex smoothness as Route K's open core ($C1\sigma$).

3.6.3 Route B inputs after `BETA_FP_U` [PROVED, UNVERIFIED; NO VERIFIER FILE]

Source: `BETA_FP_U.md` §2–§4. It reduces the two uniform hypotheses of Theorem $\mathrm{B_{PW}}$.

- (U) $\Leftarrow$ (T1) + R-025 (H2) (Theorem 2.4, odd $L \geq 3$, with A0, R-001, Lemma 2.2). (T1): $\sup_j \sup_q |c^{(1)}_{T,j}(q)| < \infty$, the plane-wave form of L-071 [ARGUED]. R-025 (H2) is the inequality of Section 3.7 (`BETA_FP_U` writes it as $\tilde{h}_{(2)} < L$).
- ($B\theta$): $\Gamma^{ex} = \gamma_{rec} + \mathfrak{d}_F$ exactly (Prop. 3.1). γ_{rec} is bounded in k given Theorem 2.4 (Prop. 3.2). $\mathfrak{d}_F$ is bounded under (DF) $\sup_k |\mathfrak{d}_F(\theta_*; L^k)| < \infty$ [HYPOTHESIS, ARGUED]. `BETA_FP_U` corrects `BETA_FP_PW` §4.3(b): $\mathfrak{d}_F$ is $O(1)$ relative at fixed θ, not $O(\theta^2)$; only boundedness is used.
- Corollary 4.1: given (T1), R-025 (H2), (DF), LW (5.7), $|S_k - ck \log L| \leq C_B$, so (5.1) holds with $R = C_B$.
- L-001 (Theorem 2w) by Route B then needs in addition R-007 (plane-wave β = Balaban's β up to the expansion point, uniform in k) [ARGUED] and H-202 (H0) (L-005 (Lemma D″) (i)).

Route B hypothesis set: (T1), R-025 (H2), (DF), LW (5.7), R-007, H-202 (H0) (plus L5, R-030 (G2), odd L, as for Route K). The fixed-level inputs R-026 (PI), (M_j), V1 of Route K do not appear; (DF) and R-007 replace them.

3.6.4 Comparison

	Route K	Route B
uniform-in-level input	(B2): $\sup_j \sup_K \lvert\sigma^{(2)}_j\rvert < \infty$, $\Leftarrow$ R-025 (H2) [PROVED+VERIFIED]	(U), ($B\theta$), and the β identification and $\mathfrak{d}_{gf}$ bound, both with k-uniform expansion-point bounds; (U) $\Leftarrow$ (T1) + R-025 (H2), ($B\theta$) $\Leftarrow$ Theorem 2.4 + (DF) [PROVED, UNVERIFIED]
single uniform inequality	R-025 (H2)	R-025 (H2)

fixed-level inputs	R-026 (PI), R-027 (F), (M_j), V1, R-030 (G2)	LW (5.7), $\mathfrak{d}_Y$, $\theta_* < 1$
uses (W), L-071	yes	no; `BETA_FP_U` uses (T1), the plane-wave analogue of R-004 [ARGUED]
verification	verified (conditional)	logic of Theorem B_{PW} verified (conditional): `BETA_FP_PW_VERIFY` §6 items 1–4 CONFIRMED [PROVED+VERIFIED, CONDITIONAL]; inputs from `BETA_FP_U` [PROVED, UNVERIFIED], no verifier file
common	H-202 (H0) at odd $L > 11$, L5, R-030 (G2), V1-type identification, Balaban's proofs, SU(2)	

Shared inputs: H-202 (H0), L5, R-030 (G2), odd L (> 11 for H-202 (H0)), and the identification of the computed $\beta_j(0)$ with Balaban's $\beta_j(0)$ (V1; in Route B also the expansion point). The uniform-in-level hypotheses ((B2) vs (U) + $(B\theta)$) are formally distinct as named hypotheses. Their content overlaps: both rest on R-025 (H2) (Section 3.7). In Route K, R-025 (H2) gives $(C1\sigma)$ (L-011) and then (B2) (L-012). In Route B, R-025 (H2) gives (U) with (T1), and the vertex bound behind $(B\theta)$. So a proof of R-025 (H2) serves both routes. One closed route suffices for L-001 (Theorem 2w).

3.7 The hypothesis (B2) of Route K, and the single uniform inequality R-025 (H2)

Both routes need a bound on the coarse symbols that is uniform in the level. This section states that bound for Route K, records what is proved and what is refuted, and reduces it to the inequality R-025 (H2). The numerical values of R-025 (H2) are in Sections 6.4–6.11. At $L = 13$, R-025 (H2) is a certified theorem (Theorem 6.1, Section 6.7).

(B2): $\sup_{j\geq 0} \sup_{K\in \mathrm{BZ}} |\sigma_j^{(2)}(K)| < \infty$. Weaker form that suffices, H-184 (B2w): $\sup_j \int_{\mathrm{BZ}} |K|^{-2} |\sigma_j^{(2)}(K)|\, d^4K < \infty$ ($\sigma^{(2)}$ enters d_j only undifferentiated; `KAPPA_BOUNDED` Remark 2.3) [PROVED+VERIFIED: NOT GRADED BY NAME; IMPLIED BY `KAPPA_BOUNDED_VERIFY` §9(IV)].

$\sigma^{(2)}$ obeys the one-step recursion (`U3_AW` (1.2)): $\sigma_j^{(2)} = L^{-2}\mathcal{A}_j \sigma_{j-1}^{(2)} + Q_j[\sigma_{j-1}^{(1)}] + g_j^{(2)}$, with $\mathcal{A}_j s = L^{-2} H_j^\dagger (\oplus_l s(k_l)) H_j$ and $H_j(K)$ the one-step gauge-fixed minimizer map. Notation: $\rho_1 := \sup_{j,K} \|H_j(K)\|/L$ (one step); $H_{(2),j}(K) := (\oplus_l H_{j-1}(k_l)) H_j(K)$ (two steps), $\rho_2 := \sup_{j\geq 2} \sup_{K \text{ real}} \|H_{(2),j}(K)\|/L^2$.

3.7.1 Proved

- $K = 0$ bound [PROVED+VERIFIED]: $\sup_j |\sigma_j^{(2)}(0)| \leq C(|e|C_A + e^2)$. L-071 + scaling give $|\partial^2 \sigma_j^{(1)}(0)| \leq 2C_A$ (polarization; `KAPPA_BOUNDED_VERIFY` §7), and the slaving identity H-178 (W-2) expresses $\sigma_j^{(2)}(0)$ through these second derivatives.

- Each fixed j [PROVED+VERIFIED]: $\sigma_j^{(2)}$ is analytic on a fixed strip (Prop. R-003), hence bounded. The content of (B2) is uniformity in j.

- Prop. 4.1 [PROVED+VERIFIED, CONDITIONAL]: if (h) $h := \sup_{j,K} \|H_j(K)\| < L^2$ and $(C1\sigma)$

$\sup_j \sup_K |\partial_K \sigma_j^{(1)}(K)| < \infty$, then (B2) holds. Proof: in the form norm $\pm s \le N(\Delta+1)$, $N_j \le (h^2/L^4)N_{j-1} + Z_2$, and Z_2 is finite by L-071, $(C1\sigma)$, and fixed f^0 vertices.

- Why the L-072 mechanism alone fails [PROVED+VERIFIED, REASON CORRECTED BY `KAPPA_BOUNDED_VERIFY` §7]: the obstruction is the missing uniform C^1 bound on $\sigma^{(1)}$, which enters twice: in Q_j (term $L^{-3}\partial_K \sigma_{j-1}^{(1)}$) and in the longitudinal part of $\sigma^{(2)}$ given by H-178 (W-2). L-071 is C^0 only.

3.7.2 Refuted, and the failed one-step bound

- (HS) $h < L\sqrt{L-1}$ (named (H′) in `U3_AW`) [REFUTED AT $L = 3$]. Measured $\max_K \|H_j(K)\|/L = 1.652, 1.736, 1.757, 1.760$ for $j = 1, 2, 3, \infty$ (`U3_AW_VERIFY` §6); the threshold is $\sqrt{2} = 1.414$. Gauge choice cannot help: H_j is already minimal-norm in its gauge orbit.

- One-step sup-norm C^1 recursion at $L = 3$ [NUMERICAL, REPRODUCED; `B2_TWOSTEP_VERIFY` GRADE 2]. Its factor is $\rho_1^2/L = 0.909, 1.0046, 1.0289, 1.0325$ for $j = 1, 2, 3, \infty$, so it exceeds 1 for $j \ge 2$ ($\rho_1 > \sqrt{3} = 1.7321$). At $K = 0$ the factor is L^{-1}; the bad region is the face $K_3 = K_4 = \pi$, $K_1 = K_2 \approx 1.86$–1.92. This shows that the one-step sup-norm bound fails at $L = 3$. It does not show that a C^1 bound fails; a weighted norm is not excluded.

3.7.3 R-025 (H2) and the two-step theorems [PROVED+VERIFIED, CONDITIONAL ON R-025 (H2)]

Source: `B2_TWOSTEP.md` §1–§3; `B2_TWOSTEP_VERIFY.md` grades 1, 3, 4 (CONFIRMED).

$$\text{R-025 (H2)} \quad \rho_2 = \sup_{j \ge 2} \sup_{K \in \mathrm{BZ\ real}} \|(\oplus_l H_{j-1}(k_l))H_j(K)\|/L^2 < L.$$

R-025 (H2) is a plain operator-norm inequality on real K. It uses no strip, no Δ-relative weight, and only $n = 2$.

Theorem 3.10 (L-011 of `B2_TWOSTEP`)**.** *R-025 (H2) ⇒ $(C1\sigma)$, with contraction factor $q_1 = \rho_2^2/L^2$ on the derivative.*

Theorem 3.11 (L-012 of `B2_TWOSTEP`)**.** *R-025 (H2) ⇒ (B2), hence H-184 (B2w), in plain sup norm, with factor $q_2 = \rho_2^2/L^4$.*

Here $(C1\sigma)$ is $\sup_j \sup_K |\partial_K \sigma_j^{(1)}(K)| < \infty$ (Section 3.7.1). The verified points are these.

- Two-step map. $\mathcal{A}_j \mathcal{A}_{j-1} s = L^{-4} H_{(2)}^{\dagger}(\oplus s) H_{(2)}$, with $dk_{l,l'}/dK = L^{-2}$ [CONFIRMED, grade 1].

- Theorem 3.10. The chain-rule gain is exactly L^{-2} per two steps on real K. So the threshold is $\rho_2 < L$, not $\sqrt{L^2-1}$: no strip-Cauchy loss enters. The large constants ($b(L)$ of O6, η_1, η_2, G_1) enter only the source term, not the contraction factor [CONFIRMED, grade 3].

- Theorem 3.11. This contraction needs only $\rho_2 < L^2$, which R-025 (H2) implies [CONFIRMED, grade 3]. In one step it is Prop. 4.1's (h).

- R-025 (H2) vs (H″). R-025 (H2) is the plain operator norm of the same matrix as `U3_AW` (H″) at $n = 2$. It is weaker: threshold L (3 at $L = 3$) against $\sqrt{L^2-1}$ (2.828), real K only, $n = 2$ only [CONFIRMED, grade 4]. (H″) ⇒ (B2) stays [ARGUED] and is no longer needed.

- Form of Q_j (G3) [CONFIRMED by read-off of `U3_AW` (1.2)/(3.3) and P10]: $\partial\sigma^{(1)}$ enters linearly. `U3_LEMMA51` (3.3) itself was not re-derived from the Moyal rule by either party.
- Not covered: C^2 and C^3 bounds on $\sigma^{(1)}$ under R-025 (H2) (`B2_TWOSTEP` §5, "look routine", not written, not graded). They are not needed for (B2).
- Submultiplicativity: $\rho_2 \le \rho_1^2$ always. At $L = 3$ the gain of the two-step map is that this is strict by a factor 1.50 (2.064 vs 3.098) [CONFIRMED].

3.7.4 One-step sufficiency [ARITHMETIC PROVED; INPUT CERTIFIED AT $L = 13$]

$\|H_{(2),j}(K)\| \le \max_l \|H_{j-1}(k_l)\| \cdot \|H_j(K)\| \le L^2\rho_1^2$. So $\rho_1^2 < L$ implies R-025 (H2), and then the one-step C^1 recursion of `B2_TWOSTEP` (2.2) contracts with factor ρ_1^2/L. At $L = 3$ the measured ρ_1^2 exceeds L, so the one-step route fails and the two-step map is needed; at $L = 5$ and $L = 7$ the measured ρ_1^2/L is below 1. The values are in Section 6.5. At $L = 13$ the input is certified: $\sup_{j,K} \rho_1^2 \le 10 < 13$ (Theorem 6.1), so R-025 (H2) holds there with $\rho_2 \le 10$.

3.7.5 The precise remaining inequality

At a fixed odd $L \ge 3$ the single uniform hypothesis is R-025 (H2).

- Route K: R-025 (H2) $\Rightarrow$ ($C1\sigma$) (L-011) $\Rightarrow$ (B2) (L-012) [PROVED+VERIFIED, CONDITIONAL ON R-025 (H2)].
- Route B: R-025 (H2) + (T1) $\Rightarrow$ (U), and R-025 (H2) $\Rightarrow$ the vertex bound for γ_{rec} in ($B\theta$) (`BETA_FP_U` Theorem 2.4, Prop. 3.2) [PROVED, UNVERIFIED].
- Status of R-025 (H2): at $L = 13$, inside Bałaban's range $L > 11$, [CERTIFIED (INTERVAL ARITHMETIC, OUTWARD ROUNDING; VERIFIED)]: $\rho_1(j, K)^2 \le 10 < 13$ for all $j \ge 1$ (including ∞) and all K, hence $\rho_2 \le 10 < 13$ (Theorem 6.1; residue in Section 6.7). At $L = 3, 5, 9$: [NUMERICAL] only (at $L = 9$ one point).
- Kept as alternatives: ($C1\sigma$) + (h) $\Rightarrow$ (B2) [PROVED+VERIFIED]; (H″) $\Rightarrow$ (B2) [ARGUED]; H-184 (B2w) directly. No per-momentum $d = 4$ constant-field code exists, so (B2) itself has not been scanned numerically (`KAPPA_BOUNDED_VERIFY` §8).

3.8 Conclusion of the reduction

L-001 (Theorem 2w) $\Leftarrow$ (Route K inputs) or (Route B inputs). Precisely:

Route K. (B2) [OPEN; REDUCES TO R-025 (H2) (VERIFIED REDUCTION, SECTION 3.7.3); R-025 (H2) CERTIFIED AT $L = 13$ (THEOREM 6.1)] + R-026 (PI), R-027 (F), (M_j), V1, R-030 (G2) [FIXED-LEVEL] + H-202 (H0) [BALABAN, ODD $L > 11$; ITS USE IS A VERIFIED REDUCTION, SECTION 5.9] + L5 [OPEN] $\Rightarrow$ L-009 (Theorem K) ($R = \kappa_*$) $\Rightarrow$ (5.1) $\Rightarrow$ L-005 (Lemma D″) $\Rightarrow$ L-001 (Theorem 2w). Status of the implication: [PROVED+VERIFIED, CONDITIONAL].

Route B. (U) + ($B\theta$) with $\theta_* < 1$ + LW (5.7) + bounded $\mathfrak{d}_Y$, $\mathfrak{d}_{gf}$ + identification of β (uniform in k) + R-030 (G2) + H-202 (H0) [ODD $L > 11$] + L5 $\Rightarrow$ Theorem $\mathrm{B_{PW}}$ ($R = C_B$) $\Rightarrow$ (5.1) $\Rightarrow$ L-005 (Lemma D″) $\Rightarrow$ L-001 (Theorem 2w). Status: Theorem $\mathrm{B_{PW}}$ logic [PROVED+VERIFIED, CONDITIONAL]; inputs [PROVED, UNVERIFIED]. With `BETA_FP_U` (Section 3.6.3): (T1) + R-025 (H2) + (DF) + LW (5.7) $\Rightarrow$ Corollary 4.1 ($R = C_B$); with R-007 + H-202 (H0) + L5 + R-030 (G2) $\Rightarrow$ L-001 (Theorem 2w). Status: [PROVED, UNVERIFIED, CONDITIONAL] (the inputs (T1), (DF), R-007 and the `BETA_FP_U` reductions are unverified).

Single uniform hypothesis. Both routes need R-025 (H2). Theorem 6.1 certifies R-025 (H2) at $L = 13$, an odd $L > 11$. This discharges (B2), the only uniform-in-level hypothesis of Route K (Theorems L-011, L-012). In Route B it reduces (U) to (T1); (DF) and R-007 remain.

Then Balaban's Theorem 1. L-001 (Theorem 2w) gives (b) and (c′). With γ shrunk so that K-018 (S) holds, every one of the 19 cross-scale uses in CMP 119 and CMP 122 I–II is covered (Section 3.4.3), and the single-scale uses need only $g_k \le \gamma$. So Theorem 1 holds for the bare coupling $g_0(K, g)$, for every K and every $g \in]0, g_1]$, without Balaban's coupling assumption. This needs L5 in the sharpened form of Section 3.4.5: β must exist on every history the shooting visits. Status: [PROVED, UNVERIFIED (SWEEP), CONDITIONAL ON ROUTE K OR ROUTE B INPUTS, ON L5, AND ON BALABAN'S PUBLISHED PROOFS]. The sweep uses Bałaban's sentence that similar inequalities hold for the later constants (ledger R-012); round 5 closes it as a verified list (Proposition 5.7), with the order entries K-029–K-031. The step to Theorem 1 also needs the finite-volume identification R-032, which rests on R-015; Proposition 5.8 gives the torus coefficient within $C_j N_j^6 e^{-m_j N_j/2}$ of the whole-lattice one, given H-141 (Loc), and exact equality cannot be derived. So L-001 (Theorem 2w) $\Rightarrow$ Theorem 1 rests on 4 Group A items (R-006, R-012, R-013, R-015), 9 Group B items (R-025–R-032, R-034), and X-005.

Order of constants. Fix $\beta_0, p_0, p_1, r, \nu, \kappa_0$ (Balaban). Fix L (odd, > 11), then R from Route K or B, then $b = c \log L$, then $\gamma \le \min(\gamma_0, \gamma_{Bal}, b/(4C_1), \gamma_*)$ with $\gamma^{-2} \ge c_{sw} \cdot 2R$. Then $g_1 = (\gamma^{-2} + 2R)^{-1/2}$. Balaban's phrase "for a sufficiently small positive g" fits $g \le g_1$.

What this does not give.

- Literal (0.31): only for windows of length $\ge 8R/b$. Literal (0.31) at every K needs (L1) or (L2) of Section 3.3, which need individual increments, not window sums. Not needed for Theorem 1.
- The value $\lim \beta_j(0)$ and any rate: Section 3.9.
- Even L, or odd $L \le 11$ without H-202 (H0) as an extra hypothesis (Balaban assumes L odd > 11 throughout CMP 109); SU(N) with $N > 2$.
- Monotone couplings: g_m can exceed g_n for $m < n$ by the bounded amount $u_m \ge u_n - D'$. Section 3.4.3 shows Balaban's proof tolerates this under K-018 (S); a line-by-line read was not done.
- A continuum limit (`ASSEMBLY.md` treats the assembly separately; not claimed here).

3.9 The convergence route: what it adds, and its status

Not needed for Theorem 1. It adds: (i) existence of $\beta_\infty = \lim \beta_j(0)$; (ii) its value $c \log L$; (iii) a rate; (iv) eventual positivity, hence literal (0.31) for $K \ge K_0$ (L-004 (Lemma D′) (d)); (v) with certified early steps, literal (0.31) at every K (Section 3.3 (L2)). It also implies (B2), since convergence of $\sigma_j^{(2)}$ implies boundedness. So it needs strictly more than Route K.

3.9.1 Components and status

- U3 Theorem 4.1 with O5: $|\partial_K^\alpha(\Delta_M - \Delta_\infty)| \le CM^{-2}$, $|\alpha| \le 2$, uniform to $K = 0$, odd L [PROVED+VERIFIED] (`U3_VERIFY`, `U3_REPAIRS_VERIFY`). Numerically $|\Delta_M - \Delta_\infty| =$ 1.007, 0.129, 0.0147 at $M = 3, 9, 27$.

- O6: $\|\mathfrak{B}_j(K)^{-1}\| \le b(L)$ for all j, K, one-step constraint, trivial background [PROVED+VERIFIED]. Crude constant ($b(3) \approx 7.5 \cdot 10^{33}$); measured corner values 103.5, 176.7, 193.0, 195.3 ($j = 0, 1, 2, \infty$, $L = 3$) use a surrogate Λ (`U3_REPAIRS_VERIFY` §5).

- Theorem 6.1: $\beta_{j+1}(0) = \Phi(\mathcal{D}_j)$ is Lipschitz in the level data [PROVED+VERIFIED, CONDITIONAL ON V1]. The norm is the Taylor-data norm (`U3_VERIFY` P8 correction).

- Lemma 5.1 (convergence of $\sigma_j^{(1)}$, $\sigma_j^{(2)}$ in C^2, $|\alpha| \le 2$):
  - neutral sector, rate L^{-4j} [PROVED+VERIFIED FOR THE VALUE BOUND; THE K-DERIVATIVE PARAGRAPH IS PROVED, UNVERIFIED];
  - (W) [PROVED+VERIFIED] (`U3_AW` L-308 (Theorem W)); it removes the marginal direction of the f^1 recursion;
  - (A) qualitative and L-071 [PROVED+VERIFIED]; (A) in full (strip/C^3 for $m = 1$, all of $m = 2$) $\Leftarrow$ (H″) [OPEN]; the route via (HS) is [REFUTED] at $L = 3$;
  - $\sigma^{(1)}$ convergence at rate jL^{-2j} [SKETCH IN `U3_AW` §4.7, UNVERIFIED]; $\sigma^{(2)}$ convergence [OPEN, NEEDS (A) IN FULL];
  - the single-block-constraint version of Lemma 5.1 is false for $m = 2$ in $d = 4$ [PROVED+VERIFIED]; Balaban's composed constraint is not affected.

- Prop. 3.6: $|\beta_{j+1}(0) - \beta_\infty| \le C(1+j)^2 L^{-2j}$ [PROVED, UNVERIFIED AS AN ASSEMBLY; CONDITIONAL ON V1, (A) IN FULL]. (The form Cj^2L^{-2j} of the first version of these notes is 0 at $j = 0$ and is corrected here.) The identification of σ-Taylor data with the data of (6.1) is cited from `U3_VERIFY` §7, not re-shown. (N) (clean rate L^{-2j}) is not needed and is judged plausibly false.

- Value of the limit. Given convergence, L-009 (Theorem K) or Theorem $\mathrm{B_{PW}}$ gives $\beta_\infty = c \log L$ at once: $S_k/k \to \beta_\infty$ by Cesàro and $S_k/k = c \log L + O(1/k)$ [PROVED+VERIFIED AS AN IMPLICATION: `THEOREM2_PROOF_VERIFY` POINT 4; `BETA_FP_PW` THEOREM 3.2(III)]. Older routes to the value: `BETA_FP_H` Theorem 3.1′ with (H) restated at the constant field, which given (W) reduces to Lemma 5.1 (1.2) alone (`BETA_FP_H_VERIFY` items (1)–(4)), plus R-026 (PI), R-027 (F), (M) [CONDITIONAL; (M) ARGUED]; and `BETA_FP_PW` Theorem 3.1 (limit form) given (U), ($C\theta$), LW (5.7) [PROVED, UNVERIFIED]. The original (H) with IR weights (`BETA_FP_LEMMA43` (4.2)) was graded UNSUPPORTED and is not used.

- Rate of $\mathcal{C}(\Lambda)$: $O(\Lambda^{-2} \log \Lambda)$ given Lemma 5.1 with rates, via `U3_AW` §3.4(b) slaving [ARGUED, `BETA_FP_H_VERIFY` §5].

3.9.2 Early steps and j_*

Under Prop. 3.6 and $\beta_\infty = c \log L$, $\beta_{j+1}(0) \ge (c \log L)/2$ for $j \ge j_* := \min\{J : C(1+j)^2 L^{-2j} \le (c \log L)/2 \text{ for } j \ge J\}$. Heuristic size [ARGUED]: $C \sim b^3$ gives $j_* \approx 110$ ($L = 3$), ≈ 70 ($L = 13$); $C \sim b^6$ (one-step inverses inside C_{data}) gives ≈ 218, ≈ 138 (`THEOREM2_PROOF_VERIFY` Point 5(e)). With the measured $\|\mathfrak{B}^{-1}\| \approx 195$: $j_* \approx 10$ ($L = 3$), ≈ 4 ($L = 13$). Certifying $\beta_1(0), \dots, \beta_{j_*}(0) > 0$ needs interval arithmetic at odd $L \ge 13$ (blocks $\approx 3.4 \cdot 10^5$ rows per momentum by dense methods; the umklapp code reduces this to 12×12 blocks at L^4 momenta). It is needed only for (L2), not for Theorem 1.

4 The finite-torus existence theorem

This section proves Theorem B. It takes the trajectory of Section 3 as the input block H-084 (B-T2). Section 4.1 fixes what "exists" means here. Section 4.2 states the uniform bound H-116 (H-U) that the blocks must deliver, and Sections 4.3–4.4 state the theorems. Section 4.5 proves them modulo the named blocks, and Section 4.6 gives the dependency table, which is the single status table of the paper. Section 6 then collects the numerical support for the hypotheses.

Source: `ASSEMBLY.md` (2026-10-27), verified in `ASSEMBLY_VERIFY.md`. The statements carry the verifier's corrections (Section 4.8). The theorem is a reduction to named blocks. It is not unconditional. The notation is that of Sections 2.1 and 2.5.

4.1 What "exists" means here

Jaffe–Witten footnote 2 [15] [VERIFIED QUOTE] (`reports/clay_ym.txt` line 619; anchor: the sentence "Such a result presumably would enable the study of the limit as $T^4 \to \mathbb{R}^4$", line 582): "We specifically exclude weak-existence (compactness) as the solution to the existence part of the Millennium problem, unless one also uses other techniques to establish properties of the limit (such as the existence of a mass gap and the axioms)."

Reading used here. "Exists" means: (i) the family $\{S_{n,\varepsilon}\}$ is relatively compact (uniform bounds); (ii) at least one limit point exists along a sequence $\varepsilon_k \to 0$; (iii) every limit point has the properties (OS0)–(OS3) below, proved on the limit, not assumed. No uniqueness is claimed. Full-sequence convergence needs E13 (comparison of two trajectories; `SOURCES.md` H-099 (E14-d), D6), which is not in this paper.

This reading is weaker than the footnote's parenthesis. The theorem below gives no gap and only the torus versions of four axioms. It is NOT an admissible Clay solution. It is the finite-torus step of Jaffe–Witten §6.5 [VERIFIED QUOTE]: "One must then verify the existence of limits of appropriate expectations of gauge-invariant observables as the lattice spacing tends to zero and as the volume tends to infinity" (only the first limit is done here).

The OS axioms are not stated in `reports/clay_ym.txt`. It cites them as OS (CMP 31 (1973), CMP 42 (1975)) [19, 20] and requires "axiomatic properties at least as strong as those cited" (lines 279–280) [VERIFIED QUOTE]. The forms used follow the template's application of OS II (`reports/o3_continuum.txt` lines 4987–5003) [VERIFIED QUOTE]: "The linear growth condition requires a bound $|S_n(F)| \le \sigma_n |F|_{s_n}$ with fixed s and $\sigma_n \le A(n!)^D$ for fixed A, D", and restriction to Schwartz tests flat on the coincidence diagonals gives the test class of that reference. OS II itself was not re-read [CITED].

4.2 Hypothesis delivered by the blocks

Definition 4.1 (H-116 (H-U) uniform bound)**.** There are $s \in \mathbb{N}$ and $A, D < \infty$ such that for all $K \ge K_0$, all $n \ge 1$, and all $F \in \mathcal{D}_n$:

$$|S_{n,\varepsilon}(F)| \le \sigma_n \|F\|_{sn}, \qquad \sigma_n := A(n!)^D, \quad \varepsilon = L^{-K}.$$

The derivative order sn grows linearly in n. The OS linear growth condition admits this. H-116 (H-U) is the output of blocks H-084 (B-T2) + R-035 (B-S) + the bridge R-045 (B-N1) of Section 4.5.

4.3 Theorem (finite-torus existence, compactness form)

Theorem 4.2 (finite-torus existence, compactness form [PROVED MODULO NAMED BLOCKS])**.** *Let $G = SU(2)$, $d = 4$, L odd and large, $\ell \in 2\mathbb{N}$ fixed. Let $\gamma > 0$ be Balaban's small constant and*

$g \in]0,\gamma]$ small. For $K \geq K_0$ let $\varepsilon = L^{-K}$ and $g_0 = g_0(\varepsilon, g)$ be a shooting solution (Lemma 3.4) whose effective couplings satisfy $g_k \in]0,\gamma]$ for $0 \leq k \leq K$ and $g_K = g$. Assume H-116 (H-U). When H-116 (H-U) is taken from the blocks (Section 4.2) rather than assumed, the chain also assumes the $\kappa \to \kappa/2$ decay recovery of D2 [R, R-046] (not stated, `SOURCES_D2_VERIFY.md` (5)), which does not follow from the other hypotheses. Then:

(a) Compactness. Every sequence $K_m \to \infty$ has a subsequence K_{m_i} such that, with $\varepsilon_i = L^{-K_{m_i}}$, the limit $S_n(F) := \lim_i S_{n,\varepsilon_i}(F)$ exists for every $n \geq 1$ and every $F \in \mathcal{D}_n$.

(b) For every limit family $(S_n)_{n\geq 1}$ obtained in (a):

- *(OS0) Regularity. S_n is linear on $\mathcal{D}_n$, and $|S_n(F)| \leq A(n!)^D \|F\|_{sn}$. In particular S_n restricted to $C_c^\infty(T^{4n} \setminus \bigcup_{i<j}\{x_i = x_j\})$ is a distribution of order $\leq sn$.*
- *(OS1) Invariance. $S_n(F \circ \tau_a^{\times n}) = S_n(F)$ for every $a \in T_\ell^4$ (all continuous translations), and $S_n(F \circ R^{\times n}) = S_n(F)$ for every R in the hypercubic group B_4 (signed permutations of the axes, acting about a lattice site).*
- *(OS2) Reflection positivity. Let $\theta(x_0, \vec{x}) = (-x_0, \vec{x})$ on T_ℓ^4 and $T_+ := \{x : x_0 \in]0, \ell/2[\}$. For every finite family $F_0 \in \mathbb{C}$, $F_n \in \mathcal{D}_n$ with $\operatorname{supp} F_n \subset T_+^n$ $(1 \leq n \leq N)$:*

$$\sum_{n,m=0}^{N} S_{n+m}(\Theta F_n \otimes F_m) \geq 0, \quad (\Theta F_n)(x_1, \ldots, x_n) := \overline{F_n(\theta x_n, \ldots, \theta x_1)}, \quad S_0 := 1.$$

- *(OS3) Symmetry and reality. $S_n(F \circ \pi) = S_n(F)$ for every permutation π of the n arguments, and $S_n(F) \in \mathbb{R}$ for real F.*

(c) Nontriviality is a separate statement, Theorem 4.3.

The sourced blocks use the unit-scale condition K-024 (U), $\rho_1 \leq c_* r_0/M^a$ at $\lambda_K = 1$ (r_0 the Cauchy margin of the constant-source family, `SOURCES_D2C.md` (3.5)), as a constant choice (K-024), made once after r_0 and before K; B-N1 meets it by the rescaling $\lambda_i = 2N_U(f_i)$ at a C^n cost, s and D unchanged (`SOURCES_UN1.md`, `SOURCES_UN1_VERIFY.md`). This K-024 (U) is not the Route B hypothesis (U). *What is claimed about measures.* No probability measure is claimed. The limit is a family of Schwinger functions on $\mathcal{D}_n$. A measure on distributions would need finite moments of smeared fields, and $\operatorname{Var} \sum_p f(x_p)\varphi_p \sim \varepsilon^{-4}$ (Section 2.5). (OS0)–(OS3) on $\mathcal{D}_n$ are the inputs of OS reconstruction.

4.4 Theorem (nontriviality), conditional on H-081 (B-NT)

Theorem 4.3 (nontriviality [CONDITIONAL ON (NT) OF SECTION 4.5.9])**.** *Under the hypotheses of Theorem 4.2 and (NT), every limit family satisfies: (i) $S_2 \neq 0$: there are $f_1, f_2 \in C^\infty(T^4)$ with disjoint supports and $S_2(f_1 \otimes f_2) \neq 0$. (ii) S_2 is not the two-point function of a scale-invariant (in particular free Maxwell) composite: $d^8 S_2(x, y)$, $|x - y| = d$, is not asymptotically constant as $d \to 0$.*

Part (ii) is stated here in the form the proof supports (two-sided envelope only); see Section 4.8, item 3. The source stated (ii) as "behaves like $c\, g(d)^4$ with $g(d)^2 \sim (\beta \log(1/d))^{-1}$", which needs $\beta_j \to \beta_\infty$ and is not proved.

4.5 Proof modulo blocks

Structure. H-084 (B-T2) gives the trajectory. R-035 (B-S) gives sourced bounds uniform in ε. R-045 (B-N1) turns them into H-116 (H-U). H-074 (B-C) gives (a). H-082 (B-REG), H-085 (B-TI), H-083 (B-RP), and (OS3) give (b). H-081 (B-NT) gives Theorem 4.3. Only H-074 (B-C), H-082 (B-REG), H-085 (B-TI), H-083 (B-RP), and (OS3) are proved in this section; they use nothing beyond H-116 (H-U) and exact lattice symmetries.

4.5.1 H-084 (B-T2) Theorem 2: the running coupling

Needed. The hypothesis of Balaban's Theorem 1 (CMP 122 II p. 355), $g_k \in]0,\gamma]$ for all k, quoted in Section 1.2 [VERIFIED QUOTE]. It was to be discharged by Theorem 2 of CMP 109 p. 259 [8]: for small g there is $g_0(\varepsilon, g)$ with all $g_k \in]0,\gamma]$, $g_K = g$, and (0.31): $1/g^2 + \beta \log(L^k\varepsilon)^{-1} \le 1/g_k^2 \le 1/g^2 + \beta' \log(L^k\varepsilon)^{-1}$. Balaban: "has not been published yet" [VERIFIED QUOTE].

Status from Section 3 (and §10 of `THEOREM2_PROOF_v4.md`). The weak form, L-001 (Theorem 2w), is [PROVED+VERIFIED, CONDITIONAL] by Route K (L-009 (Theorem K) + L-005 (Lemma D″)) on R-025 (H2) and the fixed-level inputs R-026 (PI), R-027 (F), $R - 028(M_j)$, V1, R-030 (G2), plus L5 and item 7′ (H-202 (H0) at odd $L > 11$, whose use is a verified reduction with residue N1–N9 (Section 5.9), or H-111 (G-gain) with the running-maximum lift, verified (`SOURCES_GGAIN_NONMONO_VERIFY`)); item 9 (K-independence) is closed. It suffices for Theorem 1 by the cross-scale sweep [PROVED; VERIFIED IN ROUND 7 (LEDGER R-034)]. Round 7 proves $R - 028(M_j)$ and L5 relative to listed inputs (Theorems 5.14 and 5.15); R-027 (F) is reduced but unverified. Literal (0.31) is not obtained (it needs eventual positivity of $\beta_j(0)$). Route B (L-088 (Theorem B) of `BETA_FP_PW`) is a formally distinct second route; its logic is [PROVED+VERIFIED, CONDITIONAL], its inputs (U), (DF), (T1) are [PROVED, UNVERIFIED] and rest on R-025 (H2). The reduction pieces are L-004 (Lemma D′), L-005 (Lemma D″), and Lemmas D, E (the one-argument forms in Lean).

Conclusion. At $L = 13$, R-025 (H2) is certified (Theorem 6.1); H-084 (B-T2) remains conditional on the fixed-level inputs, the inputs of L5, and item 7′; see the dependency table (Section 4.6).

4.5.2 R-035 (B-S) Sourced bounds E14-a..g

Needed. A version of Balaban's Theorem 1 with the (C1) source, i.e. with the local bare coupling $w_p = 1/g_0^2 - f(x_p)$, f real, $|\partial^m f| \le r$ for $m \le 5$ (class F_r), uniformly in ε. Output form (`SOURCES.md` H-224 (N1)): $\log Z_\varepsilon[f] - W_\varepsilon[f]$ is C^∞ in source amplitudes z with $|\partial_z^\alpha(\log Z_\varepsilon[f] - W_\varepsilon[f])| \le C^{|\alpha|}\alpha!\, r^{-|\alpha|}$ for $f = \sum_i z_i f_i$ with disjoint supports. $W_\varepsilon[f]$ is a local field-independent contact functional (allowed to diverge; it contributes zero to every mixed derivative of order ≥ 2 for disjoint supports, `SOURCES_VERIFY` item 5(a)).

Estimates needed (`SOURCES.md` §2.5): H-096 (E14-a) sourced small-field step; H-097 (E14-b) sourced large-field step and R operation; H-098 (E14-c) derivative bounds in f. Not needed: H-099 (E14-d) fixed-torus convergence (= E13); H-102 (E14-g) tensor sources. H-100 (E14-e) RP and symmetry transfer is done in this section (Sections 4.5.6, 4.5.7). H-101 (E14-f) nontriviality at a fixed layer is needed only for Theorem 4.3. The current status of each sub-block is in the dependency table (Section 4.6); R-035 (B-S) remains a [HYPOTHESIS].

4.5.3 R-045 (B-N1) Bridge: source-form bounds ⇒ H-116 (H-U) [OPEN]

`SOURCES_VERIFY` item 5(b) [VERIFIED QUOTE]: the source form "does not give $|S_n(F)| \le C^n n! \|F\|_{s,n}$ for F flat on diagonals with a fixed s … Mark N1 for general F as a gap." This paper does not

close the gap. It gives a route [DERIVED SKETCH], which shows that the gap is removed by the OS index growth sn.

1. Moments from cumulants. $\langle\prod_i \phi(f_i)\rangle = \sum_{\text{partitions}} \prod_{\text{blocks}} \kappa(\cdot)$. Singleton cumulants vanish (φ is centred). For disjoint supports the source form bounds every cumulant by $C^{|B|}|B|!\, r^{-|B|}$, so the moment by $C^n n!\, r^{-n}$.

2. Scale-δ tests. A bump of side $\delta \le 1$ and amplitude a has $|\partial^5 f| \sim a\delta^{-5}$; it lies in F_r if $a \le cr\delta^5$. So n disjoint unit-amplitude scale-δ bumps have $|\text{moment}| \le C^n n!\, \delta^{-5n}$. This is lossy but polynomial in δ^{-1}.

3. Whitney decomposition. Split T^{4n} by a smooth dyadic partition in $\rho(x) := \min_{i<j} |x_i - x_j| \in [\delta, 2\delta]$, $\delta = 2^{-m} \ge \varepsilon$. On each piece cover by cubes of side $\delta/100$; cube tuples are pairwise disjoint. Expand $F\cdot$(cutoffs) in a Fourier series on each cube tuple; the coefficients decay faster than any power; each term is a product of disjointly supported scale-δ bumps. Flatness gives $|\partial^\beta F(x)| \le C_k \|F\|_{k+|\beta|} \rho(x)^k / k!$. The count is $\le C^n \delta^{-4n}$ cube tuples. Derivatives of the cutoffs cost at most $(Cn^2/\delta)^M$; with $M = O(n)$ this is $\le C^n (n!)^c \delta^{-M}$.

4. Bound per dyadic scale: $\le C^n (n!)^{D'} \delta^{k-9n-M} \|F\|_{k+M}$. Choose $k = sn - M$ with s large enough ($s \ge 20$ suffices on this count) so the δ-exponent is $\ge n$; the sum over dyadic $\delta \in [\varepsilon, 1]$ converges uniformly in ε.

5. Sub-lattice region $\rho(x) \lesssim \varepsilon$: $\le C^n \varepsilon^{-4n}$ plaquette tuples, each moment $\le 2^n$, and $|F| \le \|F\|_k \varepsilon^k$; this is $\le C^n \|F\|_{sn}$ for $k = sn \ge 4n$.

Result (if the steps hold): H-116 (H-U) with some fixed s and D. To check: uniformity of the source bound in the position of the bumps and in ε; the C^∞ real-source form of H-098 (E14-c) with n-th derivatives for all n; the cutoff-derivative count in step 3. The verifier lists further gaps: complex Fourier modes against real sources (a factor 2^n), the extra derivatives of Fourier summation (absorbed in s), and the smooth min-distance partition. Later work (`SOURCES_D4D5`, graded in `SOURCES_D4D5_VERIFY`) confirms "(S_{src}) $\Rightarrow$ H-116 (H-U) with $s = 6$, $D > 1$" given R-043 (FX) (see Section 4.6). `ASSEMBLY_N1.md` closes the three gaps given R-045 (S) (confirmed in `SOURCES_HHIST_N1_VERIFY.md`) and, since 2026-10-28, uses the verified form: C^1 norm $\max(\|f\|_\infty/r, \|\nabla f\|_\infty/\rho_1)$, order $(n!)^{1+\eta}$, $s = 6$, $D > 1$ ($D > 2$ under the fallback order $2 + \eta$); the ∂^5, δ^{-5n}, and $s \ge 20$ counts in the sketch above are superseded.

4.5.4 H-074 (B-C) Compactness [PROVED HERE, GIVEN H-116 (H-U)]

Proposition 4.4. *Under H-116 (H-U), every sequence $\varepsilon_m \to 0$ has a subsequence along which $S_{n,\varepsilon}(F)$ converges for every n and every $F \in \mathcal{D}_n$. The limit S_n is linear and satisfies $|S_n(F)| \le \sigma_n \|F\|_{sn}$.*

Proof. 1. Separability. $C^{sn}(T^{4n})$ with $\|\cdot\|_{sn}$ is separable (trigonometric polynomials with rational coefficients are dense; Fejér means converge in C^k). A subset of a separable metric space is separable. So $\mathcal{D}_n$ has a countable subset Q_n dense for $\|\cdot\|_{sn}$. 2. Diagonal extraction. $Q := \bigcup_n Q_n \times \{n\}$ is countable; enumerate it as $(F^{(1)}, n_1), (F^{(2)}, n_2), \ldots$. Each sequence $S_{n_t,\varepsilon_m}(F^{(t)})$ is bounded by $\sigma_{n_t} \|F^{(t)}\|_{sn_t}$ H-116 (H-U), so it has a convergent subsequence. Extract successively and take the diagonal subsequence ε_i. 3. Extension. Let $F \in \mathcal{D}_n$, $\eta > 0$, and $G \in Q_n$ with $\|F - G\|_{sn} \le \eta$. By H-116 (H-U) and linearity, $|S_{n,\varepsilon_i}(F) - S_{n,\varepsilon_j}(F)| \le 2\sigma_n \eta + |S_{n,\varepsilon_i}(G) - S_{n,\varepsilon_j}(G)|$. The last term $\to 0$. So the sequence is Cauchy; call the limit $S_n(F)$. 4. Linearity passes to pointwise limits, and $|S_n(F)| = \lim |S_{n,\varepsilon_i}(F)| \le \sigma_n \|F\|_{sn}$. □

Remarks. (i) The same subsequence serves all n; this matters for (OS2). (ii) Nothing here needs E13. (iii) The bounded normalizations B_K and c_ε can be included in the extraction.

4.5.5 H-082 (B-REG) Regularity (OS0) [PROVED HERE, GIVEN H-116 (H-U)]

By step 4 of the proof of Proposition 4.4 the limit satisfies $|S_n(F)| \le \sigma_n \|F\|_{sn}$, $\sigma_n = A(n!)^D$. On the compact torus the Schwartz seminorm is replaced by $\|F\|_p$ (the weights control decay at infinity on $\mathbb{R}^{dn}$; the torus has none). This is the quoted OS II form. Distribution property: for $F \in C_c^\infty$ of the off-diagonal open set, $F \in \mathcal{D}_n$ and the bound is continuity in the C^{sn} topology, so S_n is a distribution of order $\le sn$ there.

Caveat. OS II is formulated on $\mathbb{R}^{dn}$. On T^{4n} the axioms are used in their torus form (periodic test functions). This is a change of setting, not a consequence of OS II.

4.5.6 H-085 (B-TI) Translation and hypercubic invariance (OS1) [PROVED HERE, GIVEN H-116 (H-U)]

Lattice, exact. Put the origin at a lattice site.

- Translations $a \in \varepsilon\mathbb{Z}^4$. The map $U \mapsto U^a$, $U^a(b) := U(b-a)$, permutes the links, preserves the product Haar measure and the Wilson action, and $P_p(U^a) = P_{p-a}(U)$. So $\langle \prod_i \varphi_{p_i+a} \rangle_\varepsilon = \langle \prod_i \varphi_{p_i} \rangle_\varepsilon$ and $S_{n,\varepsilon}(F \circ \tau_a^{\times n}) = S_{n,\varepsilon}(F)$. The coupling $g_0(\varepsilon)$ is one number, so nothing in μ_ε depends on position.

- Hypercubic $R \in B_4$ (signed permutation matrices, $|B_4| = 384$). R maps sites to sites, links to links (possibly reversed), plaquettes to plaquettes, and $x_{Rp} = Rx_p$. Define $(RU)(b) := U(R^{-1}b)$ with $U(-b) = U(b)^{-1}$. Haar measure is invariant under $U \mapsto U^{-1}$ and $\operatorname{Re}\operatorname{tr} W^{-1} = \operatorname{Re}\operatorname{tr} W$ for $W \in SU(2)$, so A and the measure are invariant and $P_{Rp}(RU) = P_p(U)$. Hence $S_{n,\varepsilon}(F \circ R^{\times n}) = S_{n,\varepsilon}(F)$.

- These identities hold at every ε. $F \circ \tau_a^{\times n}$ and $F \circ R^{\times n}$ are again in $\mathcal{D}_n$. They pass to the limit along the subsequence of Proposition 4.4: $S_n(F \circ R^{\times n}) = S_n(F)$, and $S_n(F \circ \tau_a^{\times n}) = S_n(F)$ for $a \in \varepsilon_i\mathbb{Z}^4$ for all large i.

Continuous translations. Let $a \in T^4_\ell$. Pick $a_i \in \varepsilon_i\mathbb{Z}^4$ with $|a - a_i| \le \varepsilon_i$ (componentwise rounding gives $\le \varepsilon_i\sqrt{4}/2 = \varepsilon_i$). Write $F_a := F \circ \tau_a^{\times n}$. The mean value theorem along the diagonal direction gives $\|F_a - F_{a_i}\|_{sn} \le 4n|a - a_i|\,\|F\|_{sn+1}$. Then

$$|S_n(F_a) - S_n(F)| \le |S_n(F_a) - S_{n,\varepsilon_i}(F_a)| + |S_{n,\varepsilon_i}(F_a) - S_{n,\varepsilon_i}(F_{a_i})| + |S_{n,\varepsilon_i}(F_{a_i}) - S_{n,\varepsilon_i}(F)| + |S_{n,\varepsilon_i}(F) - S_n(F)|.$$

The first and last terms $\to 0$ by Proposition 4.4. The third is 0 by lattice invariance. The second is $\le \sigma_n \cdot 4n\,\varepsilon_i \|F\|_{sn+1}$ by H-116 (H-U) and linearity. Let $i \to \infty$. F is smooth, so $\|F\|_{sn+1} < \infty$. Rotations outside B_4 are not available (Section 4.7).

4.5.7 H-083 (B-RP) Reflection positivity (OS2)

Lattice theorem [CITED]: Osterwalder–Seiler [21], Ann. Phys. 110 (1978) 440–471. The paper is not in the project folder; no verbatim quote. Content as usually stated: for the Wilson action with compact gauge group and any $\beta \ge 0$, and θ the reflection in a lattice hyperplane that bisects the links orthogonal to it, $\langle \Theta(\Phi)\Phi \rangle \ge 0$ for every function Φ of the link variables in the positive half-lattice, $(\Theta\Phi)(U) := \overline{\Phi(\theta U)}$. Jaffe–Witten [15] line 546 [VERIFIED QUOTE]: "Reflection positivity holds for the Wilson approximation [36], a major advantage". The template uses the same input

(`o3_continuum.txt` line 4902) [VERIFIED QUOTE]: "At every cutoff the nearest-neighbour measure is reflection positive across microscopic site and link seams." Proof idea (standard): fix the crossing links to 1 by a gauge transformation; the action splits as $A_+ + \theta A_+ + \sum_{\text{crossing}} \operatorname{Re}\operatorname{tr}(X_p \theta(X_p)^*)$, and $\exp(\beta \operatorname{Re}\operatorname{tr}(XY^*))$ is a positive-type kernel (positive combination of characters), which gives a sum of squares. On the periodic lattice there are two fixed planes, $x_0 = \varepsilon/2$ and $x_0 = \varepsilon/2 + \ell/2$; both are link planes because N_ε is even (ℓ even, Section 2.5). The periodic two-seam version is standard [CITED]; its text was not checked. For site-plane reflections see Menotti–Pelissetto [17] (Section 4.8, item 1).

Passage to the limit [PROVED HERE, GIVEN H-116 (H-U) AND THE LATTICE THEOREM]. Setup. $\theta_\varepsilon(x) := \theta(x) + \varepsilon e_0 = \theta(x - \varepsilon e_0)$, fixed planes $x_0 = \varepsilon/2,\ \varepsilon/2 + \ell/2$. The positive half Λ_+ consists of links with both ends in $\{x_0 \in \{\varepsilon, 2\varepsilon, \dots, \ell/2\}\}$. Fix $F_0 \in \mathbb{C}$ and $F_n \in \mathcal{D}_n$, $\operatorname{supp} F_n \subset T_+^n$, $1 \le n \le N$. Supports are compact in the open set, so there is $\delta > 0$ with $x_0 \in [\delta, \ell/2 - \delta]$ on all supports.

Step 1 (lattice form). For $\varepsilon < \delta/2$ every plaquette with centre in a support has all four links in Λ_+. Define $\Phi := F_0 + \sum_{m=1}^N \sum_{p_1 \dots p_m} F_m(x_{p_1}, \dots, x_{p_m}) \prod_i \varphi_{p_i}$. Φ depends only on Λ_+ links. Since φ_p is real and $P_p(\theta_\varepsilon U) = P_{\theta_\varepsilon p}(U)$, re-indexing gives $\Theta\Phi = \bar{F}_0 + \sum_n \sum_q \overline{F_n}(\theta_\varepsilon x_{q_1}, \dots, \theta_\varepsilon x_{q_n}) \prod \varphi_{q_i}$. Hence

$$0 \le \langle \Theta\Phi\, \Phi \rangle_\varepsilon = \sum_{n,m=0}^{N} S_{n+m,\varepsilon}(\Theta_\varepsilon F_n \otimes F_m), \quad \Theta_\varepsilon F_n := (\Theta F_n) \circ \tau_{-\varepsilon e_0}^{\times n},\ S_{0,\varepsilon} := 1.$$

(Argument order inside ΘF_n is immaterial by (OS3).) Each $\Theta_\varepsilon F_n \otimes F_m$ lies in $\mathcal{D}_{n+m}$: its two factors have supports at positive distance and each factor is flat on its own diagonals.

Step 2 (seam shift). $\|\Theta_\varepsilon F_n \otimes F_m - \Theta F_n \otimes F_m\|_{s(n+m)} \le 2^{s(n+m)} \cdot 4n\,\varepsilon\, \|F_n\|_{s(n+m)+1} \|F_m\|_{s(n+m)}$. By H-116 (H-U) the difference of the $S_{n+m,\varepsilon}$ values is $\le \sigma_{n+m}$ times this, which $\to 0$.

Step 3 (limit). Along the subsequence of Proposition 4.4, $S_{n+m,\varepsilon_i}(\Theta F_n \otimes F_m) \to S_{n+m}(\Theta F_n \otimes F_m)$ for each of the finitely many (n, m). So $\sum_{n,m} S_{n+m}(\Theta F_n \otimes F_m) = \lim_i \sum_{n,m} S_{n+m,\varepsilon_i}(\Theta_{\varepsilon_i} F_n \otimes F_m) \ge 0$.

Lemma 4.5 (positivity of a finite form survives pointwise limits [PROVED HERE])**.** *Let $a^{(i)} \in \mathbb{C}^{(N+1)\times(N+1)}$ with $\sum_{n,m} \bar{c}_n c_m a^{(i)}_{nm} \ge 0$ for all c and all i, and $a^{(i)}_{nm} \to a_{nm}$ for every (n, m). Then $\sum \bar{c}_n c_m a_{nm} \ge 0$ for all c.*

Proof. For fixed c the left side is a finite linear combination of the entries, so it converges to $\sum \bar{c}_n c_m a_{nm}$; $[0, \infty)$ is closed. □

Step 3 is this lemma with $a^{(i)}_{nm} := S_{n+m,\varepsilon_i}(\Theta_{\varepsilon_i} F_n \otimes F_m)$ (replace F_n by $c_n F_n$), plus Step 2.

4.5.8 (OS3) Symmetry and reality [PROVED HERE]

$S_{n,\varepsilon}(F \circ \pi) = S_{n,\varepsilon}(F)$: the φ_p are commuting real random variables; re-index the sum. For real F, $S_{n,\varepsilon}(F) \in \mathbb{R}$. Both identities are linear and pass to the limit. $S_0 = 1$ by definition.

4.5.9 H-081 (B-NT) Nontriviality: what is needed

- A two-point function alone never excludes a Gaussian: every positive-definite S_2 is the covariance of some Gaussian generalized field.
- The plaquette density is a composite ($\approx \frac{1}{4} \operatorname{tr} F^2 \times \varepsilon^4$). Even in the free theory its connected $n \ge 3$ functions are nonzero (Wick squares). So $S_n^c \ne 0$ for some $n \ge 3$ does not prove interaction.

- The meaningful test excludes (a) the zero theory and (b) the free Maxwell composite. The free composite has, at short distance, the scale-invariant form $c_0 d^{-8}$. Asymptotic freedom predicts $c\, g(d)^4 d^{-8}$ with a running coupling.

Free-theory prediction along this sequence. At tree level with the bare coupling, $\langle \varphi_p \varphi_{p'} \rangle \approx c\, \varepsilon^8 g_0^4 |x-y|^{-8}$. Since $g_0(\varepsilon)^2 \le (1/g^2 + \beta \log \varepsilon^{-1})^{-1} \to 0$ by (0.31), the bare-coupling prediction for S_2 at fixed separation tends to 0. The claim is that the true S_2 does not.

Statement (NT) needed [HYPOTHESIS]; owned by H-101 (E14-f), a corollary of H-096 (E14-a) at a fixed layer. There are $c > 0$ and C such that for all K, all $f_1, f_2 \in C^\infty(T^4)$ with disjoint supports at distance $\ge d$, $d \ge L\varepsilon$, and $k(d)$ the layer with $L^k \varepsilon \le d < L^{k+1}\varepsilon$:

$$S_{2,\varepsilon}(f_1 \otimes f_2) = c \iint f_1(x) f_2(y)\, g^4_{k(|x-y|)} |x-y|^{-8}\, dx\, dy\, (1+R), \quad |R| \le C(g^2_{k(d)} + (d/\ell)^2).$$

c is the one-loop (tree-level in the composite) coefficient: $c \propto \sum$(free field-strength covariance)^2, a sum of squares, so $c > 0$ [DERIVED, STANDARD]; its value must be computed in Balaban's normalization.

Derivation of Theorem 4.3 from (NT) and H-084 (B-T2) [PROVED HERE, GIVEN (NT)]. (i) Take $f_1, f_2 \ge 0$, not identically 0, with supports at distance $\ge d_0$. Corrected condition (Section 4.8, item 3): the support diameters are $\ll \ell$, so that $g^2 + (|x-y|/\ell)^2$ is small on the whole support pair and $|R| \le \frac{1}{2}$. By (0.31) at layer $k(d)$ with $d \in [d_0, \ell]$: $1/g_k^2 \le 1/g^2 + \beta' \log(L^k\varepsilon)^{-1} \le 1/g^2 + \beta' \log(L/d_0)$, uniformly in ε. So $S_{2,\varepsilon}(f_1 \otimes f_2) \ge \frac{1}{2} c (1/g^2 + \beta' \log(L/d_0))^{-2} \iint f_1 f_2 |x-y|^{-8} > 0$ uniformly in ε, and the limit inherits the bound. (ii) For bumps concentrated at separation $d \to 0$ (with d a fixed multiple of their width), $d^8 S_2 \le (1 + C g^2_{k(d)})\, c\, (1/g^2 + \beta \log(1/d))^{-2} \to 0$ as $d \to 0$ by the lower bound in (0.31), while $d^8 S_2 > 0$ by (i). A scale-invariant composite would give a nonzero constant limit. So the limit is neither zero nor the free composite. Only the two-sided bound (0.31) is used. Caveat. (ii) needs (NT) uniformly down to $d \ge L\varepsilon$ with the remainder controlled by $g^2_{k(d)}$, i.e. the fixed-layer expansion at every layer. If H-101 (E14-f) is proved only at a fixed layer j, (i) holds, and (ii) holds only in the weaker form "$d^8 S_2(d)$ is bounded between the two (0.31) envelopes for $d \in [L^{-j}, 1]$".

4.6 Dependency table

The table gives the status of every block of Theorem B and of every named hypothesis of Theorem A. It follows the state recorded in `PLAN.md` (through the entry `SOURCES_S2_G1_V7_VERIFY.md`, 2026-10-28) and §10 of `THEOREM2_PROOF_v4.md`. Global conditions carried by every row: H-200 (GC1) Balaban's published proofs CMP 95–122 (1984–89) are correct as published; nothing in this paper re-checks them line by line. (GC2) Balaban's inductive assumptions (CMP 109 pp. 263–264) hold with the stated uniformity. H-201 (GC3) L odd, $L > 11$, M large, γ small as Balaban requires; ℓ even (sufficient, Section 2.5). Rows "item n" are the named hypotheses of Section 3; n is their item in Section 7. Their file columns use the short names of `THEOREM2_PROOF_v2_VERIFY` §K (KAPPA = `KAPPA_BOUNDED`, H_VERIFY = `BETA_FP_H_VERIFY`, PW_VERIFY = `BETA_FP_PW_VERIFY`, v1 verifier = `THEOREM2_PROOF_VERIFY`).

Block	Content	Current tag	Files	Hypotheses carried
B-T2	L-001 (Theorem 2w) (Theorem 3.1): trajectory $g_0(\varepsilon, g)$, (b), (c′), K-018 (S)	[PROVED+VERIFIED, CONDITIONAL] (Route K = L-009 (Theorem K) + L-005 (Lemma D″)). Literal (0.31) not obtained	`THEOREM2_PROOF_v4.md`, `KAPPA_BOUNDED.md`	R-025 (H2), R-026 (PI), R-027 (F), $R - 028(M_j)$, V1, R-030 (G2), L5, item 7′ (from H-202 (H0) or H-111 (G-gain), verified (`SOURCES_GGAIN_NONMONO_VERIFY`))
B-T2.a	Lemmas D, D′, D″, E (shooting)	[PROVED+VERIFIED; LEAN] (D′, D, E, D″ one-argument form; history form open)	`LEMMA_DPRIME.md`, `lean/`	continuity of β_{k+1} in history
B-T2.b	Sweep: (c′)+K-018 (S) ⇒ Balaban's Theorem 1 inputs (Section 3.4.3; item 10)	[PROVED, UNVERIFIED]; 4 of 19 rows verified (LEMMA_DPRIME_VERIFY Steps 4–5, rows 11, 12, 16, 18)	`LEMMA_DPRIME_SWEEP.md`	Balaban readings (unchecked list)
B-T2.1	U3 Thm 4.1 (Gaussian RG, rate L^{-2j}) + O5	[PROVED+VERIFIED]	`U3_GAUSSIAN_RG.md`, `U3_REPAIRS.md`	trivial background
B-T2.2	U3 Lemma 5.1; (A), (W)	neutral sector [PROVED+VERIFIED]. (W) numerically exact; (A) confirmed qualitatively and quantitatively (`U3_AW_VERIFY.md`); (HS) [REFUTED] at $L = 3$. Not on the path of L-001 (Theorem 2w) after R-025 (H2); needed for the convergence route only	`U3_LEMMA51.md`, `U3_AW.md`	(A) in full, (N) (plausibly false)
B-T2.3	O6 bordered-matrix invertibility	one-step [PROVED+VERIFIED]; composed constraint reduced to one-step O6 + (A)	`U3_REPAIRS.md`	—
B-T2.4	$c = 11N^2/(12\pi^2)$, anchor-free identity Prop. 2.1, telescoping	[PROVED+VERIFIED]. $\beta_\infty = c \log L$ itself replaced by window sums (L-009 (Theorem K), Route B)	`BETA_FP_LEMMA43.md`, `BETA_FP_LEMMA43_VERIFY.md`	R-027 (F) [CITED, SOURCE CHECKED]

Block	Content	Current tag	Files	Hypotheses carried
B-T2.5	κ_k bounded (L-009 (Theorem K))	[PROVED+VERIFIED, CONDITIONAL ON (B2)]; (B2) [PROVED+VERIFIED, CONDITIONAL ON R-025 (H2)]	`KAPPA_BOUNDED.md`, `KAPPA_BOUNDED_VERIFY.md`, `B2_TWOSTEP.md`, `B2_TWOSTEP_VERIFY.md`	R-025 (H2), R-026 (PI), R-027 (F), $R-028(M_j)$, V1, R-030 (G2)
B-T2.H2	R-025 (H2) uniform two-step inequality $\rho_2 < L$ (Section 3.7.3; item H)	[CERTIFIED (INTERVAL ARITHMETIC, OUTWARD ROUNDING; VERIFIED)] at $L = 13$: $\rho_2 \le \sup \rho_1^2 \le 10 < 13$ (Theorem 6.1); residue: reading of (0.4), gauge choice, Prop. A at $M \ge 27$, L-169 (Lemma G) beyond point checks, `bb_arb.py` audited only. At $L = 3, 5, 9$ [NUMERICAL]; point enclosure $[1.9993, 2.1273] < 3$ at $L = 3$ [INTERVAL ENCLOSURE, VERIFIED]	`H1_ANALYTIC.md`, `H1_ANALYTIC_VERIFY.md`, `H1_ANALYTIC_VERIFY2.md`, `H2_CERTIFY_P5_VERIFY.md`, `H2_LARGE_L_VERIFY.md`, `H2_CERTIFY_VERIFY.md`; R-026_TWOSTEP §2	— (box sweep at $L = 13$ running as a cross-check)
B-T2.H1	one-step route at $L = 13$: $\sup \rho_1^2 < L \Rightarrow$ R-025 (H2)	[CERTIFIED (INTERVAL ARITHMETIC, OUTWARD ROUNDING; VERIFIED)]: $\sup_{j,K} \rho_1^2 \le 9.9995 < 13$ by the majorant B_{13} and `bb_arb.py`; factorization [PROVED+VERIFIED; LEAN] (`rho2_le`); majorant core [LEAN] (`Tower.lean`)	`H1_ANALYTIC.md`, `H1_ANALYTIC_VERIFY.md`, `H1_ANALYTIC_VERIFY2.md`, `H2_CERTIFY_VERIFY.md`	—
B-T2.B	Route B (Theorem B_{PW}; item B8)	logic [PROVED+VERIFIED, CONDITIONAL] (tag decision, `PLAN.md`; PW_VERIFY §6 items 1–4); inputs (U), (DF), (T1) [PROVED, UNVERIFIED]	`BETA_FP_PW.md`, `BETA_FP_PW_VERIFY.md`, `BETA_FP_U.md`	R-025 (H2), $(B\theta)$, R-007, LW (5.7), H-202 (H0)
item 1	(B2) / H-184 (B2w) (Section 3.7)	[VERIFIED REDUCTION] to R-025 (H2); holds at $L = 13$ (Theorem 6.1); open at other L	KAPPA §2, Rem. 2.3; KAPPA_VERIFY §7–§9	R-025 (H2)
item 1	R-025 (H2) $\Rightarrow$ $(C1\sigma)$ $\Rightarrow$ (B2) (Theorems 3.10, 3.11)	proved+verified, conditional on R-025 (H2)	R-026_TWOSTEP §2–§3; R-026_TWOSTEP_VERIFY grades 1, 3, 4	R-025 (H2)
item 1	$(C1\sigma)$ + (h) $\Rightarrow$ (B2)	implication verified; $(C1\sigma) \Leftarrow$ R-025 (H2); (h) numerical $L = 3, 5$	KAPPA Prop. 4.1; KAPPA_VERIFY §7	—

Block	Content	Current tag	Files	Hypotheses carried
item 1	(H″) ⇒ (A) ⇒ (B2)	implication argued; not needed after R-025 (H2)	U3_AW §4.7; no verifier (U3_AW_VERIFY grade 5 covers (H′) only)	—
—	(HS) (Section 3.7.2)	refuted $L = 3$	U3_AW (H′); U3_AW_VERIFY §6	—
item H	$\rho_1^2 < L \Rightarrow$ R-025 (H2) (one step, $L \geq 5$)	arithmetic; $\rho_1^2/L = 0.771$ at $L = 5$, numerical	H2_LARGE_L §3; H2_LARGE_L_VERIFY grade 4 (measurement)	—
item H	certification tail j_0	derivation confirmed; $j_0 \approx 8$ ($L = 3$), ≈ 6 ($L = 5$), corrected	H2_LARGE_L §4.1; H2_LARGE_L_VERIFY grade 6	—
item 2	R-026 (PI)	argued	BETA_FP_H §3.1; H_VERIFY §5	—
item 3	R-027 (F)	[VERIFIED REDUCTION] (round 8) to H-137 (L-LL), a kernel bound H-148 (MPT), and Step C's gauge-fixing part; round 9: H-137 (L-LL) proved+verified (Theorem 5.25); a proof of R-027 (F) without Lemma 2 or Step C, verifier running (Theorem 5.26); open tension β_1 vs $c \log L$ (Section 6.9)	BETA_FP_LEMMA43; H_VERIFY §4	—
item 4	(M_j): H-244 (S1)–H-247 (S3), Prop. 4.4	H-244 (S1), H-247 (S3), H-245 (S2)$\|_{t=0}$, H-246 (S2c) proved at fixed j given (I3); (I3) [PROVED+VERIFIED] (`SOURCES_T2_I3_VERIFY`, round 5), given the identification of the one-step objects; $R - 028(M_j)$ [VERIFIED REDUCTION] to (I3), Prop. 4.4 region 1 (argued), and the neutral sector [I]; round 6: (G'_A) [PROVED+VERIFIED] for $0 < \|f\| \leq ((1-c)/100)^2/2$ (Theorem 5.12); round 7: (M_j) [PROVED RELATIVE TO LISTED INPUTS; VERIFIED] at fixed j (Theorem 5.14), the proved f-range suffices	BETA_FP_M §4; `SOURCES_T2_S123.md`, `SOURCES_T2_S123_VERIFY.md`; H_VERIFY §5 refuted the old step; `SOURCES_T2_I3.md`, `SOURCES_T2_GAPRIME2.md`	one [I] convention (KAPPA (1.1) coefficients)

Block	Content	Current tag	Files	Hypotheses carried
item 5	V1	[PROVED+VERIFIED AT FIXED LEVEL, CONDITIONAL] (history-free)	SOURCES_T2_V1.md, SOURCES_T2_V1_VERIFY.md; R2; U3_VERIFY C4	R-030 (G2), Bałaban's inductive bounds, averaging (0.4)
item 6	R-030 (G2)	[VERIFIED REDUCTION] at fixed j (was formal)	SOURCES_T2_G2.md, SOURCES_T2_G2_VERIFY.md; BETA_FP_LEMMA43	$\bar{Q}$ onto, (Gap_f), averaging analyticity
item 7′	$\lvert\beta_{k+1}(h,g) - \beta_{k+1}(0)\rvert \le C_\beta g$ on every past (was H-202 (H0), uniform in j and history)	from H-202 (H0) [BALABAN], odd $L > 11$ (its use a [VERIFIED REDUCTION], residue N1–N9, N2 = G3 proved relative to listed inputs, Section 5.9), or from H-111 (G-gain) with the running-maximum lift [PROVED+VERIFIED]	CMP 109 (1.22); SOURCES_GGAIN_NONMONO.md, SOURCES_GGAIN_NONMONO_VERIFY.md; KUNIF_VERIFY item 3; SOURCES_T2_H0.md, SOURCES_G3_N2.md	N1, N3–N9 of H0-red; H-111 (G-gain) inputs if taken from it
item 8	L5 history form and domain (Section 3.4.5)	round 7: [PROVED RELATIVE TO LISTED INPUTS; VERIFIED], Route 2 (Theorem 5.15)	LEMMA_DPRIME Rem. 3; LEMMAS_DE_VERIFY §2.1; v2_VERIFY §F	—
item 9	$\beta_j(g)$ independent of K	closed as a reduction [VERIFIED]: volume statement, β_j whole-lattice by definition, D′/D″ at fixed K; residual volume question = ledger R-032, an input of Theorem 1; torus versus whole-lattice coefficient within $C_j N_j^6 e^{-m_j N_j/2}$ (Proposition 5.8; H-141 (Loc) proved in round 6, Theorem 5.10); R-032 replaced and proved under K-032 (Vol) (Proposition 5.11)	SOURCES_T2_KUNIF.md, SOURCES_T2_KUNIF_VERIFY.md	(K1), (K2) = item 7′, (K3)
item 10	BALABAN_CHECK item 11 readings	verified with corrections (former ledger R-048, R-049, R-051); R-012 closed as a verified list (Proposition 5.7; K-029–K-031); residue: R-020 (T-step rate loss)	BALABAN_EXCERPTS; SOURCES_BOOKKEEPING.md, SOURCES_A11_CONST.md	R-020
item 11	odd L; $L > 11$ in Balaban	input	CMP 109 pp. 251–252; v1 verifier Step 6	—
item 12	SU(2)	scope	—	—
item 13	Lean history form	open	—	—

Block	Content	Current tag	Files	Hypotheses carried
B1, B2′	Route B: R-025 (H2) + (T1) ⇒ (U) (Theorem 2.4 of BETA_FP_U)	proved, unverified	BETA_FP_U §2.4; no verifier	R-025 (H2), (T1)
B1, B2′	Route B: (U)	hypothesis; ⇐ (T1) + R-025 (H2) (BETA_FP_U); proved, unverified	BETA_FP_PW §1.2; PW_VERIFY §3	(T1), R-025 (H2)
B1	Route B: (T1)	argued	BETA_FP_U §2.3; no verifier	—
B3	Route B: $(B\theta) = \gamma_{rec} + \mathfrak{d}_F$	γ_{rec} bounded given Thm 2.4 (proved, unverified); (DF) argued	BETA_FP_PW §3.3; BETA_FP_U §3; PW_VERIFY §6 (logic of Theorem B_{PW}); none for BETA_FP_U	(DF)
B4	Route B: LW (5.7)	cited	cited; PW_VERIFY §3 (regime correct)	—
B5	Route B: R-007 (β identification, uniform in k)	argued	SOURCES_D1_VERIFY 8; PW_VERIFY §3; BETA_FP_U §4.2	—
B7	Route B: $\theta_* < 1$	trivial condition	PW_VERIFY §3	—
B-T2.6	certified early steps $\beta_j(0) > 0$	needed only for literal (0.31) at every K, not for Theorem 1; [NUMERICAL], no certificate; $\beta_1(0)$ computed to $L = 11$	`numerics/NUMERICS.md`, `UMK_BETA1.md`	$L > 11$ is Balaban's range
B-S	sourced Theorem 1 (E14-a,b,c)	[OPEN]	`SOURCES.md` §2.5	B-T2 for the sourced trajectory
B-S.D1	one sourced step, Gaussian level	[PROVED+VERIFIED] (with corrections)	`SOURCES_D1.md`, `SOURCES_D1_VERIFY.md`	—
B-S.D2	sourced small-field step H-096 (E14-a)	(a), (b), (d), (f) confirmed; (c) rejected; (e) reduced	`SOURCES_D2.md`, `SOURCES_D2_VERIFY.md`	analyticity in input activities; $\kappa \to \kappa/2$ [R, verified, R-046; uses R-020]; K-024 (U) [K-024, constant choice]
B-S.D2C	redo of (c): frozen source per domain, L-054 (Lemma A), $h \equiv 0$	L-054 (Lemma A) and recursion confirmed; R-036 (I) corrected (exact only with adapted gauge-fixing weight)	`SOURCES_D2C.md`, `SOURCES_D2C_VERIFY.md`	domain hypothesis (D); (Hβ)
B-S.Hβ	two-loop bound $\sup \sum_j \lvert B'_j \rvert \le \frac{1}{2}$	[PROVED+VERIFIED, CONDITIONAL] on R-036 (I), R-037 (H-hist)	`SOURCES_HBETA.md`, `SOURCES_HBETA_VERIFY.md`	R-036 (I), R-037 (H-hist)

Block	Content	Current tag	Files	Hypotheses carried
B-S.hist	R-037 (H-hist) history sensitivity	[VERIFIED REDUCTION] on R-044 (W4), Facts A/B, R-036 (I)/L3 and H-111 (G-gain) in both slots (`SOURCES_SM.md`; verifier: items 1–4 confirmed, 5–6 corrected); the H-111 (G-gain) chain continues in row B-S.Ggain (G3, (Q)); with the large-field half (`SOURCES_GGAIN_LF_VERIFY`) the real uniform R-037 (H-hist) is a [VERIFIED REDUCTION] in both halves (ledger R-037); L-189 (Lemma HH): `SOURCES_HHIST.md`, `SOURCES_HHIST_N1_VERIFY.md`	`SOURCES_SM.md`, `SOURCES_SM_VERIFY.md`	H-111 (G-gain), small and large field
B-S.Ggain	H-111 (G-gain): $g_k\|B\|$ gain in the small-field step	pointwise Cauchy and uniform C_β on $]0,\gamma]$ [PROVED+VERIFIED] (small-field, incl. vacuum subtraction); L-178 (Lemma G4) off-chain (`SOURCES_G4.md`); χ_k surface term = L-174 (Lemma G3), [PROVED RELATIVE TO LISTED INPUTS; VERIFIED] with (Q) assembled from R-008, R-009 (Theorem 5.9, round 5); earlier route: (Q) reduced to CMP 99 (3.132), (E_U), and the localization lemma LOC [VERIFIED REDUCTION] (`SOURCES_Q_VERIFY`, `SOURCES_CDC_VERIFY`, `SOURCES_EU_VERIFY`, `SOURCES_LOC_VERIFY`); large field [VERIFIED REDUCTION] (one Cauchy step on the large-field disc, `SOURCES_GGAIN_LF_VERIFY`); non-monotone pasts: running-maximum lift (`SOURCES_GGAIN_NONMONO.md`, verified (`SOURCES_GGAIN_NONMONO_VERIFY`))	`SOURCES_GGAIN.md`, `SOURCES_GGAIN_VERIFY.md`, `SOURCES_GGAIN_LF.md`	(M_r), K-026, K-027, CMP 102 (189), H-145 (M-B) bookkeeping; not at $g = 0$

Block	Content	Current tag	Files	Hypotheses carried
B-S.G3	L-174 (Lemma G3): χ_k surface term, uniform in g	[PROVED+VERIFIED, CONDITIONAL] on (Q) (Route 2 surface estimate; the B/B′ cut closed in `SOURCES_Q`); round 5: [PROVED RELATIVE TO LISTED INPUTS; VERIFIED] (Theorem 5.9; G_Γ bounded uniformly in M, verified)	`SOURCES_G3.md`, `SOURCES_G3_VERIFY.md`, `SOURCES_G3_N2.md`, `SOURCES_G3_GGAMMA.md`	R-008, R-009, (2.15)–(2.25) chain, K-026, K-027, N3, N6, N9, (M_r); [I] s-rule reading
B-S.Q	(Q): covariance lower bound $c_- > 0$	[VERIFIED REDUCTION] to CMP 99 (3.132) (printed; ledger R-001, kernel-decay form now a verified reduction via L-125 (Lemma CT) to CMP 99 Thms 3.1, 3.3, 3.12, `SOURCES_S2_FORM_VERIFY`), (E_U), and LOC; Woodbury identity confirmed (CDC); free-case constants [NUMERICAL] (`numerics/kappa_d.py`); round 5: (Q) in the joint (X, B) form from R-008 + R-009 (Theorem 5.9), with Λ corrected (G_Γ enters)	`SOURCES_Q.md`, `SOURCES_Q_VERIFY.md`, `SOURCES_CDC.md`, `SOURCES_CDC_VERIFY.md`, `numerics/KAPPA_D.md`	R-001 residue (round 9: H-057 (C) a verified reduction, Proposition 5.23; torus-side hypothesis discharged by K-032 (Vol); printed inputs R-021, conventions K-044, reading $a = 1$; G_1 clause closed, `SOURCES_S2_G1_V7_VERIFY`); two [I] bridges
B-S.EU	(E_U): uniform block extension	$K_0(4) = 195.19$ ($N = \infty$) [NUMERICAL, REPRODUCED]; $K \le K_0(1 + c_{2/3}(M\alpha_0)^{2/3} + c_1 M\alpha_0)$ [DERIVED SKETCH]; (E_U) [VERIFIED REDUCTION] to kernel-moment bounds Bałaban does not print	`SOURCES_EU.md`, `SOURCES_EU_VERIFY.md`, `numerics/k0_d.py`	k-uniform kernel moments; unprinted C_1'

Block	Content	Current tag	Files	Hypotheses carried
B-S.LOC	LOC: localized covariance $C^{(k)}(Z_0, 0)$	[PROVED+VERIFIED, CONDITIONAL] (short lemma; symmetry (S) not needed)	`SOURCES_LOC.md`, `SOURCES_LOC_VERIFY.md`	R_2 bound (2.16) (ledger R-003, verified reduction with the V7 repair to (W^c) in the exponential gauge + R-008 at real U + two [I]); γ_0 for general U (ledger R-004, full proof verified, $\gamma_0 \geq m_L/(48d^2L^d)$); α_0, α_1 small
B-S.thr	R-039 (H-thr) threshold shell sum	L-285 (Lemma T) and trajectory sum [PROVED+VERIFIED], conditional on R-038 (H-hist2); R-038 (H-hist2) [VERIFIED REDUCTION] to the real uniform R-037 (H-hist), no new input (`SOURCES_HHIST2`, `SOURCES_HHIST2_VERIFY`)	`SOURCES_HTHR.md`, `SOURCES_HTHR2.md`, `SOURCES_HTHR2_VERIFY.md`	R-038 (H-hist2), M3 weights, frozen integers (L3)
B-S.hist2	R-038 (H-hist2): mixed second history derivatives	[VERIFIED REDUCTION] to the real uniform R-037 (H-hist), no new input (Route 2(a); decay $\lambda_C^{m/2}$; the product form λ^m [REFUTED])	`SOURCES_HHIST2.md`, `SOURCES_HHIST2_VERIFY.md`	real uniform R-037 (H-hist)
B-S.D3	sourced large-field step + R operation H-097 (E14-b)	draft verified with corrections (`SOURCES_D3_VERIFY`, `SOURCES_D3R_VERIFY`, `SOURCES_D3W8_VERIFY`; route (a), L-164 confirmed given R-044 (W4), (Eps), F1, (D)). R-043 (FX) is a genuine new hypothesis (`SOURCES_D4D5_VERIFY.md`); its state is in row B-S.FX	`SOURCES_D3.md`, `SOURCES_FX.md`, `SOURCES_FX_VERIFY.md`	R-043 (FX), H-191 (FX2); K-024 (U) [K-024, constant choice]

Block	Content	Current tag	Files	Hypotheses carried
B-S.FX	Q-059(b), Q-060, Q-061, Q-062, R2	Q-059(b) [PROVED+VERIFIED] (Lemma [P], no additive constant); Q-060 [PROVED+VERIFIED] (the R-denominator is the bare Wilson integral (1.100)); Q-061 [PROVED+VERIFIED, CONDITIONAL ON N3] (N3 = the open HBETA strip bound $\|\delta w\| \le \frac{3}{2}\tau$); R2 ($\kappa_{fin} \ge \kappa_{final}$) [PROVED+VERIFIED] as a reduction, in volume form, given $R_{K+1} \ge 1$ and C_c independent of g; Q-062 (unbounded components in (1.103)) avoided by keeping Balaban's main form (1.98), whose fallback was Gevrey $2+\eta$. N3 reduced to R-040/R-041 (H-hol) + real R-037 (H-hist) + H-120 (H2-hist), condition (a) settled [VERIFIED REDUCTION] (`SOURCES_HBETA_N3_VERIFY`, `SOURCES_N3A_VERIFY`); R-040 (H-hol) small field [VERIFIED REDUCTION] (`SOURCES_HHOL_VERIFY`); large field [VERIFIED REDUCTION] to H-143 (M-A), H-145 (M-B), H-146 (M-C), R-038 (H-hist2) (`SOURCES_HHOL_LF_VERIFY`); H-145 (M-B) [PROVED+VERIFIED] with $p_0 \ge 17/(\gamma_0 A_1^2)$, $\theta_W + \tau_C \le 0.2$; H-143 (M-A) reduced to (H−), H-144 (M-A′) closed, (H−) rests on CMP 102 (189) [VERIFIED REDUCTION] (`SOURCES_MA_VERIFY`, `SOURCES_MA2_VERIFY`); H-146 (M-C) [VERIFIED REDUCTION] to printed CMP 116 §2 conditions (`SOURCES_MC_VERIFY`); $R_{K+1} \ge L$ and C_c free of g are verified readings (`SOURCES_BOOKKEEPING.md`)	`SOURCES_FX_O1.md`, `SOURCES_FX_O1_VERIFY.md`, `SOURCES_FX_O23.md`, `SOURCES_FX_O23_VERIFY.md`, `SOURCES_FX_R2.md`, `SOURCES_FX_R2_VERIFY.md`	residual inputs of H-146 (M-C); CMP 102 (189) (ledger R-005, verified reduction to R-001)

Block	Content	Current tag	Files	Hypotheses carried
B-S.N3	N3: HBETA strip bound $\lvert\delta w\rvert \le \frac{3}{2}\tau$	[VERIFIED REDUCTION] to R-040/R-041 (H-hol) + real R-037 (H-hist) + H-120 (H2-hist)	`SOURCES_HBETA_N3.md`, `SOURCES_HBETA_N3_VERIFY.md`	R-040/R-041 (H-hol), R-037 (H-hist), H-120 (H2-hist)
B-S.N3A	condition (a) of L-232 (Lemma N3)	settled [PROVED+VERIFIED] (slack $> 5\times$; item 1 corrected)	`SOURCES_N3A.md`, `SOURCES_N3A_VERIFY.md`	constant order $L \to$ Bałaban's constants $\to \gamma \to g \le \gamma/2 \to r_0 \to K$
B-S.hol	R-040 (H-hol), small field: β_j holomorphic on $\lvert g - x\rvert < \rho_0 x$	[VERIFIED REDUCTION] to Bałaban's conditions with modified constants	`SOURCES_HHOL.md`, `SOURCES_HHOL_VERIFY.md`	(C1′)–(C4′), H-185 (C4″), (2.24′)
B-S.holLF	R-041 (H-hol), large field (frozen thresholds at x^*)	[VERIFIED REDUCTION] to H-143 (M-A), H-145 (M-B), H-146 (M-C), R-038 (H-hist2); H-146 (M-C) itself a verified reduction	`SOURCES_HHOL_LF.md`, `SOURCES_HHOL_LF_VERIFY.md`	H-143 (M-A)–H-147 (M-D)
M-A	equipartition $\langle A - A_{\min}\rangle_u \le (n_i/u)(1+\eta_x)$	[PROVED+VERIFIED] given (H−); H-144 (M-A′) closed [PROVED+VERIFIED]; (H−) [VERIFIED REDUCTION] to CMP 102 (189), an existing trust item	`SOURCES_MA.md`, `SOURCES_MA_VERIFY.md`, `SOURCES_MA2.md`, `SOURCES_MA2_VERIFY.md`	CMP 102 (189); (C1), (C2)
M-B	(1.80)–(1.98) bookkeeping, changed constants	[PROVED+VERIFIED] with $p_0 \ge 17/(\gamma_0 A_1^2)$, $\theta_W + \tau_C \le 0.2$	`SOURCES_MB.md`, `SOURCES_MB_VERIFY.md`	H-143 (M-A), H-146 (M-C)
M-C	complex small-field step	[VERIFIED REDUCTION] to printed CMP 116 §2 (B1–B23 + 4), Lemma 2, (1.36), p. 12 (2.5)–(2.6); shrinks $\varepsilon_1/(4\lambda_V)$, $\gamma/(8\lambda_V)$; H-134 (I-X) resolved; H-133 (I-V0), H-131 (I-ord), and (I-ε) in both layers closed [PROVED+VERIFIED]; K-025 (M1) is convention K-025	`SOURCES_MC.md`, `SOURCES_MC_VERIFY.md`, `SOURCES_MC_IV0.md`, `SOURCES_T2_IEPS.md`, `SOURCES_T2_IEPS_LF.md`, `SOURCES_HIGHER_M1.md` (each + VERIFY)	[I] residue: unprinted α_2–α_4 scaling, q_0 vs p_0 (every branch closes)
M-D	frozen map vs Bałaban's map	not a convention: needs the frozen map's derivative at x^*, which is R-037 (H-hist) + R-039 (H-thr) [VERIFIED REDUCTION]	`SOURCES_HHOL_LF_VERIFY.md`	R-037 (H-hist), R-038 (H-hist2)

Block	Content	Current tag	Files	Hypotheses carried
B-S.D4	real-source derivative bounds H-098 (E14-c)	D4.1, D4.3, D4.4 and L-111 (Lemma C) (order $1+\eta$, given R-043 (FX)) [PROVED+VERIFIED]; (S_{src}) $\Rightarrow$ H-116 (H-U), $s=6$, $D>1$, confirmed given R-043 (FX)	`SOURCES_D4D5.md`, `SOURCES_D4D5_VERIFY.md`	R-043 (FX), H-191 (FX2), S1 at all scales
B-N1	source form $\Rightarrow$ H-116 (H-U) on $\mathcal{D}_n$	derived given the sourced Gevrey bound R-045 (S) (`ASSEMBLY_N1.md`; confirmed, `SOURCES_HHIST_N1_VERIFY.md`); (S_{src}) $\Rightarrow$ H-116 (H-U), $s=6$, $D>1$, confirmed given R-043 (FX); uniformity of R-045 (S) in support size and ε [VERIFIED REDUCTION] (h-independent; `SOURCES_B21_S_VERIFY`); ASSEMBLY_N1 aligned 2026-10-28 with the C^1 norm, $(n!)^{1+\eta}$, $s=6$, $D>1$ ($D>2$ if the coarsest-scale H-191 (FX2) radius fails); open: the H-098 (E14-c) clause	this paper, `SOURCES_D4D5.md`, `SOURCES_B21_S.md`	R-043 (FX) main form, coarsest-scale H-191 (FX2) radius, S1 at all scales, L-111 (Lemma C)
B-C	compactness (diagonal extraction)	[PROVED HERE]; verifier: correct	Proposition 4.4	H-116 (H-U)
B-REG	(OS0)	[PROVED HERE]; verifier: correct	Section 4.5.5	H-116 (H-U); torus form of OS II
B-TI	(OS1) all of T^4 and B_4	[PROVED HERE]; verifier: correct	Section 4.5.6	H-116 (H-U); origin at a site
B-RP	(OS2)	lattice: [CITED]; passage: [PROVED HERE]; verifier: correct	Section 4.5.7	ℓ even (sufficient); H-116 (H-U)
OS3	permutation symmetry, reality	[PROVED HERE]	Section 4.5.8	—
B-NT	nontriviality (Theorem 4.3)	[OPEN]: needs (NT) = H-101 (E14-f); derivation from (NT) [PROVED HERE], corrected statement	Section 4.5.9	(NT) at every layer for (ii); $c>0$ in Balaban's normalization
E13	full-sequence convergence	not needed here; [OPEN]	`SOURCES.md` H-099 (E14-d), D6	—

Critical path to an unconditional (modulo GC1–GC3) Theorem 4.2: R-025 (H2) is certified at $L=13$ (Theorem 6.1); B-T2 still needs R-026 (PI), R-027 (F) (the reduction verified in round 8; its input H-137 (L-LL) proved and verified in round 9; a full proof pending verification, Theorem 5.26), the inputs of $R-028(M_j)$, V1, and R-030 (G2) (fixed-level results, Sections 5.1 and 5.11), the inputs of L5 (Theorem 5.15), and item $7'$, and the step to Theorem 1 needs the sweep (R-034, cleared in

round 7; R-012 is a verified list) and the finite-volume identification R-032 (replaced form, proved relative to K-032–K-038 in round 6; R-015 a verified reduction to conventions); then B-S: the open inputs of the real uniform R-037 (H-hist) ((M_r), CMP 102 (189), H-145 (M-B) bookkeeping; the small-field (Q)/c_- is discharged relative to R-008, R-009, K-026, K-027 by Theorem 5.9), the residue under R-001–R-005 (the inputs of Theorems 5.1, 5.2, 5.3: the printed inputs of H-057 (C) (R-021), the printed sketches of CMP 99 Thms 3.7, 3.9, 3.10 (3.7 and 3.9 now written out, Theorems 5.30 and 5.29), WG-1, and [I] geometry and convention items; the p. 253 and p. 275 assertions (R-022, R-023); the inputs of Theorems 5.4, 5.5, 5.6; R-Δ2; the residue of R-018 (Proposition 5.18)), the inputs of R-039 (H-thr) (round 8: M2′ and M4 derived, M5 not needed; open: H-241 (R1′) with three [I], (R2)–(R4), the coarea Jacobian), and W4-G with the Ursell bound in R-044 (round 8: the Λ_j bookkeeping and R-044 (W4)-R proved relative to inputs; the H-098 (E14-c) clause of B-N1 reduced to R-045 (S); Section 5.12). The blocks proved in this section are short and are not on the critical path.

4.7 What the theorem does not give

Each item needs a separate argument that this paper does not give.

- Rotation invariance. Only B_4 and the torus translations are proved. Full $O(4)$ is not a symmetry of T^4_ℓ. `LEMMAS.md` ledger item 5 [VERIFIED QUOTE]: on the torus the inclined comparison "gives only the tilted-axis reflection symmetry $R_{2\theta}s$, never a rotation"; the template reaches rotations only after the torus → plane limit, which it controls with its mass gap.

- Infinite volume. ℓ is fixed. Jaffe–Witten tie the limit $T^4 \to \mathbb{R}^4$ to a volume-uniform gap [VERIFIED QUOTE]: "New ideas are needed to prove the existence of a mass gap that is uniform in the volume of space-time. Such a result presumably would enable the study of the limit as $T^4 \to \mathbb{R}^4$." No public mechanism without a gap is known to us (`LEMMAS.md` ledger item 8). Remark on scope: with the constants of this paper, a unit-scale cluster expansion uniform in ℓ would give exponential clustering at a rate of order one in unit-scale units, against the correlation length $\sim e^{1/(2b_0 g^2)}$ expected from asymptotic freedom, so such an expansion is expected to be impossible (`INFINITE_VOLUME_IV12_VERIFY.md`).

- Mass gap. Not addressed. On a finite torus a "gap" of the transfer matrix is not the Clay mass gap.

- Uniqueness. Different subsequences may give different Schwinger functions. Uniqueness needs E13, which Balaban does not have. Also not given: uniqueness of the vacuum after reconstruction (no clustering on a torus).

- Vacuum state. Time is periodic, so OS reconstruction from torus reflection positivity gives a transfer matrix on T^3 and a thermal state at inverse temperature ℓ, not a vacuum state (Section 4.8, item 4).

- Also not given: clustering (OS4); a probability measure for the density field; Wilson-loop observables (class C3); the stress tensor and the operator product expansion (class C2 and beyond); independence of the limit from the block-averaging choices of Balaban's RG (only the lattice measure enters $S_{n,\varepsilon}$, but $g_0(\varepsilon)$ is chosen through Balaban's flow).

- Status of the whole statement. Even with every block closed, the theorem is conditional on Balaban's published 1984–89 proofs H-200 (GC1). Under Jaffe–Witten footnote 2 it is a compactness ("weak-existence") result on a torus with the torus forms of OS0–OS3; it is not a solution of the Millennium problem.

4.8 Corrections after verification

The verifier confirmed the theorem statement, all in-chapter proofs (B-C, B-REG, B-TI, B_4, the RP passage, OS3, the finite-form lemma), the quotes of Jaffe–Witten, and the open status of B-N1. Four items were corrected (`ASSEMBLY_VERIFY.md`). The statements above carry the corrected wording; the original claims are listed here.

1. *ℓ even is sufficient, not necessary* (verifier §5, grade 3). The parity count is right: L odd gives $N_\varepsilon = \ell L^K$ even iff ℓ even; for odd N_ε every time reflection has one site seam and one link seam. Osterwalder–Seiler (1978) [21] treat link-plane (between-site) reflections. Site-plane reflections were added by Menotti–Pelissetto, CMP 113 (1987) 369–373 [17] [VERIFIED QUOTE] (abstract, per the verifier): "We extend the proof by Osterwalder and Seiler of reflection positivity of lattice gauge theories with fermions to the case in which the reflection is performed with respect to planes containing sites." and "Our proof applies to all observables and relies on the particular structure of the Wilson action." The mixed case (one site seam, one link seam on a periodic lattice) is very likely routine, since each mechanism is local to its seam, but no citation states it. So odd ℓ is not excluded; the hypothesis ℓ even only avoids the mixed case. The source's phrase "plausible but not cited" stands. The periodic two-seam link version attributed to Osterwalder–Seiler is [CITED], not [VERIFIED QUOTE]. Two minor points from the verifier: the support distance after the seam shift is $\geq 2\delta - \varepsilon$ (the source's $2\delta - 2\varepsilon$ is conservative); the factor $2^{s(n+m)}$ in Step 2 is unneeded but harmless.

2. *Reference title.* The Jaffe–Witten reference list gives the title "Gauge theories on the lattice"; the bibliography entry follows the Annals of Physics title.

3. *Nontriviality needs the two-sided envelope only and support diameters $\ll \ell$* (verifier §6, grade 6). In Theorem 4.3(i), $|R| \leq \frac{1}{2}$ needs $g^2 + (|x-y|/\ell)^2$ small on the whole support pair, so the support diameters must be $\ll \ell$, not only the minimum distance d_0. In Theorem 4.3(ii) the source stated the asymptotic $d^8 S_2 \sim c\, g(d)^4$ with $g(d)^2 \sim (\beta \log(1/d))^{-1}$. The proof uses only the two-sided bound (0.31); the asymptotic needs $\beta_j \to \beta_\infty$ and is not proved. The statement of Theorem 4.3(ii) above is the supported one. The overclaim was in the statement, not in the proof.

4. *Periodic time gives a thermal state* (verifier §6, grade 8). OS reconstruction from torus reflection positivity gives a transfer matrix on T^3 and a thermal state at inverse temperature ℓ, not a vacuum. This remark is added to the list of non-results (Section 4.7).

5 The sources block

This section gives one paragraph for each project file `SOURCES_*.md` that has a verifier file. Each paragraph states the result, describes the method in two sentences, gives the verifier's grade in the tags of Section 1.4, and names the inputs that stay open. Nothing here goes beyond the project ledgers: the verifier files, `TRUST_LEDGER.md`, the dependency table (Section 4.6), and `PLAN.md`. Where a verifier corrected a file, the corrected statement is given. The grouping follows §4 of `TRUST_LEDGER.md`: first the files that L-001 (Theorem 2w) (Target 1) uses, then the files of the finite-torus theorem (Target 2) in the order of its dependency chain (steps 2–9 of §4.2 there). Ledger labels An, Bn, Cn, Xn are those of `TRUST_LEDGER.md`. `SOURCES_A1_KERNEL.md` and `SOURCES_A4_D189.md` share the verifier `SOURCES_A1_A4_VERIFY.md`,

and SOURCES_HHIST.md is checked in SOURCES_HHIST_N1_VERIFY.md. Two files have no verifier and are not listed: SOURCES_G4.md, SOURCES_W4.md. SOURCES_BOOKKEEPING.md has a verifier since round 8 (Section 5.12), and two round-8 files with a running verifier are listed there as pending. Files verified after this section was assembled have no paragraph of their own: SOURCES_WC_EXPGAUGE.md and SOURCES_WC_RESIDUE.md (cited under R-002 and R-003), and SOURCES_T2_V1.md, SOURCES_T2_PI.md, SOURCES_T2_A1WO3.md, SOURCES_T2_A1WO4.md, SOURCES_T2_D189.md, SOURCES_T2_R2REM.md; their state is in TRUST_LEDGER.md and PLAN.md.

5.1 Target 1: inputs of L-001 (Theorem 2w)

SOURCES_GGAIN_NONMONO.md: item 7′, the running-maximum lift. For every step k, every past $h \in]0,\gamma]^k$, monotone or not, and every $g \in [0,\gamma]$, $|\beta_{k+1}(h,g) - \beta_{k+1}(0)| \le C_\beta g$ with $C_\beta = C_\Pi S_0$, the constant of H-111 (G-gain), uniform in k, K, and the past. This removes the monotone-history restriction of H-111 (G-gain), which blocked item 7′ on the clamped, non-monotone pasts that the shooting evaluates. The only use of monotonicity is Step 2 of L-164, where $\max_{j\le k} 4g_j^{\kappa_0} = 4g_k^{\kappa_0}$ compares bound sizes; with the running maximum $G_k = \max_{i\le k} g_i$ the induction $a_j \le 2G_j^{\kappa_0}$ closes for every past under the side condition $4\gamma^{\kappa_0} \le E_0$, which is of Bałaban's existing type. The value at zero own coupling, (K1), is history-free because the exponent vanishes at $g_k = 0$ (CMP 109 p. 268) for any input, and the endpoint $g \to 0$ is a dominated-convergence step that is not written line by line. Verifier: SOURCES_GGAIN_NONMONO_VERIFY.md, the lift [PROVED+VERIFIED], item 7′ [PROVED+VERIFIED, CONDITIONAL ON H-202 (H0) (LEDGER R-006) OR ON THE H-111 (G-GAIN) INPUTS (LEDGER R-037)]; all four items confirmed with small corrections: $C'_\beta = C_\beta$ holds because $E_0 + 4\gamma^{\kappa_0} \le 2E_0$, the input radius that H-111 (G-gain) already covers, and the possible dependence of the newest R-input on g (caveat c1) is open on the literal reading of β and void on the pure-part reading. Open inputs: H-202 (H0) (ledger R-006; its use here is a verified reduction with residue N1–N9, Section 5.9), or the H-111 (G-gain) inputs of ledger R-037 ((Q)/c_-, (M_r), CMP 102 (189), the H-145 (M-B) bookkeeping); L5, the evaluation of β off Bałaban's trajectory (item 8, ledger R-031); caveat c4, CMP 119 (2.31) and the off-trajectory small-field induction of SOURCES_SM.md §5.1 not swept for coupling-order uses.

SOURCES_T2_KUNIF.md: item 9, K-independence of $\beta_j(g)$. Item 9 is a volume statement, not a history identity, and it closes as a reduction: CMP 119 (2.24), p. 253, and p. 279 define β_j as the whole-lattice coefficient of CMP 109 (whose Π in (4.37) is a sum over $\mathbb{Z}^4$), so the β_j of the recursion is K-free by definition. The shooting needs only (K1) $\beta_j(h,0) = \beta_j(0)$ for every admissible past, (K2) $|\partial\beta_j/\partial g_{j-1}| \le C_\beta$, and (K3) the window bound $\sum_{i=k+1}^{n} \beta_i(0) \in [b(n-k) - 2R,\, b(n-k) + 2R]$ of L-009 (Theorem K); with $\gamma \le b/(4C_\beta)$ these give $T(m,n) \in [(3b/4)(n-m) - 2R,\, (5b/4)(n-m) + 2R]$ on every admissible history. Lemmas D′ and D″ run at one fixed K and never compare two trajectories or two values of K, so only their constants must be free of K. The bound $|\beta_{k+1}(h_k) - \beta_{k+1}(0)| \le C_\beta g_k$ follows from (K1) and the mean-value theorem in the own slot, and it gives the window bracket pointwise on each clamped history of the intermediate-value argument. Verifier: SOURCES_T2_KUNIF_VERIFY.md, [VERIFIED REDUCTION]; items 1, 2, 4 confirmed, item 3 corrected: H-111 (G-gain) gave C_β only on monotone slot histories, so (K2) via H-111 (G-gain) needs item 7′ (now lifted by SOURCES_GGAIN_NONMONO.md); the CMP 119 page for the definition of β_j is p. 253, not p. 249. Open inputs: item 7′ for (K2) (from H-202 (H0), ledger R-006, or from H-111 (G-gain), ledger R-037); L5, the domain and continuity of β off Bałaban's trajectory (item 8, ledger R-031; round 7: proved relative to listed inputs by nested intervals on true trajectories, Theorem 5.15); the inputs of L-009 (Theorem K) for (K3); the finite-volume identification of the step (ledger R-032, with R-015, narrowed by Proposition 5.8), an input of Theorem 1, not of the shooting.

`SOURCES_T2_V1.md`: **item 5, V1, ledger R-029.** Inside Bałaban's construction with the cut CMP 109 (2.9), the term $E^{(k+1)}$ of (2.13) vanishes identically at $g_k = 0$. So $\Pi^{(k+1)}(0)$ and $\beta_{k+1}(0)$ come from $\log Z^{(k)}$ alone: $\beta_{k+1}(0)$ is the Maxwell coefficient of the one-step Gaussian determinant, it is free of the history, and it equals the determinant of the numerics up to a constant. The file reads the definitions (1.3)–(1.22) and (2.10)–(2.15) on page images and passes from Bałaban's exponential gauge fixing to the bordered Hessian of the numerics (gauge-fixing independence, constraint Jacobian, expansion point, normalization). Verifier: `SOURCES_T2_V1_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL] at fixed level, with 14 claims confirmed and 3 corrected; the conditions are R-030 (G2), Bałaban's own inductive bounds, and a declaration that the averaging is (0.4) (the numerics use (0.4); Bałaban's step uses (0.12)). Open inputs: R-030 (G2) (item 6); the averaging declaration; Bałaban's inductive bounds.

`SOURCES_T2_G2.md`: **item 6, R-030 (G2), ledger R-030.** At fixed level j, the constrained one-loop log-determinant on a torus of side N has no zero modes, and its density per coarse cell converges to the infinite-lattice density with error $\le C_j e^{-\delta_j N}$, also for every derivative in the plane-wave amplitude. The p^2-coefficient commutes with $N \to \infty$; the f^2-coefficient at the constant field commutes with $N \to \infty$, given a gap input (Gap_f) and C^4-regularity in f, both [R]. The massless pieces of Prop. 2.1 are handled fiberwise, with an N-independent zero-mode constant. The proof uses a Combes–Thomas decay bound and the method of images; the constants are fixed-j constants ($\delta_j \gtrsim L^{-2j}$), and the j-uniform version is not proved. Verifier: `SOURCES_T2_G2_VERIFY.md`, 21 claims confirmed and 6 corrected, none affecting a conclusion; item 6 moves from [FORMAL] to [R] ([VERIFIED REDUCTION]) at fixed j. Open inputs: the surjectivity of $\tilde{Q}$, (Gap_f), and the reading that the averaging is analytic.

`SOURCES_T2_S123.md`: **(M_j), ledger R-028.** At fixed $j \ge 1$, odd $L \ge 3$, and small flux f, the lattice spectral inputs hold: H-244 (S1) $\mathcal{K}_t(f) \ge -C_1|f|$ for $t \in [0,1]$, H-247 (S3) $\pm V \le C_3[(\mathcal{K}_t + \mu)^2 + f^2]$ in the subdivided form of `KAPPA_BOUNDED.md` §1.3, H-245 (S2) at $t = 0$, and the coarse comparison H-246 (S2c) $N_0(\lambda - \eta_1) \le N_t(\lambda) \le N_0(\lambda + \eta_1)$ with $\eta_1 = O(f^2)$. Corollary 7.1 then reduces (M_j), the limit of $[\Gamma_c(\Lambda_j; f) - \Gamma_c(1; f) - (f = 0 \text{ value})]/(n_2 f^2)$ to d_j, to the near-zero input (Z_j), itself reduced to the 2D gap (G'_A), and to region 1 of Prop. 4.4 of `BETA_FP_M.md`. The proofs work in the twisted algebra of operators that commute with the magnetic translations, with Peierls products, Moyal expansions with kernel-norm remainders, a Nenciu-type inverse, and operator bounds from the vanishing order of symbols. At $t = 0$ a fibre decomposition over $k_\perp$ separates the operator, and a gauge-copy lemma (Lemma 5.8) shows that the near-zero density of states needs only (G'_A): the transverse charged 2D SU(2) Wilson Hessian at constant flux has no spectrum in $(-c|f|, c|f|)$, and numerically its gap is about $|ef|$. Verifier: `SOURCES_T2_S123_VERIFY.md`, (M_j) [VERIFIED REDUCTION], with H-244 (S1), H-247 (S3), H-245 (S2)$|_{t=0}$, H-246 (S2c) and region 2 of Prop. 4.4 [PROVED+VERIFIED AT FIXED j, CONDITIONAL ON (I3)]; 25 claims confirmed, 2 corrected (in Lemma 1.4 ∇^β is one twisted convolution; the explicit L-power of C_j in Prop. 3.2 is convention-dependent and is not quoted), and the input list of (M_j) gains the neutral-sector differentiation under the integral, an inference. Open inputs: (I3), the entrywise analyticity in f of Bałaban's linearized averaging at each of the j steps, is now proved (next paragraph); (G'_A) (open, independent of j and L, numerically confirmed at six rational fluxes with margin about $|ef|$); region 1 of Prop. 4.4 of `BETA_FP_M.md` (argued); all under ledger R-028.

`SOURCES_T2_I3.md`: **the Peierls input (I3), ledger R-028.** Let L be odd ≥ 3, and let the averaging be Bałaban's (0.4), or (0.12) with Federbush's average. Then the linearized averaging

$\tilde{Q}_f$ at the constant-flux background is entrywise analytic in complex f on $|f| < f_0$, $f_0 = \pi/8$ for (0.4) and $\min(\pi/8, 1/8)$ for (0.12), and (I3) holds at every step $i \le j$ with radius and constants independent of i and j (Theorem 7.1 there). It holds in the stated two-block range and does not use H-207 (I1) for the fixed vertices. The proof linearizes the averaging explicitly at the flux background (Prop. 5.3), bounds each charged entry of $\tilde{Q}_f$ and of D_f by its reduced kernel (Props. 5.4, 6.1), and assembles the bordered matrix of one step. Verifier: `SOURCES_T2_I3_VERIFY.md`, 15 claims confirmed, 2 corrected; (I3) moves from [I] to [P], given the identification of the one-step objects of the project with the linearization of (0.4) or (0.12) at the constant-flux background. Open inputs: that identification; for (M_j) the 2D gap (G'_A) and region 1 of Prop. 4.4 of `BETA_FP_M.md` stay. Rounds 6 and 7 settle both (Theorems 5.12 and 5.14).

5.2 Target 2, step 2: the sourced Theorem 1 (ledger R-035)

`SOURCES.md`: the E14 reduction of R-035 (B-S). The plaquette-density source is a local bare coupling, $\rho_0^f = \exp(-\sum_p (g_0^{-2} - f_p) P_p(U))$, so the Schwinger functions on test functions flat on the diagonals reduce to a sourced version of Bałaban's Theorem 1 with a pointwise coupling flow H-159 (S-A), sourced regular terms H-160 (S-B), a local contact functional $W_k[f]$ H-161 (S-C), a local β-function H-162 (S-D), and the normalization H-163 (S-E). The new estimates are H-096 (E14-a) the sourced small-field step, H-097 (E14-b) the sourced large-field step and R operation, and H-098 (E14-c) the derivative bounds in f; H-099 (E14-d) = E13 is needed only for full-sequence convergence, and H-101 (E14-f) only for nontriviality. A dimension count shows that the source of the dimension-4 density is marginal at every degree, that its field-independent part is relevant and is removed as local contact terms, and that the first field-dependent gradient term has dimension 6. One step is written out at the Gaussian level: with $K_f = (\Delta^{(1)})^{-1/2} \Delta^{(f)} (\Delta^{(1)})^{-1/2}$ and $\|K_f\| \le r$ the source enters through $-\frac{1}{2} \log\det(1 - g_0^2 K_f)$, and the linear response per step is $c_j = 1 + \frac{1}{2} g_{j-1}^3 \beta'_j(g_{j-1})$. Verifier: `SOURCES_VERIFY.md`, [VERIFIED REDUCTION] (the structure confirmed; items 5, 6, 8 corrected, item 10 confirmed with corrections); the printed CMP 119 (2.24) is $1/g_{j-1}^2(x) = 1/g_j^2(x) + \beta_j(g_{j-1})\varphi_j(x)$, and by (2.25) Bałaban evaluates β_j at the constant coupling, so H-159 (S-A) and H-162 (S-D) are new; derivatives of $\log Z$ are cumulants; $\prod_j c_j$ is uniformly bounded, not shown convergent, and assumes a one-step recursion; the uniform bound for general F flat on the diagonals is not derived from the source form. Open inputs: R-035 (B-S) as a whole, (E14-a, b, c) (ledger R-035, [OPEN]), with its parts R-036–R-044 and R-046; the H-098 (E14-c) clause of the bridge R-045 (B-N1) (ledger R-045); (NT) = H-101 (E14-f) for Theorem 4.3; E13 = H-099 (E14-d), not needed for Theorem B.

`SOURCES_D1.md`: one sourced step at one loop (D1). At one loop the plaquette-density source, a local bare coupling $w_p = 1/g_0^2 - f_p$, gives the multiplicative source renormalization $c_1 = 1$ exactly, for every background; the first correction is $c_1 - 1 = \frac{1}{2} g^3 \beta'_1(g)$, which is $\beta_1^{(2)} g_0^4 + O(g_0^6)$ if β_1 is even and $O(g_0^3)$ with a C^1 bound only. The field-independent terms are the tadpole $\frac{1}{2}(9L^4 - 9) g_0^2$ per coarse cell ($\frac{3}{4}(1 - L^{-4}) g_0^2$ per plaquette) and a contact term $C(q)$ that is analytic at $q = 0^+$, and the first field-dependent gradient terms have dimension 6. The source enters the step only through the Hessian $\Delta_f = \sum_p w_p H_p$ and a current term $\langle J_f, B \rangle$ at Bałaban's unweighted minimizer, while gauge fixing, ghosts, and the constraint carry no f; for constant f the quadratic form scales by $1 - g_0^2 f_0$, so the determinant changes by a field-independent constant over $9L^4 - 9$ modes per cell. Per-momentum formulas for the tadpole, $C(q)$, and the mixed derivative are evaluated by `numerics/d1_check.py` and checked against real-space finite differences on the periodic 6^4 lattice. Verifier: `SOURCES_D1_VERIFY.md`, [PROVED+VERIFIED] for the algebra (items 1–4, 6, 8–10 confirmed) and [NUMERICAL, REPRODUCED] for the rerun, which matched every digit; corrected: the dimension-6 count holds by index parity, the γ_2 values are unreliable, the data support $|c_1 - 1| \le$

5×10^{-4} but not geometric convergence in the grid, and the expansion-point error has the small parameter Lp. Open inputs: none at the Gaussian level; the non-Gaussian sourced step and the pointwise flow belong to D2–D3 and to R-035 (B-S), ledger R-035.

`SOURCES_D2.md`**: sourced small-field step H-096 (E14-a).** The sourced small-field step is Bałaban's step with extra source vertices of order g_k, g_k^2, g_k^3, all in his $P^{(k)}$ class (each $\le O(g_k)(1+|B|^2)$ on $|B| < \varepsilon_1/g_k$), a propagator for the sourced Hessian with Bałaban's decay and regularity bounds (constant $3c_1$ with $c_1 = O(\delta_0^{-5})c_0$, rate $\delta_0/2$), and a field-independent vacuum term W that factors out. Its step (c), the pointwise flow $1/g_{j+1}^2(x) = 1/g_j^2(x) - B_{j+1}(f(x))$ with $h \equiv 0$, is not established here and is redone in `SOURCES_D2C.md`. The source term is evaluated at Bałaban's unweighted minimizer, so his measure and propagators are used without change, and the sourced propagator follows from a weighted Neumann series in his cube-to-cube operator norms (CMP 116 (1.7)). Real-source derivative bounds are tied to polystrip holomorphy by the interpolation Lemma 7.1, and the remainder bounds are reduced to Bałaban's estimates with a freezing property and a constant-source family. Verifier: `SOURCES_D2_VERIFY.md`, steps (a), (b), (d), (f) [PROVED+VERIFIED], step (e) a [VERIFIED REDUCTION], and step (c) rejected, since its claim that CMP 109 Sects. 3–4 run verbatim is [REFUTED] ((4.34)–(4.37) use Euclidean symmetry, and the cubes are at scale k, not j); corrected: (a) needs a block-constant $\bar{f}$, the constant is $3c_1$ and not $\times 3$, W factors only in the small-field region, and the polydisc radius is $r/(2n)$. Open inputs: analyticity of the step in its input activities, R-044 (W4), and the field-dependent domains Λ_j in W (ledger R-044); the recovery $\kappa \to \kappa/2$ (ledger R-046, a verified reduction that uses R-020); (Hβ), reduced in `SOURCES_HBETA.md`; the scale condition K-024 (U) is the constant choice K-024, not an open input.

`SOURCES_D2C.md`**: step (c) redone, frozen source per domain (L-054 (Lemma A)).** Each step term $E^{(j)}(X, \cdot\,; f)$, $X \in D_j$, is frozen at $c_X = f(b_X)$, and the freezing error obeys $|D_X| \le E_0 C M^3 (\rho_1/r)(L^j\eta)^5 \lambda_k e^{-\kappa d/2}$ (L-054 (Lemma A)), irrelevant with exponent 1 and with no growth in k. With the identification R-036 (I) of the constant-source increment $B_j(c)$ with Bałaban's β_j on the trajectory from $1/g_0^2 - c$, this gives the pointwise recursion $1/g_{j+1}^2(x) = 1/g_j^2(x) - B_{j+1}(f(x))$ with $h \equiv 0$ and gradient corrections irrelevant with exponent $\min(\alpha, \beta, 1)$. The non-constant source is handled at scale j, per localization domain, as a gauge-invariant difference bounded in an axial gauge on X, where each field costs two powers of $L^j\eta$. Bałaban's CMP 109 Sects. 4–5 are then used only for constant sources, where they apply at bare coupling $1/g_0^2 - c$. Verifier: `SOURCES_D2C_VERIFY.md`, L-054 (Lemma A) and the recursion [PROVED+VERIFIED, CONDITIONAL ON (D) AND (Hβ)]; corrected: the freezing points are the points of X nearest to b_X, R-036 (I) is exact only with the adapted gauge-fixing weight $1/u_k(c)$ (otherwise up to $O(e^{-c\varepsilon_1^2/g_k^2})$), $\beta > 0$ in (4.18) must be fixed, and the exact cancellation of the cubic first-moment term of (4.34) is unsupported and not needed. Open inputs: the domain hypothesis (D) (ledger R-044; argued in `SOURCES_D2C_VERIFY.md` §2 from the CMP 109 p. 263 remark, which cites CMP 102 Prop. 9, ledger R-005); (Hβ) (L-182 (Lemma H) of `SOURCES_HBETA.md`); R-036 (I) with large-field thresholds (ledger R-036); the adapted gauge-fixing weight changes the construction (ledger K-011).

`SOURCES_D3.md`**: sourced large-field step H-097 (E14-b).** The sourced large-field step is Bałaban's step with all thresholds frozen at the reference couplings, the local coupling $w_j = w_{j-1} - B_j(f)\varphi_j$, an R operation that resets it on Z to $u_k(f(x))$ (CMP 122 II p. 365), and an adapted local gauge-fixing weight. Real sources keep the positivity of the large-field weights under $3\gamma^2 r \le \frac{1}{2}$ (revised to $5\gamma^2 r \le \frac{1}{2}$ in `SOURCES_D3R.md`), Lemma D3.1 gives the local holomorphy radius $c_\tau r/(E_0|Y|)$ of one positive denominator, and the identification R-036 (I) is off by a thresh-

old term, the new input R-039 (H-thr). The file lists the places (P1)–(P7) where Bałaban uses a constant coupling in the large-field step and states the change at each. Each change is reduced to a quoted Bałaban estimate (CMP 119 (2.23)–(2.31), CMP 122 II (1.73)–(1.75)); a line-by-line rerun of CMP 122 I–II is not done. Verifier: `SOURCES_D3_VERIFY.md`, [VERIFIED REDUCTION] with corrections: the reset term is bounded by Bałaban's sup bound (CMP 122 II p. 366), not cancelled; the t-parameter term H-240 (P8) of CMP 119 p. 271 is missing; old R-terms are merged into E_k, so freezing through R is open again; the Gevrey order is $(n!)^{2+\epsilon}$, not $(n!)^2$; and the lower bound on w_k holds relative to Bałaban's local coupling. Open inputs: R-039 (H-thr) (ledger R-039); the real uniform R-037 (H-hist) (ledger R-037); strip holomorphy of u_k, now N3 (ledger R-042); R-044 (W4) and the field-dependent domains Λ_j (ledger R-044); R-043 (FX), H-191 (FX2) for the final step (ledger R-043); K-024 (U) is the constant choice K-024.

`SOURCES_D3R.md`**: revision of D3 (H-235 (P3), H-240 (P8), freezing through R, Gevrey order).** The sourced reset coefficient differs from Bałaban's by at most $\delta_r = 4r + 2C_h M\rho_1\lambda_k$, additively and pointwise (L-245 (Lemma P3)), so his sup bound of CMP 122 II p. 366 holds with one changed constant, and the large-field factors survive with $\gamma_0 \to \gamma_0(1-\theta) \geq \gamma_0/2$, $\theta = \gamma^2\delta_r \leq 5\gamma^2 r \leq \frac{1}{2}$. The t-parameter term splits as $d_B + d_S$ with $|d_S| \leq 2r$, freezing through R splits into a pure part (L-162) and an R-induced part (L-163), and the cumulant bound has Gevrey order $(n!)^{2+\epsilon}$, admissible with $D = 3$. Positivity is kept by a split against Bałaban's local coupling, and the H-240 (P8) term is split into a piece with the support of Bałaban's deviation, booked as a boundary term, and the source weight of D2. The regular output is split by an auxiliary parameter into the pure and R-induced parts, and the Gevrey count uses a Lah-number sum over partitions. Verifier: `SOURCES_D3R_VERIFY.md`, [VERIFIED REDUCTION] with corrections: H-235 (P3) needs $\rho_1 \leq r/(4C_h M)$; in H-240 (P8) h_k is booked as irrelevant, and d_S changes the covariance $C^{(k)}$ at every t (relative size $O(\gamma^2 r)$); L-163 is a new claim, not Bałaban's bound for merged R-terms (item W8, taken up in `SOURCES_D3W8.md`); the Gevrey count is confirmed conditional on R-045 (B-N1). Open inputs: strip holomorphy of u_k (N3, ledger R-042); R-039 (H-thr) (ledger R-039); the real uniform R-037 (H-hist) (ledger R-037); R-044 (W4) and the covariance change in H-240 (P8) (ledger R-044, the latter by analogy with R-016); R-045 (B-N1) (ledger R-045).

`SOURCES_D3W8.md`**: covariance change along t and merged R-outputs (L-164).** Along the t-interpolation the measure is Bałaban's CMP 119 (3.27) with covariance $C_f^{(k)}$, independent of s and t; with $\tau \leq 3c_1 c_H \gamma^2 r \leq \frac{1}{2}$ one has $\|C_f - C_0\|_\delta \leq 2c_1\tau$, and the large-field factors survive with $\gamma_0 \to \gamma_0/4$. L-164: the R-induced part obeys $a_m \leq 32\varepsilon' g_m^{\kappa_0}$ with no compounding, and $|E_R^{(m)}(X,U) - E_R^{(m)}(X,1)| \leq CM^2 \cdot 32\varepsilon'(L^m\eta)^4 g_m^{\kappa_0} e^{-\kappa' d_m(X)}$ on regular configurations, a marginal term of Bałaban's R-type form, summable in m for $\kappa_0 > 2$. Transported inputs keep their scale label and never pass through the generated map G, so a Schwarz-lemma step on $G - G(0)$ with Bałaban's factor $\varepsilon' = O(\alpha_2^{-1}\varepsilon_1)$ (CMP 109 p. 280) gives $a_{k+1} \leq 8\varepsilon' b_k$ per generation. The covariance bounds come from the weighted Neumann series of `SOURCES_D2.md`, and positivity from domination by a rescaled C_0 Gaussian. Verifier: `SOURCES_D3W8_VERIFY.md`, route (a) and L-164 [PROVED+VERIFIED, CONDITIONAL ON R-044 (W4), (EPS), L-162, (D), AND $32\varepsilon' \leq 1$], the t-measure and its bounds corrected: the cross term $g_k^2\langle A, \mathcal{W}_S A_k\rangle$ goes to $B^{(k+1)}$, CMP 116 p. 3 covers C_f only by analogy, the domination must use the marginal region (or Anderson's inequality) to avoid a volume factor, and ε' contains Bałaban's large constant of p. 280. Open inputs: R-044 (W4) and (D) (ledger R-044; R-044 (W4) is reduced to the printed (C1)–(C5) through L-281 (Theorem SM)); (Eps), from L-281 (Theorem SM) and R-010; the extension R-016 to the fluctuation covariance; the conditions $32\varepsilon' \leq 1$, $S \geq 1$, $\kappa_0 > 2$ change the construction (ledger K-017); the

monotone-history assumption of L-164 is removed by the running maximum, with side condition $4\gamma^{\kappa_0} \le E_0$ (`SOURCES_GGAIN_NONMONO_VERIFY.md`).

`SOURCES_D4D5.md`: **RP and symmetry transfer, and the cumulant bound (S_{src}).** The sourced RG output equals the bare integral $Z_\varepsilon[f]$ (Lemma D4.1), reflection positivity and the lattice symmetries are properties of the lattice measure (Lemmas D4.2–D4.4), and given R-043 (FX) the cumulants of real sources obey (S_{src}): $|\kappa_n(f_1,\dots,f_n)| \le (C_S\sigma^{-a})^n (n!)^{1+\eta} \prod_i N(f_i)$ with $N(f) = \max(\|f\|_\infty/r, \|\nabla f\|_\infty/\rho_1)$, for $\sigma \in [16\varepsilon, \ell/8]$, supports pairwise at distance $\ge \sigma$, minimal diameter $\le \sigma$, and C_S independent of K. Then (S_{src}) $\Rightarrow$ H-116 (H-U) with $s = 6$, $D > 1$. Each RG operation is an identity of integrals (Haar invariance, δ-function blocking, partitions of unity, convergent expansions, $\int dV\, R\rho = \int dV\, \rho$), and complex sources follow by the identity theorem. L-111 (Lemma C) takes one Cauchy derivative in each of n separated source amplitudes on a polydisc whose radius does not depend on n, so the sum over polymer sizes gives a single $n!$. Verifier: `SOURCES_D4D5_VERIFY.md`, D4.1, D4.3, D4.4 and L-111 (Lemma C) [PROVED+VERIFIED] (L-111 (Lemma C) given R-043 (FX)), (S_{src}) $\Rightarrow$ H-116 (H-U) confirmed, and R-043 (FX) confirmed as a new hypothesis not contained in CMP 116–122; corrected: cite CMP 122 I (1.102) for the actual R′ operation, the block geometry is invariant only under translations by $M\mathbb{Z}^4$ or coarser, and the order $1+\eta$ needs the coarsest-scale H-191 (FX2) radius (else an extra factor $(n!)^{2p_R/p_0}$, still OS-admissible). Open inputs: R-043 (FX) in Bałaban's main form with the coarsest-scale H-191 (FX2) radius (ledger R-043; remaining input N3, ledger R-042); S1 at all scales, part of R-035 (B-S) (ledger R-035); the H-098 (E14-c) clause (ledger R-045).

`SOURCES_HBETA.md`: **L-182 (Lemma H), the two-loop bound (Hβ).** L-182 (Lemma H): given R-036 (I), (E), and R-037 (H-hist), with $x := C_1 A\gamma/\beta \le \ln(3/2)$ and $3r < g^{-2} - \gamma^{-2}$ ($g = g_K$), every real c with $|c| \le 2r$ keeps all $g_k(c) \in]0,\gamma]$ and gives $\sum_{j\le K} |B_j'(c)| \le e^x - 1 \le \frac{1}{2}$ and $\frac{1}{2} \le |v_k(c)| \le \frac{3}{2}$, uniformly in K. With $\beta = \beta_0/2$ the smallness condition reads $\gamma \le \ln(3/2)\,\beta_0/(2C_1A) \approx 0.20\,\beta_0/(C_1A)$. In $u = 1/g^2$ the chain rule gives $\partial\beta_j/\partial u_i = -\frac{1}{2} g_i^3\, \partial\beta_j/\partial g_i$, so a C^1 bound costs $O(g_i^3)$ per step, $\sum_i g_i^3 \le 2\gamma/\beta$ along the trajectory, and a Gronwall product on the variational equation gives $e^x - 1$. A bootstrap on the component of $\{|c| \le 2r : u_K(c) > \gamma^{-2}\}$ that contains 0 keeps every c-trajectory in $g \le \gamma$; the Gronwall algebra is formalized in Lean (`LemmaH`). Verifier: `SOURCES_HBETA_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON R-036 (I), R-037 (H-HIST)]; corrected: the hypothesis list adds that β_j is the infinite-volume function independent of K, R-037 (H-hist) can be weakened to a column-sum bound, the smooth-clamp remark is dropped, and the toy numbers are not reproducible (the bound holds and is nearly sharp). Open inputs: R-036 (I) (ledger R-036; it holds by construction with the adapted gauge-fixing weight K-011 and the frozen thresholds K-001, but is not written as a lemma); the real uniform R-037 (H-hist) (ledger R-037, a verified reduction with open inputs (Q)/c_-, (M_r), CMP 102 (189), and the H-145 (M-B) bookkeeping); (E), the uniform positivity of the one-loop $\beta_j(0)$; complex c is not covered and is the strip bound N3 (ledger R-042).

`SOURCES_T2_KAPPA2.md`: **ledger R-046.** The decay loss $\kappa \to \kappa/2$ of the sourced step occurs at every step in three places (the interpolated inputs of the freezing proof in `SOURCES_D2.md` §6.3, the extraction remainders of `SOURCES_D2C.md` §3, the R-induced terms of L-163 in `SOURCES_D3R.md`), and it is recovered by $L/2$-rescaling without compounding. The conditions are H-216 (K1′) $\delta \le \frac{1}{13}(1 - 2(1+4\beta)/(\theta L))$, which is $\delta \le 5/169 \approx 0.0296$ at $L = 13$, $\beta = \frac{1}{4}$, $\theta = \frac{1}{2}$ (round 7, `SOURCES_A23_TSTEP.md`; the round-6 form H-215 (K1) had $\frac{1}{10}$ and $\delta \le 0.0385$, from the small-field T step only), with the choice $\delta = 0.025$ (0.03 fails; Lean `K1_numeric_13`), and H-217 (K2)

$\theta(1-\beta)L/2 \geq 1+3\beta$; at the last scale the final expansion runs at $\kappa_{\text{eff}} = \theta\kappa/(1+4\beta) = \kappa/4$, with the M-condition H-218 (K3) $M \geq 12L(4.5\kappa/C_1)^{1/4}$, implied by the existing one of `SOURCES_FX_R2.md`. Bałaban proves only $2d_k(Z) \geq Ld_{k+1}(Z')$ (CMP 116 (2.36)), so the gain per step is $L/2 = 6.5$, and his printed choice $(1-10\delta)\frac{1}{2}L = 1$ in (2.41) spends all of it; the recovery therefore takes δ smaller, at the cost of a larger κ (a lower bound about 4.3–4.7 times the unsourced one, like for like; was 3.6 under H-215 (K1)), which moves only constants fixed after κ. The R operation drops $4\beta\kappa$ of rate (CMP 122 II (1.66)–(1.100)), the rescaling pays these drops and the T-step fraction, now at most 13δ (10δ printed for the small-field step plus at most 3δ unprinted for the large-field step, ledger R-020), and the (1.93) volume charge is paid by p_0 through the condition before (1.95). Verifier: `SOURCES_T2_KAPPA2_VERIFY.md`, [VERIFIED REDUCTION]; items 2–4 confirmed on page images, item 1 corrected: the three losses stack within one step, so a literal half per absorption would make H-215 (K1) fail at $L = 13$, and the repair reads $\theta := 1 - m_{\text{tot}}/\kappa$ as the kept fraction per cycle (each loss is a polynomial absorption with additive cost; $\theta \geq \frac{1}{2}$ for $\kappa \geq 2m_{\text{tot}}$), with which H-215 (K1)–H-218 (K3) stand and the new lower bound is $\kappa \geq 1536$ at $d = 4$, $\beta = \frac{1}{4}$. Open inputs: the unprinted large-field T-step rate loss of CMP 119 Sect. 3 (ledger R-020), on which the homogeneity of the large-field T step in the input rate rests; CMP 116 (2.36), which Bałaban asserts without proof.

`SOURCES_UN1.md`**: K-024 (U), ledger K-024.** The scale condition K-024 (U) does not conflict with the bridge R-045 (B-N1): with $\lambda_K = 1$ (physical units), the r in K-024 (U) is the Cauchy margin r_0 of the constant-source family, not a test-function amplitude, so K-024 (U) is the constant choice $\rho_1 \leq c_* r_0/M^a$, made once with $\rho_1 := \min(c_* r_0/M^a, r_0/(4C_h M))$, after r_0 and before K. The bridge evaluates the generating functional only on sources rescaled into the domain of H-191 (FX2), at the price of a factor $(4/\rho_1)^n$ per n-point bound, with s and D unchanged. The bridge uses n-linearity, $\kappa_n(f_1, \ldots, f_n) = \prod_i \lambda_i \kappa_n(f_1/\lambda_1, \ldots, f_n/\lambda_n)$, with $\lambda_i = 2N_U(f_i)$ and $N_U(f) = \max(\|f\|_\infty/r, \|\nabla f\|_\infty/\rho_1)$, and pairwise σ-separated supports make the sup and the gradient of $\sum_i z_i f_i$ a maximum over single terms, so the whole Cauchy polydisc stays in the admissible domain. Small widths $h \geq \varepsilon$ are paid by the norm growth $1/h$ per bump, which the flatness of F absorbs, while K-024 (U) costs only an h-independent factor; the scale-matched reading of `SOURCES_D2.md` l. 217 would create a real conflict and is dropped. Verifier: `SOURCES_UN1_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON THE (Hβ) GRADIENT BOUND], so K-024 (U) is a constant choice (ledger K-024), verified compatible with R-045 (B-N1); corrections: the H-191 (FX2) gradient radius is $\rho_1/2$, hence $\lambda_i = 2N_U(f_i)$, the sup radius of the activity polydisc is $\tau_m(\Gamma) \leq r$, overlapping supports never reach R-045 (S), and the (Hβ) bound $|\nabla f_k| \leq \frac{3}{2}\rho_1$ is a verified reduction for real c (L-182 (Lemma H) and the chain rule), not open. Open inputs: (Hβ) for complex c (open); for real c its inputs R-036 (I) (ledger R-036) and R-037 (H-hist) (ledger R-037); the $\kappa \to \kappa/2$ recovery (ledger R-046, a verified reduction that uses R-020).

5.3 Step 3: the R-045 (S) clauses and R-043 (FX) (ledgers R-045, R-043)

`SOURCES_B21_S.md`**: uniformity of R-045 (S), ledger R-045.** The source bound R-045 (S), $|\kappa_n(f_1, \ldots, f_n)| \leq (C_S\sigma^{-a})^n (n!)^{1+\eta} \prod_i N_U(f_i)$ for real sources with pairwise separation $\sigma \geq 16\varepsilon$ and $\min_i \operatorname{diam} \operatorname{supp} f_i \leq \sigma$, holds uniformly in the bump width $h \in [\varepsilon, \ell/512]$ and in the position of the bumps, with the C^1 norm $N_U(f) = \max(\|f\|_\infty/r, \|\nabla f\|_\infty/\rho_1)$. Every factor of $C^n(n!)^{1+\eta}$ is independent of h. The file traces one rescaled bump through L-111 (Lemma C) of `SOURCES_D4D5.md` and finds one h-dependent term, the cube count $Q_m \leq (2 + D/(M\lambda_m))^4$ at scales finer than the bump. With $D \leq \sigma$ this count is at most $(2 + 2D/\sigma)^4 u^4$, $u = \sigma/(2M\lambda_m)$, and the tree decay $e^{-3\kappa u/8}$ of the polymers that meet all supports cancels it, while the gradient cost is paid by N_U

and the flatness of the test function. Verifier: `SOURCES_B21_S_VERIFY.md`, [VERIFIED REDUCTION], corrected in scope (the inputs must also list the residue of R-043 and D2 §4.3) and in one constant (in the bridge $D/\sigma = 3/8$, not $1/16$, so the factor is $(2.75)^4 \approx 57$). Open inputs: S1 at all scales including $m = 0$; R-043 (FX) in Bałaban's main form (1.98) (ledger R-043 with the choice K-014), whose remaining input is N3 (R-042); the coarsest-scale H-191 (FX2) radius, without which the order is $2 + \eta$ and $D > 2$; D2 §4.3, itself only reduced; the H-098 (E14-c) clause of R-045, which no file maps to L-111 (Lemma C).

`SOURCES_FX.md`**: hypothesis R-043 (FX), ledger R-043.** R-043 (FX), the convergence of the final Mayer expansion of the sum over large-field geometries plus the unit-scale integral, with polymer radius $c \cdot r$ in the source H-191 (FX2), uniform in K and ε, is reduced to Bałaban's large-field bounds CMP 122 II (1.79)–(1.89) and (1.97), the convergence criterion of CMP 116 (2.38)–(2.41) and Cammarota (ledger X-004), and named local checks. The construction is a final R-step with $Z'' = \emptyset$ followed by the unit-scale cluster expansion of CMP 116 §2, as Dimock does for φ^4_3 (Dimock III, Theorem 2). The file reads CMP 122 I (0.2)–(0.6) and CMP 122 II (1.71)–(1.104) and finds that Bałaban already exponentiates the sum over renormalizable large-field components inside each R step, so that only the components that pay for later steps stay as a sum, and at the last scale there are no later steps. The fine-history sum inside a component is bounded by (1.79), because Dimock's power-decay argument ((209)–(211)) does not transfer to $d = 4$, and the source enters the large-field operations only through the upper bound (1.73). Verifier: `SOURCES_FX_VERIFY.md`, [VERIFIED REDUCTION], corrected: the per-step expansion sums only components of diameter $\leq 100MR_k$ counted by position ((1.96)), so the last step, which sums components of unbounded size, needs the tree decay Q-059; three page quotes are fixed ((1.80) has R_n^{d+1} and a local K, (1.85) has $A_1^2 p_0^2$, the condition after (1.94) ends in $-\frac{3}{2}p_0(g_k)$); the fallback order is at most $1+\eta+2/3$. Open inputs: the verifier left Q-059(b), Q-060, and Q-061 open; Q-059(b) and Q-060 are now [PROVED+VERIFIED] (`SOURCES_FX_O1.md`, `SOURCES_FX_O23.md`), R2 is [PROVED+VERIFIED] as a reduction, and Q-062 is avoided by the main form (K-014), so the remaining input of R-043 is N3 (ledger R-042), through Q-061.

`SOURCES_FX_O1.md`**: gap Q-059(b) of R-043 (FX), Lemma [P].** Lemma [P]: for a connected union Y of MR_j-cubes, $d'_{j+1}(S(Y)) \geq c_* d'_j(Y)$ with $c_* = [3 \cdot 4^d \sqrt{d}(\rho+2)^d]^{-1}$, $\rho = LR_{j+1}/R_j$, and no additive constant. With the $n = K+1$ term of CMP 122 II (1.80) this gives $\kappa_K(Y) \geq \kappa_{\text{fin}} d_K(Y)$ for every non-renormalizable final component Y, which is Bałaban's assertion of the term $-\kappa_1 d_k(X)$ on p. 387, with $\kappa_1 = \kappa_{\text{fin}}$. The proof combines three elementary facts (a tree of length ℓ meets at most $4^d(1+2\ell)$ closed unit cubes, $d'_j(Y) \leq \sqrt{d}(n_f - 1)$, and a coarse cube meets at most $(\rho+2)^d$ fine cubes) with the count of at least 21^d coarse cubes in $S(Y)$. The halo $S(Y) \setminus Y$ carries no large fields (p. 384), so nothing is absorbed by p_0, and only the first term of (1.80) is used, all its terms being non-negative. Verifier: `SOURCES_FX_O1_VERIFY.md`, [PROVED+VERIFIED], with one remark corrected (the counterexample to the form "factor $1/L$ plus a constant" needs $d \geq 2$, a constant independent of L, and L large) and R3 discharged by Bałaban's own comparison in (1.86)→(1.87). Open inputs: R2 ($\kappa_{\text{fin}} \geq \kappa_{\text{final}}$), the only point left by the verifier, is closed in `SOURCES_FX_R2.md` (choice K-015); R1 (grid alignment) changes constants only.

`SOURCES_FX_O23.md`**: gaps Q-060 and Q-061 of R-043 (FX).** Q-060: the R-denominator of CMP 122 I (1.100) is one small-field integral per renormalizable component with the bare Wilson action $-g_k^{-2} A(\zeta_i, \cdot)$ and no E-, R-, or B-terms, so it is holomorphic and non-vanishing on a source polydisc of radius $c \cdot r$, with c independent of $|\Lambda_i|$, k, and ε. Q-061: inside the large-

field operations T' the sourced weight obeys $\mathrm{Re}(w+\delta w) \geq (1-\theta_W)(g_k'')^{-2}$, so the upper bounds (1.73), (1.79), (1.89), (1.92), (1.97), (1.100) hold with $\gamma_0 \to \gamma_0(1-\theta_W)(1-\tau_C) \geq \gamma_0/4$ and $p_0 \geq 16/(\gamma_0 A_1^2)$. For Q-060 the Wilson part of the source perturbation is $|\sigma_W| \leq \tau\Phi(g_k)$ with $\Phi(g) = C(M)g^2 p_0(g)^2 R(g)^4 \to 0$, and (1.102) also allows an f-independent denominator. For Q-061 the plaquette terms $P_p \geq 0$ of the real G-valued configuration inside T' (CMP 122 II p. 379) give domination in modulus by a positive operation with a smaller coupling prefactor, while the thresholds stay fixed. Verifier: `SOURCES_FX_O23_VERIFY.md`, [PROVED+VERIFIED] for Q-060 and [PROVED+VERIFIED, CONDITIONAL ON N3] for Q-061, since the strip bound $|\delta w| \leq \frac{3}{2}\tau$ is N3; it adds the cleaner condition $p_0 \geq 8/(\gamma_0 A_1^2)$ with $\gamma_0' \geq \frac{3}{4}\gamma_0$, and corrects Q-062 (the division by $T_k'(Y_i)1$ in (1.103) has no positivity proof, but (1.103)–(1.104) is an optional final remark, so Q-062 is avoided by keeping the main form (1.98)). Open inputs: N3 (ledger R-042); the main form is the choice K-014; Q-062 binds only if (1.104) were used, with fallback order $2+\eta$.

`SOURCES_FX_R2.md`: gap R2 of R-043 (FX), $\kappa_{\mathrm{fin}} \geq \kappa_{\mathrm{final}}$. R2 holds as a reduction in volume form: $\kappa_K(Y) \geq \kappa_{\mathrm{vol}} \# M(Y)$ with $\kappa_{\mathrm{vol}} = \sqrt{d}\,\kappa_{\mathrm{fin}} = C_1 M^d R_{K+1}/(3\cdot 4^d (L+2r)^d)$, $r = R_K/R_{K+1}$, and $\kappa_{\mathrm{vol}} \geq \kappa_{\mathrm{final}} := 2\kappa + \sqrt{d}(1+2\beta)\kappa + c_d$ once $M \geq 12L(3\kappa_{\mathrm{final}}/C_1)^{1/d}$, uniformly in K, R, and g. Here $C_1 := \max(1, C_c)$ is the enlarged constant of (1.80). The file quotes Bałaban's order of constants (L first, $\kappa \geq \kappa_0$ before $M \geq M(\kappa)$, p_0 and γ last), so $\kappa_{\mathrm{final}} = \kappa_{\mathrm{final}}(d, \kappa, \beta)$ is fixed before M (route (a)), after C_1 is enlarged (route (b)). It replaces the tree rate of `SOURCES_FX_O1.md` by the volume rate, because the final expansion charges large-field domains per M-cube and the tree rate alone loses R_K^{d-1}. Verifier: `SOURCES_FX_R2_VERIFY.md`, [PROVED+VERIFIED] as a reduction, corrected in its reasoning: the 2κ is the (1.93) decay-recovery charge $(1+\beta+\beta/(3\cdot 2^d))\kappa$ per M-cube, not a Mayer-link cost, and the term $\sqrt{d}(1+2\beta)\kappa$ is surplus. Open inputs: none; the two minor inputs, $R_{K+1} \geq 1$ and C_c independent of g, are verified readings (former ledger R-049, `SOURCES_BOOKKEEPING.md`), and κ_{final} is the choice K-015, not sharp.

5.4 Steps 4–5: N3, R-040/R-041 (H-hol), and the large-field inputs (ledgers R-042, R-040, R-041)

`SOURCES_HBETA_N3.md`: N3, the strip bound, ledger R-042. With the source additive in $u = g^{-2}$ ($w_0 = g_0^{-2} - f$, τ in absolute u-units), N3 is equivalent to L-182 (Lemma H) of `SOURCES_HBETA.md` for complex c on $Q = \{|\mathrm{Re}\, c| \leq 2r,\ |\mathrm{Im}\, c| \leq \tau\}$, pointwise in x. L-232 (Lemma N3) gives $|\delta w_k| \leq \frac{3}{2}\tau$ for all $k \leq K$, uniformly in K and ε, given R-036 (I), (E), real R-037 (H-hist), R-040/R-041 (H-hol) on relative polydiscs, and H-078 (B-hol), with H-120 (H2-hist) added in the history form; the constant $\frac{3}{2}$ can be any $1+\eta$. Step 0 bounds $\partial\beta_j/\partial u$ on the thin strip by Bałaban's real C^1 bound at $\mathrm{Re}\, u$ plus a Cauchy correction of order $g^4|\mathrm{Im}\, u|$, and a complex Gronwall step with $\sum_i (\mathrm{Re}\, u_i)^{-3/2} \leq 4\gamma/\beta$ closes a bootstrap in $\mathrm{Im}\, c$. The file also shows that Cauchy estimates on relative discs alone give a factor $1+Cg^2$ per step and a bound that grows with K, so the real derivative bound is needed. Verifier: `SOURCES_HBETA_N3_VERIFY.md`, [VERIFIED REDUCTION], corrected: condition (a), $3r + \frac{3}{2}\tau < g^{-2} - \gamma^{-2}$, limits r by the gap between $g = g_K$ and γ and does not follow from γ small (settled in `SOURCES_N3A.md`); $d\beta/du = O(g^3)$, or $O(g^4)$ for even β; H-120 (H2-hist) is sufficient, not necessary, the minimal input being (H-hist$_C$) on the complex polydisc. Open inputs: R-036 (I) (ledger R-036), (E), real R-037 (H-hist) (R-037), R-040 (H-hol) in its small-field (R-040) and large-field (R-041) halves, H-120 (H2-hist) or its Route 2(a) (R-038), H-078 (B-hol), the $h \equiv 0$ booking (K-012), and the additive normalization (K-013); Step 3 in the D$''$ form is proved, unverified (R-034).

`SOURCES_N3A.md`: **condition (a) of L-232 (Lemma N3).** With the order of constants $L \to$ Bałaban's constants $\to \gamma \to g \to (r_0, \tau_0) \to K$ and $g = g_K \le \min(\gamma/2, g_1)$, condition (a) holds at the uniform radius $r_0 = \tau_0 = \min(1/8, r_0^{\mathrm{D2}}, r_0^{\mathrm{D3}}, r_0^{\mathrm{FX}}, \gamma^{-2}/10, 1/(6\gamma^2 c_1 c_H))$, since $\frac{9}{2} r_0 \le \frac{9}{16} < 3 \le g^{-2} - \gamma^{-2}$ for $\gamma \le 1$, a margin of 5.33. The radius does not depend on K, and in u-units it is the physical sup-norm radius of the source. The file checks that $g \le \gamma/2$ is admissible for every K (`LEMMA_DPRIME.md` Step 8 in the L-007 (Lemma E) form, `KAPPA_BOUNDED.md` §5.3(b) in the D″ form) and that nothing forces g_K near γ. It refutes the proposed radius $r = \tau = \gamma^{-2}/2$, which satisfies (a) but breaks conditions (b)–(d) of L-232 (Lemma N3) and the radius bounds of D3R and D3W8. Verifier: `SOURCES_N3A_VERIFY.md`, [PROVED+VERIFIED], with item 1 corrected: "u_k decreasing, g_K largest" holds only in the L-007 (Lemma E) form, while in the D″ form only $u_k \ge u_K - D'$ holds and g_0 can exceed g_K; the D″ sketch has two unnamed gaps. Open inputs: none for condition (a); the order of constants is the choice K-005; the inputs of N3 are unchanged, and N3 in the D″ form was a sketch (ledger R-034); round 7 closes both gaps (Section 5.11).

`SOURCES_HHOL.md`: **R-040 (H-hol), small field, ledger R-040.** For the small-field chain, β_j extends holomorphically to the polydisc $|g_i - x_i| < \rho_0 x_i$ with $|\beta_j| \le M$ uniformly in j, k, and volume, where $\rho_0 = \min(0.1, (LM)^{-2}(8n)^{-1/2})$, divided by $\sqrt{2}$ when the X-modes are counted (ledger K-006). This holds given Bałaban's conditions with modified constants: (C1′)–(C4′) with $\varepsilon_1 \to 1.235 c\varepsilon_1$, H-185 (C4″) $8n\rho_0^2(LM)^4 \le 1$ (with $16n\rho_0^2$ once the X-modes are counted), and (2.24′); the radius $\rho \le 1/3$ first asked for is not reached, and L-232 (Lemma N3) accepts any fixed ρ_0. In reference units $\tilde{B} = xB$ the complex coupling enters only through $w = (x/g)^2$, as the scale of the Gaussian and as the coefficient of the u-prefactored vertices, while the cut $|\tilde{B}| < \varepsilon_1$ does not depend on g. The marginal-domination lemma of the file (its Lemma D) bounds the complex Gaussian by the real one at $\operatorname{Re} w$ at a cost $(|w|/\operatorname{Re} w)^{1/2} \le e^{2\rho^2}$ per real mode, so the volume-uniform bound is made per polymer inside CMP 116 (2.14)–(2.17). Verifier: `SOURCES_HHOL_VERIFY.md`, [VERIFIED REDUCTION], corrected: the per-polymer cost misses the X-modes ($\rho_0 \to \rho_0/\sqrt{2}$), $|\beta_j| \le M$ is a reduction because the majorant covers the Gaussian only, and (C1)–(C5) are not Bałaban's printed labels; single-valuedness is confirmed with no extra assumption. Open inputs: the modified conditions (C1′)–(C4′), H-185 (C4″), (2.24′) (the X-mode question is resolved in the row M-C of the dependency table); (F1), reduced to L-281 (Theorem SM) at complex couplings, from which H-120 (H2-hist) is derived with $a_m = \lambda_C^{m/2}$; a γ-independent C_1 in real R-037 (H-hist) (R-037); the large-field half (ledger R-041, `SOURCES_HHOL_LF.md`).

`SOURCES_HHOL_LF.md`: **R-041 (H-hol), large field, ledger R-041.** The large-field half of R-041 (H-hol) holds on $P_{\rho_0}(x^*)$ around Bałaban's real trajectory x^*, with the R-ratio density, the thresholds, R_k, and N frozen at x^* (ledger K-001): the large-field integrals are entire in $u = g^{-2}$ in finite volume and are dominated in modulus with $\gamma_0 \to \gamma_0'' \ge 0.82\gamma_0$ at $\rho_0 = 0.1$, and F2′ holds on the polydisc with $32\varepsilon' \to 40\varepsilon'$. This is a reduction to H-143 (M-A), H-145 (M-B), H-146 (M-C), and H-147 (M-D). The file shows that the radius $(100MR_k)^{-4}$ of W4 is a radius in the input activities, which the real (1.100) denominator does not contain, and it removes the coupling from the denominator by freezing the ratio at x^*, at a mismatch cost of $O(1)$ per unit cube inside Bałaban's budget $p_0(g) \gtrsim R(g)^{d+2}$. F2′ compares the bound sizes $(x_j^*)^{\kappa_0}$, which are ordered on the real trajectory, so no ordering of complex couplings is needed, and a slot-dependent radius $\rho_j \propto x_j^2$ would have failed L-232 (Lemma N3). Verifier: `SOURCES_HHOL_LF_VERIFY.md`, [VERIFIED REDUCTION], corrected: R_k is unbounded, of order $(\log K)^r$, but largest at $k = 0$ and non-increasing in k (the file reverses the index); $\theta = 0.174$, not 0.26; H-147 (M-D) is not a convention, since the frozen map equals Bałaban's map only at x^* and its derivative there is R-037 (H-hist) plus R-039

(H-thr). Open inputs: H-143 (M-A), reduced to (H−) and CMP 102 (189) (ledger R-005, with R-018 and R-019), with H-144 (M-A′) closed; H-146 (M-C), a verified reduction with open (I-ε), the H-133 (I-V0) residual, and K-025 (M1); H-147 (M-D), which needs real R-037 (H-hist) (R-037) and R-039 (H-thr) (R-039) with R-038 (H-hist2) (R-038); H-145 (M-B) is closed, [PROVED+VERIFIED] with $p_0 \geq 17/(\gamma_0 A_1^2)$ and $\theta_W + \tau_C \leq 0.2$.

`SOURCES_MA.md`: **H-143 (M-A), the equipartition bound.** For the cut-off ratio density of CMP 122 I (1.100) at real weight $0 < u \leq x^{-2}$, the file proves $\langle A - A_{\min}\rangle_u \leq (n_i/u)(1+\eta_x)$, where $n_i = (N^2-1)L_i$ counts the real coordinates on the free links and $\eta_x = 0.68 M_0^2 x^2 p_0(x)^2 + 2c_H M_0^2 x\, p_0(x)^3$; the same bound with the full A is false in general, because the background minimum $A_{\min}$ is $O(\varepsilon^2)$ per block. As a corollary, the freeze-mismatch average is at most $e^{0.1906\, n_i(1+\eta_x)}$, that is $e^{0.77(N^2-1)(1+\eta_x)}$ per scale-k unit cube ($e^{2.3}$ for SU(2)), additive in the volume. A virial identity on the convex cutoff ball, centred at the true minimizer X^* of the exponent, bounds $u\langle (X-X^*)\cdot\nabla G\rangle_u$, $G = A - A_{\min}$, by n_i plus a Haar-Jacobian term (L-201 (Lemma J): $0.68N(N-1)r^2$ per link for $r \leq \frac{1}{2}$). A segment bound from the Hessian lower bound (H−) (L-268 (Lemma S)) turns this into the bound on $\langle G\rangle_u$, and integrating $\frac{d}{du}\log D^G = -\langle G\rangle_u$ gives the corollary. Verifier: `SOURCES_MA_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON (H−)]; corrections: the proved constant is $C = 1$, not $\frac{1}{2}$ (sharp when X^* lies on the boundary), the SU(2) toy value 1.87 needs a cutoff norm that the file does not state, and the bookkeeping item H-144 (M-A′) is a real gap, closed in `SOURCES_MA2.md`. Open inputs: (H−), the Hessian lower bound along segments with c_H uniform in k, volume, and background; `SOURCES_MA2.md` reduces it to CMP 102 (189) (ledger R-005) together with R-018, R-019, and its conditions (C1), (C2); ledger R-041.

`SOURCES_MA2.md`: **H-144 (M-A′) and (H−).** The bookkeeping item H-144 (M-A′) closes: every normalized ratio of CMP 122 I (1.100) sees only $G = F - A_{\min}$, and the exact identity $N_x/D_x = e^{-uF}\,[e^{uA_{\min}}/D^G(x^{-2})]\, e^{(u-x^{-2})G}$ keeps $e^{uA_{\min}}$ at the coupling u, so the mismatch costs at most $e^{1.07(N^2-1)(1+\eta_x)}$ per unit cube. The Hessian bound (H−) of `SOURCES_MA.md` is derived in the form $D^2F[h,h] \geq -c_H\varepsilon_k \sum_b \|h_b\|_2^2$, with c_H uniform in k, volume, and regular backgrounds. Since $A_{\min}$ does not depend on V', $D^A(s) = e^{-sA_{\min}} D^G(s)$, and the error in `SOURCES_HHOL_LF.md` §3.4(c) was a numerator at u over a normalizer at x. For (H−), the chain rule per plaquette is combined with one Cauchy step on $\delta\mathcal{H}/\delta B$ minus its linear part on the complex ball of CMP 102 (172), where the $1/\varepsilon$ of the radius cancels the ε of CMP 102 (187), and the kernel decay of (189) closes a Schur test. Verifier: `SOURCES_MA2_VERIFY.md`, H-144 (M-A′) [PROVED+VERIFIED] and (H−) a [VERIFIED REDUCTION] to CMP 102 (189); corrections: the first Hessian term is ≥ 0, $c_N = O(1)(B_0C_4 + C_2)B_5$, c_H depends on M^5 and M_0, and $A_{\min}$ may be only piecewise analytic in the background, so $e^{uA_{\min}}$ must stay paired with an analytic factor or inside modulus bounds. Open inputs: (C1) $M_0\varepsilon_k \leq C_1\varepsilon_1$; (C2) the localized map U_{k,X_i} obeys the CMP 102 bounds; (C3) CMP 102 (189), announced without proof and already used in Bałaban's own chain (ledger R-005, now a verified reduction to R-001); ledger R-018 (including the k-uniformity of C_2, C_4, B_0) and R-019; and the reading that Bałaban's chain removes $e^{uA_{\min}}$ by algebra.

`SOURCES_MB.md`: **H-145 (M-B), the large-field bookkeeping.** Bałaban's large-field bookkeeping CMP 122 II (1.73)–(1.99) survives the complex-coupling perturbation with thresholds frozen at the real trajectory x^*: the per-cube factors $\exp(-p_0(x_j^*))$ and the per-component factors $\exp(-2p_0(x_j^*))$ keep the same ε_j, δ_j, δ_j', R_j, and the only new condition is $p_0 \geq 17/(\gamma_0 A_1^2)$ with Bałaban's output factor kept (option B). The cost constant becomes $C_c' = C_c + C_m + C_G + C_E$, a fixed $O(1)$ per site inside the $O(1)\log g_j^{-2}$ budget, and the p. 389 condition $p_0 \geq 8\kappa(100R_k)^d + 4$ is unchanged.

Each line of (1.73)–(1.99) that carries γ_0, p_0, ε_j, δ_j, A_1, or a per-cube factor is listed (L1–L24) and redone with $\gamma_0 \to g^*\gamma_0$, which is legitimate because each factor bounds a positive Wilson action or quadratic form whose modulus at complex u is dominated with prefactor $\operatorname{Re} u \geq 0.8264\,(x^*)^{-2}$. The combined factor of p. 381 then needs $p_0 \geq c/(\gamma_0 A_1^2)$ with $c = 2/(g^* - \frac{1}{2})$, and the budget constant $C_1' = \max(1, C_c')$ (route (b) of `SOURCES_FX_R2.md`) multiplies (1.82), (1.88) by a fixed factor that p_0 large and γ small absorb. Verifier: `SOURCES_MB_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON H-143 (M-A) AND H-146 (M-C)]; corrections: the perturbation weights are additive, so $g^* = 0.82645 - \theta_W - \tau_C$ (not a product), and the constant 17 needs the margin $\theta_W + \tau_C \leq 0.2$ (strict limit 0.2088; at $\frac{1}{4}$ one needs $c = 26.2$); p. 383 prints $\frac{1}{2}\gamma_0$ and N^{β_0}, and any fixed $g^* > 0$ is accepted there. Open inputs: H-143 (M-A) (the freeze-mismatch cost C_m) and H-146 (M-C) (the complex-u activities α^c and E-terms); the link from F2′ to α stays an inference; the margin is ledger K-002, and the block feeds ledger R-041.

`SOURCES_MC.md`**: H-146 (M-C), the complex small-field step.** L-223 (Lemma MC): if Bałaban's constants satisfy the conditions B1–B23 of CMP 116 §2, then with the shrinks $\varepsilon_1' = \varepsilon_1/(4\lambda_V)$, $\gamma' = \gamma/(8\lambda_V)$, $\alpha' = \alpha/4$, $\lambda_V = \max(c, 1.235)$, the radius $\rho_0 = \min\{0.1, (LM)^{-2}(0.45a/(16n))^{1/2}\}$, and the added condition K-025 (M1), the small-field step is holomorphic in complex g with $|g/x - 1| < \rho_0$, for every real $x \leq \gamma'$ and inputs in the ball cE_0, and Lemma 3 (2.38) holds with Bałaban's $C_3\varepsilon_1$. Hence β_j is holomorphic on these discs and bounded uniformly in j, k, and volume. The 23 inequalities of CMP 116 §2 are read on the page images and typed, and each one at complex $w = (x/g)^2$ and primed constants is shown to follow from Bałaban's own. The vertex is affine in w with real-domain coefficients, so its factor λ_V is absorbed by the τ-circle (2.18) with $E_0 \to \lambda_V E_0$ and a smaller ε_1; the cut stays real, and the Gaussian cost $16n\rho^2$ per site and the α_5 items fit into the budget $(LM)^4\alpha_i \leq \frac{1}{4}$ under K-025 (M1) and (M2). Verifier: `SOURCES_MC_VERIFY.md`, [VERIFIED REDUCTION] to the printed CMP 116 §2 conditions (B1–B23 and four more), Lemma 2, (1.36), and p. 12 (2.5)–(2.6); corrections: the shrinks are $\varepsilon_1/(4\lambda_V)$ and $\gamma/(8\lambda_V)$, not ε_1/λ_V and γ/λ_V, the total output is $\leq E_0$, not $\frac{1}{2}E_0$, and the open item H-134 (I-X) is resolved by p. 12 (the X-measure is the unit Gaussian). Open inputs at the time of the verifier: (I-ε), that the smaller ε_1, α_0, α_1 keep CMP 116 §1 and the Part I induction valid; the H-133 (I-V0) residual (the $\log\sigma$ and Jacobian part); and K-025 (M1), an "M large" condition; ledger R-041. All three are now settled, see the next four paragraphs: H-133 (I-V0), H-131 (I-ord), and (I-ε) in both layers are closed and verified, and K-025 (M1) is the convention K-025.

`SOURCES_MC_IV0.md`**: H-133 (I-V0), the vertex part of H-146 (M-C).** The w-dependence of the bracket $\{\dots\}$ in CMP 116 pp. 9–10 is only the E_k difference. It is linear in w and free of w otherwise. $\operatorname{Tr}\log$ and $\log\sigma$ sit in $P^{(k)}$ with prefactor 1, and $\log\sigma$ and the Jacobian do not depend on w. The five $1/g_k^2$ terms are each of the form $u \cdot f(C\tilde{B})$, so they are affine in w, and the shift restores (2.19), (2.20), and (2.26) exactly. The file reads pp. 9–11 and 16–17 on 300 dpi page images and checks each w-dependent term against the printed display. Verifier: `SOURCES_MC_IV0_VERIFY.md`, [PROVED+VERIFIED] for the closure of H-133 (I-V0); corrections: the reading of the integral as a Haar integral is an inference [I], since the page prints only $dB'\sigma(B')$; the ε_1^2 cut factor of (2.26) belongs to (I-ε). Open inputs: none for H-133 (I-V0); H-130 (I-lin) is a mild item (per-piece balls).

`SOURCES_T2_IEPS.md`**: (I-ε) and H-131 (I-ord), small-field layer.** Every condition on ε_1, α_0, α_1 in CMP 116 §1–§2 and CMP 109 §1, §3 is an upper bound, a ratio of degree one, or a lower bound relative to a later constant; none is two-sided. So the shrinks of L-223 (Lemma MC) keep these papers valid: the ε_1^2 cut of (2.26) is restored exactly by $\gamma' = \gamma/(8\lambda_V)$, and scaling (A_0, A_1)

by $1/\theta$ restores CMP 122 II pp. 380–381 exactly. H-131 (I-ord): the order $d, L \to \delta \to \kappa \to \kappa_1 \to$ walk constants $\to M \to \beta, \alpha' \to \alpha_1 \to \alpha_0 \to \alpha_3 \to \alpha_2 \to E_0 \to \varepsilon_1 \to \rho_0 \to A_1 \to A_0, p_0 \to \gamma$ has no cycle. Verifier: `SOURCES_T2_IEPS_VERIFY.md`, [PROVED+VERIFIED] for CMP 109 and CMP 116 (five missed conditions found, all of an allowed type); [OPEN] at that time for CMP 119/122, because CMP 119 (2.28) defines $\alpha_{i,j} = g_j C_i (\log g_j^{-2})^{q_i}$ with C_0, C_1 "sufficiently large"; the restoration of (L1) by a smaller γ in the large-field layer was unsupported. H-131 (I-ord) is closed on the pages read. Open inputs: the large-field layer, closed in the next paragraph.

`SOURCES_T2_IEPS_LF.md`: **(I-ε), large-field layer.** The shrink of L-223 (Lemma MC) reaches $\alpha_{i,j}$ (CMP 119 p. 262 takes the restrictions from CMP 109). Every printed condition on C_i, q_i is a lower bound, a ratio ($q_0 \le q_1$), or an upper bound on g; none bounds C_i from above. L-223 (Lemma MC) needs only a factor-4 slack, so C_i and q_i stay fixed and γ is lowered. The order continues $\rho_0 \to$ exponents $\to A_1 \to A_0 \to C_0 \to C_1 \to \gamma$, with no cycle. Residue [I]: α_2–α_4 and γ_2 are not printed in CMP 119/122 (all three branches close); $q_0 > p_0$ versus $q_0 = p_0$ is not printed (both close). Verifier: `SOURCES_T2_IEPS_LF_VERIFY.md`, [PROVED+VERIFIED]: all four claims confirmed; corrections: the condition C3 is a printed inclusion with an unprinted constant $c(B_3)$, and $q_0 \ge p_0$ is forced; the p. 381 factor is $\exp(-\gamma_0 A_1^2 p_1^2(g_j))$ with no extra weight; one inequality on ν reverses ($\frac{1}{2}\nu \le p_0 - p_1 - 1$); $\gamma' < \min(e^{-q_0}, e^{-q_1})$ is a further upper bound on γ, harmless. Open inputs: none beyond the two [I] residues above, each of which closes on every branch.

`SOURCES_HIGHER_M1.md`: **higher derivatives, and K-025 (M1).** Part A: no consumer in the coupling and threshold chain needs derivatives of order greater than 2 in the couplings or thresholds. Part B: the condition K-025 (M1), "M large", is a block-size choice. The file lists 26 lower bounds on M (for example $M \gg \kappa_1$, $M \ge M(\kappa)$, $M \ge cL$, $M \ge 48\kappa_1/\varepsilon$, $M/M_1 > 3(6 + c_X)$) and no upper bound, and the order of constants puts M before α. So K-025 (M1) is the convention K-025: M is chosen at least the largest of these lower bounds. Verifier: `SOURCES_HIGHER_M1_VERIFY.md`, [PROVED+VERIFIED] with minor corrections: Part A is restricted to the coupling, threshold, and MA2-field variables (Bałaban's field expansion goes to order 4, and the source derivatives of D2/D3 to all orders); the lower bounds of `SOURCES_A3_SIGMA.md` 5.1 and 5.3 join the list. Open inputs: none; K-025 (M1) leaves ledger R-041 and becomes ledger K-025.

5.5 Step 6: thresholds and second history derivatives (ledgers R-039, R-038)

`SOURCES_HTHR.md`: **R-039 (H-thr), first attempt.** With the threshold mismatch $\Delta_j(c) = B_j(u(c); \bar{u}) - B_j(u(c); u(c))$ (integer parameters frozen, convention L3), the file bounds $\sup_c |\Delta_j'(c)| \le 3(1 + 3r) A\, \Phi(g_j^{-2})$ and sums this along the trajectory to $C\beta^{-1}\gamma^{-2(1+a)}(\log \gamma^{-2})^P$ times a decay factor, uniformly in K; in the verifier's corrected form the decay is $\exp(-(\log \gamma^{-2})^q)$ with $q = 2p_1 - (d+5)r_0 > p_0$, in place of $\exp(-\kappa(\log \gamma^{-2})^{2p_0})$. L-191 (Lemma H′) then splits the budget: $x \le \ln(6/5)$ for R-037 (H-hist) and $x_{\mathrm{thr}} \le \frac{1}{4}$ give $\sum_j |B_j'| \le \frac{1}{2}$ and $\frac{1}{2} \le |v_k| \le \frac{3}{2}$. Coarea writes Δ_j as the contribution of the shell between the two thresholds, which reduces Δ_j' to threshold derivatives and mixed threshold–weight derivatives of B_j, bounded in L-267 (Lemma S) under the model assumptions M1–M4. The sum along the trajectory uses $u_i \ge \gamma^{-2} + \beta m$ and an integral comparison that keeps the factor γ^{-2}/β, and L-191 (Lemma H′) is an additive Gronwall argument. Verifier: `SOURCES_HTHR_VERIFY.md`, [VERIFIED REDUCTION] of R-039 (H-thr) to M2′, M4, M5, with the shell representation, the trajectory sum, and L-191 (Lemma H′) confirmed; corrections to L-267 (Lemma S): the fluctuation cutoff is $\delta_j' = g_j A_1 p_1(g_j)$ with $p_1 < p_0$, Bałaban bounds one bond only through the block quadratic form CMP 122 II (1.78) with loss $R_j^{-(d+5)}$, so the per-link density M2

is unsupported, and the mixed derivative needs history holomorphy (M5); M1 is resolved (χ stays inside the integral, so $\Delta_j \neq 0$). Open inputs: M2′ (density form of the block bound), M4 (extraction cost), M5 (history holomorphy), M3, and the frozen-integer convention L3 (ledger R-039, K-001); `SOURCES_HTHR2.md` replaces M2′ by L-285 (Lemma T) and M5 by R-038 (H-hist2).

`SOURCES_HTHR2.md`: **R-039 (H-thr), L-285 (Lemma T).** The shell indicator is at most the large-field function $1-\chi$ at the lower threshold, so Bałaban's printed tail factor of CMP 122 II (1.77)–(1.78) (with the misprint $p_1 \to p_1^2$) bounds Δ_j itself with exponent at least $\kappa_2(\log s)^q$, $\kappa_2 = \frac{1}{2}L^{-(d+5)}$, $q = 2p_1-(d+5)r_0 > p_0$ (L-285 (Lemma T)). With a uniform bound D on Δ_j'' this gives $\sup_c|\Delta_j'| \leq 2(2AD)^{1/2}\Phi_T(u_{j-1})^{1/2} + (2A/r)\Phi_T(u_{j-1})$, a sum over j that is uniform in K under H-195 (G1′)–H-198 (G3′), and L-191 (Lemma H′) of `SOURCES_HTHR.md` unchanged. The zero-order bound on Δ_j uses L-285 (Lemma T), the history weights M3, and the extraction cost M4, a Cauchy estimate on the analyticity radius $\alpha_{1,j} = g_j C_1(\log g_j^{-2})^{q_1}$ of CMP 119 (2.28), so no history derivative enters. The derivative then comes from the interpolation inequality $\|\Delta_j'\| \leq 2(\|\Delta_j\|\,\|\Delta_j''\|)^{1/2} + \|\Delta_j\|/r$ (Lemma 7.1(a) of `SOURCES_D2.md`), which replaces the history holomorphy M5 by a uniform second-derivative bound R-038 (H-hist2). Verifier: `SOURCES_HTHR2_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON R-038 (H-HIST2)] for the summed bound, with L-285 (Lemma T) and the misprint confirmed and M4 confirmed as reduced; correction: R-038 (H-hist2) for β_j contains the mixed threshold–history derivative through the running thresholds, so it is a new hypothesis of the R-037 (H-hist) kind, not free by a Cauchy estimate in the history. Open inputs: R-038 (H-hist2) (ledger R-038, a [VERIFIED REDUCTION] to the real uniform R-037 (H-hist), ledger R-037); M3; M4 (units, and the small-field analogue of CMP 122 II (1.73)); the transfer of Bałaban's printed large-field factor to small-field shell terms; the frozen-integer convention L3; ledger R-039.

`SOURCES_HHIST2.md`: **R-038 (H-hist2), second history derivatives.** For the frozen-threshold map $F = B_j(\cdot;\bar{u})$ on $|w-\bar{u}| \leq 3r$ in u-slots, $\sum_{i,l}|\partial_{u_i}\partial_{u_l}F| \leq \varepsilon_j'' = C_H(1-\lambda^{1/2})^{-2}u_{j-1}^{-2}$, so $\sum_j \varepsilon_j'' \leq C_H(1-\lambda^{1/2})^{-2}\gamma^2/\beta$ uniformly in j, K, and volume, with product decay $\lambda^{m/2}$; the first-derivative half is $\varepsilon_j' \leq \frac{1}{2}C_1(1-\lambda)^{-1}u_{j-1}^{-3/2}$, from the real uniform R-037 (H-hist). For β_j with running thresholds the R-038 (H-hist2) step needs only a uniform bound, but L-191 (Lemma H′) still needs a summable first derivative of β_j from the real R-037 (H-hist). Route 2(a) differentiates the real chain twice: past slots stay real, the input variable is handled by Cauchy estimates (L-281 (Theorem SM)), and only the own slot is complexified, on one relative u-disc (`SOURCES_SM.md` §4.2). The Jacobians $\partial_{u_i}\partial_{u_l} = \frac{1}{4}g_i^3 g_l^3 \partial_{g_i}\partial_{g_l}$ turn the $1/x$ losses into g^4, which is summable because $u_{j-1} \geq \gamma^{-2} + \beta m$. Verifier: `SOURCES_HHIST2_VERIFY.md`, [VERIFIED REDUCTION] to the real uniform R-037 (H-hist), no new input; the Jacobian cancellation and the decay $\lambda^{m/2}$ are confirmed, the product form λ^m is [REFUTED] (toy recursion), and $\varepsilon'' \propto u^{-2}$ follows from the u-disc alone; correction: "β_j needs only a uniform bound" holds for the R-038 (H-hist2) step only, since L-191 (Lemma H′) needs a summable β_j'. Open inputs: the real uniform R-037 (H-hist) (ledger R-037: (Q)/c_-, (M_r), CMP 102 (189), H-145 (M-B) bookkeeping); L-281 (Theorem SM) and the joint u-disc and input-ball holomorphy [R]; the large-field part of the frozen map; for β_j, the mixed threshold–history derivative through L-133 (Lemma D) and M3 (ledger R-038).

`SOURCES_SM.md`: **L-281 (Theorem SM), the step-map Lipschitz bound.** The small-field step map factors as $G = \hat{G} \circ \mathcal{L}$ with $\mathcal{L}$ linear, and L-281 (Theorem SM) gives $|DG(I_0)[W]| \leq \theta_0 \sum_{l\leq k} L^{-\alpha_*(k-l)}\|W^{(l)}\|\, e^{-\kappa' d_{k+1}(X)}$ with $\theta_0 = 2K_H K_V \varepsilon_1$ and $\alpha_* = \min(1,\alpha)$. With it L-190 (Lemma HH) holds at rate $\lambda = L^{-\alpha_*} + C_V\theta_0$ when $C_V\theta_0 < 1-L^{-\alpha_*}$, and (Eps) of `SOURCES_D3W8.md` becomes a corollary with $\varepsilon' = \theta_0/(1-L^{-\alpha_*})$. The per-input bound CMP 116 (1.24) gives the

linear stage a fluctuation factor $\alpha_2^{-1}\varepsilon_1$ and a label weight $L^{-(k-l)}$ (L-050 (Fact A)). The map $\hat{G}$ is holomorphic on a ball of radius $R_V = \alpha_4\alpha_6 e^{-5\kappa}/(4(L+2)^4 O(1))$, free of ε_1 and E_0, by CMP 116 (C1)–(C5) (L-082 (Fact B)), and a Cauchy estimate on this ball gives the contraction. Verifier: `SOURCES_SM_VERIFY.md`, [VERIFIED REDUCTION] (items 1–4 confirmed on page images, items 5–6 corrected); the own-slot constant C_β has the same $1/g$ gap as S_0, so H-111 (G-gain) must be applied in both the hop and the own-slot bound, and the per-label decay of the lower-order terms stays reduced. Open inputs: L-050 (Fact A) for the lower-order bracket terms (ledger R-010, now Theorem 5.6 in the corrected form R-011; the per-label exponent is $\alpha_* = \beta_H$, a free parameter, and (Eps) reads $\varepsilon' = \theta_0(1 - L^{-\beta_H})^{-1}$); L-082 (Fact B), that is R-044 (W4) on CMP 116 (C1)–(C5) with α_4, α_6 fixed before ε_1; R-036 (I)/L3 (ledger R-036); H-111 (G-gain) in both slots, carried by `SOURCES_GGAIN.md` and `SOURCES_GGAIN_LF.md` (ledger R-037).

5.6 Step 7: history bounds and the gain (ledger R-037)

`SOURCES_HHIST.md`: **L-189 (Lemma HH), R-037 (H-hist) in column-sum form.** Given CMP 109 (0.29), uniform own-slot bounds $\|\partial_{g_{l-1}} V^{(l)}\|_l \le S_0$ and $|\partial\beta_l/\partial g_{l-1}| \le C_\beta$, and a step-map Lipschitz bound (SM) with $\theta < 1 - L^{-\alpha}$, L-190 (Lemma HH) gives $|\partial\beta_j/\partial g_i| \le (C_\Pi/C_V) S_0 \lambda^{j-1-i}$ for $i \le j-2$, with $\lambda = L^{-\alpha} + \theta$. The column sum is then at most $C_\beta + (C_\Pi/C_V) S_0 \lambda/(1-\lambda)$, uniformly in i, j, K, and the volume. The past enters step $k+1$ only through the curly bracket of CMP 109 (2.12) and only through the irrelevant remainders $V^{(l)}$ (the marginal parts cancel, p. 298), and a step-l remainder costs $L^{-\alpha(k-l)}$ per π_k cube by (0.29)–(0.30). An induction on $s_l = \|\partial_{g_i} V^{(l)}\|_l$ with one hop per step then gives the rate λ in place of the one-hop rate $L^{-\alpha m}$. Verifier: `SOURCES_HHIST_N1_VERIFY.md` (Part 1), [VERIFIED REDUCTION]: the quotes, the dependence chain, the cube count, and the slot definition are confirmed; the rate is sharp, but the numerical claim "ratio 1.0" is false ($s_{i+1+n} = (\theta/\lambda)\lambda^n S_0$, so the bound is loose by θ/λ), and Bałaban gives no per-hop Lipschitz constant θ. Open inputs: the own-slot bounds S_0 and C_β (C_β is Bałaban's inductive assumption, ledger R-006) and (SM), now carried by `SOURCES_GGAIN.md` and `SOURCES_SM.md`; validity of the step bounds off the trajectory, close to R-036 (I) (ledger R-036); the large-field contributions, now in `SOURCES_GGAIN_LF.md`; the whole chain is ledger R-037.

`SOURCES_GGAIN.md`: **H-111 (G-gain), small field.** For every step k and $g_k \in]0, \gamma]$ the own-slot derivatives obey $\|\partial_{g_k} V^{(k+1)}\| \le C_V S_0$ and $|\partial\beta_{k+1}/\partial g_k| \le C_\beta := C_\Pi S_0$, with S_0 free of $1/g_k$ and of ε_1 (L-098). With L-281 (Theorem SM) the column sum of L-190 (Lemma HH) is uniform, and in u-slots it is $O(g_i^3)$, which meets the threshold $C g_i^{2+\delta}$ of `SOURCES_HBETA.md` with $\delta = 1$. By CMP 116 (1.34) the vertex $V_k'(Y, \cdot, B)$ is an analytic function of $x = g_k B$ on a polydisc of g-free radius ε_1, so a pointwise Cauchy estimate in x costs $|B|_Y/\varepsilon_1$, the ε_1 cancels against the vertex size, and the one polynomial insertion is absorbed by the Gaussian at strength α_4. The $P^{(k)}$ part is cubic in B' by (1.39)–(1.40), so $|\partial_g| \le C_P |B|^3$ there, and in the hop the factor $g_k|B|$ of (1.22)–(1.24) gives $\theta_0 \to \theta_g g_k$. Verifier: `SOURCES_GGAIN_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON L-174 (LEMMA G3) AND THE MARGIN (M_r)], for the small-field step on $]0, \gamma]$ including vacuum subtraction (items 1, 2, 4, 5, 6 confirmed, 2 and 6 with corrections, item 3 corrected); (M_r) is also needed for the $P^{(k)}$ part, L-178 (Lemma G4) fails as stated and is off-chain, the sharp-cut surface constant is $O(\gamma_2^{-(n_b+1)/2}\varepsilon_1^{-1})$, and Bałaban's p. 264 assumption is proved only in its first-derivative part, on $]0, \gamma]$, for the small-field step. Open inputs: L-174 (Lemma G3), hence (Q)/c_- (round 5: proved relative to listed inputs, Theorem 5.9); the margin (M_r), which shrinks the χ_k threshold (ledger K-009); the large-field half (`SOURCES_GGAIN_LF.md`); the endpoint $g = 0$ (ledger K-020); all inside ledger R-037.

`SOURCES_GGAIN_LF.md`: **H-111 (G-gain), large field.** With the thresholds frozen at Bałaban's trajectory x^*, the own-slot derivatives of the large-field outputs are uniform in $g_k \in]0,\gamma]$ with $|g_k - x_k^*| \le \rho_0 x_k^*/2$, in k, K, and the volume, for example $|\partial_{g_k} R'^{(k)}(X)| \le 20\gamma^{2A_0-1} e^{-\kappa d_k(X)}$ and $|\partial_{g_k} T_k(Y_i)1| \le 20\gamma^{A_0-1} e^{-\frac{1}{2}p_0(x_k^*)}$ (L-097). With `SOURCES_GGAIN.md` and L-281 (Theorem SM) this makes the real uniform R-037 (H-hist) a reduction in both halves. On the disc $|g-x| < \rho_0 x$, $\rho_0 = 0.1$, of `SOURCES_HHOL_LF.md` the printed bounds hold with $\gamma_0 \to 0.826\gamma_0$, and one Cauchy step gives $|\partial_g F| \le 20x^{-1} \sup|F|$. The factor $1/x$ is absorbed by the large-field suppression, since $p_0(x) = A_0(\log x^{-2})^{p_0}$ with p_0 a positive integer and A_0 large (CMP 119 p. 244) gives $x^{-1}e^{-\theta p_0(x)} \le x^{2\theta A_0 - 1}$. Verifier: `SOURCES_GGAIN_LF_VERIFY.md`, [VERIFIED REDUCTION] (items 1, 2, 4, 5 confirmed, item 3 corrected); the mismatch average sits at $u' = 2x^{-2} - u \in [0.892, 1.093]x^{-2}$, which gives the constant $2.48 \le 2.6$, the Gaussian normalization adds a factor $e^{0.01n}$ per unit cube inside H-145 (M-B), and "β from E-class terms only" is upgraded to a verified reading of CMP 119 p. 279. Open inputs: (Q)/c_-, the margin (M_r), CMP 102 (189) through H-143 (M-A) (ledger R-005), and the H-145 (M-B) bookkeeping (ledger R-037); not covered: (1.103) at complex u, C^1 at $g = 0$, higher derivatives.

`SOURCES_G3.md`: **L-174 (Lemma G3), the χ_k surface term.** Given (Q) with constants c_- and Λ, for every k and $g_k \in]0,\gamma]$ the g_k-derivative acting on the sharp cut $\chi_{k,Y_0}\chi^c_{k,P}$ of CMP 116 (2.3) in a term of (2.14) is at most $c_1' U(g_k) e^{\delta\kappa d_k(Z_0)} T_{\mathrm{maj}}$, with $U(g) = A_n D_n \varepsilon_1^{-1} e^{-\varepsilon_1^2/(8\Lambda g^2)}$ and $n = N^2 - 1$. The cut adds $S_\chi = c_2 A_n D_n \varepsilon_1^{-1} e^{-\varepsilon_1^2/(8\Lambda\gamma^2)}$ to S_0, uniform in g_k, X, k, and the volume, and $O(\gamma^p)$ for every p. Route 2 keeps Bałaban's cut, a round sphere $|B(b)| = \varepsilon_1/g_k$ per bond, and computes the surface term in polar coordinates, where the Gaussian marginal turns $g^{-1}(\varepsilon_1/g)^n$ into $\varepsilon_1^{-1} t^{n+1} e^{-ct^2}$, $t = \varepsilon_1/g$, with no $1/g$ left. The bond and polymer sums use (2.30), and the smooth cut $\hat\chi$ (Route 3, $\varepsilon_1 \to \varepsilon_1/2$, $\gamma \to \gamma/2$) is the fallback. Verifier: `SOURCES_G3_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON (Q)] (items 2, 4, 5 confirmed, items 1 and 3 corrected); the surface term is real in the fixed-B computation that `SOURCES_GGAIN.md` uses (finite-difference check to 10 digits at $n = 8$), (Q) must be read on the joint (X, B) majorant with $a = 0$, the bond count is $c_d|Y_0| + |P|$, the printed γ_k fails under both parses, and the B versus B' cut question the verifier added is closed in `SOURCES_Q.md`. Open inputs: (Q)/c_- (row B-S.Q; ledger R-037), now discharged for the small field by Theorem 5.9; numerical values of c_- and Λ; higher derivatives.

`SOURCES_Q.md`: **input (Q), the covariance lower bound $c_- > 0$.** The $B(b)$-block of the covariance of Bałaban's joint Gaussian majorant CMP 116 (2.23) is bounded below by $c_- > 0$ and above by $\Lambda \le 2(1 + \gamma_0^{-1})$ for $\alpha_5(1 + \gamma_0^{-1}) \le \frac{1}{2}$, uniformly in k, the volume, and regular U; after the localization step of `SOURCES_LOC.md`, $c_- = (\gamma_1 + \varepsilon_M)^{-1}$, and in the free case $c_- \ge (a\kappa_d L^{d+1})^{-1}$. The cut of CMP 109 (2.9) is $|B(b)| < \varepsilon_1/g_k$ on every bond except b_0, a round sphere with no Jacobian. The precision of (2.23) is $Q = Q_0 - \alpha_5 \operatorname{diag}(Z, Z_0) \le Q_0$, and inversion reverses order, so the $B(b)$-block of Q^{-1} dominates that of Q_0^{-1}, which is $C_0 + C_0\Gamma\Gamma^* C_0 \ge C_0$ with $C_0 = C^{(k)}(Z_0, 0)$. The remaining norm input $\|C^*\Delta_k C\| \le \gamma_1$, which CMP 116 (2.7) itself uses, is traced to CMP 99 (3.132), and the free-case constants come from CMP 95 (1.100) and CMP 96 (2.125). Verifier: `SOURCES_Q_VERIFY.md`, [VERIFIED REDUCTION] to CMP 99 (3.132), (E_U), and LOC (items 1, 3, 4 confirmed, 1 and 3 with corrections, item 2 corrected in favour of the reduction); Q_0 is coupled, so the block identity is only an inequality, the operator is the $\sigma = 0$ localized one (handled in `SOURCES_LOC.md`), CMP 99 p. 422 prints the kernel bound (3.132), and κ_d depends on a and L unless $a = 1$. Open inputs: the R-001 residue (torus-side hypothesis, two conventions, reading $a = 1$); the two [I] bridges $Q(U) = Q_{\mathrm{free}}(1 + O(\alpha_0))$ and $\|C(U)\|^2 \le L^{d+1}(1 + O(\alpha_0))$ (ledger R-018); (E_U), reduced to k-uniform kernel moments that Bałaban does not print; the inputs of LOC.

Round 5 assembles (Q) from R-008 and R-009 instead (Theorem 5.9); its Λ corrects 4.3 of this file, because G_Γ enters the B-block of Q_0^{-1}.

`SOURCES_LOC.md`: **L-213 (Lemma LOC), the localized covariance $C^{(k)}(Z_0,0)$.** Given $\gamma_0 \le C^*\Delta_k C \le \gamma_1$ and $C_0^{-1} = Z_0C^*\Delta_kCZ_0 - R$ with $\|R\| \le \varepsilon_M \le \gamma_0/2$, the localized covariance $C_0 = C^{(k)}(Z_0,0)$ of CMP 116 (2.23) has $C_0(b,b) \ge c_-I$ on every bond $b \subset Z_0$ and $\|C_0\| \le 2/\gamma_0$. Here $c_- = (\gamma_1+\varepsilon_M)^{-1}$ with the symmetry (S) of the truncated walk sum and $c_- = (\gamma_0-\varepsilon_M)/(\gamma_1+\varepsilon_M)^2$ without it, and $\varepsilon_M = S_d(O(1)e^{-\delta_0 M/3}+O(\alpha_0+\alpha_1))$ with $S_d \le d\coth(\delta_0/(4\sqrt{d}))^d$. The printed (2.5) is conditioning on Z_0^c, so $C^{(k)}(Z_0)$ is the inverse of the compression of $C^*\Delta_kC$ and inherits its bounds, while $C^{(k)}(Z_0,0)$ is the walk sum with the walks meeting σ_0 deleted and is not a compression. Its distance to the compression is Bałaban's bound (2.16) applied to R_2 at $\sigma \equiv 1$, turned into the norm bound ε_M by the Schur test. Verifier: `SOURCES_LOC_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON THE R_2 BOUND (2.16), γ_0, AND γ_1] (item 1 confirmed, items 2–4 corrected); the symmetry (S) is not needed, $\varepsilon_M \to 0$ needs $\alpha_0,\alpha_1 \to 0$ as well as $M \to \infty$, γ_0 sits in the prose after CMP 99 (3.158) and is proved there only at $U = 1$, and the replacement point is the real configuration U. Open inputs: the R_2 bound (2.16), which Bałaban states with "similarly" (ledger R-003, now a verified reduction to (W^c), R-008, and two [I] items); the upper bound γ_1 through CMP 99 (3.132) and (E_U); γ_0 for general U (ledger R-004, now a verified full proof, $\gamma_0 \ge m_L/(48d^2L^d)$, with residue R-$\Delta$2); α_0,α_1 small after M is fixed.

5.7 Step 8: Bałaban-level propagator inputs (ledgers R-001–R-005)

`SOURCES_A1_KERNEL.md`: **ledger R-001, kernel-decay form.** Given (K) and H-057 (C) below, the file proves CMP 99 (3.132) in its printed two-scale form, $|(QGQ^*)^{-1}(y,y')| \le 2K_{\text{ext}}L^{d/2+1}(L^j\eta)^{-2}(L^{j'}\eta)^{-d}e^{-\delta_1 d(y,y')}$ for $y \in \Lambda_j$, $y' \in \Lambda_{j'}$, uniformly in j, k, η, the volume and regular U, and the same for $(QG_1Q^*)^{-1}$ plus a diagonal a-term. The rate comes from $\varepsilon = \min\{\delta/2,\, c_-/(2c_+\Sigma(\delta/2))\}$ and is much smaller than Bałaban's rate. L-125 (Lemma CT), a Combes–Thomas argument on $\hat N = S^{-1/2}(QGQ^*)S^{-1/2}$ on $\ell^2(\mathfrak{B},\mu)$, turns kernel decay (K) of QGQ^* and a lower bound H-057 (C) $\hat N \ge c_-$ into decay of the inverse, with no Fourier analysis and no use of R-004. (K) is taken from CMP 99 Theorems 3.3 and 3.12 for G, H-057 (C) from one feasible extension in the variational form $\langle B, N^{-1}B\rangle = \min\{\langle A, G^{-1}A\rangle : QA = B\}$, and the global perturbation from $U = 1$ and the Demko–Moss–Smith route are shown not to work. Verifier: `SOURCES_A1_A4_VERIFY.md` (Part A), [VERIFIED REDUCTION]; it confirmed L-125 (Lemma CT), the two-scale form, the non-circular order of the G_1 clause, and both refutations, and it found the file's face-midpoint bumps for H-057 (C) not feasible ($Q_j(U)$ is supported on both whole blocks, CMP 98 (14), CMP 99 (3.102)), so its own grade was "reduced to H-057 (C), open"; the newer `PLAN.md` entry regrades R-001 to a verified reduction once H-057 (C) is supplied by `SOURCES_QE_INV.md`, `SOURCES_QE_KE_HQ.md`, and `SOURCES_S2_FORM.md`. Open inputs: the residue of ledger R-001, namely the torus-side hypothesis (side $\ge c \cdot RM$ coarsest blocks), two unprinted conventions, and the reading $a = 1$ of CMP 99 Theorem 3.1 [I]; the acceptance of the small rate δ_1 by each downstream consumer (S4) is not re-checked (low risk).

`SOURCES_QE_KE_HQ.md`: **H-136 (KE′) and H-123 (H_Q), inputs of H-057 (C).** H-136 (KE′): Bałaban's H of CMP 96 Cor. 2.8 already satisfies $Q_{98}(1)H = I$ on all of $\mathfrak{B}$, interfaces and Λ_0 included, with the decay of CMP 96 (2.151) for H and ∇H, because CMP 96 uses the straight-line average CMP 95 (1.11)/(1.18). H-123 (H_Q): $|K_j(U;c,b) - q_j(c,b)\,\mathrm{Ad}\,U(\Gamma_{c_-\leftarrow b})| \le c_H\alpha_0\ell_j^2\eta_j^d$ on $B^j(c_-)\cup B^j(c_+)$, $c_H = 2C_1' + c_T(d)L$, uniformly in $j \le k$. For H-136 (KE′) the file reads the pages: the invariance group $Q_j'\lambda = 0$ of CMP 96 (2.7) fits only the straight-line average, and CMP 98 p.

27 states that the propagators hold for the average (1.11). For H-123 (H_Q) it applies Bałaban's induction (144)–(145) to the difference between $Q_j(U)$ and the product of the main terms, and it compares the averaged background links with fine straight-line transports and fixed axial contours by a cone-area bound. Verifier: `SOURCES_QE_KE_HQ_VERIFY.md`, [PROVED+VERIFIED], H-136 (KE′) by citation on the page images (CMP 95 (1.11), (1.12), (1.18); CMP 96 (2.7), (2.20), (2.35); CMP 98 p. 27 verbatim), so the gap of `SOURCES_QE_INV_VERIFY.md` item 2 was a misreading; corrected: the verifier proved the one [R] step (iii) itself (ribbon of $\le (d-1)L^2$ plaquettes, lasso decomposition, error $\le 9.6(d-1)\alpha_0\ell_i^2$ for $\alpha_0 \le \min(c_2, 1/(8(d-1))))$, the cited (112) is the wrong reference and is not needed, and the corridor fallback $H + d\psi$ is exact within layers but not across interfaces (not needed). Open inputs: the unvalued constants $C_1'(d,L)$ and $c_{\mathrm{cone}}(d)$, and two unprinted conventions ($U(c)$ the straight segment; translation-covariant axial contours), both in the residue of ledger R-001.

`SOURCES_QE_INV.md`**: L-252 (Theorem QE), input H-057 (C) of L-125 (Lemma CT).** For the free multiscale minimizer dressed with axial transports, $\mathcal{E}_U$, the file proves, given (KE) and H-123 (H_Q), $\|Q(U)\mathcal{E}_U - I\|_S \le c_Q M\alpha_0 + c_{\mathrm{far}} e^{-\delta_H RM/(16L^2)}$ in the S-weighted norm on $\mathfrak{B}$, uniformly in j, k, the volume, $\{\Omega_j\}$, and regular U. Hence $K_Q \le 2$, and $A = \mathcal{E}_U(Q(U)\mathcal{E}_U)^{-1}B$ is an exact extension, $Q(U)A = B$ and $A = 0$ on Λ_0, with $\langle A, G^{-1}A\rangle \le K_{\mathrm{ext}}\|B\|_S^2$, $K_{\mathrm{ext}} = 4(1 + O(M\alpha_0))K_{\mathcal{E}} + a$, which gives H-057 (C) with $c_- = 1/K_{\mathrm{ext}}$. The defect of $Q(U)\mathcal{E}_U$ is written exactly as a $q|E|$-weighted sum of loop holonomies minus one, and a cone-area lemma bounds each holonomy linearly in the distance. A Schur test with the decay (KE), a near/far split across layers with far terms e^{-cRM}, and a Neumann series then give $K_Q \le 2$, and a crude absolute-value Schur bound gives the energy. Verifier: `SOURCES_QE_INV_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON H-136 (KE′), H-123 (H_Q), S2, AND A TORUS SIDE $\ge c{\cdot}RM$]; numerics [NUMERICAL, REPRODUCED] (defect linear in the curvature, slope 0.010 at $d = 2$ and 0.019 at $d = 4$, rigorous Schur constant 41–76 times the true slope); corrected: ε'' is of order $M\alpha_0$, not α_0, $\alpha_F = 2 \cdot O(1)M\alpha_0$ with $O(1) \ge 10$ (so the Neumann threshold reads $O(1)M\alpha_0 \lesssim 0.08$), and a torus-size hypothesis must be added. Open inputs: H-136 (KE′), H-123 (H_Q), and S2 for G have since been closed (`SOURCES_QE_KE_HQ_VERIFY.md`, `SOURCES_S2_FORM_VERIFY.md`); what stays open is the torus-side hypothesis and the two unprinted conventions of ledger R-001.

`SOURCES_S2_FORM.md`**: S2, the form bound for** G**.** For every bond field A, $|\langle A, \Delta_\pi' A\rangle| \le C_\Pi' M\alpha_0(\|DA\|^2 + \|D^*A\|^2 + \|\ell^{-1}A\|^2)$, hence $G^{-1} \le (1 + O(M\alpha_0))(D^*D + DD^*) + O(M\alpha_0)\ell^{-2} + Q^*aQ$ and $\langle \mathcal{E}_U X, G^{-1}\mathcal{E}_U X\rangle \le K(d,L,a)\|X\|_S^2$, uniformly in j, η, the volume, $\{\Omega_j\}$, and regular U. The route through CMP 99 (3.131) stalls, because (3.131) is printed for one-block supports and a tiling then forces sharp cut-offs that cost η^{-1} on the sup side. Route Π instead uses the exact identity (3.119), $\langle A, \Delta_\pi A\rangle = \langle \tilde{A}, \Delta\tilde{A}\rangle$ with $\tilde{A} = A - D\lambda$, $\lambda = G'RD^*A$, the curl of a gradient $\eta^{-2}(\mathrm{Ad}\,U(\partial p) - 1)\lambda$, and the block L^2 bounds (3.46) of Theorem 3.1 with CMP 96 Lemma 2.1 and a Schur test, which give $\|\ell^{-2}\lambda\|, \|\ell^{-1}D\lambda\| \le B_0 c_1 \|D^*A\|$. Verifier: `SOURCES_S2_FORM_VERIFY.md`, [PROVED+VERIFIED, CONDITIONAL ON H-136 (KE′), $a = 1$ IN THEOREM 3.1, AND THE BLOCK GEOMETRY], together with §5.1 and §5.3 of `SOURCES_QE_INV.md`; corrected: Δ' of (3.10) has a second term $\eta^d \,\mathrm{tr}(DA)^2(\mathrm{Re}\,U(\partial p) - 1)$, bounded by $O(M\alpha_0)\|DA\|^2$, so $C_\Pi' = C_\Pi + 2c_R(1 + \kappa^2 a_0^2)$ and c_- drops only by a factor $1 - O(M\alpha_0)$; (3.120) carries a minus sign; the normalization of Q^*aQ is settled by (3.16) (one constant a). Open inputs: the reading $a = 1$ of Theorem 3.1 and the block-geometry item (both in the residue of ledger R-001); the G_1 analogue, left at sketch level here, is closed in `SOURCES_S2_G1_V7.md`.

`SOURCES_S2_G1_V7.md`: **S2 for G_1, and the V7 repair of ledger R-003.** Part A proves $|\langle B, \Delta^{(2)} B\rangle| \le c_2 M\alpha_0 \|\ell^{-1}B\|^2$, $c_2 = \dim \mathfrak{g} \cdot C_{137}$, and $|\langle A, \Delta_\pi^{(2)} A\rangle| \le 2c_2 M\alpha_0(\|\ell^{-1}A\|^2 + B_0^2 c_1^2 \|D^*A\|^2)$, so $\langle \mathcal{E}_U X, G_1^{-1}\mathcal{E}_U X\rangle \le K_1 \|X\|_S^2$ with $K_1 = K(1 + O(M\alpha_0))$. Part B proves $|R_2(b,b')| \le (O(1)e^{-\delta_0 M/3} + O(\alpha_0+\alpha_1))e^{-(1-\theta)\delta_A r}$ for $M \ge \max\{M_*, 96\kappa_1/\delta_0\}$ and $\alpha_0+\alpha_1 \le \alpha_*$. In Part A, (3.135) is the sandwich $(I - DG'RD^*)^*\Delta^{(2)}(I - DG'RD^*)$, so the form is taken at the $\tilde{A}$ of (3.119), and the kernel of $\Delta^{(2)}$ is local by (3.137) and symmetric as a Hessian by (3.136), which allows a Schur test with AM–GM. In Part B, $C(\sigma; U_c) = A_Z(U)^{-1} - W'$ with $W' = W_\sigma(U_c) + [C(1;U_c) - C(1;U)]$, the bracket is bounded by a one-variable Cauchy estimate on the complex walk bound, and a Neumann series around the real $A_Z(U)$ lets $A_Z(U)$ cancel in R_2. Verifier: `SOURCES_S2_G1_V7_VERIFY.md`, Part A [PROVED+VERIFIED, CONDITIONAL] on the conditions of `SOURCES_S2_FORM.md`, Part B [VERIFIED REDUCTION]; corrected: symmetry of $\Delta^{(2)}$ is load-bearing for the column sums, the Cauchy route needs (W^c) on the α'-space ($\alpha' \gg \alpha$), not only on the α-space that W_σ needs (already in the ledger, no new item), the two-constant theorem weakens the complex input to a sup bound but does not remove it, touching σ_0 cubes occur generically, not always, and the two covariances do not coincide on a gauge-fixed subspace; the lower bound for G_1 (A3.5) was not checked. Open inputs: for R-003, (W^c) on the α'-space, now proved in the exponential gauge without the Thm 3.15 transfer (`SOURCES_WC_EXPGAUGE.md`, `SOURCES_WC_RESIDUE.md`, both verified) relative to the inputs listed under `SOURCES_A2_R2.md`, the real decay R-008 [R], the boundary-contact reading of item (i) [I-forced], and $\mathrm{dist}(Z_0, \Lambda_k^c) > RM$ [I]; for Part A, the inherited reading $a = 1$ and the block geometry.

`SOURCES_A1_WALK.md`: **ledger R-002, walk-expansion form of R-001.** L-063: for regular U, $(QGQ^*)^{-1} = \sum_\omega T_\omega$ with factors localized in $X \times X$, U-dependence on $\tilde{X}_0^5 \cup \cdots \cup \tilde{X}_n^5$, and $|T_\omega(y,y')| \le (4/c_-)L^{d/2+1}(L^j\eta)^{-1-d/2}(L^{j'}\eta)^{-1-d/2}(K_W/M_1)^{|\omega|}e^{-\frac{1}{2}\varepsilon d(\omega,y,y')}$, absolutely convergent; this is the (3.108) form at rate $\delta_W = \frac{1}{2}\varepsilon \ll \frac{1}{2}\delta_0$, and the same holds for $(QG_1Q^*)^{-1}$ with trees. The local operator on a cube is G's walk sum truncated to $\square^+$, its positivity $\frac{1}{2}c_-$ comes by restriction of the global H-057 (C) (not R-004), and an accretive L-125 (Lemma CT) gives $\|C_\square\|_\varepsilon \le 4/c_-$, as in Bałaban's own local step (CMP 96 p. 237). The parametrix $P = \sum h_\square C_\square h_\square$ satisfies $\hat{N}P = I + \sum F_\square$ with three remainders (separation, truncation, commutator), each $\le K_F/M_1$, and the Neumann series is the walk series. Verifier: `SOURCES_A1_WALK_VERIFY.md`, [VERIFIED REDUCTION] (construction verified); corrected: (L1) must read "kernel vanishes within about M_1 of ∂X", CMP 96 (2.84) has $e^{-\frac{1}{8}\delta_0 M}$ and (2.82) three error sums, the $\breve{G}_2$ citation is p. 432, the RM condition (2.59) at the walk rate joins the constant order, (W$_G$) has proof sketches printed only for G', $(Q'G'^2Q'^*)^{-1}$ and G_0 (Theorems 3.7, 3.9, 3.10), and H-057 (C) is conditional, not proved. Open inputs: (W$_G$) [R] (itself reduced in `SOURCES_WG.md` to Bałaban's printed sketches, the Neumann series (3.130), and CMP 96 Lemma 2.1, with WG-1 in the M_1 form $\alpha_0 \le c'M^{-1}M_1^{-10}$), H-057 (C) with the residue of ledger R-001, CMP 96 Lemma 2.1, the [I] items (Lipschitz cut-offs H-002, block-locality of Q, κ_1 independent of M), and O3 (Z_0-compressed variant) and O4 (complex α'-space), both now closed in `SOURCES_WC_EXPGAUGE.md` and `SOURCES_WC_RESIDUE.md` (verified), O4 relative to CMP 99 Thm 3.4 [V] and the complex walk form of G [R]. The construction is now written out in full as Theorem 5.1 (Section 5.8); its statement and constants supersede the ones of this paragraph.

`SOURCES_WG.md`: **(W$_G$), walk expansions of G and G_1.** The file writes out Bałaban's walk expansions (W$_G$) of the final propagators G and G_1 of CMP 99 Sect. E: for regular U, $M \ge M_*(d,L)$ and $M\alpha_0 \le a_*(d,L)$, $G = \sum_\omega T_\omega$ is a walk expansion with bounded branching and rate $(1-\theta)\frac{1}{2}\delta_0$, and G_1 has a tree expansion with rate $(1-\theta)\frac{1}{2}\varepsilon$, set by the walk expansion of $(QGQ^*)^{-1}$. The literal (3.108) form, with a factor M^{-1} per step also at the Δ'_π junctions, needs in addition $\alpha_0 \le c'M^{-11}$

(WG-1; the M_1 form is in Theorem 5.2). The identity $G^{-1} = G_0^{-1} - \Delta'_\pi$ and the Neumann series (3.130) $G = \sum_n G_0(\Delta'_\pi G_0)^n$ are expanded with the printed walk expansions of G_0, G' and $(Q'G'^2Q'^*)^{-1}$ (CMP 99 Thms 3.10, 3.7, 3.9), one local J-insertion of size $O(M\alpha_0)$ per Δ'_π, and a concatenation lemma that splits the exponent at each junction and sums the junction points by CMP 96 Lemma 2.1. For G_1 the same scheme runs on $G_1 = \sum_n G(\Delta_\pi^{(2)}G)^n$ with $\Delta_\pi^{(2)} = (I - DRG'D^*)\Delta^{(2)}(I - DG'RD^*)$, whose side branches use the walk expansion A1-W of $(QGQ^*)^{-1}$, and the order $G_0 \to G \to (QGQ^*)^{-1} \to H \to G_1$ has no loop. Verifier: `SOURCES_WG_VERIFY.md`, [VERIFIED REDUCTION]: (W_G) for G reduces to the printed proof sketches of CMP 99 Thms 3.7, 3.9, 3.10, the Neumann series, CMP 96 Lemma 2.1 and [I] items, and the G_1 clause to the same plus A1-W (Bałaban himself writes "assuming (3.132)" on p. 426); corrections: WG-1 needs $\alpha_0 \le c'M^{-11}$, not M^{-8}, because a Δ'_π insertion has up to $N_V = 9$ walk blocks, and "after adjusting a definition of δ_0" is on p. 427. Open inputs: WG-1 (admissible in Bałaban's constant order, not implied by $M\alpha_0 \le a_0$); A1-W for the G_1 clause, with H-057 (C) conditional on the torus-side hypothesis and two conventions; the [I] items (Q-locality, two-sided derivative bounds, $C^{(2)}$-locality, block counts); all under ledger R-002 (O3 and O4 are closed, see `SOURCES_A1_WALK.md` above). The write-out `WG_FULL.md` (Theorem 5.2, Section 5.8) supersedes the statement and constants of this paragraph: the walks live on the M_1-cover, and WG-1 reads $\alpha_0 \le c'M^{-1}M_1^{-10}$ for G and $\alpha_0 \le c''M^{-1}M_1^{-12}$ for G_1.

`SOURCES_A2_R2.md`**: ledger R-003, R_2 and R_3 of CMP 116 (2.16).** The precision difference $R_2 := C_0^{-1} - \mathrm{Re}\, C^{(k)}(Z_0,\sigma)^{-1}$ obeys $|R_2(b,b')| \le (O(1)e^{-\delta_0 M/3} + O(\alpha_0+\alpha_1))e^{-\rho|b-b'|}$ for $|\sigma(\Delta)| \le e^{\kappa_1}$, with the printed rate $\rho = \frac{1}{2}\delta_0$ when the decay rate δ_A of $C^*\Delta_k C$ exceeds $\frac{1}{2}\delta_0$, and a smaller rate $\rho < \delta_A$ otherwise, which (2.21) and LOC tolerate. R_3 holds in two weaker forms H-242 (R3a), H-243 (R3b), sufficient for its only printed use (2.21), and LOC's input (E) follows directly at $\sigma = 0$ with $\varepsilon_M = O(e^{-\delta_0 M/2})$. A Neumann series around the compression $C(1)^{-1} = Z_0 A Z_0$, $A = C^*\Delta_k C$ (case (a), (2.5)), treats $W_\sigma = C(1) - C(\sigma)$, the walks that meet σ_0, as the perturbation: each such walk makes a round trip to distance $> \frac{2}{3}M$ from Z_0, which gives $O(e^{-\delta_0 M/3})$ once the weight $e^{16\kappa_1}$ is absorbed for $M \ge 96\kappa_1/\delta_0$. The $O(\alpha_0+\alpha_1)$ part comes from a one-variable Cauchy estimate on the analyticity space, and the verifier's repair V7 expands around the real $A_Z(U)$, so that decay of A is needed at real U only. Verifier: `SOURCES_A2_R2_VERIFY.md`, [VERIFIED REDUCTION] (with `SOURCES_S2_G1_V7.md` and its verifier); corrections: the walk bounds (W) are CMP 99 (3.107)–(3.108), not CMP 109; L-228 (Lemma N) and the Cauchy disc needed decay of A at complex U, repaired by V7; the Cauchy radius must respect the plaquette condition; and "not needed: R-001" is wrong, since (W) for $C^{(k)}(Z_0)$ contains the unprinted walk expansion of $(QGQ^*)^{-1}$ (R-002). Open inputs: (W^c), the walk bounds on the complex α'-space, is no longer open: it is proved directly in the exponential gauge, real and complex, with no use of the Thm 3.15 transfer (`SOURCES_WC_EXPGAUGE.md`, `SOURCES_WC_RESIDUE.md`, both verified; ledger R-003 regraded to a verified reduction); it rests on the G_1 tree expansion (CMP 99 Thm 3.10, printed sketch), the conditional H-057 (C) of A1-W, R2 and R-Δ2 of R-004, CMP 109 p. 254 [R], CMP 99 Thm 3.4 [V], and [I] items; it is now written out in full as Theorem 5.3 (Section 5.8); the decay of $C^*\Delta_k C$ at real U (R-008); the boundary-contact reading of "intersect" [I-forced]; $\mathrm{dist}(Z_0, \Lambda_k^c) > RM$ [I]; all under ledger R-003.

`SOURCES_A3_GAMMA0.md`**: ledger R-004, the Kato route.** The lattice Kato (diamagnetic) route to $C^*\Delta_k C \ge \gamma_0$ is refuted, and R-004 is reduced instead to Bałaban's two-step mechanism at $U = 1$ made gauge covariant, with $\gamma_0 \ge (48d^2)^{-1}L^{-d-1}$ independent of k, volume and M. This is the verifier's sharpened constant; for $L \ge 10$ it carries the factor Lm_L found later in

SOURCES_A3_GAMMA0_FULL.md. Step (i) bounds the minimal-extension energy below by the unit-lattice curl energy, $\langle B, \Delta_k B\rangle \geq \|\partial_1 B\|^2$ at $U = 1$ by exact lattice Stokes and Cauchy–Schwarz, and step (ii) is CMP 96 Lemma 2.4, a unit-lattice Poincaré inequality in axial gauge. Both steps are made covariant in a local regular gauge on each region, with the gauge-fixing part removed through (3.115) and relative errors $O(M\alpha_0 + L^2\alpha_0)c(d, L)$ for $M \geq cL$. Verifier: SOURCES_A3_GAMMA0_VERIFY.md, [VERIFIED REDUCTION] at sketch level, not a proof; corrections: the example $A = d\delta_x$ fails once $\|RD^*A\|^2$ is included and is replaced by an unfixed mode $\lambda \in \mathrm{Ran}(\Delta^{-2}Q'^*)$ (0 against 4.62), so the Kato route stays refuted; $c_* \leq 1$, not 4, so the free constant $(12d^2)^{-1}L^{-d-1}$ stands and the bound sharpens from $(192d^2)^{-1}L^{-d-1}$ to $(48d^2)^{-1}L^{-d-1}$; R1 must be CMP 99 Thms 3.1, 3.3, 3.11, 3.12. Open inputs: R1 with a uniform bound on the oblique projection $\kappa_\pi = \|I - DG'RD^*\|^2$; R2, the k-step averaging comparison (an R-018 piece); the small items R3–R5; the σ-localized clause [I]; all under ledger R-004, and taken up in SOURCES_A3_GAMMA0_FULL.md and SOURCES_A3_SIGMA.md.

SOURCES_A3_GAMMA0_FULL.md: **ledger R-004, full proof.** L-069: for U obeying (3.35)–(3.36) with $M\alpha_0$ small, $\langle \tilde{B}, C^*\Delta_k C\tilde{B}\rangle \geq \gamma_0\|\tilde{B}\|^2$ with $\gamma_0 = m_L/(48d^2L^d)$, $m_L = \min(2(1 - \cos(\pi/L)), 1/L)$, uniform in k, volume and regular U; this equals $(48d^2)^{-1}L^{-d-1}$ for $L \leq 9$ and is 0.7555 times it at $L = 13$. The proof step inside CMP 96 (2.126)–(2.127) is false for $L \geq 10$ (a lost factor L), while the statement of Lemma 2.4 is not refuted and stands with the corrected constant. The failure of (2.127) for every $L \geq 10$ is checked in Lean (SlabCounterexample.lean, Appendix A). Lemma 5 shows that the G_1-minimizer already satisfies $RD^*A_* = 0$, because $A \mapsto A - DG'RD^*A$ keeps QA and lowers $\langle A, G_1^{-1}A\rangle$ by exactly $\|RD^*A\|^2$, so the gauge-fixing part drops out and no bound on κ_π is needed; Lemma 5 is checked in Lean from (F1), (F3), (F4) as hypotheses (GaugeMin.lean). The two steps of SOURCES_A3_GAMMA0.md are then written out in full (step (i) with $c_* = 1$ and a covariant relative error $2(1 + \delta^{-1})\varepsilon_2^2$, step (ii) as CMP 96 Lemma 2.4 reproved with m_L in place of $1/L$ and made covariant in one regular gauge per corridor), and the assembly gives $\gamma_0 = \gamma_L/4$ with $\gamma_L = m_L/(12d^2L^d)$. Verifier: SOURCES_A3_GAMMA0_FULL_VERIFY.md, [PROVED+VERIFIED, CONDITIONAL ON R-Δ2 AND R3–R5], with CMP 99 Thms 3.11–3.12 published and H-123 (H_Q) [PROVED+VERIFIED]; the slab claim against CMP 96 was checked first and confirmed (lowest eigenvalue $L^{-d}\min(2(1-\cos(\pi/L)), 1/L)$); corrections: (2.127) fails from $L = 10$ with free block bonds and from $L = 15$ in axial gauge ($d = 2$), not only from $L = 30$; fact (E) needs a factor $\sqrt{N}$; the reason for R-Δ2 must be the form (3.136) plus a block-average bound on $C^{(2)}$. Open inputs: R-Δ2 [R]; the Schur step of R1 from CMP 99 Thm 3.12, not written out; R3, R4, R5, R-Δ' and R2$'$; the σ-localized clause, reduced in SOURCES_A3_SIGMA.md; all under ledger R-004.

SOURCES_A3_SIGMA.md: **ledger R-004, σ-localized clause.** For real $s \in [0, 1]^{\sigma_0}$ the s-localized covariance $C_s = C^{(k)}(Z_0, s)$ obeys $\|C_s - C_1\| \leq \eta_M = S_d B_W e^{-\frac{1}{3}\delta_0 M}$ and, with no symmetry assumption, $\langle v, C_s^{-1}v\rangle \geq (\gamma_0 - 2\gamma_1^2\eta_M)\|v\|^2 \geq \frac{1}{2}\gamma_0\|v\|^2$ for $\eta_M \leq \gamma_0/(4\gamma_1^2)$, uniformly in s, Z, Z_0, k, volume and real regular U, with no α-condition. The resolvent bound $\|C_1^{-1} - C_s^{-1}\| \leq 2\gamma_1^2\eta_M$ also gives LOC's input (E) in norm at real points without (2.16). The walk sum of CMP 116 p. 3 expands the covariance, $C_s = \sum_\omega s^\omega T_\omega$, so $C_s - C_1 = -\sum_{\omega\in W_\sigma}(1 - s^\omega)T_\omega$ is dominated termwise by one s-free absolute majorant over the walks that meet σ_0, each of which reaches distance $> \frac{2}{3}M$ from Z_0. A Schur bound gives η_M, and the resolvent identity with $\gamma_0 \leq Z_0AZ_0 \leq \gamma_1$ turns it into the lower bound on the precision. Verifier: SOURCES_A3_SIGMA_VERIFY.md, [VERIFIED REDUCTION] to (W) and (N); corrections: (W) is CMP 99 (3.107)–(3.108) with the Cor. 3.8 counting, not CMP 109; the bound without symmetry is the verifier's strengthening of the author's bound $\frac{2}{3}\gamma_0$ under (S_s), so (S_s) is optional; the uniformity of B_W in Z_0 uses part of the exponential decay (constants only). Open inputs: (W), Bałaban's assertion (p. 13) that the [13] expansion of $C^{(k)}(Z_0)$ with the

large domain X_0 obeys (3.108) uniformly in Z_0 (R-002 enters here [I]; in the exponential gauge this form is Theorem 5.3(ii), relative to its inputs); the interior reading of "intersect" for the zero-length walk, forced by (2.16); (N) $\gamma_0 \le A \le \gamma_1$, with γ_0 from ledger R-004 and γ_1 through R-008; complex σ stays with R-003.

`SOURCES_A4_D189.md`**: ledger R-005, CMP 102 (189).** CMP 102 (189) bounds the two-point Hessian kernel of the nonlinear action part V at fixed A', and the file derives $|V''(A';b,b')| \le c_V \varepsilon_3 e^{-\frac{1}{2}(1-\alpha)^5\delta_0 d(b,b')}$, rate $\ge \frac{1}{4}\delta_0$ for $\alpha \le 0.129$, volume-free and on the complex domain (77); (190) follows through the Neumann series (188) at rate $\frac{1}{8}\delta_0$ ($\alpha \le 0.159$). Every operator that CMP 102 inverts has the form I plus a small decaying kernel, so a weighted Schur Neumann series (CMP 96 Lemma 2.1) suffices, without Demko–Moss–Smith and without a Hessian lower bound. L-139 (Lemma D2) shows that the second derivative of the linearizing map solves $(I+\mathfrak{N})\mathfrak{D}_2 = C''[(I-H\mathfrak{D})\cdot,(I-H\mathfrak{D})\cdot]$ with the operator of (68), with two-sided decay at $\frac{1}{2}(1-\alpha)^4\delta_0$, and composition then gives V''. Verifier: `SOURCES_A1_A4_VERIFY.md` (Part B), [VERIFIED REDUCTION] to R-001 (kernel-decay form), published CMP 102 statements, R-018, and one bookkeeping item; corrected: the rates are relabelled ($\frac{1}{2}(1-\alpha)^4\delta_0$ for $\mathfrak{D}_2$, $\frac{1}{2}(1-\alpha)^5\delta_0$ for V''), the name $\mathfrak{D}_2$ clashes with Bałaban's (first derivative minus linear term), V'' uses (3.132) directly through (87)–(88) and the decay of $\tilde{G}$ depends on R-001 through (3.137) and (3.133), and the CMP 102 rates are fractions of the common $\delta_0 \le \delta_1$. Open inputs: R-001 (itself a verified reduction), R-018, the second-derivative bookkeeping of the V_0 terms (90)–(97), and the ∇, Hölder and Δ versions of (190) [I].

5.8 Step 8′: the three write-outs of the propagator expansions

Three files write out in full the constructions that the paragraphs on `SOURCES_A1_WALK.md`, `SOURCES_WG.md`, and `SOURCES_A2_R2.md` reduce to: `A1_WALK_FULL.md`, `WG_FULL.md`, and `WC_FULL.md`. Each was checked by an independent verifier file, fixed, and regraded. Each is paper-grade relative to its tagged inputs. None of them removes an input: they replace "reduced to a printed sketch" by a written proof that names its inputs. So the tag stays [VERIFIED REDUCTION]. The inputs common to all three are the lower bound H-057 (C), Bałaban's printed proof sketches of CMP 99 Thms 3.7, 3.9, and 3.10, and a short list of geometry and convention items [I]. Notation: M_1 is the cube size of the walks (CMP 116 p. 3), M is the regularity parameter of CMP 99 (3.36) and the cube size of the s-rule, and $R_1 M_1 \le M$. A walk expansion is a sum $T = \sum_\omega T_\omega$ over walks of domains, with (L1) each factor kernel vanishing within $\frac{1}{3}M_1$ of the boundary of its domain (the band of CMP 116 p. 3), (L2) U-dependence of each factor only on $\tilde{X}^5$, (B1) a per-walk bound $B\rho^{|\omega|}e^{-\delta d(\omega,y,y')}$ with the path distance (3.93), and (B2) bounded branching.

Theorem 5.1 (L-017, `A1_WALK_FULL.md` §0.5 [VERIFIED REDUCTION])**.** *Let $\hat{N} = S^{-1/2}QGQ^*S^{-1/2}$. Assume H-204 (H1) $\hat{N}$ has a walk expansion with (L1), (L2), $|\hat{n}_\omega(y,y')| \le b_G \rho_G^{|\omega|} m(y,y') e^{-\delta_G d(\omega,y,y')}$, $\rho_G = c_G/M_1$, at most $C_{\mathcal{D}}$ domains meeting a set of diameter $\le D_+ M_1$, and $\operatorname{diam}\tilde{X}^5 \le c_X M_1$; (H2) the lower bound H-057 (C), $\operatorname{Re}\langle u, \hat{N}u\rangle \ge c_-\|u\|^2$, uniform; (H3) the cube geometry of Bałaban's cover; H-206 (H4) the constant order $(d, L, a, a_0) \to \theta \to c_\pm, \delta_G, \Sigma, b_G, c_G, C_{\mathcal{D}}, c_X \to \varepsilon \to M_1 \to M \to \alpha_0$, with WG-1 in the M_1 form $\alpha_0 \le c' M^{-1} M_1^{-10}$ for G and $\alpha_0 \le c'' M^{-1} M_1^{-12}$ for G_1. Put $c_+ = C_{\mathcal{D}} b_G$ and $\varepsilon = \min\{\frac{1}{2}\delta_G,\ c_-/(8c_+\Sigma(\frac{1}{2}\delta_G))\}$. Then for $M_1 \ge M_W(d, L, a, a_0)$: (i) $\hat{N}^{-1} = \sum_\omega T_\omega$ is a walk expansion; (ii) $|T_\omega(y,y')| \le (4/c_-)(c_1\rho)^{|\omega|} m(y,y') e^{-\frac{1}{2}\varepsilon d(\omega,y,y')}$ with $\rho \le K_W/M_1$; (iii) $\sum_\omega |T_\omega(y,y')| \le (8C_{\mathcal{D}}/c_-) m(y,y') e^{-\frac{1}{2}\varepsilon d(y,y')}$; (iv) $\|\hat{N}^{-1}\|_{\frac{1}{2}\varepsilon} \le 8/c_-$. All constants are uniform in U, j, k, η, $\{\Omega_j\}$, and the volume.*

The file adds two variants. The Z_0-compressed variant (§7): for an operator A with a walk expansion and $\mathrm{Re}\, A \geq \gamma$, the inverse of $A_Z = 1_{Z_0} A 1_{Z_0}$ has a walk expansion with prefactor $4/\gamma$, and for $q = M/M_1 > 10$ and $q \geq 3(7 + c_X)$ it regroups into the CMP 116 p. 13 form (one domain X_0 with $\tilde{X}_0^5 = \tilde{Z}_0$, all other domains at distance $> \frac{2}{3}M$ from Z_0, the band kept, and a merged-slot cost $8C_W/\gamma$ per walk); for $(x + A_Z)^{-1}$, $x \geq 0$, the prefactor is $4/(\gamma + x)$. The complex variant (§8): Lemmas CP and AW and a four-level chain of complex domains give the expansion on the α'-space. The chain constants $\Gamma_3 = 2C_{\mathcal{D}} B_1'''\Sigma(\delta_1''')$ and $\Gamma_1'' = 2C_{\mathcal{D}} B_A''\Sigma(\delta_A'')$ carry the factor $C_{\mathcal{D}}$, so k_1, k_2 are smaller by at most $C_{\mathcal{D}}$. The bound $\sum_{\square} \varphi_{\square}^2 \geq 1$ on Z_0, needed in §7.3, is proved by the verifier (tent family on a grid of spacing $M_1 L^k$). The tree form of H-204 (H1), which the G_1 clause uses, is stated explicitly in `A1_WALK_FULL.md` 6.9. Verifier: `A1_WALK_FULL_VERIFY.md`, then a regrade: paper-grade relative to its inputs H-296 ((W$_G$) = Theorem 5.2), H-298 (H-057 (C), conditional), H-303–H-307 [I] (cube geometry, the Lipschitz partition H-002, Q-locality, the WG-1 condition, κ_1 independent of M), and H-313 [R] ((W$_G$) at complex points, from CMP 99 Thms 3.4 and 3.10).

Theorem 5.2 (L-018 (Theorem WG), `WG_FULL.md` 0.7 [VERIFIED REDUCTION]). *Let U satisfy CMP 99 (3.35)–(3.36) and fix $\theta \in (0,1)$. (G) (L-316 (Theorem WG-G).) There are $M_*(d, L, \theta)$ and $a_*(d, L, \theta) > 0$ such that for $M_1 \geq M_*$, $M\alpha_0 \leq a_*$, and RM with CMP 96 (2.59) at rate $\theta\frac{1}{2}\delta_0$, $G = \sum_\lambda T_\lambda$ is a walk expansion on the M_1-cover, with constants $(B_0', \rho_*, (1-\theta)\frac{1}{2}\delta_0)$, $\rho_* = \max\{c_0 M_1^{-1}, (K_V M\alpha_0)^{1/10}\}$; the bound holds also with one covariant derivative on either side. If in addition $\alpha_0 \leq c' M^{-1} M_1^{-10}$ H-179 (WG-1), then $\rho_* \leq c_0 M_1^{-1}$, the CMP 116 (1.7) form. (G1) (L-317 (Theorem WG-G1).) Assume in addition Theorem 5.1 for $(QGQ^*)^{-1}$. There are M_W and $a_{**} > 0$ such that for $M_1 \geq \max\{M_*', M_W\}$, $M\alpha_0 \leq a_{**}$, and RM $\geq C(d, L, \theta, \varepsilon)$, $G_1 = \sum_\tau T_\tau$ is a tree expansion with rate $(1-\theta)\frac{1}{2}\varepsilon$. The per-step form needs $\alpha_0 \leq c'' M^{-1} M_1^{-12}$. All constants are uniform in U with (3.35)–(3.36).*

The proof expands $G = \sum_n G_0(-VG_0)^n$, $V = -\Delta_\pi'$, with the walk expansions of G_0, G', and $(Q'G'^2Q'^*)^{-1}$. L-228 (Lemma N) shows that the Neumann series gives the inverse in finite volume, so no separate existence of G is needed. V splits into 24 patterns with at most $N_V = 9$ walk blocks, and one local J-insertion of size $O(M\alpha_0)$. L-115 (Lemma C) concatenates walks: it splits the exponent at each junction and sums the junction point by CMP 96 Lemma 2.1, so the rate $(1-\theta)\frac{1}{2}\delta_0$ is lost once. The 10 junction steps of a V-insertion give the exponent of WG-1. To obtain the band (L1), every domain is enlarged by one layer of M_1-cubes; this changes the branching count to $C_{\mathcal{D}} = 5^d c(d) C_{\mathcal{D}}^{\mathrm{in}}$ and c_X to $c_X + 2$, and adds $M_1 \geq 3c_{\mathrm{loc}}$ to M_*. The output satisfies H-204 (H1) of Theorem 5.1. Inputs: CMP 99 Thm 3.10 and Thm 3.9 (printed proof sketches), Thm 3.7 with Cor. 3.8 (proof by reference to CMP 95 Prop. 1.2 and CMP 96 Prop. 2.2), CMP 96 Lemma 2.1 under (2.59), the bound $|J| \leq O(1) M\alpha_0 (L^j \eta)^{-3}$ [V]; for (G1) also Theorem 5.1, so H-057 (C); [I] items: the scale bookkeeping C0 (written out for the vector $\hat{J} = S^{3/2}J$), locality of Q, Q', D, and the parallel transports, the assignment of ι-domains, the locality of $C^{(2)}$, and the M_1-cover (asserted on CMP 116 p. 3). Verifier: `WG_FULL_VERIFY.md`, then a regrade: paper-grade relative to these inputs; the verifier found that every consumer uses the M_1 form, so the corrected WG-1 replaces the earlier $\alpha_0 \leq c' M^{-11}$; the band enlargement passed the band regrade.

Theorem 5.3 (L-019 (Theorem WC), `WC_FULL.md` 0.10 [VERIFIED REDUCTION]). *Fix $d \geq 2$ and odd $L \geq 3$. There are M_1, M with $M/M_1 > 3(6 + c_X)$ and $M/M_1 > 10$, a smallness α_*, a four-level chain of complex domains $\alpha' \subset \alpha'' \subset \alpha''' \subset$ the a_1-domain of CMP 99 Thm 3.4, and constants $\gamma_{\mathrm{exp}} = \frac{1}{4}\gamma_e(d, L) > 0$, $\varepsilon'' > 0$, $\rho > 0$, such that for every real regular U and every Z_0: (i) $\langle \tilde{B}, A_Z \tilde{B} \rangle \geq \gamma_{\mathrm{exp}} \|\tilde{B}\|^2$ on the reduced space of Z_0; (ii) $C^{(k)}(Z_0) = \sum_\omega T_\omega$ with (L1), (L2), $|T_\omega(b, b')| \leq (4/\gamma_{\mathrm{exp}})(c_1\rho)^{|\omega|} e^{-\frac{1}{2}\varepsilon d(\omega, b, b')}$, absolutely convergent, and after regrouping in the CMP 116 p. 13 form; (iii) at every U_c in the α'-space, $\mathrm{Re}\langle u, A_Z(U_c) u \rangle \geq \frac{1}{2}\gamma_{\mathrm{exp}} \|u\|^2$, $C^{(k)}(Z_0; \cdot)$ is analytic,*

and (ii) holds with $4/\gamma_{\exp} \to 8/\gamma_{\exp}$ and $\varepsilon \to \varepsilon''$; (iv) (i)–(iii) hold for $(x+A_Z)^{-1}$, $x \geq 0$, uniformly in x, with the per-walk prefactor $4/(\gamma_{\exp}+x)$ (real) and $8/(\gamma_{\exp}+x)$ (complex), so the $x^{-1/2}$ integral for $(C^{(k)})^{1/2}$ converges walk by walk. The constants are uniform in k, η, Z_0, $M \geq M_1$, and the background. The proof does not use CMP 99 Thm 3.15 or any transfer from the axial gauge.

The lower bound comes from L-153 (Lemma E), $S_y \leq K_b(P_y + G_y)$ on each block, with $K_b = 8dL^d(1+16dc_{\mathcal{G}}) \leq 64d^3L^{2d+1}$ under $16d^3L^dc_{\mathcal{G}}\tau_p^2 \leq 1$, $\tau_p = 16\sqrt{N}c_4M\alpha_0$; then $\gamma_e^{-1} = K_b(1+6d/m_L)+6/m_L$ and $\gamma_{\exp} = \frac{1}{2}\min(\frac{1}{2}, c_G)\gamma_e$ with $c_G = 1$. For Federbush's mean (CMP 109 (0.10)) the exact weights give $c_M \leq 4\sqrt{N}$, and $\kappa \leq \sqrt{2}$ for $w \leq 0.054/\sqrt{N}$. The walk expansion is the Z_0-compressed variant of Theorem 5.1 applied to the tree expansion of $A_{\exp}$, which rests on Theorem 5.2(G1). The Combes–Thomas Schur sums carry the factor of $|e^x-1| \leq |x|e^{|x|}$: $\Sigma(\frac{1}{2}\delta) \leq (1+4/(e\delta))S(\frac{1}{4}\delta)$, which changes only ε, ε^a, and ε''. Inputs: [R] Theorem 5.2(G1) (printed sketches), the complex walk form of G (CMP 99 Thms 3.4 and 3.10), the inputs of ledger R-004, and CMP 109 p. 254; conditional: H-057 (C) and $c_{-,1} \geq c_-(1-O(M\alpha_0))$; [I] items I1–I9 of `WC_FULL.md` §9 (among them the term match of CMP 109 (2.11) with CMP 99 (3.156), the locality of $\tilde{C}^{(2)}$ and Q, the identification $\tilde{Q} = Q_1(U_1)$, $\mathrm{dist}(Z_0, \Lambda_k^c) > RM$, and the tilde convention). Verifier: `WC_FULL_VERIFY.md` (needs fixes, none structural), then two regrades: after the fixes and after the band fix, paper-grade relative to these inputs.

5.9 Step 8″: round 5, the remaining Group A rows and L-174 (Lemma G3)

Round 5 treats the Group A rows that had neither a verified proof nor a verified reduction: R-008, R-009, R-010, R-012, and R-015. It also proves L-174 (Lemma G3), the residue N2 of the reduction of R-006. Each result below has its own project file and an independent verifier file. Each holds relative to the inputs that it names. None of the rows leaves the ledger, because those inputs are not proved here. The tag of each item therefore stays [VERIFIED REDUCTION] or [PROVED RELATIVE TO LISTED INPUTS]; nothing in this step makes Theorem A or Theorem B unconditional.

The operator of R-008. Bałaban's C in $C^*\Delta_kC$ is the constraint map $I - hL\tilde{Q}$, not a square root of the covariance. The operator $C^*\Delta_kC$ is the *precision*: CMP 116 (2.7) reads $(C^{(k)})^{1/2} = (C^*\Delta_kC)^{-1/2}$. So the lower bound $A_Z \geq \frac{1}{4}\gamma_e$ of Theorem 5.3(i) is the γ_0 side (row R-004) and cannot give γ_1. The old reduction "R-008 ⇐ R-001 + R-018" of `SOURCES_CDC.md` is superseded; R-018 now enters through A3-R4 and the item I3 of `WC_FULL.md`.

Theorem 5.4 (R-008, `SOURCES_A7_CDC.md` [PROVED RELATIVE TO LISTED INPUTS; VERIFIED])**.** *Let U be real and regular, $s \equiv 1$, and let Λ be as in `WC_FULL.md` 0.8. (N) (L-226 (Lemma N).) $\|C^*\Delta_kC\| \leq \gamma_1^N$ with $\gamma_1^N = \|C\|^2\max\{K_{\mathrm{ext},1} - a + c_J + 2c_{\mathcal{G}},\, a + c_J\}$, uniform in k, the volume, and U. (D) (L-138 (Theorem D).) $|(C^*\Delta_kC)(b,b')| \leq 2B_Ae^{-\delta_A|b-b'|}$ with $\delta_A = (1-\theta)\min\{\frac{1}{2}\varepsilon, \delta_1\}$, and the series (2.7) for $(C^{(k)})^{1/2}$ converges walk by walk for $\gamma_1 := 4B_AS(\theta'\delta_A)$. Both bounds hold for real $s \in [0,1]$ with the same constants.*

Inputs of (N): C2 of `WC_FULL.md` (the bound H-057 (C) for G_1, conditional on the torus-side hypothesis and two conventions, plus one [I] step for G_1), A3-R4, I1, and I3. Inputs of (D): W_{G_1} [R] (Theorem 5.2(G1)), Theorem 5.1 [C], I2, and I9. The clause for complex s with $|s| \leq e^{\kappa_1}$ and complex U on $\mathcal{U}(\alpha'')$ is at proof-sketch level and needs M_1 large after κ_1. The value $\gamma_1^N \approx 243(K_E + 648)$ at $d = 4$, $L = 3$ is [I]. Verifier: `SOURCES_A7_CDC_VERIFY.md`; claims (1), (3), (4), (5) confirmed, claim (2) confirmed with the input list corrected (the two-sided route of §2.3 is an alternative to L-208 (Theorem L)).

The positivity device of R-009. Only $H(K) = \operatorname{sym}\operatorname{Re} K$ enters the modulus of the Gaussian measure, $K = C^{(k)}(Z_0, \sigma)^{-1}$. Bałaban states that $C^{(k)}(Z_0, 0)$ is symmetric. That claim is not proved: the parametrix walk series is not reversal-symmetric, and the truncation need not keep symmetry. The claim is also not needed. The majorant uses $H_0 = \operatorname{sym} C_0^{-1} \geq \frac{3}{4}\gamma_{\exp}$ in its place.

Theorem 5.5 (R-009, `SOURCES_A8_POS.md` [PROVED IN CORRECTED FORM RELATIVE TO LISTED INPUTS; VERIFIED]). *(R) (Real case.) At $(U, 0)$ with $s \equiv 1$, $K = A_Z$ is real symmetric (L-266 (Lemma S)) with $\gamma_{\exp} \leq A_Z \leq \Gamma_1$, so $d\mu_{C^{(k)}(Z_0)}$ is a positive Gaussian measure. (D) (Lemma D.) For $|\sigma| \leq e^{\kappa_1}$ the σ-truncation obeys $\|C^{(k)}(Z_0, \sigma) - A_Z^{-1}\| \leq \eta_M = (16/\gamma_{\exp})e^{2\cdot 3^d \kappa_1} W e^{-\varepsilon'' M/8}$. (P) (L-242 (Lemma P).) The complex Gaussian exists; its modulus is $r_\sigma\, d\mu_{\Gamma_\sigma}$ with $1/(2\Gamma_1'') \leq \Gamma_\sigma \leq 4/\gamma_{\exp}$, which is Bałaban's (2.15). (M) (Majorant.) With $H_0 = \operatorname{sym} C_0^{-1} \geq \frac{3}{4}\gamma_{\exp}$ the majorant (2.23)–(2.25) holds at a cost $\exp(O(\eta_M^2)|Z_0|)$ and an extra term $O(\eta_M^2)\|ZX\|^2$, absorbed in (2.25).*

Inputs: Theorem 5.3(i)–(iii) with their residue (row R-003), and five readings I-a–I-e (quadratic forms versus symmetric operators; the σ-weighted walk sum by the CMP 116 p. 3 rule; three geometry facts; $W < \infty$ at rate $\varepsilon''/8$; "$\operatorname{Re} C^{-1}$" as the symmetric real part). The M-large conditions are in the order of constants (K-025). Verifier: `SOURCES_A8_POS_VERIFY.md`; Lemmas S and P confirmed; Lemma D corrected (a tree-distance slip: the exponent is $e^{-\varepsilon'' M/8}$, not $e^{-\varepsilon'' M/4}$); the sign of the extra (2.24) term corrected (it is a cost, not a gain). `SOURCES_G3.md` reads R_2 of (2.16) as $H_0 - H_\sigma$.

The form R-011 of R-010. Bałaban writes that the lower-order bracket terms are "bounded similarly as in (1.24)" (CMP 116 p. 8). For the raw order-n Taylor terms, $n \leq 4$, this is false as a bound: the cube count $(6L)^4(L^j\eta)^{-4}$ leaves a sum over j that diverges in k. The sentence holds for the terms after the transformations of CMP 109 Sect. 4.

Theorem 5.6 (R-011, `SOURCES_A10_LOWER.md` [PROVED IN CORRECTED FORM RELATIVE TO LISTED INPUTS; VERIFIED]). *After the transformations of CMP 109 Sect. 4: the order-0 term is removed by the $t_\square$-derivative; the order-1 term is zero by (4.14); the marginal parts of (4.34) and (5.43) cancel exactly against the counterterm, because β_j is defined as the coefficient β in (5.43) (p. 298). Every other term obeys the (1.24) form with the factor $8B_0C_1e^{16\kappa_1}\alpha_2^{-1}g_k|B|$ and field power 5, or $4 + \beta_H$. The "other terms" reach (1.29): F with exponent $(1 - 2\delta)\kappa$; R and C by the same mechanism.*

Residue: R1 unprinted constants (finite, fixed before ε_1); R2 the R- and C-terms by mechanism, not written out; R3 Bałaban's Sect. 4–5 bounds as printed; R4 the tail constant of (4.5) [I]. The per-label exponent of L-050 (Fact A) in L-281 (Theorem SM) is the Hölder exponent β_H of CMP 109 (3.31), which Bałaban does not print (see the paragraph on β_H below). Transcription fix: (1.29) reads $O(M^q)$ and $(1/16)\kappa_1$. Verifier: `SOURCES_A10_LOWER_VERIFY.md`; claims (1), (2), (4), (6) confirmed; (3) corrected (a factor 2); (5) corrected (β_H is unprinted and needs its own symbol; L-186 (Lemma HH)'s θ has no numerical value, so $\theta < 1 - L^{-\beta_H}$ holds by ε_1 small).

The free exponent β_H. The consumers of R-011 change as follows (correction blocks in `SOURCES_SM.md` and `SOURCES_HHIST.md`): L-050 (Fact A) has $\alpha_* = \beta_H$; (Eps) reads $2\varepsilon' = 2\theta_0(1 - L^{-\beta_H})^{-1}$; L-189 (Lemma HH) needs $\theta < 1 - L^{-\beta_H}$, with column sum $\leq C_\beta + 2C_\Pi\theta_0 S_0(1 - L^{-\beta_H})^{-1}$; and $\lambda_g = L^{-\beta_H/2} + \theta_g\gamma$ in §4.3 of `SOURCES_SM.md`. The assessment `SOURCES_GGAIN_BETAH.md` ([UNVERIFIED ASSESSMENT]) finds that every consumer needs only $\beta_H > 0$, chosen before ε_1 and γ. Each condition is an ε_1- or γ-smallness condition. So $\beta_H \in (0, 1)$, $\beta_H \leq \beta_0 < 1$, is a free parameter, first in the order of constants (ledger K-028). The cost is the factors $(1 - L^{-\beta_H})^{-1}$, a larger $B_4(\beta_0)$,

and K_T fixed before ε_1. At $L = 13$ the H-111 (G-gain) margin $1 - 13^{-\beta_H/2}$ is 0.120 at $\beta_H = 0.1$ and 0.685 at $\beta_H = 0.9$; the L-186 (Lemma HH) margin is 0.901 at $\beta_H = 0.9$. The reading of CMP 102 Thm 1 (9) "$\beta_0 = 1$" as a misprint is [I]. This β_0 is not the β_0 of CMP 119 p. 255.

Proposition 5.7 (R-012, `SOURCES_A11_CONST.md` [CLOSED AS A VERIFIED LIST])**.** *The "other constants" of CMP 119 p. 256 are further scale-indexed numbers $X_j = f(g_j)$: $\alpha_{0,j}$, $\alpha_{1,j}$ ((2.28)), $g_j^{\kappa_0}$ ((2.31)), $\delta_k = (A_1/A_0)\varepsilon_k$ (p. 265), $p_1(g_k)$ ((3.43)), and $g_j(x)$ ((2.24)). Every similar inequality for them follows from (2.6)–(2.9). The three unprinted cross-scale steps ($R_k \le LR_{k+1}$ on p. 268, the monotonicity of k_1 on p. 270, and $\varepsilon_{k+1} \le (1+\beta_0)\varepsilon_k$) also follow from (2.9). The side conditions H-255–H-265 are admissible and acyclic.*

The file reads 43 introduction points (pp. 255–280 as images, pp. 281–284 as OCR). Three entries join the order of constants: K-029 ($M_1 < M_2 < M$, order $M_1 \to M_2 \to M$), K-030 ($M_2 \ge 1/\delta$), and K-031 ($A_1/A_0 \le 1/(80B_3)$, A_1 after A_0). The residue $r > 1$ is closed by "$r \ge 2$" (CMP 119 p. 245). Non-blocking residue: the form of $p_1(g)$ inherited from CMP 122 I, CMP 109 restrictions outside the file, and an OCR slip on p. 278. Verifier: `SOURCES_A11_CONST_VERIFY.md`; claims (1), (4) confirmed; (2), (3) corrected without a change of conclusion (three list omissions, a $(1 + \beta_0)^2$ slip fixed by monotonicity, the order $M_1 \to M_2 \to M$ with $M > M_2$ added).

The whole-lattice coefficient (R-015). $\beta_{j+1}(g_j)$ is defined by CMP 109 (1.20)–(1.22), p. 264: the mixed second moment $\sum_x \Pi_{\mu\nu}(x)x_\mu x_\nu$ of the vacuum-polarization tensor at $U = 1$, taken as the limit of periodic-torus kernels $T \nearrow \mathbb{Z}^d$. Bałaban states that this limit exists "by the localized representation (1.7)". (1.7) and (1.9) say only that each term depends on U restricted to X. They do not say that each term is the same function on every torus and on $\mathbb{Z}^d$. So the existence of the limit stays Bałaban's bare assertion unless one adds the hypothesis H-141 (Loc) below.

Proposition 5.8 (Prop. A15-N, `SOURCES_A15_WHOLE.md` [PROVED GIVEN H-141 (LOC), (BD), (H), (CNT); VERIFIED])**.** *Fix j, $g \in [0, \gamma]$, and $U = 1$. Assume H-141 (Loc): for every non-wrapping polymer X the torus term equals the $\mathbb{Z}^d$ term of its lift; (Bd): CMP 109 (1.18); (H): the linearized background map decays exponentially, at the same rate on the torus and on $\mathbb{Z}^d$; and (Cnt): the standard polymer count. Let N_j be the number of level-j cells per side and $m_j = \frac{1}{2}\min(\delta_H, \kappa - \kappa_0)$. Then the limit in (1.22) exists, and for the torus coefficient $\beta_j^{(N)}$ in moment form*

$$|\beta_j^{(N)}(g) - \beta_j(g)| \le C_j N_j^6 e^{-m_j N_j/2},$$

with C_j independent of N. The Fourier form adds $O(N_j^{-2})$.

Exact equality $\beta_j^{(N)} = \beta_j$ cannot be derived in general. Image terms and wrapping polymers are real error sources; no explicit N with $\beta_j^{(N)} \ne \beta_j$ was computed, and Bałaban never claims coefficient equality. Equality holds term by term only for non-wrapping interior X under H-141 (Loc). The hypothesis H-141 (Loc) is not implied by (L2) of Theorem 5.2, which covers walk factors only. The case $g = 0$ is in `SOURCES_T2_G2.md` (ledger R-030). Verifier: `SOURCES_A15_WHOLE_VERIFY.md`; (1) confirmed (definition on the page image; the "(1.41)–(1.43)" citation on CMP 119 p. 253 is Bałaban's slip); (2) corrected (three unstated steps: the first-order term drops by gauge invariance, the linearized map includes the J channel, and m_j changes by a fixed factor; the power N^6 is loose); (3) corrected ("refuted" becomes "cannot be derived"); (4) corrected (the binding-set move below). Consequence for the binding sets: L-001 (Theorem 2w) uses only the definition of β_j (through R-029, R-030, and R-006), so R-015 leaves its set. Bałaban's assertion that the $\mathbb{Z}^d$ limit of (1.22)

exists moves into row R-006. R-015 stays in the set of L-001 (Theorem 2w) ⇒ Theorem 1 and of Theorem B, through R-032 and R-013: H-141 (Loc) for $g > 0$, the reading of $\partial\Lambda_{k+1}$ on CMP 119 p. 278, and the R-term part, which cannot hold literally for terms built from merged R′ inputs.

The reduction of R-006 and L-174 (Lemma G3). `SOURCES_T2_H0.md` (verified: 47 confirmed, 10 corrected) proves L-183 (Proposition H0-red): the use of R-006 in L-001 (Theorem 2w), item 7′, reduces to a residue N1–N9 in which no constant depends on j or on the past. The residue item N2 is L-174 (Lemma G3). Bałaban's C^∞ and all-derivative claims are not used.

Theorem 5.9 (L-174 (Lemma G3) = N2, `SOURCES_G3_N2.md` [PROVED RELATIVE TO LISTED INPUTS; VERIFIED])**.** *Assume Theorem 5.4 in its two-sided form, Theorem 5.5, Bałaban's chain CMP 116 (2.15)–(2.25) read as pointwise exponent bounds (including R-003), and the conditions (Hα) and (Hγ) below. Then (Q) holds in the joint (X, B) form with*

$$c_- = (\gamma_1^N + \varepsilon_0)^{-1} \geq \tfrac{4}{5\gamma_1^N}, \qquad \Lambda = 2\Big(1 + \tfrac{4}{3\gamma_{\exp}} + \tfrac{16}{9\gamma_{\exp}^2} G_\Gamma^2\Big),$$

$\varepsilon_0 = 2\Gamma_1^2\eta_M^0 \leq \frac{1}{4}\gamma_{\exp}$*, and L-174 (Lemma G3) holds: the cut adds S_χ to S_0, and $C_\beta = C_\Pi(S_0^{\mathrm{GGAIN}} + S_\chi)$, uniform in j, the past, the volume, and regular U.*

The one-bond-fixed form of (Q) is a corollary: fixing $B(b)$ integrates the compression exactly, with Schur complement $(\Gamma_{bb})^{-1}$ and no cost factor ($a = 0$). The proof uses no symmetry of A_Z and no L-207 (Theorem L). It corrects `SOURCES_Q.md` 4.3: the B-block of Q_0^{-1} is $H_0^{-1} + H_0^{-1}\Gamma\Gamma^* H_0^{-1}$, so $G_\Gamma = \|\Gamma_k(Z_0, 0)\|$ enters Λ. Conditions. (Hα): $(\alpha_5 + G_\Gamma^2\delta_S)\Lambda_0 \leq \frac{1}{2}$, with $\delta_S = O(\eta_M^2)$ (ledger K-026). (Hγ): $\frac{1}{20}\gamma_2\varepsilon_1^2/\gamma^2 \geq 4\kappa + 1$, and the condition $\frac{1}{20}x - 4M^4 e^{-x/20} \geq 0$ at $x = \gamma_2\varepsilon_1^2/g_k^2$ that the verifier added for (2.31) (ledger K-027). Both are smallness conditions of Bałaban's type, one unit stronger than his own. The bound on G_Γ (`SOURCES_G3_GGAMMA.md`, [VERIFIED], all four claims confirmed): $\Gamma_k(Z_0, 0) = 1_{Z_0} A_{\exp}(0) 1_{Z_0^c} R(0)$ with $R(0) = (C^{(k)})^{1/2}(0)$, and the Schur test gives $G_\Gamma \leq \bar{G} := \Gamma_1 K_{1/2}$, $\Gamma_1 = 2B_A S(\delta_A)$, $K_{1/2} = 8\gamma_{\exp}^{-1/2} S(\frac{1}{2}\varepsilon) + 4\gamma_1^{-1/2} S((1-\theta')\delta_A)$. So G_Γ is bounded uniformly in M (verified), and in the volume, j, Z_0, and regular U. With this bound (Hα) is the single block-size threshold

$$M \geq \max\{M_\Gamma, M_{\alpha 5}\}, \quad M_\Gamma = \tfrac{4}{\varepsilon''} \log\big[16h^3\Gamma_1^4\bar{G}^2\bar{\Lambda}_0(16W/\gamma_{\exp})^2\big], \quad M_{\alpha 5} = \tfrac{3}{\delta_0}\log(8c_5\bar{\Lambda}_0),$$

$h = 4/(3\gamma_{\exp})$, together with Bałaban's smallness of $\alpha_0, \alpha_1, \alpha_4, \gamma_2$; κ_1 does not enter M_Γ, and there is no cycle. One reading residue stays [I]: the s-rule acts on Bałaban's resolvent expansion (3.185) of [13] (CMP 116 p. 13), and the bound identifies $R(0)$ with the walk sum of Theorem 5.3(iv) with the σ_0-walks deleted (the reading I-b of R-009). Other inputs of Theorem 5.9: N3 (the Kotecký–Preiss derivative step, the margin (M_r) of R-017 with K-009, dominated differentiation), N6 (Bałaban's sup induction; past-uniformity of E_0 is reduced, not proved), N9 (R-015, definition only), and the reading that the pointwise bounds hold on the closed small-field set. Verifier: `SOURCES_G3_N2_VERIFY.md`; (1), (2), (4) confirmed; (3) corrected (the $e^{|P|}$ payment comes from the fifth $e^{-(1/10)x|P|}$ factor via (Hγ), which adds the second γ condition); 2000 random numerical instances pass. Consequence: the use of R-006 in L-001 (Theorem 2w) is a [VERIFIED REDUCTION] with residue N1–N9, and N2 is proved relative to the inputs above. The small-field "(Q)/c_- open" of ledger R-037 is discharged. R-006 stays a row: N5 (continuity at $g = 0$, [R]), N6, N8 (readings [I]), and N9 are not proved here, and N7 is R-031.

5.10 Step 8‴: round 6, locality, the boundary, the 2D gap, and the order of constants

Round 6 treats the Bałaban-level residue of row R-015, the R-term row R-032, and the open 2D gap of row R-028. It also fixes one acyclic order of constants. Each result has its own project file and an independent verifier file. The exception is the check file `SOURCES_T2_VOL.md` for K-032 (Vol), which has no verifier file. Each result holds relative to the inputs that it names. The tags stay [VERIFIED REDUCTION] or [PROVED RELATIVE TO LISTED INPUTS]. Nothing in this step makes Theorem A or Theorem B unconditional.

Locality of torus activities. Work at level k on $T_N = \mathbb{Z}^4/N\mathbb{Z}^4$ in level-k units, with projection π. A set is *liftable* if it lies in $\pi(B)$ for a box $B \subset \mathbb{Z}^4$ of side $< N/2$. A domain X is *non-wrapping* if its collar $X^\sharp$ (X plus one layer of width c_k, the largest cube side at level k) is liftable; $\operatorname{diam}_\infty X \le N/2 - 3c_k$ suffices. $\hat{X}$, $\hat{U}$, $\hat{J}$ denote lifts.

Theorem 5.10 (L-026 (Theorem LOC), `SOURCES_A15_LOC.md` §7.5 [PROVED RELATIVE TO LISTED INPUTS; VERIFIED])**.** *For every k, every coupling history in Bałaban's range, and every non-wrapping $X \in \mathcal{D}_{k+1}$,*

$$E^{(k+1)}_{T_N}(X, g, U, J) = E^{(k+1)}_{\mathbb{Z}^4}(\hat{X}, g, \hat{U}, \hat{J}) \quad \textit{for all } (U, J) \in \mathcal{U}^c_{k+1}(X, \alpha_0, \alpha_1).$$

There is no error term.

Method. The mechanism is the s-rule of CMP 116 p. 3. At $s = 0$ off Y_0, every walk-expanded operator is a finite walk sum inside Y_0, and the background equations (1.3)–(1.4) decouple by components. A bijection between the surviving torus walks and $\mathbb{Z}^4$ walks gives equal kernels (L-305 (Lemma W)). Uniqueness gives equal background fields (L-224 (Lemma N)). The cluster assembly and the subtractions are cube geometry (Lemmas A, S). The image terms of a periodized kernel sit in winding walks, so they enter only wrapping activities. The induction on k uses that β_{k+1} is the same $\mathbb{Z}^d$ number in both volumes; a torus coefficient would break exactness at the next step. Inputs: H-103 (FI) each walk factor is given by the same local formula on T_N and $\mathbb{Z}^4$ (supported by CMP 99 Sect. C); (U) uniqueness of the solutions of (1.3)–(1.4) (printed in CMP 102); and the Group C rows K-033–K-037. These are K-033 (Cube): $U_j(\square_0, \cdot)$ of [14] uses only data on $\square_0$ [I]; K-034 (Cases): the other term classes are localized "in exactly the same way" as the typical term (CMP 116 p. 2) [R]; K-035 (Conv): cube sides divide N, the partitions are π-images of the $\mathbb{Z}^4$ partitions, and $M = KR_0M_0$ with K a power of L; K-036 (Det′): a domain-assignment rule for the closed-walk terms of the determinant expansion of CMP 102; and the reading of "H_j with free boundary conditions" (CMP 109 p. 290) as the $\mathbb{Z}^4$ operator [I]. Verifier: `SOURCES_A15_LOC_VERIFY.md` (page images); confirmed with minor corrections. (i) L-167 (Lemma G)(b) had the split of CMP 116 (2.7) reversed: for large x the resolvent is a power series in $C^*\Delta_k C$, and the integral with $\tilde{G}_3(x)$ covers $x \in [0, \gamma_1]$; the conclusion is unchanged. (ii) The determinant expansion of CMP 102 does not follow the s-rule. It is a trace of walk expansions, and a trace splits exactly by walk; this needs the convention K-036 (Det′). (iii) The object table omitted two block-local terms (a Jacobian and $\log \sigma$); they are intrinsic by the same argument. H-103 (FI) and (U) are supported on the pages. Consequences. With Theorem 5.10, Proposition 5.8 holds relative to (Bd), (H), (Cnt), and the inputs above; H-141 (Loc) is no longer a hypothesis. Bałaban's assertion that the $\mathbb{Z}^d$ limit of CMP 109 (1.22) exists at $g > 0$ follows from Proposition 5.8. So row R-006 has no bare assertion left.

The boundary of the small-field region.

Proposition 5.11 (L-104 (Proposition BDRY), `SOURCES_A15_BDRY.md` [PROVED RELATIVE TO LISTED INPUTS; VERIFIED WITH CORRECTIONS]). *(f) The small-field terms $E_0^{(k+1)}(\Lambda_{k+1}, X, z)$ of CMP 119 p. 278 with $X \subset \Lambda_{k+1}$, also those with X next to $\partial\Lambda_{k+1}$, equal the whole-lattice terms exactly. (Thr) Let N_k be the number of level-k cells per side of the torus. If $N_k > 22M$, every wrapping or image term of the coefficient analysis at step k lies in Bałaban's irrelevant class $X \not\subset \tilde{\square}^2$ (CMP 109 (4.36), CMP 119 (3.48)), and the torus correction to β_j is at most $\varepsilon(M) = C_j(22M)^6 e^{-11 m_j M}$, valid for $m_j M \geq 1$.*

Method. On the image of CMP 119 (1.20), $S_1 \cap \Lambda_1 = \emptyset$. The change of variables (1.22) acts only on S_1^* bonds, so H, the Jacobians, and $V(S)$ reach only terms that meet Λ^c, and these go to the large-field part. Step $k+1$ (p. 279) keeps every $X \subset \Lambda_{k+1}$. L-305 (Lemma W)$_\Lambda$: with $s = 0$ off $Y_0 \subset X \subset \Lambda$, every surviving walk of $C(\Lambda)$ lies inside Y_0 and does not see $\partial\Lambda$. Inputs: the inputs of Theorem 5.10; K-038 (Enl): the s-rule localization domain of a walk is $\bigcup X_i^5$ as a whole-lattice set, not $\bigcup X_i^5 \cap \Lambda$; H-139 (L2$_\Lambda$) through (L2) of Theorem 5.1; (Dom); H-091 (Conv$_\Lambda$) [I]; and the nesting of the Λ_j [I]. Verifier: `SOURCES_A15_BDRY_VERIFY.md` (page images). (1) Corrected: Bałaban prints neither form of the walk domains, so K-038 (Enl) is a convention, admissible. Under the Λ-restricted convention, (f) holds with a margin of at most one M-cube layer, also admissible. $C^*\Delta_k C$ is walk-expanded, not finite range (CMP 116 (2.7), CMP 119 p. 275); the s-truncation of the Δ_k walks follows the same convention. (2) Corrected: see the R-032 paragraph below. (3) Corrected: the local set is $\tilde{\square}^5$, of side $11M$, so the threshold is $N_k > 22M$, not $10M$. (4) Corrected, admissible: see K-032 (Vol) below.

Row R-032 replaced. The old row R-032 asked that the finite-volume R-term increment equal the infinite-volume one. That form is ill-posed: the whole lattice of CMP 109 has no R-terms, and no infinite-volume large-field construction exists to compare with. (The author called it false; the verifier corrected this to ill-posed.) CMP 119 p. 279 prints instead that the R″ terms with $X \subset \Lambda$ "do not depend on Λ_{k+1}". L-209 (Claim L$_G$) of `SOURCES_A15_BDRY.md` proves this: the step output on X is one map $\mathcal{G}_X$ of the inputs on X, the same in every volume. R-032 is replaced by three statements: (a) β_j has no R part, because it is subtracted from E_{pure} (p. 279, confirmed on the image); (b) E_{pure} on a non-wrapping $X \subset \Lambda$ equals the $\mathbb{Z}^4$ term exactly; (c) E_R is uniform in K through F2′, with the threshold $N_k > 22M$. All three are [PROVED RELATIVE TO LISTED INPUTS; VERIFIED]: the inputs of Theorem 5.10 and Proposition 5.11, the F2′ inputs (R-044 (W4), (Eps), (D); ledger R-044), Bałaban's (1.18), (3.48), (4.36), and K-032 (Vol). R-032 is an input of L-001 (Theorem 2w) ⇒ Theorem 1, not of L-001 (Theorem 2w).

The volume hypothesis K-032 (Vol). Proposition 5.11 needs $N_k > 22M$ at every step $k \leq K$. N_k is smallest at $k = K$, where $N_K = \ell$. So the torus must have side $\ell > 22M$ final-scale cells. This is K-032 (Vol) of Section 2.1 (ledger K-032; `THEOREM2_PROOF_v4.md` §1.1). The verifier graded K-032 (Vol) admissible with a real condition. CMP 109 prints the torus exponent (p. 251, $L_\mu = L^m$) and the cube exponent (p. 257, $M = L^m$) with the same letter m. If the two were equal, then $\ell = 2M$, and K-032 (Vol) would fail at the last steps. `SOURCES_T2_VOL.md` reads this as reuse of a symbol, and this paper fixes the two exponents independently. K-032 (Vol) does not depend on K, the proof of L-001 (Theorem 2w) does not use it, and it excludes only tori with $\ell \leq 22M$.

The 2D gap ($\mathbf{G}'_A$). Let $A' = H' + D'D'^\dagger$ be the charged block of the SU(2) Wilson Hessian at constant flux f on the coarse $\mathbb{Z}^2$ lattice, in Landau gauge, as in `SOURCES_T2_S123.md` §5. Write $\alpha = ef$, so $|\alpha| = 2|f|$.

Theorem 5.12 ((G'_A), `SOURCES_T2_GAPRIME2.md` Theorem 5.1 and Corollary 5.2 [PROVED+VERIFIED]). *Let $0 \le c < 1$. For every real f with $0 < |f| \le ((1-c)/100)^2/2$, the operator A' has no spectrum in $(-2c|f|, 2c|f|)$, and neither has H' restricted to $\ker D'^{\dagger}$. For example, $c = \frac{1}{2}$ gives a gap $|f|$ for $0 < |f| \le 1.25 \cdot 10^{-5}$. The constants do not depend on j, L, or the averaging.*

Method. A lattice ladder operator $\Pi = \nabla_0 + i\,\mathrm{sgn}(\alpha)\nabla_1$ gives $X(X - 2|\alpha|) \ge -157|\alpha|^3$ on low states. So the scalar magnetic Laplacian $\mathfrak{L}$ has no spectrum in $[(1+\theta)|\alpha|, (3-\theta)|\alpha|]$ for $\theta > 26|\alpha|^{1/2} + 157|\alpha|$. Then $A' = \mathfrak{L} \otimes 1 - 2\alpha\sigma_y + R$ with $\|R\psi\| \le 48|\alpha|^{3/2}$ near 0, and a contradiction argument gives the gap. The $\ker D'^{\dagger}$ form follows from $A'D' = D'\mathfrak{L}$. No Harper-band or Helffer–Sjöstrand result is used. Verifier: `SOURCES_T2_GAPRIME2_VERIFY.md`; all four claims confirmed. The constants 26, 157, and 48 are rederived. The lattice commutator is tracked exactly through $2|\sin(\alpha/2)|$. No invariant subspace is needed. The midpoint-transport identification moves from [I] to [P]. The verifier restated the range in units of f (as above) and reproduced the numerics (141 values).

Remark 5.13 (the proved range is narrow). The proof covers only $|ef| \le 10^{-4}$. Numerics show the gap on all of $\alpha \in (0, 3.02]$, with physical ratio ≥ 0.95: at 11 rational fluxes with denominators up to 2213, the smallest |eigenvalue| of A' is $|\alpha|(1 - 0.125|\alpha|)$, and a sparse shift-invert run at $\alpha = 10^{-4}$ gives the ratio 1.0000125. So the proof is valid but far from sharp, and the numerics are not a proof. The Nielsen–Olesen mode is the band at $-|ef|(1 + 0.12|ef|)$, not a zero mode (numerical). The upper edge $3|ef|$ of H' is not proved and is not needed. Open: Prop. 5.6 of `SOURCES_T2_S123.md` uses (G'_A) for $|f| \le \min(f_0, f_1(j))$, where $f_1(j)$ comes from its size condition $g_f \ge 4\eta_1 + 4C_M^2|f|^3$. Round 7 (`SOURCES_T2_FRANGE.md`): this f-range need not fit a fixed window, because (M_j) is a limit $f \to 0$; the proved range suffices (Section 5.11).

Consequence. Open item Q-059 of `SOURCES_T2_S123.md` is closed. Prop. 5.6 and (Z_j) hold given the inputs of that file. (M_j) (row R-028) reduces to (I3) [P], Prop. 4.4 region 1 of `BETA_FP_M.md` [argued], and the neutral sector [I]. Both residues are under attack. Row R-028 is a verified reduction, subject to the f-range check of Remark 5.13. Round 7 proves both residues and settles the f-range (Theorem 5.14).

The canonical order of constants. `SOURCES_ORDER_CANON.md` gives one acyclic order of all constants of the source files. It was verified twice, with amendments E1–E8, and it supersedes the orders of rounds 4 and 5 (ledger K-005). `SOURCES_ORDER_CHECK.md` found two nominal cycles in the orders as earlier write-outs print them. Cycle A, on the M_1 side (`WC_FULL.md` 6.9, `A1_WALK_FULL.md` 9.1): $M_1 \to M \to \alpha''' \to \alpha'' \to B_{A''}, \delta_{A''} \to M_1$. Cycle B, on the M side: $M \to \alpha''' \to \alpha'' \to \alpha' \to \varepsilon'' \to M$. Neither cycle is real. A split resolves both: the level constants (rates, prefactors, Γ_n, k_n, ε'') are bounds on domains capped by a_1, a_{**}, and $M\alpha_0 \le a_0$, so they are free of M and of the α's and come before M_1; the domain sizes α''', α'', α' are smallness choices and come after M. The verifier also found forward references and a double count of M in the J scale. The amendments fix them, with two new entries: real $\alpha_0 \le \alpha'_0/(CM)$ with $C > c_J$, and $M \ge 1/c_J$. The block size M has 32 lower bounds and no upper bound (ledger K-025), so the admissible set of M is an up-set. G_Γ is bounded uniformly in M (`SOURCES_G3_GGAMMA.md` + VERIFY; Section 5.9), so (Hα) is one more lower bound on M and adds no cycle. One caveat stays: the plaquette row of L-220 (Lemma M1) is nested by reading only (amendment E6).

5.11 Step 8′′′′: round 7, the implicit Group A rows, (M_j), L5, and the sweep

Round 7 treats the implicit or unprinted Group A rows R-016–R-019 and R-020, the Group B rows R-028, R-031, R-034, and R-041, the certification of R-025 (H2) at more values of L, and the canonical order of constants. Each result has its own project file and an independent verifier file. The exception is `SOURCES_T2_FRANGE.md`, a short check with no verifier file, and `SOURCES_T2_F_STEPB.md`, whose verifier is still pending. Each result holds relative to the inputs that it names. Nothing in this step makes Theorem A or Theorem B unconditional.

The fixed-level input (M_j) (ledger R-028).

Theorem 5.14 ((M_j), `SOURCES_T2_MJ_REST.md` Theorem 4.2 and §5 [PROVED RELATIVE TO LISTED INPUTS; VERIFIED]). *Fix $j \geq 1$ and odd $L \geq 3$. The one-loop increment of level j at constant abelian flux f satisfies $\delta\Gamma_j(f) - \delta\Gamma_j(0) = n_2 d_j f^2 + o(f^2)$ as $f \to 0$, with d_j the formal coefficient of `KAPPA_BOUNDED.md` (1.1). The charged-sector error is $O(|f|^{9/4}) + o(1)f^2$. The neutral sector contributes its formal coefficient exactly in the limit.*

Method. Region 1 of Prop. 4.4 of `BETA_FP_M.md` is not an inequality. It is an asymptotic identity: the f^2 coefficient of a spectrally localized trace equals the per-momentum formal integral. The file replaces the region split by a split of $\log|x|$ in three pieces with a shifted logarithm. The gapped piece F_s has shift $s = |f|^a$, $a = \frac{3}{4} \in (\frac{2}{3}, 1)$, so $f/s \to 0$ and a Moyal expansion has a small parameter (L-056 (Lemma A), with an exact two-factor trace identity and an approximate Moyal inverse to order $M = 14$). The Nielsen–Olesen window piece is the term II of `SOURCES_T2_S123.md` Prop. 5.5. The localized remainder is $O(s^3 + |f|^3)$. The neutral sector (N2) is proved by an induction on bordered Hessians: $\pm\partial^p \Delta_i \leq C_p \Delta_i(0)$ for $|f| \leq f_*$. On a finite torus the d constant 1-forms are zero modes; they are not gauge directions, and they form a null set of the per-cell trace, so the infinite-lattice statement does not see them. Inputs: (G'_A) (Theorem 5.12); (I3) [P] (`SOURCES_T2_I3.md` + VERIFY); the results of `SOURCES_T2_S123.md` that its verifier confirmed; Prop. 2.2 of `U3_LEMMA51.md`; and one low-risk convention [I]: the coefficients $b_{t,m}$ of `KAPPA_BOUNDED.md` (1.1) are the Moyal coefficients of §3.1 at $\sigma = 0$. Verifier: `SOURCES_T2_MJ_REST_VERIFY.md`; all four claims confirmed (L-056 (Lemma A) Steps 2–4, the exponent count, the (N2) induction, and the numerics; Section 6.8). Only the $o(f^2)$ form is proved. The printed rate $O(\varepsilon|f|^3 \log(1/|f|))$ of Prop. 4.4 is not proved, and no consumer uses it. The f-range. `SOURCES_T2_FRANGE.md` shows that the proved range of Theorem 5.12 suffices. (M_j) is a limit $f \to 0$ at fixed j, so the gap is needed only on some punctured neighbourhood $0 < |f| \leq f_{\lim}$; its size plays no role. The size condition $g_f \geq 4\eta_1 + 4C_M^2|f|^3$ of Prop. 5.6 of `SOURCES_T2_S123.md` has a left side linear in $|f|$ and a right side $O(f^2)$ ($\eta_1 = C_\eta f^2$), so it holds for all small $|f|$. Usable range: $0 < |f| \leq \min(f_0(c), f_1(j, L), f_*(j, L))$. The numerical gap up to $\alpha \approx 3$ stays supporting evidence only. Consequence. Row R-028 is proved relative to listed inputs. The chain from `SOURCES_T2_S123.md` through (M_j) to Prop. 1.1 of `KAPPA_BOUNDED.md` is closed up to conventions and readings.

History dependence L5 (ledger R-031).

Theorem 5.15 (Lemmas D″-nest and D′-nest, `SOURCES_T2_B7_CLAMP.md` Route 2 [PROVED RELATIVE TO LISTED INPUTS; VERIFIED]). *The conclusions (a″), (b), (c′) of Lemma 3.4, with the same constants $\beta = 3b/4$, $B' = 5b/4 + 2R$, $D' = 2R$, $g_1 = (\gamma^{-2} + 2R)^{-1/2}$, hold for true (unclamped) trajectories, with no clamp. Every evaluation of β lies on a true trajectory with all $u_i \geq \gamma^{-2}$, inside Bałaban's domain. The same holds for Lemma 3.2 with its constants.*

Method. Nested intervals. The envelopes $p_k, q_k = t+T_0(k,K)\mp(b/4)(K-k)$ subtract $\beta_{k+1}(0)$ exactly, and the slack shrinks by $b/4$ per step, which equals the worst single-step error $C_1\gamma$. So negative increments are absorbed. At each step the intermediate-value theorem selects a closed subinterval of u_0 on which u_{k+1} sweeps the next target interval. The window bound (5.1) is used at every endpoint $n \le K$. Inputs: $7'$ on trajectories; H-089 (C-tr), the continuity of one step in its inputs (the single CMP 116 step, L-281 (Theorem SM), `SOURCES_HHOL.md` §2–§3, `SOURCES_GGAIN_NONMONO.md` or R-015, and the enlarged input list of N6); and one reading [I]: Bałaban's step $k+1$ uses only $g_0, \dots, g_k$. Route 2 still evaluates the step map at one off-trajectory mixed point per step, inside the proof of H-089 (C-tr). Verifier: `SOURCES_T2_B7_CLAMP_VERIFY.md`; Route 2 confirmed with the constants of `THEOREM2_PROOF_v4.md`. Route 1 (continuity on the clamped cube) holds, but its γ condition uses an undefined constant γ_{hol}, so it is dropped. The history form of N3 needs Route 1; the one-argument form does not. The quote of CMP 109 p. 298 is confirmed on the image; "β_j only on the trajectory" is a correct reading of the construction.

The cut and the analyticity domain (ledgers R-017, R-019).

Proposition 5.16 (R-017, `SOURCES_A17_A19.md` [PROVED RELATIVE TO THE PRINTED TEXT AND ONE READING; VERIFIED]). *On every bond b that the cut χ_k of CMP 109 (2.9) covers, $B'(b) = g_k B(b)$ exactly. Hence* $\operatorname{supp}\chi_k \subset \{g_k|B|_Y < \varepsilon_1\}$*, the analyticity domain of CMP 116 (1.34), with margin $r = 1$ and no slack.*

Method. CMP 109 (2.9) takes the product over $T^{(k)} \setminus \{b_0(c)\}$; $hB = 0$ off the bonds $b_0(c)$ (p. 267); CMP 116 (2.3) writes the cut directly as $|B(b)| \ge \varepsilon_1/g_k$ on the same bonds. The bound (1.20) of CMP 116 gives only the converse inclusion. The one reading [I, strong]: C is the identity on the remaining bonds; the norm on Y is the per-bond sup of CMP 99 (3.41).

Proposition 5.17 (R-019, `SOURCES_A17_A19.md` §4–§5 [REDUCED TO READINGS; VERIFIED]). *The bound (190) of CMP 102 holds at complex B on the ball $|B| < 2C_1\varepsilon_1$, given three readings [I]: the complex clauses of (189) and (73), which Bałaban does not print, and the identification of the set (172) with this ball. Fallback without the ball reading: the two-constants theorem gives the bound on $|B| < \frac{1}{2}C_1\varepsilon_1$ with harmonic measure $\omega = (4/\pi)\arctan\frac{1}{3} = 0.4097$, so the decay rate is multiplied by* 0.59.

Four of the five named inputs carry a printed complex clause (Prop. 4, Prop. 6, (55), (80) of CMP 102); (130) is linear. The fallback still needs a complex sup bound on kernel entries, from (187) and (55). Verifier: `SOURCES_A17_A19_VERIFY.md` (page images). R-017 confirmed. The option K-010 (keep the cut at ε_1 and run CMP 116 §1 with domain constant ε_1/r) is confirmed for CMP 116 §1, with the condition $4B_0C_1e^{16\kappa_1}\varepsilon_1 \le \frac{1}{4}\alpha_2$ (the file had $\frac{1}{2}\alpha_2$). K-009 is no longer needed at $r = 1$. R-019 corrected from "proved" to "reduced to readings". The Cauchy step of `SOURCES_MA2.md` §4.4 reaches the boundary of the open ball, but a limit $\rho \uparrow C_1\varepsilon_1$ keeps its constant c_N/C_1.

The regular class (ledger R-018).

Proposition 5.18 (R-018, `SOURCES_A18_REGULAR.md` [VERIFIED REDUCTION, WITH CORRECTIONS]). *Relative to printed results: (i) the background domains of CMP 109, 116, and 122 lie in the regular class of CMP 99 (3.35)–(3.36); (ii) L-250 (Lemma Q): $Q_k(U) = Q_{\text{free}} + O(\alpha)$ in a per-row axial gauge, uniformly for $L \ge 3$, and $O(M\alpha_0)$ at operator level; (iii) L-109 (Lemma C): $\|C(U)\|^2 \le c_C(1)(1+O(\alpha))$; (iv) $c_J = O(\alpha_0)$, with no factor M; (v) σ-localized operators add nothing.*

Method. Every explicit use of (3.36) in CMP 99 goes through the current J, so a bound on J replaces (3.36) up to $O((M\alpha_0)^2)$. L-250 (Lemma Q) telescopes $Q_k(U) - Q^k$ over levels on CMP 98 (124)–(125), (139)–(146). L-109 (Lemma C) uses the printed definition of $b_0(c)$: each row of the linear one-step map contains exactly one eliminated variable. Verifier: `SOURCES_A18_REGULAR_VERIFY.md` (page images). Corrections: the inclusion constant is CMP 109's unnamed cube constant $c_\square^{109}$, and 5.2(d) needs two factors L; the telescoping sum of L-250 (Lemma Q) misses a factor 2^{k-j-1}, so the correct sum is $c\alpha L/(L-2)$, k-uniform for $L \geq 3$; the CMP 122 cube patching is [UNSUPPORTED] and is replaced by CMP 99 at cube scale $M_\square^{99} = CM/10$, with the new canon condition $M \geq 5M_{99}$ ($10 \mid C^{122}M$). Residue: [R] CMP 102 Thm 1, CMP 98 Prop. 2 and (139), CMP 99 (3.133), CMP 109 p. 263; [I] I2, I3, $U_{k,Z}$ obeys CMP 102 Thm 1, and B, B_5 are fixed numbers of (d, L). Conditions H-063 (A18-1)–H-066 (A18-4): ledger K-042. The item (C2) of `SOURCES_MA2.md` is merged into this row.

The large-field T-step rate loss (ledger R-020).

Proposition 5.19 (R-020, `SOURCES_A23_TSTEP.md` [VERIFIED REDUCTION (PROOF SKETCH)]). *In one large-field T step the decay rate κ loses at most the fraction 13δ: 10δ printed for the small-field step (CMP 116 (2.28)–(2.41)) and at most 3δ unprinted for the Z_0-marked cluster expansion of CMP 119 (3.44)–(3.47). No loss inside T is an additive constant in κ.*

Method. "CMP 109 Sects. 6–7" is CMP 116 Sects. 1–2 in a preprint numbering, so the two deferral targets of row R-020 are one printed source. Each merging step costs a δ-fraction of the current rate (one δ for a sum over pieces, one for a volume factor by (2.30)). The tree bound is printed as CMP 116 (2.27). Verifier: `SOURCES_A23_TSTEP_VERIFY.md` (page images); confirmed. The verifier counts 2δ for the unprinted part, so 13δ is a safe upper bound. Open: the count R1 (a sketch) and R3, Bałaban's assertion that $H'(Z_0)$ obeys (2.38). Consequence for ledger R-046: H-215 (K1) becomes H-216 (K1′) $\delta \leq \frac{1}{13}(1 - 2(1+4\beta)/(\theta_K L))$ with $\theta_K \geq \frac{1}{2}$ and $\kappa \geq 2m_{\mathrm{tot}}$; at $L = 13$, $\beta = \frac{1}{4}$ this is $\delta \leq 5/169 \approx 0.0296$, so $\delta = 0.025$ is the choice (0.03 fails; Lean `K1_numeric_13` in `lean/YmLean/K1Prime.lean`, no `sorry`, standard axioms). H-217 (K2), H-218 (K3), κ_{eff}, $\kappa \geq 1536$ are unchanged. The paragraph on `SOURCES_T2_KAPPA2.md` in Section 5.2 carries the new form.

The propagator replacement (ledger R-016).

Theorem 5.20 (R-016, `SOURCES_A16_REPLACE2.md` [PROVED RELATIVE TO LISTED INPUTS; VERIFIED WITH MINOR CORRECTIONS]). *The sourced covariance $C_f = (A_{\exp} - g^2\mathcal{W}_S)^{-1}$, $\mathcal{W}_S = C^*\mathcal{V}_S C$, has the properties H-281–H-291 that CMP 116 uses for $C^{(k)}$, and the uses H-292 (U-a) whole-volume C_f and $(C_f)^{1/2}$, H-293 (U-b) σ-differences of $(C_f)^{1/2}$ and Δ^f, and H-294 (U-c) the tree form of Δ^f, under the tree conditions K-041.*

Method. The sentence of CMP 116 p. 3 licenses only the replacement in (1.3)–(1.4); it is not cited for C_f. The properties are proved directly. $\mathcal{W}_S$ is written as a tree sum of the type of L-312 (Proposition W_N) of `WC_FULL.md`, on the complex chain of `WC_FULL.md` §6, with at most 12 junctions per term and $K_W^T = 2C_{\mathcal{D}}(C'_{\mathcal{D}})^{12}B_{\mathcal{W}}$. The conditions H-060 (A16-a^T) and H-062 (A16-b^T) are linear in $g^2 r$. History. The first proof (`SOURCES_A16_REPLACE.md`) had a false locality step: it treated each $\mathcal{W}$ term as a one-domain block, but $\mathcal{W}_S$ is not finite range, because $H_1 = G_1Q^*(QG_1Q^*)^{-1}$ is nonlocal (`SOURCES_A16_REPLACE_VERIFY.md`). The repaired proof uses the tree form. Inputs: Theorem 5.3 (with W_N and §6), Theorem 5.2 (L-115 (Lemma C), WG-G1, 5.2, 5.5), Theorem 5.1 for G_1 (conditional on H-057 (C)), Theorem 5.5, `SOURCES_G3_GGAMMA.md` 2.1, and `SOURCES_D2.md` (2.a), `SOURCES_D3W8.md` §1.1, §1.4. Residue [I]: H-035 (M1) the completeness of the property list

(CMP 116 pp. 12–17); H-166 (S_H) the scale normalization of D_S, J_S; the volume convention of the CMP 122 Ω-setting; the s-version choice for $(C_f)^{1/2}$. Euclidean covariance of C_f is false for non-constant f; no consumer may use it. Verifier: `SOURCES_A16_REPLACE2_VERIFY.md` (page images); claims (1)–(3) confirmed; H-293 (U-b) corrected without a change of constant order (R_2 is anchored at both ends in Z_0; the product form needs concatenation at the level of elementary blocks).

The large-field holomorphy chain (ledger R-041, chain item 5). `SOURCES_T2_HMINUSC.md` closes the two items of `SOURCES_MA2.md`. (C1) is [PROVED GIVEN R-019 AND R-018; VERIFIED]: every condition on CMP 102's ε_1 and C_1 is an upper bound (and $C_1 \geq 1$), so ε_1 can be raised per scale to give Cauchy room. The four cases K1–K4 of Proposition 5.17 give radii $\kappa_H \in \{2, \frac{1}{2}, \frac{1}{2}, \frac{1}{8}\}$ and constants c_N/C_1 to $16c'_N/C_1$, all free of C_1. The price is the condition (C1*) $c_m M_0\, g_k p_0(g_k) \leq \frac{1}{2}\kappa_H C_1^{102} a_*^{102}$ (ledger K-039), with M_0 placed in the canon before γ (K-040). (C2) is [REDUCED] and merged into R-018, with two readings [I]: the averaging map $m = M^\bullet \circ Q_k^{s*}$ is local and analytic, so V' always enters through m; and $R_1 < L$ suffices for condition (1) of CMP 102. Verifier: `SOURCES_T2_HMINUSC_VERIFY.md` (page images); the map of (H−) is CMP 122 I (1.74), not (1.20), and "~ 4" means four layers of M-cubes. So chain item 5 reads: R-041 $\Leftarrow$ H-143 (M-A) $\Leftarrow$ (H−) $\Leftarrow$ R-005 + R-019 (K1–K4) + R-018 + (C1*). R-041 stays a verified reduction.

The sweep and N3 in the D″ form (ledger R-034). `SOURCES_B10_SWEEP.md` with `SOURCES_B10_SWEEP_VERIFY.md` (page images, three readers over 63 pages) clears the four items of R-034. (i) All 19 rows of the sweep of CMP 119 (2.6)–(2.9) now have a second reader. The page images of CMP 122 I–II Sect. 1 show about ten cross-scale uses and five exponent or threshold conditions that the list missed. They are now listed in the addendum of `SOURCES_A11_CONST.md` as H-271–H-280 and H-266–H-270 (ledger K-043). None is a new type: each is a H-156 (P-up) window, a H-155 (P-low) window with a factor $(1+\beta_0)^a \operatorname{poly}(n-m)$, or a single-scale γ-smallness threshold. The form $p_1(g) = (\log g^{-2})^{p_1}$, $p_1 < p_0$, is confirmed on CMP 122 I p. 183. (ii) Prop. 4.3 of `BETA_FP_M.md` is confirmed given H-244 (S1)–H-247 (S3), for $j \geq j_1$ only; below j_1 it needs the open H-248 (S3w), which L-001 (Theorem 2w) does not use. (iii) The Prop. 3.6 assembly is confirmed. (iv) N3 in the D″ form: L-192 (Lemma H″) (real base) takes J as the component of 0 in $\{c \in [-2r, 2r] : u_k(c) > \gamma^{-2}\ \forall k \leq K\}$ and gives $S_3 \leq (D'/\beta + 1)\gamma^3 + 2\gamma/\beta$; N3″ (complex) splits the bracket at $\beta/2$, gives $\operatorname{Re} u_i \geq \gamma^{-2} + \max(0, (\beta/2)(K-i) - D')$, and has slack 6.7 at the N3A radius. The "$+\eta$" of `SOURCES_N3A.md` is not needed. The history form of N3 needs Route 1 of Theorem 5.15.

The canonical order, section F. `SOURCES_ORDER_CANON.md` section F adds 28 condition entries from the round-7 files (new lines N-260, N-300, N-320, N-380, N-460) and resolves the symbol clashes by renaming (a_*^{102}, C_1^{102}, ε_1^{102}, κ_H, θ_H, θ_K, $M_\square^{99}$, and others). A topological pass finds the order acyclic. It removes two forward references: H-216 (K1′) uses the bound $\theta_K \geq \frac{1}{2}$, so δ is a canon constant (line N-060) and $\kappa \geq 2m_{\text{tot}}$ moves to line N-070; and the cube cap of CMP 102 Thm 1 is booked on ε_1^{102} and ε_0^{109}, not on M, so M stays one-sided. Four constants are two-sided, and each has an existence condition: p_1 (exists under the N-410 condition on p_0); ε_0^{102} (nonempty by CMP 102 Thm 1); ε_1^{102} per scale (exists if and only if (C1*) holds and $\varepsilon_1^{(0)} \leq a_*^{102}$); and N per scale (exists for γ small; $N = R_j$ is admissible). The budget α_5 is shared between (Hα) and the entries of line N-490. Not yet in the canon: the constant C_1 of H-218 (K3) (to be named C_1^{FX}) and $\kappa \geq 1536$.

The lattice log coefficient (ledger R-027) [REDUCED [R], UNVERIFIED]**.** `SOURCES_T2_F_STEPB.md` narrows Step B of R-027 (F). Theorem 3.1 of `BETA_FP_H.md` uses

R-027 (F) only through (3.1), which is equivalent to $G(\varphi) - G(\lambda\varphi) \to n_2 c_2 \ln(1/\lambda)$ as $\varphi \to 0$ for $G = F_W(\varphi)/\varphi^2$ and $\lambda = L^{-2k}$. The file reduces this to H-137 (L-LL), the existence of the limit of $H_\varphi(\mu) = \varphi^{-2}[W(\varphi;0) - W(\varphi;\mu|\varphi|)]$ as $\varphi \to 0$ (its value is not needed; it holds the Nielsen–Olesen zero), to a gauge-covariant kernel bound (Lemma 1, cited as standard and not checked), and to the gauge-fixing part of Step C [I] (the ghost part is proved). The continuum side is X-005. The verifier of this file has not reported, so R-027 stays on the list of rows without a proof or a verified reduction. (Round 8: the reduction is verified, and H-137 (L-LL) has a proof pending verification; Section 5.12.) Open numerical tension. The single-step values β_1 at $L = 5, 7, 9, 11$ do not follow the slope $c \log L$, $c = 0.3715$ (Section 6.1). The open question is whether 0.3715 is compared with the right quantity (the single-step β_1 against the fixed-point β_∞), or whether the gap sits in the coarse bracket, in the dependence on the family, or in the $N_g = 8$ grid at $L \geq 9$. The tension may also bear on R-028 through Theorem 3.1 of `BETA_FP_H.md`. The claim of `NUMERICS` §7 that the first-step constant decreases "toward the universal slope" is under verification. (Round 8: that claim is withdrawn; Section 6.9.)

R-025 (H2) at $L = 15, 17, 21$ (ledger R-025). The certificate of Theorem 6.1 runs unchanged at these L and is now certified with the same standing as at $L = 13$ (Section 6.8). Other L are not used by the targets: K-003 fixes the certification target at $L = 13$.

5.12 Step 9: round 8, the sub-items of R-039, R-044, R-045, the condition H-057 (C), and H-137 (L-LL)

Round 8 narrows the open sub-items of rows R-039, R-044, R-045, and R-027, checks the page readings of `SOURCES_BOOKKEEPING.md`, and drafts a proof of the condition H-057 (C). Every result below has an independent verifier file with a final grade. The verifiers of `SOURCES_C_CONDITION.md` and `SOURCES_T2_LLL.md` reported in round 9: H-057 (C) is a verified reduction, and H-137 (L-LL) is proved and verified. Nothing in this step makes Theorem A or Theorem B unconditional.

The threshold mismatch R-039 (H-thr) (ledger R-039). `SOURCES_B15_M245.md` replaces the input list M2′, M4, M5 of `SOURCES_HTHR2.md`. M2′ is derived: the estimate of CMP 122 II p. 383 is pointwise in B (page image), so the one-bond factor gives a Schur-complement density bound. M4 is derived with $P_D = np_1 + dr_0$. M5 is not needed: the mismatch splits into a weight part W_j and a threshold part T_j, and the weight part is bounded by a directional Cauchy estimate at thresholds frozen at a real s', with an extra factor $30r\gamma^2$. Summability and the Gronwall step are proved. Verifier: `SOURCES_B15_M245_VERIFY.md` (page images), [VERIFIED REDUCTION], with corrections. M2′: the density step must divide by the old bond-b Haar factor $c_H = O(1)$, and the density $\sigma(g_k B)$ was omitted; only constants change. M4: the term $s^{-1/2}$ was missing. The W_j/T_j split is confirmed in structure, and (G2‴) is corrected. The verifier finds one new input, H-241 (R1′): the sup and small-difference bounds of `SOURCES_HHOL_LF.md` at thresholds frozen at a real $s' \neq x^*$ within $3r$. R-038 is no longer an input of R-039.

Proposition 5.21 (H-241 (R1′), `SOURCES_HHOL_LF_SPRIME.md` [VERIFIED REDUCTION, WITH CORRECTIONS]). *On the box $S = \{|s_i' - \bar{u}_i| \leq 0.1\bar{u}_i\}$ the holomorphy and sup bounds of R-041 (H-hol) in the large-field layer hold at thresholds frozen at s', with $\gamma_2, c'' \to 0.895\times$, $\gamma_0 \to 0.73\gamma_0$, $C_i \to 1.06\, C_i$, and one more lowering of γ. The difference between the objects frozen at s' and at x^* has the tail form $\leq C_T \sum_i \lambda_C^{j-1-i} (\log \bar{u}_i)^{dr_0} e^{-\kappa_3 (\log \bar{u}_i)^q}$, with $\kappa_3 \geq 0.734\kappa_2$ on S and $\kappa_3 \geq 0.80\kappa_2$ on the $3r$-set of R-039.*

Method. The threshold s' enters only through the cut radii $\delta(s') = s'^{-1/2}A(\log s')^p$ and the split that they induce; the vertex domains stay frozen at x^*. On S the cut ratios lie in $e^{\pm 1/18}$ and the large-field exponents in $e^{\pm 1/9}$. The proof of `SOURCES_HHOL.md` and `SOURCES_HHOL_LF.md` then runs uniformly on S. The tail form suffices for the weight part W_j of R-039. Verifier: `SOURCES_HHOL_LF_SPRIME_VERIFY.md`. Claims (1) and (4) confirmed; (3) confirmed with two repairs; (2) corrected: besides the inclusion C3 there are three more conditions that compare a cut radius with a frozen domain radius (from `SOURCES_T2_IEPS_LF.md`), and each is restored by the joint scaling $(C_0, \dots, C_4) \to 1.06(C_0, \dots, C_4)$. Grade: H-241 (R1′) is reduced, not proved, relative to H-143 (M-A)–H-146 (M-C), (R2), and L-285 (Lemma T) of `SOURCES_HTHR2.md`, plus three readings [I]: (i) C_2–C_4 are free upward; (ii) the R-ratio cutoff in the numerator and denominator of `SOURCES_HHOL_LF.md` 3.4 telescopes only with a lower bound on the denominator, which is not written (or the cutoff is frozen at x^*); (iii) the inter-step cut-nesting slack is not checked on the full box S. On the $3r$-set of R-039 the move is $e^{\pm O(r\gamma^2)}$, so (iii) does not affect R-039. Open inputs of R-039: the three [I] of H-241 (R1′); (R2)–(R4); the large-field coarea Jacobian (`SOURCES_G3_N2.md` 2.6); the convention items [I] (integers and partitions frozen at x^*, K-001).

The R operation and the domains Λ_j (ledger R-044).

Proposition 5.22 (R-044 (W4)-R, `SOURCES_B20_W4R.md` [PROVED RELATIVE TO LISTED INPUTS; VERIFIED WITH CORRECTIONS])**.** *The output of Bałaban's R operation is holomorphic in the input activities on the set $3\times$(inputs), which contains the $2E_0$ ball, and obeys $|R'(X)| \le O(1)c_1' e^{-(1+\frac{1}{2}\beta)\kappa d}$, under the changes $\varepsilon_1 \to \varepsilon_1/3$ in (C1)–(C4), $\alpha \to c'\alpha$, and $A_0 \to \sqrt{c'}A_0$.*

Method. `SOURCES_W4.md` §6.2 rested on a wrong premise, an input-dependent denominator. The denominators of CMP 122 II (1.1)–(1.2) and CMP 122 I (1.100) contain only the Wilson action, (1.10) turns the division into a product with an input-free exponential, and the final R operation (1.71)–(1.72), (1.98) divides by nothing. The inputs enter affinely. $T_k'(X)$ is holomorphic by domination under a positive input-free measure. The bracket is a finite sum of products and real t-integrals, and the logarithm in (1.98) is the Ursell series, which converges under (1.97) with changed constants. Verifier: `SOURCES_B20_W4R_VERIFY.md` (page images CMP 122 II pp. 356–358, 379–381, 388–391). Claims (1) and (3) confirmed; (2) corrected (minor): T_k' is holomorphic on the complex domain $\mathfrak{A}_c$, not entire; Bałaban's positivity step (1.73) fails at complex inputs, and the domination bound replaces it; C_0, C_1 rise with A_0 when $q_0 = p_0$. The line check of `SOURCES_D3W8.md` §1.4(i) is corrected: the positivity display is CMP 122 II (1.73) on p. 380, and the repair of item L5 is $A_0, A_1 \to 2A_0, 2A_1$, because a fixed shrink of g cannot pay a factor 4 in $\gamma_0 p_0^2$ ($p_0(cg)/p_0(g) \to 1$). Residue: W4-G [R], the holomorphy of $V(Y)$ in the inputs (Bałaban's sketch of the preparatory integrations in CMP 122 I Sect. 1), and the Ursell bound (7.13) [I]. The Λ_j sub-item. `SOURCES_B20_LAMBDAJ.md` shows that Λ_j is a summation index, not a domain that moves with the field. Bałaban's expansion is a finite sum over region sequences outside the T operations (CMP 119 (1.25), (1.28)), and each term has fixed regions. So R-044 (W4) holds at fixed regions on a fixed finite slot space, and H-240 (P8) holds for each Λ with $p_0(g_j) \ge 16/(\gamma_0 A_1^2)$. Verifier: `SOURCES_B20_LAMBDAJ_VERIFY.md` (page images CMP 119 pp. 252–253, CMP 122 I pp. 182–183), [PROVED RELATIVE TO LISTED INPUTS; VERIFIED WITH CORRECTIONS]: the transcription of (1.25) is fixed (sum over R_1; second factor on $R_1^c \cap \Lambda_1^c$); several cutoff factors sit inside the integral (harmless: sharp, free of inputs and sources); and $E_R(X, 1; f)$ on $X \subset \Lambda_m$ must be booked into e_k^f. It is not a new open item. So R-044's residue is W4-G [R] and the Ursell bound [I].

The H-098 (E14-c) clause (ledger R-045) [REDUCED TO R-045 (S); VERIFIED]**.** `SOURCES_B21_E14C.md` shows that the present chain of Theorem B uses no interpolation lemma

and no source derivative at $t \neq 0$. The only source-derivative object is $\kappa_n(f_1, \dots, f_n) = \partial_{t_1} \cdots \partial_{t_n} \log Z_\varepsilon[\sum_i t_i f_i]|_{t=0}$, with pairwise distinct, separated, real arguments (`ASSEMBLY_N1.md` §4). So the consumed real-source form of H-098 (E14-c) is R-045 (S) at the real point. Verifier: `SOURCES_B21_E14C_VERIFY.md`; claims (1), (3), (4) confirmed; (2) confirmed in substance and corrected in wording: the $n!$ comes from the polymer-size sum of L-111 (Lemma C) Step 4 and $(n!)^\eta$ from Step 8, and "verified inputs" overstates, because S1 and R-042 are open and K-014 is a choice. Two more stale C^5 strings were found; neither is load-bearing. The clause closes as a reduction to R-045 (S). It carries S1, R-043 (FX) with R-042, K-014, the coarsest-scale H-191 (FX2) radius, and N2/N3 of Q-060.

The page readings R-047, R-048, R-049, R-051 [VERIFIED READINGS]**.** `SOURCES_BOOKKEEPING_VERIFY.md` checks the four readings of `SOURCES_BOOKKEEPING.md` on page images and confirms all four: R-047 (CMP 98 pp. 36, 39–40; C_1' is unprinted and not derivable from the printed lines), R-048 (CMP 119 p. 255; β_0 arbitrarily small, fixed before γ), R-049 ($R_{K+1} \geq L$ for $\gamma < e^{-1/2}$; "C_c free of g" is supported by the printed lines; the $O(1)$ constants are unprinted), and R-051 (every clause on CMP 109 pp. 251, 259, 291–292 and CMP 119 pp. 256, 258, 261). The S1 fix pass of the ledger had downgraded these readings because no verifier existed. This file reverses the downgrade, and the round-7 binding sets stand. The "10δ" of `SOURCES_BOOKKEEPING.md` §4 is superseded by the bound $\leq 13\delta$ of R-020. The verifier also settles a rate question: the printed rate is $e^{-\frac{1}{2}\delta_0 d}$ (CMP 99 (3.108)); `WC_FULL.md` uses a fixed rate $\delta_A < \frac{1}{2}\delta_0$ and says so; no consumer needs the printed rate.

The condition H-057 (C) (ledgers R-001, R-002, R-003, R-008, R-009, R-016). Notation (`SOURCES_C_CONDITION.md` §1). $\mathfrak{B} = \bigcup_{j=0}^{k} \Lambda_j$ is the multiscale coarse lattice, $\mathcal{H} = \ell^2(\mathfrak{B}, \mu)$ with $\mu(y) = \ell_y^d$ and $\ell_y = L^j \eta$ for $y \in \Lambda_j$, and S is multiplication by ℓ_y^2. $Q = Q(U)$ is the multiscale averaging, $G = (\Delta_\pi + DRD^* + Q^* a Q)^{-1}$ with $a = 1$ is the fluctuation propagator of CMP 99 Thm 3.3, and $G_1 = (G^{-1} - \Delta_\pi^{(2)})^{-1}$ (CMP 99 (3.128), (3.135), (3.138)). Put $\hat{N} = S^{-1/2} Q G Q^* S^{-1/2}$ and $\hat{N}_1 = S^{-1/2} Q G_1 Q^* S^{-1/2}$.

Proposition 5.23 (H-057 (C), `SOURCES_C_CONDITION.md` §8 [VERIFIED REDUCTION])**.** *There are $c_- > 0$ and $c_{-,1} > 0$ such that*

$$\langle B, \hat{N} B \rangle_{\mathcal{H}} \geq c_- \|B\|_{\mathcal{H}}^2, \qquad \langle B, \hat{N}_1 B \rangle_{\mathcal{H}} \geq c_{-,1} \|B\|_{\mathcal{H}}^2 \qquad \text{for every } B \text{ on } \mathfrak{B},$$

uniformly in k, η, the sequence $\{\Omega_j\}$, the volume, and every real U with (3.35)–(3.36), $M \geq M_1$, $M\alpha_0 \leq a_0$. Equivalently, $\min\{\langle A, G^{-1} A\rangle : QA = B\} \leq c_-^{-1} \sum_j \sum_{y \in \Lambda_j} \ell_y^{d-2} |B(y)|^2$ for every B (and the same for G_1). The proof gives $c_- = 1/(4K^D)$ and $c_{-,1} = c_-/(1 + \delta_1)$, under $M\alpha_0 \leq a_0(d, L)$, $RM \geq \max(4L^2 D_0(d, L), c(d, L))$, $\theta = \log L/(RM) \leq \delta_0/2$, $\theta_S \leq \delta_H/8$, $M \geq c_T(d, L)/22$, and K-032 (Vol) $\ell > 22M$.

H-057 (C) is an upper bound on a constrained energy, the γ_1 direction. It is not the lower bound γ_0 of R-004: R-004 bounds a different operator, the constrained precision $C^* \Delta_k C$ on unit fields.

Lemma 5.24 (L-039 (Lemma R), `SOURCES_C_CONDITION.md` §3.2 [PROVED+VERIFIED])**.** *Put $\varepsilon = M\alpha_0$ and $g(A) = \langle A, G^{-1} A\rangle$, and assume R1(G): $\|\ell^{-1} A\|^2 \leq K_G g(A)$ for all A. If $2B_0^2 c_1^2 \kappa' \varepsilon \leq \frac{1}{2}$, then $|\langle A, \Delta_\pi^{(2)} A\rangle| \leq \delta_1 g(A)$ for every fine A, with $\delta_1 = 4c_2 \varepsilon (K_G + B_0^2 c_1^2)$. Hence $(1 - \delta_1) G^{-1} \leq G_1^{-1} \leq (1 + \delta_1) G^{-1}$; for $\delta_1 < 1$, $G_1 > 0$ and $c_{-,1} = c_-/(1 + \delta_1)$.*

L-039 (Lemma R) removes the [I] Schur step for G_1 (`SOURCES_A7_CDC_VERIFY.md` (b2)). Its new input R1(G) rests on CMP 99 Thm 3.3 (3.46), Thm 3.11, and CMP 96 Lemma 2.1 [R]. As a side result, the input R1 of R-004 for G_1 reduces to R1(G). The periodic-lattice seam. The dressed extension of `SOURCES_QE_INV.md` transports along axial paths $y_- \to b$. On the periodic lattice every single-valued path rule has a seam. For adjacent bonds on opposite sides of the seam, the thin loop of `SOURCES_QE_INV.md` 5.3 winds once around the torus and carries a Polyakov loop, which the regularity (3.35) does not bound (example: constant abelian links in one direction; every plaquette is 1, and the holonomy around the torus is arbitrary). The energy defect at the seam is about $4C_H^2 N_T^{d-1} e^{-\delta_H N_T} L^k$, so it is not uniform in k for any fixed torus side, and the hypothesis "side $\geq c \cdot RM$" does not repair it. So the reading [I-c] of `SOURCES_QE_INV.md` 5.3 is false on the periodic lattice, and `SOURCES_S2_FORM.md` 4.3 inherits the defect; their verifiers did not check the energy at the seam. The repair is L-284 (Lemma T) of `SOURCES_C_CONDITION.md`: the dressed kernel is cut off by $\chi(d(b,y)/D)$. If $RM \geq 4L^2 D$ and the torus side in coarsest-block units exceeds $2L(D + 3(d+1)) + 2$, every support embeds in a box, no path crosses a seam, and the cutoff defect is controlled through $Q(1)E = I$. The side condition follows from K-032 (Vol) $\ell > 22M$ with $M \geq c_T(d, L)/22$. So the separate torus-side hypothesis is discharged by K-032 (Vol), and the two unprinted conventions ($U(c)$ the product over the straight segment; translation-covariant axial Γ) change only constants. Inputs (`SOURCES_C_CONDITION_VERIFY.md` §8): H-136 (KE′) [V, by citation], with C_H, δ_H uniform in the torus side; CMP 99 Thms 3.1 (at $a = 1$), 3.3 (3.46), 3.11, and the constant C_{137} of (3.137) [R] (ledger R-021); CMP 96 Lemma 2.1; the verified project steps H-123 (H_Q), L-252 (Theorem QE), and S2; L-039 (Lemma R) and L-284 (Lemma T); readings [I-a1], [I-geom], the path metric d, and the symmetry reading [I-conv] of $\Delta^{(2)}$; the conventions K-044 (Cv1), K-044 (Cv2) (ledger K-044). Free case [N]: $c_-^{\text{free}} = 1/(1+K_0)$ (Section 6.9). Verifier: `SOURCES_C_CONDITION_VERIFY.md`, all five claims confirmed. H-057 (C) is proved relative to the inputs above, for G and for G_1. L-039 (Lemma R) and L-284 (Lemma T) are verified. The seam defect is confirmed by explicit numerics (Section 6.10), and errata to the two verified files are recorded in `SOURCES_QE_INV_VERIFY.md` and `SOURCES_S2_FORM_VERIFY.md`. No torus-side hypothesis remains beyond K-032 (Vol) with $M \geq c_T(d, L)/22$ (ledger K-045). Minor omissions, none of which changes a conclusion: [I-conv] and the two θ conditions were missing from the lists (they are in the lists above); the claim that no other choice of $U(c)$ is possible holds for contour choices only; the seam count N_T^{d-1} is an upper count, and the leading term is $c(\delta_H) N e^{-\delta_H N_T}$. So H-057 (C) is no longer an open sub-item. Its residue is the printed CMP 99 material of row R-021: Thm 3.3 (3.46) for G and Thm 3.1 at $a = 1$ rest on walk sketches, and the torus-side uniformity of H-136 (KE′) is a reading of a printed corollary.

The lattice log coefficient (ledger R-027). The reduction of round 7 (Section 5.11) is now verified. `SOURCES_T2_F_STEPB_VERIFY.md` grades R-027 [VERIFIED REDUCTION], with corrections: R-027 (F) $\Leftarrow$ Lemma 2 (massive log coefficient equals the continuum one, given Step C) + H-137 (L-LL) + Lemma 1 (a gauge-covariant kernel bound from magnetic Weyl bounds, cited as standard and not checked) + the gauge-fixing part of Step C [I]; the ghost part of Step C is proved. The verifier graded H-137 (L-LL) unsupported as a theorem.

Theorem 5.25 (H-137 (L-LL), `SOURCES_T2_LLL.md` Theorem 5.1 [PROVED+VERIFIED])**.** *Let $W(\varphi; m^2)$ be the per-cell charged bracket $-\frac{1}{2}\log|\det|(A_\varphi + m^2) + \log|\det|(\mathcal{G}_\varphi + m^2)$ minus its value at $\varphi = 0$, for the constant abelian background of plaquette angle φ, and $H_\varphi(\mu) = \varphi^{-2}[W(\varphi; 0) - W(\varphi; \mu|\varphi|)]$. For every $\mu > 0$, as $\varphi \to 0$ through real values,*

$$H_\varphi(\mu) \to h(\mu) = -4\big[I_{\text{heat}}(\mu/2) + g(1, \mu/2)/(8\pi^2)\big].$$

Here $h(3) = 0.116322$, $h(5) = 0.169938$, *and* $h(\mu) = \frac{11}{12\pi^2}\ln\mu + 0.023444 + o(1)$ *as* $\mu \to \infty$. *The coefficient* $11/(12\pi^2) = n_2c_2$ *with* $n_2 = \frac{1}{2}$ *is the one that Step B of* `SOURCES_T2_F_STEPB.md` *§2.6 needs.*

Method. Proper time instead of band-by-band semiclassics. In the fine bracket the ghost cancels the (b_2, b_3) block exactly, so W reduces to the 2D operator A' integrated over $k_\perp$. The negative Nielsen–Olesen modes split off exactly; they give a term of order φ^2 with no $\log\varphi$. The positive part is controlled by heat traces: L-272 (the second moment of the lattice Lévy area is $\le Ct$); L-301 ($|\tau e^{-tA'_\alpha} - \tau e^{-tA'_0}| \le C\alpha^2 t$ for $t \le 1/\alpha$, by a Dyson expansion, S1, and charge-conjugation evenness); L-275 (the scaled heat trace converges to the Mehler kernel, through a Feynman–Kac representation and the invariance principle for random-walk bridges); L-304 (the scaled integrated density of states converges to the spin-split Landau measure, through Lemma 4.1 of `SOURCES_T2_GAPRIME2.md`). Inputs: textbook facts (the invariance principle, the continuity theorem for Laplace transforms, counting monotonicity in the magnetic algebra, Bessel asymptotics) and verified project lemmas (Lemmas 1.1 and 4.1 of `SOURCES_T2_GAPRIME2.md`; Lemmas 5.1 and 5.2 of `SOURCES_T2_S123.md`). No Harper band-width theorem is used, and the gap theorem (Theorem 5.12) is not needed. No rate is proved. Numerics: Section 6.9. Verifier: `SOURCES_T2_LLL_VERIFY.md`, all five claims confirmed. An independent quadrature with a closed form for the Nielsen–Olesen constant gives $h(3) = 0.1163219$ and $h(5) = 0.1699377$, and the lattice values are extended to $\varphi = 0.00125$ (Section 6.10). Grade: H-137 (L-LL) proved relative to the invariance principle for random-walk bridges (to be cited by name), Laplace continuity, counting monotonicity, and the identification step of Lemma 5.1 of `SOURCES_T2_S123.md` [I]. No rate is proved, and `SOURCES_T2_F_STEPB.md` §2.6 needs none. Effect: R-027 is a verified reduction to Lemma 1 H-148 (MPT) and the gauge-fixing part of Step C, plus X-005. Step 10 gives a proof of R-027 (F) that needs neither (Theorem 5.26, pending verification). The numerical tension of the single-step β_1 (Section 6.9) is not affected.

5.13 Step 10: round 9, a proof of R-027 (F), H-168 (S′), the s-rule, the p. 254 transfer, and CMP 99 Theorems 3.7 and 3.9

Round 9 adds the results below. Each result has an independent verifier file unless its tag says that the verifier is running. Nothing in this step makes Theorem A or Theorem B unconditional.

A proof of R-027 (F) (ledger R-027).

Theorem 5.26 (R-027 (F) by proper time, `SOURCES_T2_MPT_STEPC.md` §4–§6 [PROVED, UNVERIFIED; VERIFIER RUNNING]). *Let* $G(\varphi) = F_W(\varphi)/\varphi^2$, *where* $F_W(\varphi)$ *is the per-cell lattice bracket at the constant abelian background of plaquette angle* φ. *For every fixed* $\lambda > 0$, $G(\lambda\varphi) - G(\varphi) \to n_2c_2\ln(1/\lambda)$ *as* $\varphi \to 0$, *with* $n_2c_2 = 11/(12\pi^2)$. *So the form (3.1′) that* `SOURCES_T2_F_STEPB.md` *needs holds, and with it R-027 (F) at constant field.*

Method. L-060 (Theorem A) of the file writes $W(\varphi; m^2)$ as a proper-time integral. It proves H-148 (MPT), Lemma 1 of `SOURCES_T2_F_STEPB.md`, for $m^2 \ge 6|\alpha|$ with $\alpha = 2|\varphi|$: $W = \alpha^2\tilde{w}_2(m^2) + R$ and $|R| \le C_A\alpha^4 m^{-4}$. The main step is a fourth-order Dyson bound, L-303: $|D_\alpha(t) - \alpha^2 c_A(t)| \le C\alpha^4 t(1+t)^2$ for $t \le 1/|\alpha|$. Charge-conjugation evenness removes the orders α and α^3. The proof uses no Weyl symbol classes and no Combes–Thomas bound. L-112 (Lemma C) gives $c_A(t)/t \to 11/(12\pi)$ from L-304 of `SOURCES_T2_LLL.md` and L-303. A Frullani-type argument with H-137 (L-LL) then gives the theorem. This route uses neither Lemma 2 of `SOURCES_T2_F_STEPB.md` nor Step C. The gauge-fixing part of Step C is also proved (§6): the midpoint-gauge symbols are exact in α, and all corrections are even in K. Numerics (20 s): the finite-difference Hessian matches

the predicted kernel to 10^{-15} at $\alpha = 0.3,\ 1,\ 2$; $c_A(t)/t \to 0.29178 = 11/(12\pi)$; the fourth-order coefficient divided by t^3 tends to 0.0561, against the continuum value 0.05615. Inputs: textbook facts (the invariance principle for random-walk bridges, Laplace continuity, counting monotonicity, Bessel bounds) and verified project lemmas (Lemmas 1.1 and 4.1 of `SOURCES_T2_GAPRIME2.md`, Lemmas 5.1 and 5.2 of `SOURCES_T2_S123.md`, and the lemmas of Theorem 5.25). Readings [I]: the identification step of Lemma 5.1 of `SOURCES_T2_S123.md`, and the midpoint convention of U3 Lemma 3.4. `SOURCES_T2_F_STEPB.md` §1 exchanges λ and $1/\lambda$; this is a notation slip with no change of content. Status. The verifier (`SOURCES_T2_MPT_STEPC_VERIFY.md`) is running. Until it reports, ledger R-027 stays a verified reduction. The numerical tension of the single-step β_1 (Section 6.9) is not affected, because it concerns a different family.

R-045 (S) at every scale (ledgers R-045, R-043).

Proposition 5.27 (H-168 (S′), `SOURCES_B21_S_PRIME.md` [PROVED+VERIFIED, RELATIVE TO LISTED INPUTS]). *For $\sigma \geq 16\varepsilon$ and minimal support diameter $D \leq \sigma$, with no upper limit on σ,*

$$|\kappa_n| \leq \big(C_{S'} \max(1, \sigma^{-a})\big)^n (n!)^{1+\eta} \prod_i N(f_i), \qquad C_{S'} = 16 C_A C_1 C_B C_2 C_\eta,$$

and $C_{S'}$ does not depend on ℓ. At $\sigma = 16\varepsilon$, H-168 (S′) is R-045 (S).

Method. The ℓ-dependence of R-045 (S) sits in three inequalities of `SOURCES_D4D5.md` §5.1. Two of them fail for $\sigma \gg 2M$, and their left sides are $O(1)$. L-127 (Lemma C′) redoes Steps 6–7 of L-111 (Lemma C) with $\log_L^+$ and $\Lambda_1(\min(\sigma, 2M))$. Inputs: S1(b), R-043 (FX) with ℓ-free constants, and the ℓ-free trajectory [R]. Verifier: `SOURCES_B21_S_PRIME_VERIFY.md`; claims (2)–(4) confirmed; claim (1) corrected: the real defect is that L-111 (Lemma C) itself is false for $\sigma \geq 2ML$ and undefined for $\sigma \geq \sigma_0$. L-127 (Lemma C′) repairs it, so the R-045 (S) clause of R-045 now rests on L-127 (Lemma C′). The bridge R-045 (B-N1) uses the whole ladder σ_m, with top rung $d_B/4$ in N1-B, so H-168 (S′) is used inside N1-B. The text fixes in the consumer files are applied. `SOURCES_FX_D.md` shows that the item H-105 (FX-D) of the project notes is §6.3(b) of the R-043 (FX) file, that is Q-060, which `SOURCES_FX_O23.md` closed by another route. The source radius is $c \cdot r$ at every scale relative to R-042, N2, S1(c), Q-061, and K-014. The verifier (`SOURCES_FX_D_VERIFY.md`) finds no new open item beyond R-042.

The s-rule convention (ledger K-046). The s-versions of the propagators in the cluster expansions were a reading [I]. `SOURCES_SRULE_CONVENTION.md` replaces the reading by a convention: the s-versions are defined per distinct σ_0-cube met on the project's expansions (the walks of `WC_FULL.md` 5.3 and 5.5, the trees of $\mathcal{W}_N$, of Theorem 5.1, and of WG-G1, and the composites of R-016). The file finds twelve consumers. Each uses only bounds: normalization at $s \equiv 1$, (L1)–(L2) with the band, per-walk bounds, analyticity on $|s| \leq e^{\kappa_1}$, the anchored difference, and multi-affinity. No consumer needs the form (3.185) of CMP 99. Verifier: `SOURCES_SRULE_CONVENTION_VERIFY.md`, [ADMISSIBLE, VERIFIED], relative to H-103 (FI) ("the same formula on T_N and $\mathbb{Z}^4$") for tree factors [R], WG-G1 [R], I-a, and K-035. Three edits: H-103 (FI) must cover the admissible walk set; the loc set is $\bigcup_i \tilde{X}_i^5$; "C^1" becomes mixed partials up to order $|\sigma_0|$, from joint analyticity. The symmetry of $C(Z_0, 0)$ comes from R-009 §6, not from the convention.

The completeness item H-035 (M1) of R-016 [CLOSED AS A LIST; VERIFIER RUNNING]. `SOURCES_M1_SWEEP.md` reads CMP 119 Sect. 3 and CMP 122 II on page images. Every use of the

fluctuation covariance falls in the classes H-281–H-291, H-292 (U-a), H-294 (U-c), Q-047, and Q-048. Q-047 is the gauge and Euclidean covariance of $C^{(k)}$: Bałaban uses only block translations and the hypercubic point group about block points, and C_f satisfies $O_r C_f(U; d_S) O_r^{-1} = C_f(rU; r \cdot d_S)$ [P] jointly with the source d_S. Q-048 is the large-field Gaussian with covariance $C^* \Delta_{\Lambda^c}^{(j)} C$; its only use (CMP 122 II p. 383) follows from H-281 and the Schur complement bound $S \geq \frac{3}{4}\gamma_{\exp}$ [P]. So H-035 (M1) is closed as a list. The sweep records two items outside H-035 (M1): CMP 119 pp. 275–276 use the walk expansions in the multiscale Ω-geometry, while `WC_FULL.md` and `SOURCES_A16_REPLACE2.md` are single-scale; and CMP 122 II uses a lower bound on the R-operation Hessian, which needs a smallness check if the source enters the R-operation Wilson term.

The transfer of CMP 98 results (ledger R-003, input R4; new rows R-022, R-023).

Proposition 5.28 (p. 254 transfer, `SOURCES_P254_TRANSFER.md` [PROVED RELATIVE TO LISTED INPUTS; VERIFIED WITH CORRECTIONS]). *The sentence of CMP 109 p. 254 that "all results of [12]" (CMP 98) stay valid for the new averages holds for Propositions 1–2 of CMP 98 and for their other uses U3–U9, relative to the CMP 98 proofs [R], R-018, and the analyticity axiom for the mean M.*

Verifier: `SOURCES_P254_TRANSFER_VERIFY.md` (page images). Claim (1) confirmed; the gauge-covariance reading of CMP 98 (64)–(108) is settled, so U3, U4, U6 are [P]. Claim (2) corrected: Bałaban's own average already carries axial chain terms, so $\tilde{Q}$ and Q_1 differ in chain layout only; a cap on c_2 was misquoted; the factor S_b is $\leq 2(1 + dL^{d+1})$. Claim (4) corrected: S_b lowers $\gamma_{\exp}(d, L)$. The verifier also finds a gap in `WC_FULL.md` §3.1(2): that step applies A3-4.3 in the exponential gauge, where the axial condition fails, so its sentence "A3-4.3 uses no condition inside the blocks" is false. The repair is the same (d, L) constant change (a smaller $\gamma_{\exp}$). The fix pass is running; until it lands, Theorem 5.3 is paper-grade up to this constant. The p. 254 sentence does not cover the CMP 99 operators G_1, H_1, $(QG_1Q^*)^{-1}$, and Δ_J; they rest on the p. 253 assertion that the results are "valid universally" (new row R-022). The p. 275 assertion that the new backgrounds have "approximately the same regularity properties" is new row R-023; R-018 does not cover it. The write-outs of R-022 and R-023 are running.

CMP 99 Theorems 3.7 and 3.9 written out (ledger R-002). Theorems 5.1 and 5.2 rest on Bałaban's printed sketches of CMP 99 Theorems 3.7, 3.9, and 3.10. Round 9 writes out the first two in full and corrects printed errors.

Theorem 5.29 (T39, `THM39_FULL.md` [PAPER-GRADE RELATIVE TO TAGGED INPUTS; VERIFIED]). *CMP 99 Theorem 3.9 holds in walk-expansion form for $\hat{N}' = S^{-1} Q' G'^2 Q'^* S^{-1}$. The expansion is built from the walk expansion of G' (Theorem 5.30) with step factor c'/M_1, $b_G = B'^2 L^{2+d/2} c_1(\frac{1}{4}\delta')$, and rate $\delta_G \geq \frac{7}{16}\delta'$. L-210 (Lemma LB): $Q'G'^2Q'^* \geq \gamma_N^2 S$ for every real U with (3.35), with $\gamma_N(d, L, a) > 0$; this bound has no further condition. The (3.99) rate is $\frac{1}{2}\varepsilon - |2 - \frac{d}{2}|\theta_L$ ($= \frac{1}{2}\varepsilon$ at $d = 4$), not the printed $\frac{1}{2}\delta_0$.*

Method. L-017 is applied to $\hat{N}'$. L-210 (Lemma LB) uses a covariant bump trial function and Cauchy–Schwarz, and needs only $|A|$ and $|\nabla A|$ on single blocks. The same argument does not give H-057 (C) for QGQ^*: the corridor average has no diagonal dominance at $p = \pi$. The domain-change estimate, which the print cites to CMP 89, becomes a walk-truncation bound. Corrections of the print: three misprints ((3.96) drops a square; p. 411 prints the reversed inequality $e^{-x} \geq 3! x^{-3}$; p. 412 prints Q' for Q'^*) and four gaps (the smallness of $M\alpha_0$ is not stated; the domain-change estimate is not written; (3.99) is not proved; the local inverses rest on Cors. 3.5–3.6 and the $U = 1$

Fourier bound). Verifier: `THM39_FULL_VERIFY.md` (page images), needs minor local fixes; fix pass T-0127 applied, so the write-out is paper-grade relative to its tagged inputs. Tagged inputs: the block-diagonality of Q'_j [V] (CMP 98 (77)–(80)); a small-field bound on averaged links [R/I], reconstructed by the fixer and not yet verified. Consequence: the common rate of `WG_FULL.md` H-209 (I3) must be $\le \frac{1}{2}\varepsilon - |2 - \frac{d}{2}|\theta_L$, and M_*, a_* must be recomputed (fix pending; no circularity).

Theorem 5.30 (T37, `THM37_FULL.md` [PROVED RELATIVE TO TAGGED INPUTS; VERIFIED, LOCAL FIXES RUNNING]). *CMP 99 Theorem 3.7 and Corollary 3.8 hold with rate $(1-\theta)\delta_0$ for every $\theta \in (0,1)$, with explicit constants $K_1 = 2(3d+1)c_h L^2 B_0$ (times $e^{c\delta_0}$) and $M_{37} = \max\{M_{\mathrm{th}}, 3L + 1, 2C_{\mathcal{D}}^{\mathrm{in}} K_1 c_1(\theta)\}$, $C_{\mathcal{D}}^{\mathrm{in}} \le 3(2L+2)^d$. G' exists by a finite-dimensional Neumann series. The theorem includes an M_1 version and a complex form on the (3.37) domain.*

Corrections of the print. H-180 (X-PU): the printed partition of unity (CMP 96 (2.36) with CMP 95 (1.118), on the cover of CMP 96 p. 229) gives $\sum h^2 = 2$ in a band near level interfaces, and the agreement step (L-073 (Lemma AG)) fails where a cube crosses an interface. This is an error in the print, not a sharpening. The repair $c_h \to L\, c_h$ (a factor L in K_1) works on flat faces; corners stay a reading [I-geo]. The repair is a constant change in Theorems 3.7, 3.9, 3.10 and in Theorems 5.1, 5.2, 5.3; it is not yet applied. Corollary 3.8 ("similar estimates for other norms") holds in three forms, and the rate cost $\log L/(RM)$ of the scale shift is not printed. Verifier: `THM37_FULL_VERIFY.md` (page images), needs fixes, all local: K_1 needs the factor $e^{c\delta_0}$; the complex form has $K_1^c = 8K_1$ and a radius with a factor $1/L^2$; one citation of R-018 points to an unsupported item. Fix pass T-0128 is running. The write-out of Theorem 3.10 (`THM310_FULL.md`) exists; its verifier is running.

Order of constants and a new condition (ledgers K-005, K-047). `SOURCES_ORDER_CANON.md` sections E–F add the constants of R-018, (C1*), the R-012 addendum, R-016, R-020, and L-174 (Lemma G3), and the topological pass stays acyclic. Ledger K-047, $\gamma^2\beta' \le 1/L$, is an unprinted side condition of CMP 119 (2.9), line 2. The Lean file ScaleChain (`eq29_line2_full`) found it. It is an upper bound on γ after L and β', of Bałaban's type, and it adds no cycle.

5.14 Step 11: the alternative ($\mathbf{E}_U$) route

`SOURCES_EU.md`: **($\mathbf{E}_U$), uniform block extension (alternative route).** (E_U) asks for a fine field A with $Q(U)A = B$ and covariant energy $\le K\|B\|^2$ for every block field B, uniformly in k; in the free case $K_0(4) = 103.25, 146.73, 185.64, 195.19$ at $N = 3, 5, 13, \infty$ ($= \kappa_4 - 1$ at $a = 1$), and for regular U the dressed extension $A = \mathcal{E}_U(Q(U)\mathcal{E}_U)^{-1}B$ gives $K \le K_0(1 + c_{2/3}(M\alpha_0)^{2/3} + c_1 M\alpha_0)$. The claim that transport and back gives $Q(U)A = B$ exactly is false, and the linear form $K \le K_0(1 + cM\alpha_0)$ is not proved. K_0 is computed in momentum space by `numerics/k0_d.py` from the CMP 98 (124)–(125) form of $Q(V_0)$, an equal-weight average over L^d straight lines with an $O(L^2\alpha_0)$ remainder, and is checked in real space at $d = 2$. For regular U the free extension kernel is dressed by parallel transport, the constraint error $\|Q(U)\mathcal{E}_U - I\|$ is bounded linearly in the plaquette bound $\alpha_F \le 20M\alpha_0$ by lattice Stokes, and IMS patching over flat frames of radius $R \propto \alpha_F^{-1/3}$ gives the power 2/3. Verifier: `SOURCES_EU_VERIFY.md`, K_0 [NUMERICAL, REPRODUCED], the bound on K [DERIVED SKETCH], and (E_U) [VERIFIED REDUCTION] to kernel-moment bounds Bałaban does not print; corrections: the $N = \infty$ tail is $\propto 1/M^3$, not $1/M$; (139) is on journal p. 39; K_0 belongs to the straight-line Q_{98}, not to the corridor Q_{95}; $K \to K_0$ is indicated, not proved; the k-step remainder constant is $2C_1'$ by CMP 98 (143). Open inputs: R-050 (k-uniform moments and decay of the free kernel E_0); the unprinted constant C_1' (R-047, a verified reading); the four [I] items (a)–(d) of `SOURCES_EU_VERIFY.md` §5 (uniform moments Γ_j, ν_j, lattice-sum factors, the boundary cutoff at

$\partial\Lambda$, the gauge-fixing term); Q_{95} versus Q_{98}; the linear form. The route is an alternative to R-001 and is not binding. R-050 is now closed, see the next paragraph.

`SOURCES_T2_EUMOM.md`**: ledger R-050,** k**-uniform moments of** E_0**.** The free minimal-extension kernel E_0 of the straight-line average $Q_{98}(1)$ (composition of k one-step averages, $N = L^k$) obeys $|E_0(y,c)|, |\nabla_\nu E_0(y,c)| \le Ce^{-\delta|y-c|}$, with C, δ depending on d only. All moments ν_j, γ_j, the block sums, and Γ_j are bounded uniformly in k. The proof uses symbol identities in momentum space, lower bounds on the symbols, a weighted L^2 Combes–Thomas bound uniform in η, and pointwise regularity from the symbol; it uses no Bałaban estimate. Verifier: `SOURCES_T2_EUMOM_VERIFY.md`, [PROVED+VERIFIED], 31 claims confirmed, 3 corrected, 1 unsupported (a side remark); it closes open item (i) of `SOURCES_EU_VERIFY.md`. Open inputs: the other [I] items of the (E_U) route listed above.

`SOURCES_CDC.md`**: the norm bound** γ_1 **(ledger R-008).** The norm bound $\|C^*\Delta_k C\| \le \gamma_1$, uniform in k and regular U, is reduced: by Woodbury $(QG_1Q^*)^{-1} = N_1^{-1} + a$ exactly, so the term $-a\langle B, B\rangle$ of CMP 99 (3.156) cancels and $\gamma_1 = L^{d+1}[(1+\varepsilon)K + \varepsilon a + c_J]$ with $\varepsilon, c_J = O(M\alpha_0)c(d,L)$ and K the constant of (E_U); in the free case $K \le a(\kappa - 1)$, which is 195.19 at $d = 4$, $a = 1$. The brief's mechanism, a local difference operator for which unitary U suffices, is [REFUTED]: the leading term $(QG_1Q^*)^{-1}$ is an inverse whose naive coefficients scale as L^{2k}, so the regularity (3.35)–(3.36) is needed for the upper bound. Comparing (3.155) with (3.156) identifies $\langle B, (QG_1Q^*)^{-1}B\rangle$ as the minimum of $\langle A, G_1^{-1}A\rangle$ over $QA = B$, and the split $G_1^{-1} = M_1 + aQ^*Q$ of (3.128) gives the exact cancellation of a. The remaining minimal energy is bounded by a trial extension of B (the lemma (E_U)), the Δ'_π and J parts by $O(M\alpha_0)$ perturbation, and the free constant by the CMP 95 Fourier formula (1.99)–(1.101). Verifier: `SOURCES_CDC_VERIFY.md`, [VERIFIED REDUCTION]: the Woodbury cancellation is confirmed (the variational form needs only $G_1 > 0$ and Q onto), but (E_U) is not the last input, since CMP 99 (3.132) prints a k-uniform kernel bound on $(QG_1Q^*)^{-1}$ that the file omits; corrections: $\|C\|^2 \le L^{d+1}$, not $1 + L^{d+1}$; K_0 is an upper bound only; the free case is proved for single-scale $U = 1$ only; J is a second derivative of A; the negative-mode branch is dropped, and "regularity is needed for γ_0" is unsupported. Open inputs: CMP 99 (3.132) (ledger R-001), or in its place (E_U) (R-050, alternative route); R-018 for the J-term and $\|C(U)\|$; the multiscale correction $e^{-\delta RM}$ to the free constant [I]; the s-localized operator $C^{(k)}(Z_0, 0)$ that L-174 (Lemma G3) uses, handled by LOC (R-003, R-004). Through this file ledger R-008 was reduced to R-001 + R-018. Round 5 supersedes this reduction: Theorem 5.4 proves R-008 relative to the inputs of Theorem 5.3, without the kernel form of R-001.

6 Numerical and certified results

This section collects the numerical results that support the hypotheses of Section 3. Sections 6.1–6.3 give the one-loop increments and the per-level quantities of the Route K proof. Sections 6.4–6.6 give the values of R-025 (H2), the one-step values, and one rigorous point enclosure. Section 6.7 proves R-025 (H2) at $L = 13$ (Theorem 6.1, certified). Section 6.8 adds the certification at $L = 15, 17, 21$, the round-7 checks of the 2D gap and the neutral sector, and one open tension. Section 6.9 rewrites that tension, reports the first cross-check of the cell certificate, and gives the numerics of H-137 (L-LL) and of the free constant of H-057 (C). Section 6.11 gives the original plan for a cell certificate, now an independent cross-check at $L = 13$. SU(2), infinite volume, Balaban's normalized-trace units. Except for Sections 6.6, 6.7, and the first paragraph of Section 6.8, none of this is certified. Only odd L is inside the proofs; $L = 2, 4$ use corner blocks and are outside them.

6.1 First increment $\beta_1(0)$

Sources: `NUMERICS` §6–§7; `UMK_BETA1.md`.

L	$\beta_1(0)$	how	$b = c \log L$	$\beta_1 - b$	status
2	0.630	$N_g = 12$, $p \to 0$ fit	0.2575	+0.372	[NUMERICAL, REPRODUCED] (`KAPPA_BOUNDED_VERIFY` §4: 0.6296)
3	0.664	$N_g = 8$, $p \to 0$ fit (0.6625–0.6661)	0.4082	+0.256	[NUMERICAL, REPRODUCED]; N_g $6 \to 8$ shift ≈ 0.015
4	0.757	$N_g = 6$, single $p = 0.2$; $O((Lp)^2)$ bias ≈ 0.01	0.5150	+0.242	[NUMERICAL]
5	0.749 ($N_g = 12$, $p = 0.1$); fit 0.743	umklapp code	0.5979	+0.145	[NUMERICAL]
7	0.803	umklapp code, $N_g = 8$, $Lp = 0.5$	0.7229	+0.080	[NUMERICAL]
9	0.833 ($Lp = 0.5$); 0.846 ($Lp = 0.9$), 0.851 ($Lp = 1.0$)	umklapp code, Windows, $N_g = 8$	0.8163	$\approx +0.02$	[NUMERICAL]
11	0.850 ($Lp = 0.5$); 0.868 ($Lp = 1.0$)	umklapp code, Windows, $N_g = 8$; $N_g = 8$ grid adequacy at $L = 11$ unchecked	0.8908	≈ -0.04	[NUMERICAL]
13	$0.857293 \pm 1.5 \cdot 10^{-5}$ ($Lp = 0.5$); $0.8757528 \pm 3.1 \cdot 10^{-6}$ ($Lp = 1.0$)	arb port `numerics/umk_beta1_arb.py`, $N_g = 8$; the ball covers the rounding of the $N_g = 8$ sum only	0.9529	−0.096	[NUMERICAL]; rounding certified (round 8)

The first-step constant $\beta_1 - c \log L$ is +0.37, +0.26, +0.24, +0.15, +0.08, +0.02, −0.04 for $L = 2, 3, 4, 5, 7, 9, 11$. It decreases with L and changes sign between $L = 9$ and $L = 11$; at $L = 13$ it is −0.096 (round 8). $\beta_1(0)$ itself stays positive at every L computed. The umklapp code was validated against `ym_momentum` to 10^{-12} per K by its author; no independent verifier for the β_1 values at $L \geq 5$. Round 8: the value at $L = 13$ is certified for rounding only, and a converged-grid run at $L = 9, 11$ is running on Windows (Section 6.9). Round 7: the decrease does not level off, and the slope of β_1 in $\log L$ is well below c; this is an open tension (Section 6.8).

6.2 Later increments ($L = 2$ only)

quantity	value	status
$\beta_1 + \beta_2$ (composed, $N_g = 4/6/8$)	1.148 / 0.967 / 0.953	[NUMERICAL]; single $p = 0.2$, Lp bias label

per-level $\beta_2(0)$	≈ 0.32 ($0.953 - 0.630$). Per-level recursion 0.317–0.341 ($N = 8$–12, single $p = 0.2$). Symmetric Nyquist: 0.33334 ($N = 8$, $p_1 = 0.2$), 0.31679 ($N = 12$, $p_1 = 0.2$), 0.32283 ($N = 12$, $p_1 = 0.4$) (`BETA_FP_VERIFY` §7; `BETA_FP_PW_VERIFY` §5–§6). Level-1 values carry N-aliasing of about $\pm 5\%$.	[NUMERICAL]; per-level recursion reproduced in `BETA_FP_VERIFY` §7
$\beta_3(0)$	0.40 ($N = 8$), 0.286 ($N = 12$), 0.261 (verifier, symmetric Nyquist, $N = 8$)	unconverged; not usable
abelian sector S_k^{ab}	$(1 - L^{-4k})/4$, matched to all digits	[PROVED+VERIFIED]; [LEAN] counting part (`AbelianCount.lean`)
(U) test	$\lvert\beta_{j+1}(p) - \beta_{j+1}(0)\rvert \le 0.2p^2$ at every computed point ($L = 2$ levels 0–1, $L = 3$ level 0, $p \le 0.8$); uniformity in j not testable with two levels	[NUMERICAL, RUN BY THE VERIFIER] (`BETA_FP_PW_VERIFY` §6)

All computed increments are positive.

6.3 Quantities of the Route K proof

quantity	value	status
$d_1 = b - \beta_1$, $L = 3$	-0.256 ± 0.01 (fits $-0.254 \cdots - 0.258$)	[NUMERICAL, REPRODUCED]
d_1, κ_1, $L = 2$	-0.372, 1.114	[NUMERICAL, REPRODUCED]
d_2, d_3, $L = 2$	≈ -0.44, ≈ -0.45 (d_3 uses unconverged β_3)	[NUMERICAL]; bounded-looking, not a test
$\rho_1 = \max_K \lVert H_j(K)\rVert / L$, $L = 3$, $j = 1, 2, 3, \infty$	1.652, 1.736, 1.757, 1.760	[NUMERICAL, RUN BY THE VERIFIER, `U3_AW_VERIFY` §6]; refutes (HS); $h^2/L^4 \approx 0.34 < 1$; one-step C^1 factor $\rho_1^2/L > 1$ for $j \ge 2$
ρ_1, $L = 5$, $j = 1, 2, \infty$	1.860, 1.9565, 1.963; $\rho_1^2/L = 0.771$	[NUMERICAL]; $j = 1$ and the bound measured by the verifier (`H2_LARGE_L_VERIFY` §6)
ρ_2 (R-025 (H2)), $L = 3, 5, 7, 9$	see Section 6.4: 2.0634, 2.108, $\ge$ 2.1131, $\ge$ 2.1142 at $j = \infty$ against thresholds 3, 5, 7, 9	[NUMERICAL, REPRODUCED] at $L = 3, 5$; [NUMERICAL, RUN BY THE VERIFIER] at $L = 7, 9$
a_*, $L = 5$	0.0107 (binding mode near $K = 0$, gauge/ghost sector)	[NUMERICAL, RUN BY THE VERIFIER] (`H2_LARGE_L_VERIFY` §6)
(H″), $n = 2$, $L = 3$	$2.056 < 2.828$	[NUMERICAL, RUN BY THE VERIFIER], coarse grid, one level pair
$\sigma_j^{(1)}(0)$ J-part, $j = 0, 1, 2$ ($d = 2$ proxy)	0.750000, 0.75015, 0.75051 vs 0.75	[NUMERICAL, RUN BY THE VERIFIER] supports L-308 (Theorem W)

$\sigma_j^{(2)}(K)$ scan	not run: no $d = 4$ constant-field per-momentum code	—

6.4 Values of R-025 (H2) [NUMERICAL; R-025 (H2) AT $L = 13$ IS CERTIFIED IN SECTION 6.7]

Code: `numerics/h2_scan.py` (umklapp Schur pieces, one colour, Gram form; `H2_LARGE_L` §1). Code structure CONFIRMED (`H2_LARGE_L_VERIFY` grade 1): the one-colour reduction is exact at $U = 1$, the H and Δ_{out} blocks and the Gram identity are correct, the separable Prop. A form is exact. Sign slip in `H2_LARGE_L` §4 ($H = +[\mathfrak{B}^{-1}]_{f,c}$); harmless.

L	j	ρ_2 (scan max)	threshold L	status
3	2	2.00857	3	[NUMERICAL, REPRODUCED] (`B2_TWOSTEP_VERIFY` §4; `H2_LARGE_L_VERIFY` §1)
3	3	2.05601	3	[NUMERICAL, REPRODUCED] (spot value 2.05599 at the argmax)
3	∞	2.0634 (2.06332–2.06377 over l_{max} 2–5)	3	[NUMERICAL, REPRODUCED]; margin 31%; $q_1 \le 0.473$, $q_2 \le 0.053$
5	2, 3, 4, ∞	2.098, 2.108, 2.108, 2.108	5	[NUMERICAL, REPRODUCED]: independent dense code gives 2.098283 at $j = 2$; the verifier's own search gives 2.107921 at $j = \infty$, same argmax; margin 58%
7	∞	$\ge$ 2.1131 (face diagonal only)	7	[NUMERICAL, RUN BY THE VERIFIER] (spot 2.11309)
9	∞	$\ge$ 2.1142 (one point, $(2.6, 2.6, \pi, \pi)$)	9	[NUMERICAL, RUN BY THE VERIFIER]; the $j = 2$ point of `H2_LARGE_L` gives $\ge$ 2.1104

- Shape. The argmax lies on the face $K_3 = K_4 = \pi$, on the diagonal $K_1 = K_2$: $\approx 2.22 \to 2.40$ ($L = 3$, $j = 2 \to \infty$), ≈ 2.54 ($L = 5$). Off-face points are lower: best off-face 1.995 ($L = 3$, $j = 2$), 2.004 ($L = 3$, $j = \infty$), 2.088 ($L = 5$, $j = \infty$). The corner (π, π, π, π) stays below the face peak, but the gap shrinks: 0.0070, 0.0028, ≥ 0.0025, ≥ 0.0023 ($L = 3, 5, 7, 9$). The corner is the obvious place for a competing maximum at large L.

- Level dependence. At $L = 5$, $\rho_2(\infty) - \rho_2(j) = 9.64 \cdot 10^{-3}$, $4.0 \cdot 10^{-4}$, $1.6 \cdot 10^{-5}$ ($j = 2, 3, 4$); ratios 24.1, 25.0, against $L^2 = 25$. At $L = 3$ the ratios are 7.5, 8.7. Levels $j = 4, \dots, \infty - 1$ are not sampled at $L = 3$.

- L-trend. $j = \infty$: 2.0633, 2.1079, $\ge$ 2.1131, $\ge$ 2.1142 ($L = 3, 5, 7, 9$). This is consistent with a bounded ρ_2 near 2.11–2.14. The rate claim "at least as fast as L^{-2}" is CORRECTED (`H2_LARGE_L_VERIFY` grade 5): it used a lower bound at $L = 7$ as if it were the sup. "R-025 (H2) gets easier with L" is a measured trend for $L \le 9$, not a result for $L > 11$.

- All values are lower estimates of a sup (grid plus local search).

6.5 One-step values

The arithmetic of Section 3.7.4 shows that $\rho_1^2 < L$ implies R-025 (H2). The measured one-step values are as follows.

- $L = 3$: $\rho_1^2 = 3.10 > 3$. The one-step route fails (Section 3.7.2); the two-step map is needed.
- $L = 5$: $\rho_1 = 1.860$ ($j = 1$, Wilson fine kernel, dense, measured by the verifier), 1.9565 ($j = 2$), 1.963 ($j = \infty$). So $\rho_1^2 \le 3.853 < 5$ and $\rho_1^2/L = 0.771$ [NUMERICAL, CONFIRMED AS A MEASUREMENT, UNCERTIFIED; `H2_LARGE_L_VERIFY` GRADE 4].
- $L = 7$: $\rho_1 \ge 2.032$ (diagonal only), $\rho_1^2/L \approx 0.59$ [NUMERICAL].
- $L = 13$: measured $\sup \rho_1^2 = 4.37$; certified bound $\sup_{j,K} \rho_1^2 \le 10 < 13$ (Theorem 6.1).

So at $L \ge 5$ the one-step map suffices for L-011, as far as measured. For certification this replaces L^8 blocks per K (two-step map) by L^4 blocks per K (625× fewer at $L = 5$).

6.6 A rigorous point enclosure at $L = 3$ [INTERVAL ENCLOSURE, VERIFIED]

Source: `H2_CERTIFY.md` with the prototype `numerics/h2cert_proto.py`; verifier `H2_CERTIFY_VERIFY.md`. At $L = 3$, $j = \infty$, and the face point $K^* = (2.4035, 2.4035, \pi, \pi)$ near the argmax of Section 6.4, ball arithmetic (python-flint) gives

$$\rho_2(\infty, K^*) \in [1.9993,\ 2.1273] < 3.$$

The width comes from the umklapp tail and scales as N^{-4} in the truncation N. The verifier reran the code at $N = 12$ and obtained [1.5475, 2.5792], which contains the floating-point value 2.063361 and confirms the N^{-4} scaling; at $N = 6$ the code stops with an assertion instead of returning a wrong interval. The verdict is CONFIRMED given the premises P1–P3 of `H2_CERTIFY_VERIFY`; P1 (the loop-family coefficients of the (0.4) row are exact rationals $k/648$, k even) is proved there, and the fidelity of the loop family to Bałaban's paper stays inside P3. Further results of the same two files:

- One-step factorization: $\rho_2(j, K) \le \rho_1(j, K) \cdot \max_l \rho_1(j-1, k_l)$, exact [PROVED+VERIFIED]. So two-step work is needed only where the right side is $\ge L$, near the face diagonal.
- Oishi–Rump bound $\|\mathfrak{B}^{-1}\| \in [42.7,\ 43.6]$ at K^* (bounds CONFIRMED). The values 177 and 195 of Section 3.9 use the full Δ instead of $\frac{1}{4}\Delta$; this factor reconciles them. No $\|\mathfrak{B}^{-1}\|$ enters ρ_2.
- The umklapp tail enclosure is CONFIRMED for $M = \infty$. For finite M it is CORRECTED: its lower bound must be 0, with class-wise factors $(\pi/2)^{12}$ for two or more large components.
- A uniform Lipschitz grid is refuted as a route: $\Lambda_K \approx 2.8 \cdot 10^4$ (CONFIRMED) would need about 10^{13} cells. The replacement is a per-cell invertibility test with a mean-value enclosure.
- Cost: `H2_CERTIFY` estimates about $1.5 \cdot 10^4$ core-hours for a full certification at $L = 3$. The verifier grades this estimate UNSUPPORTED: at the tested sizes the arb overhead dominates, and the box-sum speedup is overstated.

This is one point at one level. It does not certify the supremum over the Brillouin zone or over j. Section 6.7 does this at $L = 13$.

6.7 R-025 (H2) at $L = 13$ [CERTIFIED (INTERVAL ARITHMETIC, OUTWARD ROUNDING; VERIFIED)]

Sources: `H1_ANALYTIC.md` §0–§8 (statement, lemmas, majorant); verifier `H1_ANALYTIC_VERIFY.md` V0–V9 with `numerics/h1analytic/bb_arb.py`; operator check `H1_ANALYTIC_VERIFY2.md` with `numerics/h1analytic/indep_H.py`; symmetry check `H2_CERTIFY_P5_VERIFY.md`. Here $\rho_1(j, K) = \|H_j(K)\|/L$ with $H_j(K)$ the one-step gauge-fixed minimizer map of Section 3.7.

Theorem 6.1 (R-025 (H2) at $L = 13$)**.** *For $L = 13$, every level $j \geq 1$ including $j = \infty$, and every coarse momentum $K \in \mathrm{BZ}$,*

$$\rho_1(j, K)^2 \leq 10 < 13.$$

Hence $\rho_2 \leq \sup_{j,K} \rho_1(j, K)^2 \leq 10 < L$ by the factorization $\rho_2 \leq \rho_1(j) \max_l \rho_1(j-1, k_l)$ (Lean `rho2_le`, Section 3.7.4). That is, R-025 (H2) holds at $L = 13$.

Proof sketch. Write $s = L^{-2}$, $k_l = (K + 2\pi l)/L$, $M = L^{j-1}$, d^M for the Prop. A symbol at level M, $P_\perp$ for the projection onto $\hat{K}^\perp$, and T, T_2, σ, $Z = (P_\perp T P_\perp)^+$ as in `H1_ANALYTIC` §1.

1. Exact split. $\rho_1(j, K)^2 L^2 = \lambda_{\max}\big(ZT_2Z + cc^\dagger/(\sigma|\hat{K}|^4)\big)$, $c^\dagger = \hat{K}^\dagger(1 - TZ)$: a transverse part and an orthogonal pure-gauge part.

2. Three exact lemmas. L-264 (Lemma S): the averaging symbol is $E_q(k_l) = E_l^0 - \hat{K} g_l^\dagger$, with E_l^0 diagonal (block average times straight path), $|E^0_{l,\mu\mu}|$ built from $a_{l,\rho} = \sin(K_\rho/2)/(L \sin(k_{l,\rho}/2))$, and $\sum_l \prod_\rho a_{l,\rho}^2 = 1$ (Fejér). Tower identity: $d_\mu^{ML}(K) = L^{-2} \sum_l w_{l,\mu} d_\mu^M(k_l)$, hence $P_\perp T P_\perp = P_\perp \operatorname{diag}(d^{ML}(K)) P_\perp$ and Z is the next-level kernel of Prop. A. L-166 (Lemma G): a closed form of the gauge row g_l from the path average, so the averaging operator is explicit.

3. Majorant. L-283 (Lemma T) (transverse): $\lambda_{\max}(ZT_2Z)/L^2 \leq \gamma_L F(K)$, $F(K) = 1/(|K|^2 \underline{d}(K))$, $\gamma_L = \operatorname{sinc}(\pi/2L)^{-2}$, from $T_2 \leq d_{\max} T$, $ZTZ = Z$, $d_{\max} \leq L^2 \gamma_L/|K|^2$, and $\|Z\| \leq 1/\underline{d}$. L-180 (Lemma Gauge): the gauge part is $\leq (1+\tau)^2/(|\hat{K}|^2 S_L)$ with explicit sums V, U and $\tau = (V + |\hat{K}|U)/\underline{d}$. So $\rho_1(j, K)^2 \leq B_L(K) := \gamma_L F(K) + (1+\tau)^2/(|\hat{K}|^2 S_L)$. By Cauchy–Schwarz $F \leq F_{CS}$, a closed form made of one-dimensional factors in each K_ρ.

4. Uniformity in j. B_L does not depend on j. For every $M \in \{1, L, L^2, \ldots, \infty\}$ the fine symbol satisfies $d^M \leq 1/|\tilde{q}_0^M|^2 \leq 1/|\hat{k}|^2$, because the Fejér weights sum to 1. The coarse symbol d^{ML} dominates the truncated $M = \infty$ sum termwise. No monotonicity in j and no O5 rate is used.

5. Branch-and-bound. B_L is invariant under the 384 signed permutations of K by construction, so the sup over the sorted chamber $0 \leq K_1 \leq \cdots \leq K_4 \leq \pi$ is the sup over BZ. The program `numerics/h1analytic/bb_arb.py` encloses $\gamma_L F_{CS} + \bar{G}_L$ on boxes $K = \pi u$ with exact dyadic u, evaluates every one-dimensional value in arb (python-flint), and rounds outward by one ulp after every floating-point operation, with Higham bounds on every sum. Result: $\sup_{\text{chamber}} B_{13} \leq 9.9995277$, 25,169 boxes, 8,075 gauge evaluations, 0 failed leaves, 58 s. Boxes that contain $K = 0$ are enclosed directly; $\rho_1(j, 0) = 1$.

Margins. The measured $\sup \rho_1^2$ is 4.37. The majorant is tight to 2.6% at $|K| \approx 1.5$ ($\min B_{13}/\rho_1^2 = 1.026$ over 896 cases and a local search). The certificate refines boxes only until each is ≤ 10, so its margin to 10 is $5 \cdot 10^{-5}$; its margin to the threshold 13 is 23%.

Verification. `H1_ANALYTIC_VERIFY` rederives the split, Lemmas S, G, the tower identity, L-283 (Lemma T), and L-180 (Lemma Gauge) on paper and checks them against the production

objects to $\le 5.3 \cdot 10^{-13}$ (V1–V2); it checks $B_{13} \ge \rho_1^2$ at 896 (j, K) cases (V3–V4); it grades the original float program CORRECTED (no directed rounding) and replaces it by `bb_arb.py` (V5–V8). `H1_ANALYTIC_VERIFY2` rebuilds $H_j(K)$ from the definitions (position-space (0.4) paths, sparse KKT, ghost-domain gauge) and reproduces the production ρ_1 to $5.8 \cdot 10^{-9}$ (10^{-14} with the same d symbol); the two parts of the split agree to $3 \cdot 10^{-8}$, and the L-264 (Lemma S) conventions are derived by hand and confirmed to $1.2 \cdot 10^{-13}$. `H2_CERTIFY_P5_VERIFY` proves and checks the invariance of ρ_1 under the 384 signed permutations (P5a, P5b); the analytic certificate does not need it, the box sweep does. Lean: `lean/YmLean/Tower.lean` formalizes the algebraic core of the majorant (tower identity in weighted form, Rayleigh and rank-one bounds, `majorant_core`, `lemmaT_core`); `BlockNorm.rho2_le` is the one-step $\Rightarrow$ two-step factorization.

Residue. The theorem is certified given the following, which no second source has checked.

- The reading of Bałaban's (0.4) (the path family that defines the averaging operator) and the gauge choice (`NUMERICS` §5). The certificate is a statement about H_j so defined; both implementations use the same reading.
- Prop. A at $M \ge 27$ and $M = \infty$: checked only by a second implementation of the same formula (independent recursion at $M = 3, 9$).
- L-166 (Lemma G) and the bound chain: proved on paper and rederived by the verifier, checked numerically at points only. The Fourier/umklapp derivation, Lemmas S and G, and the identification of the split are not in Lean.
- `bb_arb.py`: audited line by line and checked against the float program on 30 boxes; not formally verified. It trusts python-flint arb and IEEE correctly rounded $+, -, \times, \div, \sqrt{}$.
- Independent cross-check: the box sweep of the cell certificate (Section 6.11). Round 8: mode $M = 1$ is partial; it covers the chamber outside $|K|_\infty \le 0.32$ at $j = 1$ (Section 6.9).

6.8 Round 7: more certified L, the gap and neutral-sector checks, and an open tension

R-025 (H2) at $L = 15, 17, 21$ [CERTIFIED (INTERVAL ARITHMETIC, OUTWARD ROUNDING; VERIFIED)]. The program `numerics/h1analytic/bb_arb.py` of Theorem 6.1 was run with target $L-1$ on the initial grid $n_0 = 16$ of the sorted chamber (3,876 boxes, no refinement). The lemmas behind the majorant B_L are stated for odd $L \ge 3$, and B_L does not depend on j. A full rerun on the Mac covered all boxes, with 0 failed leaves.

L	$\sup_{j,K} \rho_1^2 \le$	target	$F_{\max}$	$G_{\max}$	rerun time
15	13.737723	14	10.020	4.005	33 s
17	12.826432	16	10.020	3.061	51 s
21	11.776985	20	10.020	1.951	113 s

So $\rho_2 \le \sup \rho_1^2 < L$ by `rho2_le`, and R-025 (H2) holds at $L = 15, 17, 21$, with the same standing and the same residue as at $L = 13$ (Section 6.7). Sources: `numerics/H1_OTHER_L.md`; verifier `H2_CERTIFY_L15_17_21_VERIFY.md`. The verifier reproduced the worst cell, $F_{\max}$, and $G_{\max}$ at each L to the printed digits, checked 9 spot cells, and found $B_L \ge \rho_1^2$ at 5 points of an independent H_j (ratio ≥ 1.63). The full sweep of 896 cases $B_L \ge \rho_1^2$ and the production cross-checks were done at $L = 13$ only; neither affects the proof logic. These values of L are not used by the targets (ledger K-003).

The 2D gap (G'_A) [NUMERICAL]**.** The proof of Theorem 5.12 covers only $|ef| \le 10^{-4}$. Numerics (`numerics/t2_gaprime2_check.py`; reproduced by the verifier, 141 values) show the gap on all of $\alpha = ef \in (0, 3.02]$: at 11 rational fluxes with denominators up to 2213, the smallest |eigenvalue| of A' is $|\alpha|(1 - 0.125|\alpha|)$, so the ratio to $|\alpha|$ is ≥ 0.95; a sparse shift-invert run at $\alpha = 10^{-4}$ gives the ratio 1.0000125. Round 7 (`SOURCES_T2_FRANGE.md`) shows that (M_j) needs the gap only on a punctured neighbourhood of $f = 0$. So these values are supporting evidence, not proof inputs.

The neutral sector of (M_j) [NUMERICAL, REPRODUCED]**.** Model: the 2D $N \times N$ torus of `SOURCES_T2_S123.md` §6, Landau gauge, flux $\varphi = 2\pi m/N$ per plaquette, the operator $4\,\mathrm{Hess} + D_r D_r^T$ restricted to the neutral (colour-3) block. Prediction: the spectrum is $\bigcup_K \{l(K), \cos\varphi\, l(K)\}$, with exactly $d = 2$ zero modes (the constant 1-forms). Author's run (`numerics/t2_mjrest_neutral.py`): $N = 12$–48, $m = 0$–2, maximal deviation from the prediction $\le 4.9 \cdot 10^{-8}$ (the finite-difference floor), two zero modes at every flux, gap $\cos\varphi \cdot 4\sin^2(\pi/N)$ to relative error $\le 8 \cdot 10^{-8}$. Verifier's run (`numerics/t2_mjrest_verify_neutral.py`): the author's JSON is reproduced bit for bit, and at new sizes

N	m	φ	zero modes	first positive	$\cos\varphi \cdot 4\sin^2(\pi/N)$
36	3	0.5236	2	0.02631374	0.02631374
20	3	0.9425	2	0.05753652	0.05753652
64	2	0.1963	2	0.00944548	0.00944550

The zero modes span the two constant 1-forms (residual $\le 5.6 \cdot 10^{-7}$), and the charged–neutral leak is 0. Limits: 2D, Wilson level ($t = 0$, no Bałaban averaging), rational fluxes. The level-j neutral symbol is not checked numerically; Theorem 5.14 uses only the bounds (N1), (N2).

Open: the single-step β_1 against $c \log L$. `SOURCES_T2_F_STEPB.md` §3.3 takes the extrapolated single-step values $\beta_1(Lp \to 0) = 0.743, 0.797, 0.827, 0.844$ at $L = 5, 7, 9, 11$. The constants $\beta_1 - c \log L$ are 0.145, 0.074, 0.010, -0.047. The local slopes $d\beta_1/d\log L$ are 0.161 ($5 \to 7$), 0.119 ($7 \to 9$), and 0.084 ($9 \to 11$), and they decrease with L. Fits give slopes 0.129 ($a + s\log L$), 0.052 ($a + s\log L + d/L^2$), and -0.18 ($a + s\log L + d(\log L)^2/L^2$). None is near $c = 0.3715$. The first-step constant falls by about 0.06 per step and does not level off at $L \le 11$. So the reading of `NUMERICS` §7, "decreasing toward the universal slope", is not supported by these data, and that claim is under verification. Open: whether 0.3715 is compared with the right quantity (a single-step β_1 at each L is a different family from the fixed-L sequence $\Lambda = L^j$ of Theorem 3.1 of `BETA_FP_H.md`), or whether the gap sits in the coarse bracket, in the dependence on the family, or in the $N_g = 8$ grid at $L \ge 9$ (unchecked). This tension is [OPEN]. It bears on ledger R-027 and possibly on R-028 through Theorem 3.1 of `BETA_FP_H.md`. Round 8 supersedes this paragraph (Section 6.9): `NUMERICS` §7 is rewritten from the converged values, and the old claim is withdrawn.

6.9 Round 8: the β_1 tension rewritten, the cell cross-check, H-137 (L-LL), and H-057 (C)

The single-step β_1 against $c \log L$ [OPEN]**.** `NUMERICS` §7 is rewritten (source: `SOURCES_T2_F_STEPB_VERIFY.md` §2, §4, §6–§8). Its earlier sentence that the first-step constant decreases "toward the universal slope" was wrong as written and is withdrawn.

- The quantity. The single-step $\beta_1(L)$ is one step from the unit lattice with block size L, for SU(2) at $p \to 0$. Each L is a different constraint family. This is not the fixed-L composed

family of Theorem 3.1 of `BETA_FP_H.md`, which concerns $\beta_\infty = \lim_k [\mathcal{C}(L^{k+1}) - \mathcal{C}(L^k)]$ at fixed odd L.

- The comparison value. $c = 11N^2/(12\pi^2) = 0.3715$ at $N = 2$ is the theoretical coefficient, not a fit. No run computed β_∞.
- The expected form is $\beta_1(L) = c \log L + C + O(L^{-2} \log L)$, so the local slope should rise toward c from below. The measured slopes fall away from c, and no fit of the form $c \log L + C + d/L^2$ or $+d/L$ matches.

L	$\beta_1(0)$	grid	local slope from the previous L
5	0.74233	$N_g = 16$ (converged)	—
7	0.79024	$N_g = 12$ (converged)	0.142
9	0.8267	$N_g = 8$	0.119
11	0.8436	$N_g = 8$	0.084
13	0.8511	$N_g = 8$; $Lp = 0.5$, 1 certified for rounding; the $Lp \to 0$ fit is not certified	0.045

$\beta_1(0) = (4\beta(0.5) - \beta(1))/3$, quadratic in Lp; the extrapolation holds to $2 \cdot 10^{-4}$ at $L = 5$. At $N_g = 8$ the $5 \to 7$ slope is 0.161, and grid convergence lowers it to 0.142. The value at $L = 13$ continues the fall. The likely cause [P] is the residual gauge, which fixes $\lambda = 0$ at one anchor per block: the anchored ghost then has an infrared scale $2.54/L$ in coarse units (0.51, 0.36, 0.28, 0.23, 0.195 at $L = 5, 7, 9, 11, 13$), and the shifted $N_g = 8$ grid has no point with $|K| < 0.785$. The $N_g = 8$ grid error was measured at -0.0044 ($L = 5$) and about -0.006 ($L = 7$), growing with L, so $\beta_1(13)$ at $N_g = 8$ is not grid-converged, and its uncertified $Lp \to 0$ value 0.8511 is likely low. The $L = 13$ point strengthens the tension at $N_g = 8$ but does not test the grid cause. The tension neither refutes nor supports Theorem 3.1 of `BETA_FP_H.md`, because the family is different.

Decisive test (running; `numerics/BETA1_BOX_RUN.md`). $L = 9$ at $N_g = 24$ and $L = 11$ at $N_g = 28$ ($N_g \geq 2.5L$), each at $Lp = 1$, 0.5, 0.25, with the extrapolation in $(Lp)^2$. The run writes one result per point (skip if done) and a partial checkpoint every 60 s. It started on the CPU of the Windows box and was restarted on its RTX 4090 GPU (jax with CUDA, float64). The GPU smoke test at $L = 5$, $N_g = 12$ agrees with the Mac to 10^{-15}. The GPU runs at 238 μs per (K, l) block against 543 μs on the loaded CPU (2.3×); the estimate for all six points is about 150 h (about 340 h on the CPU). A port of the remaining numpy part to jax is planned for a larger speed-up. No result is in yet. Reading rule: if the converged slopes still fall, the tension stands; if they rise toward c, the deficit was the grid. The slope targets are $c \log(9/7) = 0.0934$ and $c \log(11/9) = 0.0745$.

The cell certificate at $L = 13$, mode $M = 1$ [PARTIAL CROSS-CHECK]. Source: `numerics/H1CERT2_M1_RESULT.md` (the box run `numerics/h1cert_drive2.py`, level $j = 1$, sorted chamber, initial half-width $h = 0.08$, bisection depth at most 2). Cells: 8,855 top cells, 14,423 cell records; 13,872 ok, 348 split, 200 FAILED, 3 near0. The largest ok bound is $\rho_1 \leq 3.59685 < \sqrt{13} = 3.60555$, so $\rho_1(1, K)^2 \leq 12.937$ on every ok leaf. The margin is 0.0087 in ρ_1 (0.24%); it reflects the loose depth-0 enclosure, not the true ρ_1. The 200 FAILED cells are non-certifications, not violations. All have depth 2, half-width 0.02, no exception, and a finite bound between 3.621 and 9.936. All descend from two top cells next to $K = 0$ ($|K|_\infty \leq 0.32$), where the Taylor and Oishi–Rump enclosures grow as the cell width approaches $|K|$. There the true value is small: the analytic majorant gives $\rho_1^2 \approx 2$ near $K = 0$, and $\rho_1(j, 0) = 1$. The two certificates have their difficulty at opposite ends

of the chamber: the analytic one is tightest near the face diagonal, and the cell one near $K = 0$. Result: at $j = 1$, the cell certificate covers the sorted chamber outside $|K|_\infty \le 0.32$, independently of the analytic majorant, under the premises P1′, P2, P5b. Theorem 6.1 does not depend on it. A completion run (depth 3 on the 200 cells, about 27 core-hours) is not launched. Modes $M = 13$ and M_0:169 of the same launcher are not reported here.

H-137 (L-LL) [NUMERICAL]**.** Source: `SOURCES_T2_LLL.md` §6 (`numerics/t2_lll_window.py`, 13 s on the Mac). The operator A'_α comes from the builder of `numerics/t2_gaprime2_check.py`, the code of the (G'_A) numerics of Section 6.8, at flux $\alpha = 2\pi/q$ and $\varphi = \pi/q$. $H_\varphi(\mu)$ is computed directly from its definition, not from the proof.

φ	0.03977	0.02001	0.01001	0.00500	limit h (Theorem 5.25)
$H_\varphi(3)$	0.124869	0.121263	0.119109	0.117872	0.116322
$H_\varphi(5)$	0.183646	0.177929	0.174470	0.172467	0.169938

Checks: the Mehler identity of L-275 to relative error $9 \cdot 10^{-16}$; the evenness $A'_{-\alpha} = CA'_\alpha C$ exact; the Nielsen–Olesen count $\tau(P_-)/(\alpha/2\pi) = 1.00000$ at every q; the k-grid stable to all printed digits. The error behaves as $H - h \approx \varphi(0.07 + 0.046\ln(1/\varphi))$ at $\mu = 3$ and $\varphi(0.10 + 0.077\ln(1/\varphi))$ at $\mu = 5$, so the observed rate is $O(\varphi\ln(1/\varphi))$; no rate is proved. In the continuum, $h(\mu) - \frac{11}{12\pi^2}\ln\mu \to 0.023444$. The proof of H-137 (L-LL) does not use the gap theorem, so the (G'_A) values of Section 6.8 stay supporting evidence for (M_j) only.

The free constant of H-057 (C) [NUMERICAL]**.** Source: `SOURCES_C_CONDITION.md` §6 (`numerics/t2_ccond_free.py`). For $U = 1$ and a single layer, $c_-^{\text{free}} = 1/(1 + K_0)$ equals $5.10 \cdot 10^{-3}$ ($d = 4$), $7.82 \cdot 10^{-3}$ ($d = 3$), and $1.37 \cdot 10^{-2}$ ($d = 2$), independent of L as $N \to \infty$. The script reproduces $K_0(4, N = 3) = 103.248$, with a real-space check 48.198346 against 48.198347. This is part of the proof of H-057 (C) (Proposition 5.23), now a verified reduction (Section 6.10).

6.10 Round 9: the seam test of H-057 (C), H-137 (L-LL) at smaller φ, and the β_1 run

The free constant and the seam test of H-057 (C) [NUMERICAL; VERIFIED]**.** The verifier of H-057 (C) reproduces the free constant $c_-^{\text{free}} = 1/(1 + K_0) = 5.10 \cdot 10^{-3}$ at $d = 4$ (Section 6.9). Its seam test (`SOURCES_C_CONDITION_VERIFY.md` §7, `numerics/c_verify_seam.py`, 2 s) uses $d = 2$, a torus of $N_T = 6$ unit blocks with N fine steps per block, and flat SU(2) links $U_1 = \exp(i\theta\sigma_3/N)$ in direction 1. Every plaquette is 1, and the Polyakov angle is θ. The field is the dressed extension with model kernel $E = e^{-d_1}$ and the axial path rule in the fundamental domain.

N	$\theta = 1$: total energy	seam row	cut-off $\chi(d/2.5)$: total	seam row
3	2.075	0.032	2.500	0
27	2.187	0.196	2.483	0
81	2.565	0.574	2.485	0
243	3.699	1.709	2.486	0

At $\theta = 1$ the seam energy grows by a factor 2.98 per factor 3 in N, so it is linear in $N = L^k$. The predicted value at $N = 243$ is $N|\mathrm{Ad}P - 1|^2 e^{-\delta N_T}\int e^{-2|t|}dt = 1.70$; the measured value is 1.709. At $\theta = 0$ the seam row tends to 0 like $1/N$. With the cut-off of L-283 (Lemma T) inside the fundamental domain ($D < N_T/2$) the energy is uniform in N. A cut-off with $D = 4 > N_T/2$ still meets the seam, and its energy still grows, so the threshold (T2) is needed.

H-137 (L-LL) at smaller φ [NUMERICAL; VERIFIED]. The verifier reruns `numerics/t2_lll_window.py` and reproduces every value of Section 6.9. Its new script `numerics/t2_lll_verify_window.py` (41 s) extends the lattice values to $\varphi = 0.00125$; k-grids $n_k = 1$ and $n_k = 3$ agree to 6 digits at $q = 628$.

φ	0.03977	0.02001	0.01001	0.00500	0.002501	0.001251
$(H_\varphi(3) - h(3))/\varphi$	0.215	0.247	0.278	0.310	0.341	0.372
$(H_\varphi(5) - h(5))/\varphi$	0.345	0.399	0.453	0.506	0.558	0.609

$H_\varphi(3) = 0.117174$, 0.116787 and $H_\varphi(5) = 0.171333$, 0.170699 at $\varphi = 0.0025$, 0.00125. The step per halving is constant, 0.031 at $\mu = 3$ and 0.054 to 0.051 at $\mu = 5$, so $H - h \approx \varphi(a + b\ln(1/\varphi))$ holds down to $\varphi = 0.00125$, with $b \approx 0.045$ ($\mu = 3$) and $b \approx 0.075$ ($\mu = 5$). An independent continuum quadrature gives $h(3) = 0.1163219$, $h(5) = 0.1699377$, and $h(\mu) - \frac{11}{12\pi^2}\ln\mu = 0.0142853$, 0.0204567, 0.0234371, 0.0234442 at $\mu = 3$, 5, 10^2, 10^4. A no-log test of the Nielsen–Olesen term shows that its limit is finite, with a correction of order $\alpha\ln(1/\alpha)$; this explains the observed rate $\varphi\ln(1/\varphi)$.

The converged-grid β_1 run [RUNNING; RESULTS PENDING]. The numpy part of the run of Section 6.9 is ported to jax (a full jit, float64). The port agrees with the old path to $4.4 \cdot 10^{-16}$ on the GPU and $7.8 \cdot 10^{-16}$ on the Mac. It needs 4–7 μs per (K, l) block on the RTX 4090, against 238 μs before (about 55×). The estimate for all six points is 3–5 h. The run was restarted on the port and resumes from its per-point checkpoints. No result is in yet; the reading rule of Section 6.9 applies.

6.11 Certification plan and j_0 [DESIGNED; SUPERSEDED AT $L = 13$ BY SECTION 6.7]

At $L = 13$ the analytic route of Section 6.7 replaces this plan: it needs no tail level j_0, no $\|\mathfrak{B}^{-1}\|$ bound, and no Lipschitz constant. The cell certificate below is run at $L = 13$ only as an independent cross-check (box sweep, in progress). `B2_TWOSTEP_VERIFY` grade 6 (CORRECTED in size): certifying R-025 (H2) is a computer-assisted proof project, not a check. `H2_LARGE_L` §4 gives the design; `H2_LARGE_L_VERIFY` grades 6–8 check it. Three parts.

(a) Tail $j \geq j_0$. From the resolvent identity, $\rho_2(j) \leq R_\infty + (\rho_1^2/a_*)(\delta_{j-2} + \delta_{j-1})$, where $R_\infty = \rho_2$ at $j = \infty$, $a_* = \inf_K a_j(K)$ with $\|[\mathfrak{B}^{-1}]_{f,f}\| = 1/a_j(K)$ exactly, and $\delta_m = \sup_K \|\Delta_{L^m} - \Delta_\infty\|$ (O5 decay) [derivation CONFIRMED]. Inputs CORRECTED by the verifier: the 2-norm sup of δ_m is at the corner, $\delta = 179$, 92.0, 18.7, 2.39, 0.317 ($M = 1, 3, 9, 27, 81$), 45–90× the assumed value at small m; $a_*(L = 5) = 0.0107$, not 0.05. With these inputs, $j_0 \approx 8$ ($L = 3$) and ≈ 6 ($L = 5$). With the O6 coercivity constant, $j_0 = 25$, which cannot be used in practice (`B2_TWOSTEP_VERIFY` §5 estimated $j \gtrsim 45$ with $\|\mathfrak{B}^{-1}\| \leq b(L) \sim 10^{19}$).

(b) Levels $2 \leq j < j_0$ and $j = \infty$. A certified sup over the BZ of $\|H_{(2),j}(K)\|$ (at $L \geq 5$: of $\|H_j(K)\|$, Section 3.7.4). Plan: interval arithmetic with local mean-value enclosures on boxes in the sorted chamber (1/384 of the BZ), 4×4 and 5×5 verified inverses, one 4×4 λ_{max} per K; about $2.5 \cdot 10^3$ boxes at half-width 0.05 at $L = 3$; the analytic form of Δ near $K = 0$, where $\rho_2(0) = 1$. At $j = \infty$ also a certified tail bound for the Prop. A sum (l_{max} truncation; observed effect $4 \cdot 10^{-4}$). There is no explicit global Lipschitz constant today: the A0/O5 strip width is existential, and the constant that can be made explicit is $\sim 10^{40}$. Monotonicity in j is not proved and is not used.

(c) Inputs to certify first: $\|\mathfrak{B}_j(K)^{-1}\| \leq B_{num}$ for all j, K (the O6 constant $b(L)$ is unusable), the O5 constant C_0 explicitly (or certified δ_m), certified a_*, ρ_1, R_∞.

The run plan (`H2_LARGE_L_VERIFY` grade 8) uses atomic JSON writes and pre-copied profile caches; the profile build time at $L \geq 11$ is not budgeted. At $L = 13$ the cell certificate runs as the box sweep (`numerics/h1cert_cell2.py`, sorted chamber, valid by P5), an independent cross-check of Theorem 6.1; round 8 reports mode $M = 1$ as partial (Section 6.9).

7 What remains

This section lists every open item of the paper in one place. The items are grouped by the kind of work they need: certification, fixed-level and structural inputs of L-001 (Theorem 2w), items of the sources block, and verification. The item labels H, 1–13, and B1–B8 are those of the dependency table (Section 4.6). For L-001 (Theorem 2w) $\Rightarrow$ Theorem 1 the list covers §10 of `THEOREM2_PROOF_v4.md`. The trust ledger (Section 7.7, `TRUST_LEDGER.md`) supersedes this list as the complete record of what the two main theorems take on trust: Bałaban-level statements, external results, our open items, and conventions.

7.1 Certification

Single uniform hypothesis (both routes):

H. R-025 (H2) $\rho_2 = \sup_{j\geq 2} \sup_{K \text{ real}} \|(\oplus_l H_{j-1}(k_l))H_j(K)\|/L^2 < L$. [DONE AT $L = 13$: CERTIFIED (INTERVAL ARITHMETIC, OUTWARD ROUNDING; VERIFIED)], Theorem 6.1: $\rho_1(j, K)^2 \leq 10 < 13$ for all $j \geq 1$ (including ∞) and all K, hence $\rho_2 \leq 10 < 13$ by `rho2_le`. Route K: R-025 (H2) $\Rightarrow$ $(C1\sigma)$ $\Rightarrow$ (B2) [PROVED+VERIFIED, `B2_TWOSTEP_VERIFY` GRADE 3]. Route B: R-025 (H2) + (T1) $\Rightarrow$ (U) [PROVED, UNVERIFIED, `BETA_FP_U` THEOREM 2.4]. Cross-check in progress: the box sweep of the cell certificate at $L = 13$ (Section 6.11). Round 7: R-025 (H2) is also certified at $L = 15, 17, 21$ (full reruns, verified; Section 6.8). At $L = 3, 5, 9$ R-025 (H2) stays [NUMERICAL].

H′. Residue of Theorem 6.1 (Section 6.7): the reading of Bałaban's (0.4) and the gauge choice (single source); Prop. A at $M \geq 27$ and $M = \infty$ (second implementation of the same formula only); L-166 (Lemma G) and the bound chain (point checks only beyond the paper proof); `bb_arb.py` audited, not formally verified.

At $L = 3$ one point is enclosed: $j = \infty$, $K^* = (2.4035, 2.4035, \pi, \pi)$, $\rho_2 \in [1.9993, 2.1273]$ [INTERVAL ENCLOSURE, VERIFIED] (Section 6.6). Certified early steps $\beta_1(0), \ldots, \beta_{j_*}(0) > 0$ at odd $L > 11$ are needed only for literal (0.31) at every K (Section 3.9.2).

7.2 Fixed-level and structural inputs of L-001 (Theorem 2w)

Route K hypothesis set (all must hold):

1. (B2) $\sup_j \sup_K |\sigma_j^{(2)}| < \infty$ (or H-184 (B2w)). Reduced to H [VERIFIED REDUCTION]. Older reductions kept: $(C1\sigma)$ + (h) [PROVED+VERIFIED]; (H″) [ARGUED]. (HS) is refuted.

2. R-026 (PI) probe independence, at each fixed level [VERIFIED REDUCTION] (`SOURCES_T2_PI.md` + VERIFY), given H-207 (I1) massiveness at $U = 1$, H-208 (I2) gauge covariance, and (I3).

3. R-027 (F) lattice = continuum log coefficient at constant field [VERIFIED REDUCTION] (ledger R-027; the continuum side is cited, Savvidy [22], ledger X-005). Round 8: the reduction

to H-137 (L-LL), Lemma 1 H-148 (MPT), and the gauge-fixing part of Step C is verified (Section 5.12). Round 9: H-137 (L-LL) is [PROVED+VERIFIED] (Theorem 5.25), and a proof of R-027 (F) that needs neither Lemma 2 nor Step C is [PROVED, UNVERIFIED; VERIFIER RUNNING] (Theorem 5.26). Open numerical tension: the single-step β_1 at $L = 5$–13 does not follow $c \log L$ (Section 6.9); it concerns a different family.

4. (M_j) formal = true f^2 coefficient at each fixed level: H-244 (S1) $\mathcal{K}_t(f) \geq -C_1|f|$, H-245 (S2) density of states $\leq C_2|f|$ near 0, H-247 (S3) relative bound of the level difference, all on the lattice: [VERIFIED REDUCTION] (`SOURCES_T2_S123.md` + VERIFY, 25 confirmed, 2 corrected): H-244 (S1), H-247 (S3), H-245 (S2)$|_{t=0}$, H-246 (S2c) are proved at fixed j given the Peierls input (I3). Round 5 proves (I3) at every step, with constants independent of the step (`SOURCES_T2_I3.md` + VERIFY, 15 confirmed, 2 corrected; given the identification of the one-step objects with the linearized averaging). Round 6 proves (G'_A) for $0 < |f| \leq ((1-c)/100)^2/2$, gap $2c|f|$ (Theorem 5.12) [PROVED+VERIFIED]. So $R - 028(M_j)$ reduces to (I3), the region-1 step of Prop. 4.4 [ARGUED], and the neutral sector [I]. The proved f-range is narrow. Round 7: (M_j) is [PROVED RELATIVE TO LISTED INPUTS; VERIFIED] at fixed j (Theorem 5.14); region 1 and the neutral sector are proved, only the $o(f^2)$ form, and the proved f-range suffices (`SOURCES_T2_FRANGE.md`).

5. V1 order-g^0 completeness of CMP 109 (1.3)–(1.5): $\beta_{j+1}(0)$ is the Maxwell coefficient of the per-step Gaussian determinant [PROVED+VERIFIED AT FIXED LEVEL, CONDITIONAL ON R-030 (G2), BAŁABAN'S INDUCTIVE BOUNDS, AND THE DECLARATION THAT THE AVERAGING IS (0.4)] (`SOURCES_T2_V1.md` + VERIFY). It is history-free.

6. R-030 (G2) per-cell infinite-volume densities [VERIFIED REDUCTION] at fixed j (`SOURCES_T2_G2.md` + VERIFY; was [FORMAL]): the torus densities converge exponentially, and the p^2- and f^2-coefficients commute with $N \to \infty$, modulo the surjectivity of $\tilde{Q}$, (Gap_f), and the analyticity reading of the averaging. The j-uniform version is not proved.

Route B hypothesis set (alternative to 1–5; per `BETA_FP_U` §4; all [PROVED, UNVERIFIED] as implications, no verifier file for `BETA_FP_U`):

R-025. (T1) $\sup_j \sup_{q \text{ real}} |c^{(1)}_{T,j}(q)| < \infty$, the plane-wave form of L-071 [ARGUED; THE LANDAU-TO-SYMMETRIC GAUGE CONVERSION AND STEPS 2–5 OF R-004 FOR (2.1) ARE NOT WRITTEN].

B2′. R-025 (H2) = item H.

R-027. (DF) $\sup_k |\mathfrak{d}_F(\theta_*; L^k)| < \infty$, the gauge-fixing difference at the plane wave [HYPOTHESIS, ARGUED]. With the exact split $\Gamma^{ex} = \gamma_{rec} + \mathfrak{d}_F$ (`BETA_FP_U` Prop. 3.1) there is no Y-defect; all non-stationarity sits in $\mathfrak{d}_F$. This replaces the earlier $\mathfrak{d}_Y$, $\mathfrak{d}_{gf}$ items for this route (Section 3.6.2).

R-028. Lüscher–Weisz (5.7) [CITED].

R-029. R-007 (the label of v4; not ledger R-007): the plane-wave increment equals Balaban's β up to the expansion point, uniformly in k [ARGUED]. A uniform-in-k input, of the same kind as (DF).

R-030. H-202 (H0) = item 7′ (L-005 (Lemma D″) (i)).

R-031. $\theta_* < 1$, so that $c_2 \ln(1/\theta_*^2) \geq 0$ in C_B [TRIVIAL CONDITION ON THE CHOICE OF θ_*].

Route B also needs 6 (R-030 (G2)), item 7′, and the items 8 and 10–13 (item 9 is closed). It does not need R-026 (PI), (M_j), V1; it needs (DF) and R-007 instead.

Common to both routes:

7′. The own-slot bound $|\beta_{k+1}(h,g)-\beta_{k+1}(0)| \le C_\beta g$ on $]0,\gamma]^k \times [0,\gamma]$, for every past h, monotone or not; the lift [PROVED+VERIFIED] (`SOURCES_GGAIN_NONMONO_VERIFY`), 7′ itself conditional on H-202 (H0) or on the H-111 (G-gain) inputs. Item 7 asked for H-202 (H0), Bałaban's smoothness in g with bounds uniform in j and history (CMP 109 inductive assumption at odd $L > 11$; "uniform in j" is our reading of p. 264, not printed, `R2_VERIFY` row 12; uniformity in the history is not printed; ledger R-006). The shooting evaluates β on clamped, non-monotone pasts, and `SOURCES_T2_KUNIF_VERIFY` found that H-111 (G-gain) proves C_β only on monotone pasts. `SOURCES_GGAIN_NONMONO.md` lifts the restriction with no change of constant: the only use of monotonicity is a comparison of bound sizes in L-164, and the running maximum $G_k = \max_{i\le k} g_i$ closes the induction for every past. So 7′ follows from H-202 (H0) or from H-111 (G-gain) with this lift. The proof uses the Lipschitz form on $]0,\gamma]$ only. Round 5: the use of H-202 (H0) here is a [VERIFIED REDUCTION] (L-183 (Proposition H0-red), `SOURCES_T2_H0.md` + VERIFY) with residue N1–N9; N2 is L-174 (Lemma G3), proved relative to listed inputs (Theorem 5.9); the rest of the residue (N5 [R], N6, N8 [I], N9 definition only, N7 = R-031) is not proved, so R-006 stays a ledger row.

8. L5 history dependence. Round 7: [PROVED RELATIVE TO LISTED INPUTS; VERIFIED] by Route 2, nested intervals on true trajectories with no clamp (Theorem 5.15); (a)–(c) below are kept as the history of the item. (a) The clamp proof of Lemmas D′/D″ is valid only if $\beta_{k+1}(g_0,\dots,g_k)$ is defined and continuous on the whole cube $]0,\gamma]^{k+1}$; Balaban does not supply this. (b) Balaban's β_{k+1} exists only on histories where his induction has run; that domain must carry (c′) + K-018 (S), not only $g_i \in]0,\gamma]$. Clamped histories can violate the growth side of (c′). (c) True trajectories with all $u_i \ge \gamma^{-2}$ inherit (c′) from (a″). A nested-interval variant of L-005 (Lemma D″) on these trajectories would avoid the gap [NOT WRITTEN]; the variant is written only for L-006 (Lemma D) (`LEMMAS_DE_VERIFY` §1.5). Lean covers one-argument β only (Lemmas D′, D″, D, E).

9. K-independence of $\beta_j(g)$ [CLOSED AS A REDUCTION; VERIFIED] (`SOURCES_T2_KUNIF.md` + VERIFY). Item 9 is a volume statement, not a history identity: K enters β_j only through the torus size. CMP 119 (2.24), p. 249, and p. 279 define β_j as the whole-lattice coefficient, so the β_j of the recursion is K-free by definition. Lemmas D′/D″ run at one fixed K and never compare trajectories. The window bound needs only (K1) (history-free at zero own coupling), (K2) C_β (item 7′), and (K3) the window bound (5.1) of L-009 (Theorem K). The residual volume question (the finite-volume step equals the whole-lattice one, ledger R-015, R-032) is an input of Theorem 1, not of the shooting. Round 5 narrows it: given H-141 (Loc), the torus coefficient is within $C_j N_j^6 e^{-m_j N_j/2}$ of the whole-lattice one (Proposition 5.8); exact equality cannot be derived in general. Round 6 proves H-141 (Loc) (Theorem 5.10), so this bound no longer assumes it, and proves the boundary reading and the replaced R-032 under K-032 (Vol) (Proposition 5.11).

11. Odd L. All one-loop proofs need odd $L \ge 3$ (centered blocks: U3 Lemma 1.2, O5/Prop. A, evenness, Lemma 4.1 of LEMMA43). Balaban assumes L odd > 11 throughout CMP 109, so H-202 (H0) and $\beta_j(g)$ at $g > 0$ exist only there. L-001 (Theorem 2w) is stated for odd $L > 11$, or with H-202 (H0) at this L as a hypothesis. Numerics: $\beta_1(0)$ up to $L = 11$ (Section 6.1); R-025 (H2) at $L = 3, 5$ (scans), 7 (face diagonal), 9 (one point); ρ_1 at $L = 3, \dots, 13$. Only $L = 13$ is inside Bałaban's range; there R-025 (H2) is certified (Theorem 6.1).

12. SU(2) only. The charged/neutral split of Lemma 5.1 and (W) are SU(2) statements; $c = 11N^2/(12\pi^2)$ is general but the chain is not checked for $N > 2$.

7.3 Items of the sources block

These items stand between the trajectory of Section 3 and the bound H-116 (H-U) of Section 4.2. Tags and files are those of the dependency table.

- R-035 (B-S) The sourced version of Bałaban's Theorem 1, (E14-a, b, c), is [OPEN] as a whole.
- R-037 (H-hist) via H-111 (G-gain). The small-field step is done: pointwise Cauchy and uniform C_β on $]0, \gamma]$ [PROVED+VERIFIED]. L-174 (Lemma G3) (the χ_k surface term) holds given (Q). Round 5 proves G3 relative to listed inputs, with (Q) assembled from R-008 and R-009 in the joint (X, B) form (Theorem 5.9); its conditions are K-026 and K-027, and G_Γ is bounded uniformly in M (verified). The large-field half is a [VERIFIED REDUCTION] (`SOURCES_GGAIN_LF.md` + VERIFY): one Cauchy step on the large-field disc, with the factor $1/g$ absorbed by the exponent of p_0 (CMP 119 p. 244). The real uniform R-037 (H-hist) is therefore reduced [R] in both halves; open inputs: the margin (M_r), CMP 102 (189), and the H-145 (M-B) bookkeeping; the small-field (Q)/c_- is discharged relative to the inputs of Theorem 5.9. L-178 (Lemma G4) is off-chain (`SOURCES_G4.md`). Second derivatives: R-038 (H-hist2).
- Bałaban-level core (round 3). The propagator and covariance inputs R-001–R-005 of the ledger reduce to published theorems:
  - R-001, kernel-decay form: [VERIFIED REDUCTION] via L-125 (Lemma CT) to the printed CMP 99 Theorems 3.1, 3.3, and the G part of 3.12, with H-136 (KE′) proved by citation (CMP 95 (1.11)/(1.18), CMP 96 (2.35)), H-123 (H_Q) proved, and S2 closed for G. Round 9: the condition H-057 (C) is a verified reduction (Proposition 5.23); L-284 (Lemma T) discharges the torus-side hypothesis under K-032 (Vol), and the two unprinted conventions are row K-044. Residue: the printed CMP 99 inputs of H-057 (C) (row R-021), the reading $a = 1$ of Theorem 3.1, [I-geom], the path metric, and [I-conv]; S2 for G_1 is closed (`SOURCES_S2_G1_V7.md` + VERIFY).
  - R-002, walk-expansion form of R-001 (CMP 99 Theorem 3.15, "analyzed in the same way"): [VERIFIED REDUCTION], now written out in full. Theorem 5.1 (`A1_WALK_FULL.md`) gives the walk expansion of $(QGQ^*)^{-1}$ with (3.108)-form bounds at rate $\frac{1}{2}\varepsilon$, the Z_0-compressed variant (O3), and the complex α'-space (O4). Its input (W_G) for the final propagators is Theorem 5.2 (`WG_FULL.md`): the walk expansion of G and the tree expansion of G_1, built from Bałaban's printed proof sketches of CMP 99 Thms 3.7, 3.9, 3.10, the Neumann series (3.130), and CMP 96 Lemma 2.1. Both are paper-grade relative to their tagged inputs. What remains: the lower bound H-057 (C) (the residue of R-001), the three printed sketches [R], (W_G) at complex points [R], the condition WG-1 in the M_1 form $\alpha_0 \le c' M^{-1} M_1^{-10}$ ($c'' M^{-1} M_1^{-12}$ for G_1), admissible in Bałaban's order, and geometry and convention items [I] (cube geometry, Lipschitz partition, Q-locality, $C^{(2)}$-locality, the scale bookkeeping C0, the M_1-cover, κ_1 independent of M). Round 9: Theorem 3.9 is written out and verified (Theorem 5.29, paper-grade relative to tagged inputs); Theorem 3.7 is written out with local fixes running (Theorem 5.30); the write-out of Theorem 3.10 waits for its verifier. The printed partition of unity fails near level interfaces (H-180 (X-PU)); the repair is a factor L in a constant, not yet applied in the three write-outs.

– R-003 (R_2, R_3 of CMP 116 (2.16)): [VERIFIED REDUCTION] (`SOURCES_A2_R2.md`, `SOURCES_S2_G1_V7.md` + VERIFY): R_2 by a Neumann series around the compression with the complex-U repair V7, to the complex walk bounds (W^c) on the α'-space, now Theorem 5.3 (`WC_FULL.md`, paper-grade relative to its tagged inputs): in the exponential gauge, $A_Z \geq \frac{1}{4}\gamma_e$ at real U, the walk expansion of $C^{(k)}(Z_0)$ in the CMP 116 p. 13 form, $\operatorname{Re} A_Z \geq \frac{1}{2}\gamma_{\exp}$ on the α'-space, and the resolvent bound with per-walk prefactor $4/(\gamma_{\exp} + x)$, with no use of Thm 3.15; its inputs are Theorem 5.2(G1), H-057 (C), the inputs of R-004, CMP 109 p. 254 [R], and [I] items; the decay of $C^*\Delta_k C$ at real U (R-008), the boundary-contact reading (item (i), [I-forced]), and $\operatorname{dist}(Z_0, \Lambda_k^c) > RM$ [I]; R_3 in weaker forms sufficient for (2.21). Round 9: CMP 109 p. 254 is proved for the CMP 98 results (Proposition 5.28); the CMP 99 operators rest on the p. 253 assertion (row R-022); a gap in `WC_FULL.md` §3.1(2) costs a (d, L) constant in $\gamma_{\exp}$ (fix pass running).
– R-004 ($C^*\Delta_k C \geq \gamma_0$): full proof, verified (`SOURCES_A3_GAMMA0_FULL.md` + VERIFY), from the printed CMP 99 Theorems 3.11–3.12, H-123 (H_Q) [PROVED+VERIFIED], R-Δ2 [R], and items R3–R5: $\gamma_0 \geq m_L/(48d^2L^d)$ with $m_L = \min(2(1-\cos(\pi/L)), 1/L)$, uniform in k, volume, and regular U, for $M \geq L/2$ (this equals $(48d^2)^{-1}L^{-d-1}$ for $L \leq 9$ and is 0.7555 times it at $L = 13$); the σ-localized clause reduces to (W) + (N), without symmetry. The proof step inside CMP 96 (2.126)–(2.127) is false for $L \geq 10$ (a lost factor L); the statement of Lemma 2.4 stands with the corrected constant.
– R-005 (CMP 102 (189)): [VERIFIED REDUCTION] to R-001 + published CMP 102 statements + R-018 + one bookkeeping item.
– Round 5 (Section 5.9). R-008: proved relative to listed inputs (Theorem 5.4); C is the constraint map and $C^*\Delta_k C$ the precision; the norm bound by L-226 (Lemma N) from C2 of `WC_FULL.md`, A3-R4, I1, I3; the decay from W_{G_1}, Theorem 5.1, I2, I9; complex s at proof-sketch level. R-009: proved in corrected form (Theorem 5.5); Bałaban's symmetry claim is not proved and not needed; the majorant uses $H_0 = \operatorname{sym} C_0^{-1}$; Lemma D with $e^{-\varepsilon'' M/8}$; readings I-a–I-e. R-010: proved in the corrected form R-011 (Theorem 5.6); the raw Taylor terms do not obey (1.24); the transformed terms do, after the exact marginal cancellation; residue R1–R4; per-label exponent β_H. R-012: closed as a verified list (Proposition 5.7). R-015: narrowed (Proposition 5.8); H-141 (Loc) for $g > 0$, the $\partial\Lambda_{k+1}$ reading, and the R-term part (R-032) stay asserted. R-006: its use in L-001 (Theorem 2w) is a verified reduction, with N2 = L-174 (Lemma G3) proved (Theorem 5.9).
– Round 6 (Section 5.10). R-015: H-141 (Loc) proved exact for non-wrapping polymers relative to K-033–K-037 (Theorem 5.10); the $\partial\Lambda_{k+1}$ reading (f) proved exact relative to K-038 (Proposition 5.11); threshold $N_k > 22M$, $\varepsilon(M) = C_j(22M)^6 e^{-11 m_j M}$. R-015 is a verified reduction to conventions. R-006: the $\mathbb{Z}^d$-limit assertion now follows, so R-006 has no bare assertion left. R-032: ill-posed as stated; replaced by (a)–(c), proved relative to listed inputs and K-032 (Vol).
– Round 7 (Section 5.11). R-016 proved relative to listed inputs (Theorem 5.20; the first proof's locality step was false). R-017 proved relative to the printed text and one reading (Proposition 5.16). R-018 (Proposition 5.18), R-019 (Proposition 5.17), and R-020 (Proposition 5.19) are verified reductions. Conditions K-039–K-043.
– Round 8 (Section 5.12). The condition H-057 (C) is stated exactly in row R-001. A draft proof (Proposition 5.23, Lemma 5.24) closes the [I] Schur step for G_1, finds a seam defect on the periodic lattice in two verified files, repairs it, and replaces the torus-side hypothesis by K-032 (Vol); its verifier is running, so the ledger keeps the round-7 form. K-009 is unused.

– Round 9 (Sections 5.12 and 5.13). H-057 (C) is a verified reduction, with residue R-021; H-137 (L-LL) is proved and verified; a proof of R-027 (F) is pending verification; H-168 (S′) is proved and verified; the s-rule is convention K-046; H-035 (M1) is closed as a list (verifier running); the p. 254 transfer is proved and verified, and R-022, R-023 are new rows; CMP 99 Theorems 3.7 and 3.9 are written out with corrections of printed errors.

So the Bałaban-level core now rests on published propagator theorems, Bałaban's printed proof sketches of CMP 99 Thms 3.7, 3.9, 3.10 (through Theorems 5.1, 5.2, 5.3), the printed inputs of H-057 (C) (row R-021), the p. 253 and p. 275 assertions (rows R-022, R-023), the inputs of Theorems 5.4, 5.5, 5.6, R-018, R-Δ2, two [I] items of R-003, the geometry and convention items of the write-outs, the conventions K-044 and K-046, and, since round 6, the locality conventions K-033–K-038 and K-032 (Vol) (K-032). This is a verified reduction, not a proof.

- R-038 (H-hist2), on which the threshold-shell bound R-039 (H-thr) depends: [VERIFIED REDUCTION] to the real uniform R-037 (H-hist), no new input; with it the M3 history weights and the frozen-integer convention (L3).

- R-043 (FX). Q-059(b), Q-060, and R2 are [PROVED+VERIFIED]; Q-061 is [PROVED+VERIFIED, CONDITIONAL ON N3]; Q-062 is avoided by Bałaban's main form (1.98). N3, the HBETA strip bound $|\delta w| \le \frac{3}{2}\tau$, is reduced to R-040/R-041 (H-hol) + real R-037 (H-hist) + H-120 (H2-hist) (condition (a) settled); R-040/R-041 (H-hol) is a verified reduction to H-143 (M-A) (via (H−) and CMP 102 (189)), H-145 (M-B) (verified), and H-146 (M-C). H-146 (M-C) is a [VERIFIED REDUCTION] (`SOURCES_MC.md` + VERIFY) to the printed conditions of CMP 116 §2 (B1–B23 and four more), Lemma 2, (1.36), and p. 12 (2.5)–(2.6), with shrinks $\varepsilon_1/(4\lambda_V)$, $\gamma/(8\lambda_V)$. Its residual inputs are now settled: H-133 (I-V0) is closed [PROVED+VERIFIED] (`SOURCES_MC_IV0.md`); (I-ε) is closed [PROVED+VERIFIED] for CMP 109/116 (`SOURCES_T2_IEPS.md`) and for the large-field layer of CMP 119/122 (`SOURCES_T2_IEPS_LF.md`; residue [I]: the unprinted scaling of α_2–α_4 and q_0 versus p_0, each branch closes); H-131 (I-ord) is closed; K-025 (M1) is the convention K-025 (`SOURCES_HIGHER_M1.md` + VERIFY). No consumer needs coupling or threshold derivatives of order greater than 2. The two minor inputs of R2 ($R_{K+1} \ge L$, C_c independent of g) are verified readings (`SOURCES_BOOKKEEPING.md`). H-191 (FX2) is conditional on Q-061, so also on N3.

- (B-S.D2C, B-S.Hβ) The domain hypothesis (D), and the conditions R-036 (I) and R-037 (H-hist) of the two-loop bound.

- (B-S.D2, B-S.D3) The recovery of the decay loss $\kappa \to \kappa/2$ [VERIFIED REDUCTION] by $L/2$-rescaling, $\kappa \ge 1536$ at $d = 4$ (ledger R-046; it uses the unprinted large-field T-step rate loss, ledger R-020, now a verified reduction with loss $\le 13\delta$, so H-216 (K1′) and $\delta = 0.025$ at $L = 13$). The scale condition K-024 (U) ($\rho_1 \le c_* r_0/M^a$ at the unit scale) is a constant choice (ledger K-024), verified compatible with R-045 (B-N1) (`SOURCES_UN1.md`), and does not appear in Theorem B.

- R-045 (B-N1) The bridge from the source form to H-116 (H-U) derived given R-045 (S) (`ASSEMBLY_N1.md`, confirmed); the uniformity of R-045 (S) in support size and in ε is a [VERIFIED REDUCTION] (`SOURCES_B21_S.md` + VERIFY): every factor of $C^n(n!)^{1+\eta}$ is independent of h, given L-111 (Lemma C), S1 at all scales, R-043 (FX) in the main form, and the coarsest-scale H-191 (FX2) radius; the H-098 (E14-c) clause is reduced to R-045 (S), verified (round

8, Section 5.12). Given R-043 (FX), the step (S_{src}) ⇒ H-116 (H-U) with $s = 6$, $D > 1$ is confirmed.

- (NT) = H-101 (E14-f), for nontriviality (Theorem 4.3); at every layer for part (ii). The constant $c > 0$ must be computed in Bałaban's normalization.
- E13, full-sequence convergence: not needed for Theorem B; [OPEN].

7.4 Verification debt

10. Balaban readings: the sweep (Section 3.4.3) is grep plus page images, not line-by-line, with 4 of 19 rows checked by a second reader; "Similar inequalities hold for other constants, which will be introduced later" (CMP 119 p. 256) is closed as a verified list (Proposition 5.7, ledger R-012); "β_0 can be chosen arbitrarily small" is on CMP 119 p. 255, and β_0 is fixed before γ, as used (verified reading, `SOURCES_BOOKKEEPING.md`; former ledger R-048); the readings of `BALABAN_EXCERPTS_CHECK` item 11 (Π (4.37)/(5.1), CMP 119 (2.42), (2.12)/(2.23), the "L odd" page) are verified, with corrections, in `SOURCES_BOOKKEEPING.md` (former ledger R-051); the CMP 119 Sect. 3 residue is the unprinted T-step rate loss (ledger R-020); Theorem 1 itself rests on Balaban's published proof, which uses the Group A items of Section 7.7.

R-032. Verification: `BETA_FP_U` has no verifier file. `BETA_FP_PW_VERIFY` §6 items 1–4 grade the logic of L-088 (Theorem B) [PROVED+VERIFIED, CONDITIONAL] (tag decision, `PLAN.md`); the inputs (U), (DF), (T1) stay [PROVED, UNVERIFIED].

13. Lean formalization of the history form of Lemmas D′/D″ [OPEN]. L-005 (Lemma D″) in the one-argument form is done (Section 3.4.2).

- Results tagged [PROVED, UNVERIFIED] need a verifier pass; Appendix B lists them. They include the `BETA_FP_U` reductions and H2_CERTIFY_L13. Round 7 cleared the sweep, `BETA_FP_M` Prop. 4.3 (for $j \geq j_1$), the Prop. 3.6 assembly, and N3 in the D″ form (ledger R-034, Section 5.11); this also covers the sweep in the step L-001 (Theorem 2w) ⇒ Theorem 1. H2_CERTIFY_L13 (its cell plan is superseded at $L = 13$ by Theorem 6.1; the box sweep runs as a cross-check). H2_CERTIFY_P4 has partial support: identity (D) is proved in Lean (`P4_divergence`), L-264 (Lemma S) is rederived in `H1_ANALYTIC_VERIFY.md` V1, and the rebuilt H_j of `H1_ANALYTIC_VERIFY2.md` confirms it numerically; the Fourier step has no independent verifier.
- The paper is not verified as one document. Its statements carry the tags of the project files.

7.5 Open for stronger statements

Not needed for Theorem 1, open for the stronger statements: (A) in full and Lemma 5.1 for $m = 2$ (convergence, rate); C^2, C^3 of $\sigma^{(1)}$ under R-025 (H2) (not written); certified $\beta_1(0), \ldots, \beta_{j_*}(0) > 0$ at odd $L > 11$ (literal (0.31) at every K); explicit j_*; (N).

7.6 Why the mass gap is not on this route

Write β_W for the coefficient of $\sum_p (1 - \frac{1}{N} \operatorname{Re} \operatorname{Tr} U_p)$ in the Wilson action. Bałaban's action is $g^{-2} \sum_p (1 - \operatorname{Re} \operatorname{tr} U_p)$ with the normalized trace (CMP 109 (0.2)), so $\beta_W = 1/g_B^2$. Shen, Zhu, and Zhu [24] prove a mass gap for SU(N) lattice Yang–Mills with action $N\beta \operatorname{Re} \sum_p \operatorname{Tr} Q_p$ (their (1.2)) when $|\beta| < 1/(16(d-1))$ (their (1.3)); with $\beta_W = N^2\beta$, $N = 2$, and $d = 4$, this is $\beta_W < 1/12$.

Bałaban's construction requires $-2(1+\beta_0)^{-1}p_0(g_k)+1+2\kappa(100R_k)^d \le -\frac{3}{2}p_0(g_k)$ (CMP 122 II p. 389). With $d=4$ and $R_K \ge 1$, this gives $p_0(g_K) > 4\kappa\cdot 10^8 + 2$ for every $\beta_0 > 0$ (twice this at Bałaban's $\beta_0 = 1/7$). The unit-scale threshold $g_K p_0(g_K)$ must be below 2, the largest value of $\|U-1\|$ on SU(2); with $\kappa \ge 1$ this gives $\beta_W = 1/g_K^2 > p_0(g_K)^2/4 > 4\times 10^{16}$ at the unit scale. The two regimes are therefore separated by at least seventeen orders of magnitude in β_W, and no choice of L narrows this, because L does not enter the condition. A proof of a mass gap needs a mechanism outside the renormalization-group construction, and Bałaban's papers do not address it.

7.7 Trust ledger

The file `TRUST_LEDGER.md`, corrected by `TRUST_LEDGER_VERIFY.md` (Steps 1–28), lists every statement that the two main theorems use and that this project does not prove with a verifier's confirmation. The three tables below condense it to one row per item. Group A holds Bałaban-level statements and external results (X), Group B holds our own open items, and Group C holds conventions and choices. The IDs are those of the ledger. Ledger R-025–R-046 are not the Route B items B1–B7 or item B8 of this section, and ledger R-007 is not the item "A6" of Route B (that label comes from v4 §4.2a). Sources are cited by paper and page; "[I]" marks an inference step of ours that no verifier proved.

Group A: Bałaban-level statements and external results

ID	statement	source	status
R-001	Kernel bounds on $(QGQ^*)^{-1}$, $(QG_1Q^*)^{-1}$, uniform in k and regular U	CMP 99 p. 422 (3.132)	asserted, "not repeated"; kernel-decay form a verified reduction via L-125 (Lemma CT) to CMP 99 Thms 3.1, 3.3, the G part of 3.12 + H-136 (KE′) + S2; residue: the condition H-057 (C), stated exactly (round 8): $\hat{N} = S^{-1/2}QGQ^*S^{-1/2} \geq c_-$ and $\hat{N}_1 = S^{-1/2}QG_1Q^*S^{-1/2} \geq c_{-,1} \geq \frac{1}{2}c_-$ on $\ell^2(\mathfrak{B}, \mu)$, uniform in regular U, j, k, η, $\{\Omega_j\}$, and volume, R-004's γ_0 does not imply it; round 9: H-057 (C) a verified reduction (Proposition 5.23), $c_- = 1/(4K^D)$, $c_{-,1} = c_-/(1 + \delta_1)$ by L-039 (Lemma R); the torus-side hypothesis discharged by L-284 (Lemma T) under K-032 (Vol) with $M \geq c_T/22$ (K-045); the seam defect of `SOURCES_QE_INV` 5.3 repaired; residue: the printed inputs R-021, the reading $a = 1$, [I-geom], the path metric, [I-conv], conventions K-044

Group A: Bałaban-level statements and external results

ID	statement	source	status
R-002	Walk-expansion form of R-001 (builds the walk expansion of $C^{(k)}(Z_0)$, bounds (3.107)–(3.108))	CMP 99 Thm 3.15, p. 416	asserted, "analyzed in the same way"; verified reduction, written out as Theorems 5.1 and 5.2 (paper-grade): rests on H-057 (C), CMP 99 Thms 3.7, 3.9, 3.10 (printed sketches), CMP 96 Lemma 2.1, WG-1 ($\alpha_0 \le c' M^{-1} M_1^{-10}$), and [I] geometry/convention items; O3, O4 closed; round 9: Thm 3.9 written out, verified, paper-grade (Theorem 5.29; L-210 (Lemma LB) unconditional; averaged-link input; (3.99) rate $\frac{1}{2}\varepsilon - \lvert 2 - \frac{d}{2}\rvert\theta_L$); Thm 3.7 written out, local fixes running (Theorem 5.30); Thm 3.10 verifier running; H-180 (X-PU) a factor L, not yet applied; the `WG_FULL` H-209 (I3) rate fix pending
R-003	R_2 and R_3 obey (2.16)	CMP 116 p. 16	asserted, "similarly"; verified reduction (V7 repair) to (W^c) = Theorem 5.3 (paper-grade, no Thm 3.15 transfer) + R-008 at real U + item (i) [I-forced] + RM distance [I]; R_3 in weaker forms; round 9: input R4 (CMP 109 p. 254) proved and verified for the CMP 98 results (Proposition 5.28); the CMP 99 operators rest on R-022; gap in `WC_FULL` §3.1(2), a (d, L) constant in γ_{exp}, fix pass running; H-180 (X-PU) note
R-004	$C^* \Delta_k C \ge \gamma_0 > 0$, uniform in k and U	CMP 99 p. 428	asserted, "we can prove"; full proof, verified, from CMP 99 Thms 3.11–3.12 + H-123 (H_Q) + R-Δ2 [R]; $\gamma_0 \ge m_L/(48 d^2 L^d)$; σ-localized clause to (W) + (N); CMP 96 (2.126)–(2.127) proof step false for $L \ge 10$

Group A: Bałaban-level statements and external results

ID	statement	source	status
R-005	Kernel decay (189) of $\delta^2 V/\delta A'^2$ and its consequence (190)	CMP 102 p. 308, Prop. 9	announced; verified reduction to R-001 + published CMP 102 + R-018 + one bookkeeping item
R-006	H-202 (H0): $E^{(j)}$, β_{j+1} smooth in g with bounded derivatives; includes the assertion that the whole-lattice limit of (1.22) exists	CMP 109 pp. 263–264	inductive assumption; L-001 (Theorem 2w) uses only item $7'$ (Lipschitz form on $]0,\gamma]$, every past), which also follows from H-111 (G-gain) with the running-maximum lift [verified]; round 5: that use is a verified reduction (H0-red), residue N1–N9, N2 = L-174 (Lemma G3) proved relative to listed inputs (Theorem 5.9); round 6: the $\mathbb{Z}^d$-limit assertion follows from Proposition 5.8 and Theorem 5.10, so no bare assertion is left; for L-001 (Theorem 2w) only the H0-red residue remains
R-007	$E^{(j)}$, β_j analytic in the couplings	CMP 109 pp. 264, 266	asserted; not used
R-008	$\|C^*\Delta_k C\| \le \gamma_1$ and uniform decay of $C^*\Delta_k C$	CMP 116 p. 13 (2.7); CMP 99 p. 428	premise; round 5: proved relative to listed inputs, verified (Theorem 5.4): C = constraint map, $C^*\Delta_k C$ = precision; inputs C2 [C + one I step], A3-R4, I1, I3 (norm), W_{G_1} [R], A1-W [C], I2, I9 (decay); complex s at sketch level; old R-001 + R-018 route superseded
R-009	Positive Gaussian measure for the real pair $(U,0)$; complex case "perturbative"	CMP 116 p. 15	asserted; round 5: proved in corrected form relative to Theorem 5.3 and readings I-a–I-e, verified (Theorem 5.5); symmetry of $C^{(k)}(Z_0,0)$ not proved and not needed ($H_0 = \operatorname{sym} C_0^{-1}$); Lemma D at $e^{-\varepsilon'' M/8}$

Group A: Bałaban-level statements and external results

ID	statement	source	status
R-010	Lower-order bracket terms "bounded similarly as in (1.24)"	CMP 116 pp. 8–9	asserted, "similarly"; round 5: proved in corrected form R-011 relative to CMP 109 Sect. 3–5, verified (Theorem 5.6); raw Taylor terms do not obey (1.24) (the j-sum diverges); transformed terms do after the marginal cancellation; residue R1–R4; exponent β_H (K-028)
R-012	"Similar inequalities hold for other constants"	CMP 119 p. 256	asserted; round 5: closed as a verified list (Proposition 5.7); all follow from (2.6)–(2.9); order entries K-029–K-031
R-013	Theorem 1	CMP 122 II p. 355; CMP 116, 119, 122 I–II	proved, conditional on $g_k \in]0, \gamma]$; uses R-001–R-005, R-002, R-008, R-009, R-010, R-014, R-015, R-017, R-018
R-014	Induction closure recovers E_0	CMP 109 p. 280	asserted, constant unspecified; inside R-013 only
R-015	Small-field terms and β_j are whole-lattice quantities, independent of Λ_{k+1}	CMP 119 p. 278, (2.24), p. 249; CMP 109 (4.35)–(4.37)	asserted, small-field part only; round 6: verified reduction to conventions: H-141 (Loc) proved exact relative to K-033–K-037 (Theorem 5.10), so Proposition 5.8 needs only (Bd), (H), (Cnt); (f) proved exact relative to K-038 (Proposition 5.11); threshold $N_k > 22M$, $\varepsilon(M) = C_j(22M)^6 e^{-11 m_j M}$; exact equality only term by term for non-wrapping X; R-term part is R-032; not in the set of L-001 (Theorem 2w)
R-016	Propagators may be replaced by operators with the same bounds	CMP 116 p. 3	asserted for (1.3)–(1.4); round 7: for C_f proved relative to listed inputs, verified (Theorem 5.20; tree form of $\mathcal{W}_S$; the first proof's locality step was false); residue four [I]: H-035 (M1) completeness, H-166 (S_H), the Ω-setting volume convention, the s-version of $(C_f)^{1/2}$; conditions K-041

Group A: Bałaban-level statements and external results

ID	statement	source	status
R-017	$\operatorname{supp}\chi_k$ lies in the analyticity domain of (1.34)	used in CMP 116 Lemma 1	implicit; inside R-013; round 7: proved relative to the printed text and one reading, verified (Proposition 5.16): $B' = g_k B$ on every cut bond, $r = 1$; K-009 no longer needed
R-018	Backgrounds lie in the regular class of CMP 99/102; C-maps agree	used in CMP 109, 116, 122 I–II	implicit; round 7: verified reduction with corrections (Proposition 5.18): inclusions, L-250 (Lemma Q), L-109 (Lemma C), $c_J = O(\alpha_0)$; cube patching replaced, $M \geq 5M_{99}$ (K-042); residue [R] CMP 102 Thm 1, CMP 98, CMP 99 (3.133), [I] I2, I3; absorbs (C2) of MA2
R-019	(190) at complex B	CMP 102 Prop. 9 (silent)	implicit; round 7: reduced to three readings [I], verified (Proposition 5.17): complex (189), (73), and the (172) ball; fallback without the ball reading: two constants, $\omega = 0.4097$, half ball, rate $\times 0.59$
R-050	Uniform-in-k moments of the free extension kernel E_0	not printed	proved+verified (`SOURCES_T2_EUMOM`; constants depend on d only); left the ledger in round 4; row kept for reference
R-020	Large-field T-step rate loss	CMP 119 Sect. 3 (defers to CMP 109 Sects. 6–7 = CMP 116 Sects. 1–2)	not printed; round 7: verified reduction (proof sketch, Proposition 5.19): loss $\leq 13\delta$; open R1 (count sketch), R3 ($H'(Z_0)$ obeys (2.38)); used by R-046 through H-216 (K1′)
R-021	Printed CMP 99 inputs of H-057 (C): Thm 3.1 (3.46) at $a = 1$; Thm 3.11 ($G > 0$); the constant C_{137} of (3.137); the H-136 (KE′) constants uniform in the torus side	CMP 99 pp. 399, 416, 423; CMP 96 Cor. 2.8	printed statements [V], taken as [R] inputs; Thm 3.1 at $a = 1$ and Thm 3.3 rest on walk sketches, Thm 3.11 has a printed proof; new in round 9; no reduction of ours
R-022	"Valid universally for all averages" applied to the CMP 99 operators G_1, H_1, $(QG_1Q^*)^{-1}$, Δ_J under $\tilde{Q}$	CMP 109 p. 253	asserted; new in round 9; partial: the p. 254 transfer covers the CMP 98 results only; $\tilde{Q}$ and Q_1 differ in chain layout only; write-out pending

Group A: Bałaban-level statements and external results

ID	statement	source	status
R-023	$U_{k+1}(\square_0, M'(U))$ has "approximately the same regularity properties"	CMP 109 p. 275	asserted; new in round 9; not covered by R-018; write-out pending
X-001	Lüscher–Weisz (5.7) one-loop coefficient	Lüscher–Weisz [16]	cited; Route B only
X-002	OS II growth condition and reconstruction from torus data	OS II [20]	cited
X-003	Reflection positivity of the Wilson action, link planes	Osterwalder–Seiler [21]	cited, text not fetched
X-004	Convergence criterion of the final cluster expansion	Cammarota, CMP 85 (1982)	cited
X-005	(F): continuum log coefficient at constant field	Savvidy [22]	cited, source checked

Group B: our own open items

ID	statement	source	status
R-025	(H2) $\rho_2 < L$	none	certified at $L = 13$ (Theorem 6.1) and, round 7, at $L = 15, 17, 21$ (full reruns, verified; Section 6.8); residue: reading of (0.4), gauge choice; other L are a K-003 cost, not an open item
R-026	(PI) probe independence	none	verified reduction [R] at fixed j (`SOURCES_T2_PI` + VERIFY) given H-207 (I1) massiveness at $U = 1$, H-208 (I2) gauge covariance, (I3) [P]
R-027	(F) lattice coefficient = continuum coefficient	X-005	round 8: verified reduction, with corrections (`SOURCES_T2_F_STEPB` + VERIFY) to H-137 (L-LL), Lemma 1 (MPT, cited, unchecked), and the gauge-fixing part of Step C [I]; round 9: H-137 (L-LL) proved+verified (Theorem 5.25); R-027 (F) proved by proper time without Lemma 2 or Step C, verification pending (Theorem 5.26); grade stays verified reduction until the verifier reports; open numerical tension: single-step β_1 at $L = 5$–13 does not follow $c \log L$ (Section 6.9)

Group B: our own open items

ID	statement	source	status
R-028	(M_j) formal = true f^2 coefficient, H-244 (S1)–H-247 (S3)	none	round 7: proved relative to listed inputs, verified, at fixed $j \geq 1$, odd $L \geq 3$ (Theorem 5.14): region 1 as an asymptotic identity, neutral sector (N2); only the $o(f^2)$ form; the proved (G'_A) range suffices; residue one [I] convention
R-029	V1: $\beta_{j+1}(0)$ is the Maxwell coefficient	CMP 109 (1.3)–(1.5)	proved+verified at fixed level, conditional on R-030, Bałaban's inductive bounds, and the averaging (0.4)
R-030	(G2) per-cell infinite-volume densities	none (definition of β_j only)	verified reduction [R] at fixed j (was formal); modulo $\tilde{Q}$ onto, (Gap_f), averaging analyticity
R-031	L5: β_{k+1} on the clamped cube, or nested-interval L-005 (Lemma D″)	CMP 109 p. 298	round 7: proved relative to listed inputs, verified, by Route 2 (Theorem 5.15): nested intervals on true trajectories, no clamp, v4's constants; one reading [I] (causality)
R-032	K-independence of $\beta_j(g)$, R-term part; the original form is ill-posed (the whole lattice has no R-terms); replaced by (a) β_j has no R part, (b) E_{pure} on non-wrapping $X \subset \Lambda$ = the $\mathbb{Z}^4$ term, (c) E_R K-uniform through F2′	→ R-015; CMP 119 p. 279	round 6: proved relative to listed inputs, verified (Section 5.10): K-032–K-038, F2′ inputs (R-044), CMP 109 (1.18), (3.48), (4.36); input of Theorem 1, not of the shooting
R-033	Route B inputs (T1), (DF), v4 "A6", `BETA_FP_U` Theorem 2.4	none	argued; Route B only
R-034	Proved, unverified results in the chains (sweep, `BETA_FP_M` Prop. 4.3, Prop. 3.6, N3 in D″ form)	→ R-012	round 7: cleared, verified (Section 5.11): all 19 sweep rows second-read, CMP 122 additions H-271–H-280, H-266–H-270 (K-043); Prop. 4.3 for $j \geq j_1$ only; L-192 (Lemma H″), N3″
R-035	(B-S): sourced Theorem 1, (E14-a, b, c)	→ R-013	open as a whole; parts R-036–R-044, R-046
R-036	(I): $B_j(c) = \beta_j(u(c))$	none	by construction (K-001, K-011); not written as a lemma

Group B: our own open items

ID	statement	source	status
R-037	Real uniform R-037 (H-hist)	→ R-001–R-004, R-002, R-008, R-009, R-010, R-017, R-018	verified reduction, both halves (large field: `SOURCES_GGAIN_LF`); round 5: small-field (Q)/c_- discharged by Theorem 5.9 (with K-026, K-027); open inputs (M_r), N3 Kotecký–Preiss step, CMP 102 (189), H-145 (M-B) bookkeeping; L-281 (Theorem SM) rests on R-011 with β_H; G4 off-chain
R-038	(H-hist2)	none	verified reduction
R-039	(H-thr) threshold mismatch	CMP 122 II p. 383	round 8: verified reduction, with corrections (`SOURCES_B15_M245` + VERIFY): M2′, M4 derived, M5 not needed; H-241 (R1′) a verified reduction (Proposition 5.21) with three [I]; open: (R2)–(R4), the large-field coarea Jacobian; R-038 no longer an input
R-040	(H-hol), small field	conditions (C1)–(C5)	verified reduction
R-041	(H-hol), large field	→ R-005, R-018, R-019	verified reduction; H-146 (M-C) reduced to printed CMP 116 §2 conditions + Lemma 2; H-133 (I-V0), H-131 (I-ord), (I-ε) (both layers) closed and verified; K-025 (M1) → K-025; round 7, chain item 5: (H−) ⇐ R-005 + R-019 (K1–K4) + R-018 + (C1*); (C1) proved given R-019, R-018; (C2) merged into R-018
R-042	N3 strip bound	none	proved+verified given R-036–R-041
R-043	(FX) final Mayer expansion	not in CMP 116–122	reduced; remaining input R-042; round 9: the item H-105 (FX-D) of the project notes is Q-060, already closed (`SOURCES_FX_D` + VERIFY); H-168 (S′) uses R-043 (FX) with ℓ-free constants

Group B: our own open items

ID	statement	source	status
R-044	Sourced step bookkeeping: R-044 (W4), H-240 (P8), F2′, (D), Λ_j	→ R-005, R-010, R-016	reduced; round 8: R-044 (W4)-R proved relative to inputs, verified with corrections (Proposition 5.22); Λ_j sub-item proved relative to inputs, verified with corrections; residue W4-G [R] and the Ursell bound (7.13) [I]
R-045	(B-N1) bridge and R-045 (S)	none	R-045 (S) clauses: verified reduction, h-independent; H-098 (E14-c) clause reduced to R-045 (S), verified (round 8); (S_{src}) ⇒ H-116 (H-U) proved+verified; compatible with K-024 (U) (K-024); round 9: H-168 (S′) proved+verified (Proposition 5.27), ℓ-free, no upper limit on σ; L-111 (Lemma C) is false for $\sigma \geq 2ML$ and is replaced by L-127 (Lemma C′)
R-046	Recovery of $\kappa \to \kappa/2$ in the sourced step (only; K-024 (U) moved to K-024)	none	verified reduction [R] by $L/2$-rescaling, $\kappa \geq 1536$; uses R-020; round 7: H-216 (K1′) $\delta \leq \frac{1}{13}(1 - 2(1+4\beta)/(\theta_K L))$, $\theta_K \geq \frac{1}{2}$, $\kappa \geq 2m_{\text{tot}}$; $\delta = 0.025$ at $L = 13$ (0.03 fails; Lean `K1_numeric_13`)

Group C: conventions and choices

ID	choice	source	status
K-001	Freeze thresholds and partitions at Bałaban's trajectory x^*	`SOURCES_D3`, `SOURCES_HTHR`	free; derivative needs R-039
K-002	Option B margin, $\theta_W + \tau_C \leq 0.2$	`SOURCES_MB_VERIFY`	free; costs a smaller γ
K-003	L odd, $L > 11$; target $L = 13$	v4 §1.1	loses generality
K-004	$G = SU(2)$, $d = 4$	Theorem 2	loses generality
K-005	Order of constants: the canonical acyclic order of `SOURCES_ORDER_CANON` (amended E1–E8), β_H (K-028) first	`SOURCES_ORDER_CANON` (+ VERIFY), `SOURCES_N3A`	settled; verified twice; two nominal cycles resolved by the split (Section 5.10)
K-006	Holomorphy radius ρ_0	`SOURCES_HHOL`	free
K-007	Cut on B, not B'	CMP 109 (2.9), (2.12)	reading; closed
K-008	Sharp χ_k; smooth fallback	`SOURCES_G3`	free

Group C: conventions and choices

ID	choice	source	status
K-009	Shrunk χ_k threshold	`SOURCES_GGAIN_VERIFY`	unused (round 8): R-017 holds with $r = 1$, so no consumer uses K-009; before round 8 it changed the construction
K-011	Gauge-fixing weight $1/u_k(c)$	`SOURCES_D2C_VERIFY`	changes the construction; R-013 must hold for it
K-012	$h \equiv 0$ booking; freezing set	`SOURCES_D2C`	free
K-013	Additive source normalization	`SOURCES_HBETA_N3`	free
K-014	Main R-step form (1.90)–(1.98)	`SOURCES_FX_O23_VERIFY`	free; removes Q-062
K-015	κ_{final} fixed before M	`SOURCES_FX_R2`	free; not sharp
K-016	$2p_1 - (d+5)r_0 > p_0$	CMP 122 II p. 383	condition (Bałaban's own)
K-017	$32\varepsilon' \le 1$, $S \ge 1$, $\kappa_0 > 2$	`SOURCES_D3W8_VERIFY`	changes the construction; R-013 must hold for it
K-018	(S): $\gamma^{-2} \ge c_{sw} D'$	v4 §1.2	condition; needs β_0 fixed before γ (verified reading)
K-019	ℓ even; thermal state; plaquette observable; test space D_n; origin at a site; compactness form	`ASSEMBLY` §0–§1	loses generality
K-020	Coupling domain $]0, \gamma]$	`SOURCES_GGAIN_VERIFY`	free
K-021	Route K primary, Route B alternative	v4 §1.3	free
K-022	$\theta_* < 1$ in Route B	`BETA_FP_PW_VERIFY`	free; trivial
K-023	$g_0(K, g)$: any shooting solution	L-006 (Lemma D)	loses generality
K-024	(U): $\rho_1 \le c_* r_0 / M^a$ at $\lambda_K = 1$	`SOURCES_UN1` (+ VERIFY)	constant choice, not a hypothesis on test functions; verified compatible with B-N1; C^n cost per n-point bound
K-025	(M1): $M = L^m$ at least the largest of 32 lower bounds; no upper bound; real $\alpha_0 \le \alpha_0'/(CM)$ with $C > c_J$	`SOURCES_ORDER_CANON` (+ VERIFY), `SOURCES_HIGHER_M1` (+ VERIFY)	convention, free choice; the admissible set of M is an up-set

Group C: conventions and choices

ID	choice	source	status
K-026	(Hα): $(\alpha_5 + G_\Gamma^2\delta_S)\Lambda_0 \le \frac{1}{2}$	`SOURCES_G3_N2`, `SOURCES_G3_GGAMMA` (+ VERIFY)	condition, not choice; stronger than Bałaban's $\alpha_5\|C^{(k)}\| < \frac{1}{2}$; $G_\Gamma \le \bar{G}$ uniformly in M (verified), so (Hα) is the single threshold $M \ge \max\{M_\Gamma, M_{\alpha 5}\}$; closed as an M threshold (round 6); one [I] residue: the s-rule acts on Bałaban's (3.185) resolvent expansion (CMP 116 p. 13)
K-027	(Hγ): $\frac{1}{20}\gamma_2\varepsilon_1^2/\gamma^2 \ge 4\kappa+1$, and the (2.31) condition $\frac{1}{20}x - 4M^4e^{-x/20} \ge 0$ at $x = \gamma_2\varepsilon_1^2/g_k^2$	`SOURCES_G3_N2` (+ VERIFY)	condition, not choice; one unit stronger than Bałaban's $\ge 4\kappa$; γ-smallness after M, κ, γ_2
K-028	$\beta_H \in (0,1)$, $\beta_H \le \beta_0 < 1$: the unprinted Hölder exponent of CMP 109 (3.31), first in the order	`SOURCES_A10_LOWER_VERIFY`; `SOURCES_GGAIN_BETAH` (unverified assessment)	free; every consumer needs only $\beta_H > 0$; cost $(1 - L^{-\beta_H})^{-1}$ factors, larger $B_4(\beta_0)$
K-029	$M_1 < M_2 < M$, order $M_1 \to M_2 \to M$	CMP 119 p. 245; `SOURCES_A11_CONST_VERIFY`	free; printed order
K-030	$M_2 \ge 1/\delta$	`SOURCES_A11_CONST` (+ VERIFY); CMP 119 (3.7), (3.17)	free; unprinted lower bound
K-031	$A_1/A_0 \le 1/(80B_3)$, A_1 after A_0	`SOURCES_A11_CONST` (+ VERIFY); CMP 119 p. 266	free; Bałaban: "sufficiently small"
K-032	(Vol): torus side $\ell > 22M$ final-scale cells; even form $\ell = 2L^{m_T}$, $m_T \ge m_M+1$	`SOURCES_A15_BDRY` (+ VERIFY), `SOURCES_T2_VOL`	condition, admissible; K-free; not used by L-001 (Theorem 2w); used by R-032 and Theorem B
K-033	(Cube): $U_j(\square_0, \cdot)$ of [14] uses only data on $\square_0$	CMP 109 p. 276; `SOURCES_A15_LOC` (+ VERIFY)	reading [I]
K-034	(Cases): the other term classes are localized like the typical term	CMP 116 p. 2	Bałaban's printed sketch [R]
K-035	(Conv): cube sides divide N; partitions are π-images of the $\mathbb{Z}^4$ partitions; $M = KR_0M_0$, K a power of L	`SOURCES_A15_LOC` (+ VERIFY)	convention on an unprinted part, admissible
K-036	(Det′): domain-assignment rule for the closed-walk terms of the determinant expansion	CMP 102 pp. 271–272; `SOURCES_A15_LOC_VERIFY`	convention on an unprinted part, admissible
K-037	"H_j with free boundary conditions" is the $\mathbb{Z}^4$ operator	CMP 109 p. 290	reading [I]
K-038	(Enl): the s-rule domain of a walk is $\bigcup X_i^5$ as a whole-lattice set	CMP 116 p. 3; `SOURCES_A15_BDRY` (+ VERIFY)	convention on an unprinted part, admissible; the Λ-restricted form costs at most one M-cube layer

Group C: conventions and choices

ID	choice	source	status
K-039	(C1*): $c_m M_0\, g_k p_0(g_k) \le \frac{1}{2}\kappa_H C_1^{102} a_*^{102}$, uniform in k	`SOURCES_T2_HMINUSC` (+ VERIFY); `SOURCES_ORDER_CANONF`	condition, not choice; γ-smallness; ε_1^{102} per scale exists iff (C1*) and $\varepsilon_1^{(0)} \le a_*^{102}$
K-040	Slot of M_0 and a_*^{102} (with the cube cap $\varepsilon_1^{102} \le R_1 M_1 a_1^{102}/M$) after M, before γ	`SOURCES_T2_HMINUSC_VERIFY`; `SOURCES_ORDER_CANONF`	free constant-order entry; the cap is booked on ε_1^{102}, so M stays one-sided
K-041	R-016 tree conditions H-060 (A16-a^T), H-062 (A16-b^T), H-126 (H_κ^f), (H_η^f), $\bar\Gamma_1^f \eta_M^f \le \frac{1}{2}$, $C_{\mathcal{D}}'^f$, the $O(\gamma^2 r)$ budget entries	`SOURCES_A16_REPLACE2` (+ VERIFY); `SOURCES_ORDER_CANON` F	conditions, verified; linear in $g^2 r$ or M thresholds; α_5 slack shared with K-026
K-042	H-063 (A18-1)–H-066 (A18-4) and $M \ge 5M_{99}$ with $10 \mid C^{122}M$	`SOURCES_A18_REGULAR` (+ VERIFY); `SOURCES_ORDER_CANONF`	conditions, constant-order entries; acyclic; ε_0^{102} two-sided but nonempty
K-043	CMP 122 I–II threshold and exponent conditions H-266–H-270, with the cross-scale list H-271–H-280	`SOURCES_A11_CONST` addendum; `SOURCES_B10_SWEEP_VERIFY`	conditions, Bałaban's own (printed); exponents before γ; N two-sided per scale, exists for γ small
K-044	(Cv1) $U(c)$ = product of the level-$(i-1)$ links on the straight segment; K-044 (Cv2) translation-covariant axial Γ	CMP 98 (42); `SOURCES_C_CONDITION` §5 (+ VERIFY)	conventions on an unprinted part, admissible; enter only H-123 (H_Q)(iii) of H-057 (C); other contour choices change only constants
K-045	L-284 (Lemma T) cut-off $\chi(d/D_0)$ with threshold (T2): torus side $> c_T(d,L)$ coarsest blocks; $\theta_S \le \delta_H/8$, $\theta \le \delta_0/2$, $RM \ge \max(4L^2 D_0, c(d,L))$, $M\alpha_0 \le a_0(d,L)$, $M \ge c_T(d,L)/22$	`SOURCES_C_CONDITION` §2 (+ VERIFY)	conditions, not choices; $M \ge c_T/22$ is a new lower bound on M
K-046	(K-046) s-rule: s-versions per distinct σ_0-cube on the project's expansions; H-103 (FI) on the admissible walk set; loc set $\bigcup \tilde X_i^5$	`SOURCES_SRULE_CONVENTION` (+ VERIFY)	convention on an unprinted part, admissible, verified (round 9); replaces the reading H-022; consumers use only bounds
K-047	$\gamma^2\beta' \le 1/L$, side condition of CMP 119 (2.9) line 2	Lean `ScaleChain`; `SOURCES_ORDER_CANON`; `SOURCES_A11_CONST` addendum	condition, not choice; γ-smallness of Bałaban's type; no cycle

Verified readings. Four former Group A rows are now verified readings and appear in no set (`SOURCES_BOOKKEEPING.md`): R-047 (the constant $C_1'(d,L)$ is unprinted and not derivable from the printed lines; its value is not used), R-048 (β_0 "arbitrarily small" is on CMP 119 p. 255; β_0 is fixed before γ), R-049 ($R_{K+1} \ge L$ follows from CMP 119 (2.5); C_c is free of g), and R-051 (Π (4.37)/(5.1), CMP 119 (2.42) corrected, (2.12)/(2.23), the "L odd" page; the Sect. 3 residue is R-020). Round 8: `SOURCES_BOOKKEEPING_VERIFY.md` confirms all four on page images (Section 5.12).

The free parameter β_H. The per-label exponent of L-050 (Fact A) (L-281 (Theorem SM)) is the Hölder exponent β_H of CMP 109 (3.31), which Bałaban does not print (Theorem 5.6). It is a

free parameter, $0 < \beta_H \le \beta_0 < 1$, and it comes first in the order of constants: $\beta_H \to d, L \to \cdots \to \varepsilon_1 \to \gamma$ (K-005, K-025, K-028). Every consumer needs only $\beta_H > 0$, because each condition it adds is an ε_1- or γ-smallness condition (`SOURCES_GGAIN_BETAH.md`, [UNVERIFIED ASSESSMENT]). The cost is the factors $(1 - L^{-\beta_H})^{-1}$, a larger $B_4(\beta_0)$, and K_T fixed before ε_1. Margins at $L = 13$: the H-111 (G-gain) margin $1 - 13^{-\beta_H/2}$ is 0.120 at $\beta_H = 0.1$ and 0.685 at $\beta_H = 0.9$; the L-186 (Lemma HH) margin is 0.901 at $\beta_H = 0.9$.

Binding sets (round 9; ledger round 10). *L-001 (Theorem 2w)* (Theorem A(i), Route K) rests on one Group A item, R-006, on R-025–R-031, and on X-005; Group C: K-003, K-004, K-018, K-023. No genuine hypothesis remains; the chain rests on Bałaban's printed material, verified project work, conventions, readings, scope restrictions, and constant conditions. R-006 has no bare assertion left (Proposition 5.8, Theorem 5.10); its use is a verified reduction with residue N1–N9. If R-006 is taken through this reduction, L-001 (Theorem 2w) also rests on R-008, R-009, R-017, the inputs of Bałaban's sup induction, K-026, K-027, and K-033–K-037; these are not counted in the set. Of R-025–R-031: R-025 is certified at $L = 13$ (and at 15, 17, 21); R-026, R-029, R-030 are verified reductions or proofs relative to named inputs; R-028 (Theorem 5.14) and R-031 (Theorem 5.15) are proved relative to listed inputs and verified; R-027 is a verified reduction whose input H-137 (L-LL) is proved and verified, and a full proof of R-027 (F) is pending verification (Theorem 5.26); R-027 also carries an open numerical tension. R-031 adds `SOURCES_HHOL.md` §2–§3 and the causality reading [I] to the inputs of L-001 (Theorem 2w). Route B: {R-006} + {R-025, R-030, R-031, R-033} + X-001.

L-001 (Theorem 2w) ⇒ Theorem 1 (Theorem A(ii)) rests on 4 Group A items (R-006, R-012, R-013, R-015), 9 Group B items (R-025–R-032, R-034), and X-005; Group C adds K-029–K-038, K-043, and K-047. R-034 is cleared (verified). Inside the proof of Theorem 1 (R-013), Bałaban also uses R-001–R-005, R-002, R-008, R-009, R-010, R-014, R-017, R-018, R-021–R-023, and the conventions K-044–K-046.

Theorem B rests on 20 Group A items (R-001–R-006, R-002, R-008, R-009, R-010, R-012, R-013, R-015–R-019, R-020, R-021–R-023), 4 externals (X-002–X-005), and 21 Group B items (R-025–R-032, R-034–R-046); Group C adds K-026–K-047. Of the Group A items that Bałaban prints without a proof, R-002, R-003, R-004, R-008, R-009, and R-011 carry full proofs relative to listed inputs, R-012 is a verified list, and R-001, R-005, R-006, R-015 carry verified reductions; R-001 now includes the condition H-057 (C). Of the implicit or unprinted items, R-016 and R-017 are proved relative to listed inputs, and R-018, R-019, and R-020 are verified reductions. Group A rows without a verified proof or reduction: R-021, R-022, R-023. R-021 is a set of printed CMP 99 statements taken as [R] inputs; R-022 and R-023 are printed CMP 109 assertions whose write-outs are pending. Group B rows without a proof or a verified reduction: none. R-014 (inside R-013 only) stays asserted. This does not make Group A or Group B proved. Every row stays in the ledger, because its residual inputs are not proved: Bałaban's printed sketches [R], conventions, readings [I], K-032 (Vol), and K-039–K-047. Sub-items open after round 9: the gap of `WC_FULL.md` §3.1(2) (R-003; constant fix running); H-180 (X-PU), a factor L (R-002, R-003); the rate fix of `WG_FULL.md` H-209 (I3) (R-002); the verifiers of Theorems 5.30 (fix pass) and 3.10 (R-002, R-003); the averaged-link input of Theorem 5.29; the write-outs of R-022 and R-023; the verifier of Theorem 5.26 (R-027); the verifier of the H-035 (M1) sweep (R-016). Carried: the three [I] of H-241 (R1′) and the large-field coarea Jacobian (R-039); W4-G [R] with the Ursell bound [I] (R-044); the β_j part of R-038; the other [I] of R-016; the three readings of R-019; the numerical tension of R-027. Closed in round 9: the condition H-057 (C) under R-001, R-002, R-003, R-008, R-009, R-016; H-137 (L-LL); H-168 (S′); the s-rule reading (now K-046); input R4 of R-003 for the CMP 98 results; H-035 (M1) as a list

(verifier running).

Counts: Group A 22, Group B 22, Group C 46, externals X 5. Group A has 22 items (R-050 left the ledger in round 4; its row is kept above for reference; R-047, R-048, R-049, R-051 are verified readings). 17 are printed by Bałaban: the 14 of round 8, R-021 (printed statements, partly with sketch proofs), and R-022, R-023 (assertions without proof). Group C gained K-026–K-031 in round 5, K-032–K-038 in round 6, K-039–K-043 in round 7, and K-044–K-047 in round 9. Of Group C, K-003, K-004, K-019, K-023 lose generality; K-024 (K-024 (U)) and K-025 (K-025 (M1)) are constant choices; K-011, K-017 change Bałaban's construction, and K-009 is unused; K-016, K-018, K-026, K-027, K-032, K-039, K-041, K-042, K-043, K-045, K-047 are conditions; K-028–K-031 and K-040 are free entries of the order of constants; K-033 and K-037 are readings; K-034 is a Bałaban sketch [R]; K-035, K-036, K-038, K-044, K-046 fix unprinted parts of the construction. This is still a verified reduction, not a proof.

Lean support (round 9). Lean certifies finite or algebraic steps only, and no row changes grade because of Lean (`TRUST_LEDGER.md` §7). `lean/YmLean/` has 45 files (45 imports in `lean/YmLean.lean`); `lake build` passes with no `sorry`, and the axioms are propext, Classical.choice, and Quot.sound (`CanonOrder` uses none). Statement checks are the files `lean/VERIFY_*.md`. Waves 2–3 added ten files: Concatenation, ComplexGaussian, NeutralSpectrum, QConstants, TheoremL, ScaleChain, ParametrixCore, WGAlgebra, MagneticShift, and H1Majorant.

row	Lean files	what is formalized
R-001	CombesThomas, BlockSchur, Schur, KernelRow	L-125 (Lemma CT) in kernel form; the row sum for `Fin n` only, not the block-lattice sum
R-002	ParametrixCore, WGAlgebra, Concatenation, WalkAffine	the A1-W core (μ-weighted L-124 (Lemma CT), the parametrix identity, the Neumann step, entry decay); WG L-228 (Lemma N), the G_1 resolvent and its two-sided Loewner bound, the (3.135) four-term expansion, the WG-1 exponent counts 10 and 12, pattern counts; L-115 (Lemma C) concatenation. Not: the chain expansions as matrix identities, the walk distance
R-003, WC chain	WeightedNorm, Resolvent, BlockInequalityE, CauchyPerturbation, TheoremL, Concatenation	the weighted norm and L-225 (Lemma N); L-153 (Lemma E); L-122 (Lemma CP); L-207 (Theorem L) assembly, $\gamma_{\exp} = \gamma_e/4$, $K_b \le 64d^3L^{2d+1}$ (the compression step not formalized)
R-004	SlabCounterexample, GaugeMin, Interp, WalkAffine, Resolvent	the CMP 96 slab step fails for $L \ge 10$; Lemma 5; `SOURCES_A3_SIGMA.md` 3.2 and §4
R-008, R-009	QConstants, TheoremL, ComplexGaussian	L-226 (Lemma N) for rectangular C; $\gamma_{\exp}$; L-242 (Lemma P) core and `lemmaD_core`
R-010 (=R-011)	ElemBounds	the raw j-sum diverges (`a10Sum_*`)
R-012, R-034, K-047	ScaleChain, ElemBounds	(2.6)–(2.9), (T1)–(T4), H-156 (P-up)/H-155 (P-low), `eq29_line2_full` with $\gamma^2\beta' \le 1/L$; monotone maps

row	Lean files	what is formalized
R-006, R-037 (G3, (Q))	SchurMarginal, QConstants, ComplexGaussian	block inverse; the B-block $H_0^{-1} + H_0^{-1}\Gamma\Gamma^* H_0^{-1}$; Schur positivity; one-bond compression; `c_minus_ge`; Λ and c_- bounds; the Lemma S form. Not: the Gaussian integrals of `SOURCES_G3_N2.md` §2.4, $\Gamma_{bb} \geq H_0^{-1}$
R-025	BlockNorm, OishiRump, CertBridge, Tower, Divergence, H1Majorant	`rho2_le`; the certifier bridge; the majorant core; identity (D) of P4; L-283 (Lemma T) bounds, the F_{CS} core, L-166 (Lemma G), the gauge step, L-264 (Lemma S) / P5 invariance. Not: an evaluation of B_L, the end-to-end $F \leq F_{CS}$ over $\mathbb{Z}^4$
R-026	TheoremC1	the two-step bound (B2)
R-028	NeutralSpectrum, MagneticShift	the neutral sector (spectrum completeness not formalized); the magnetic-shift algebra on $\mathbb{Z}^2$ with exact commutators (the analytic norm forms not formalized)
R-031	NestEnvelope	`nest_DDoublePrime`, `nest_DPrime`; the induction needs the continuous family $u_0 \mapsto u_k$ and the IVT, with hypotheses on the Good-k set; `env_of_terminal` is the single-sequence form
D′, D″, K	LemmaDPrime, LemmaDE, LemmaDDoublePrime, TheoremK, Gronwall	the shooting lemmas (one-argument form) and L-009 (Theorem K)
R-042 (N3)	StripBound, LemmaH	the strip bootstrap; L-182 (Lemma H)
R-043 (FX_O1)	CubeCount	the cube count
R-046	K1Prime, Interp	`K1_numeric_13` (13δ form, $\delta = 0.025$); `K1_iff`
canon (K-005, K-025), R-012	CanonOrder, ElemBounds	50 lines, 170 edges, `id_eq_rank`, `canon_acyclic`, `canon_wf` (the transcription is acyclic; the mathematics is not certified); monotonicity of the constant maps
none	Telescoping, DetIdentities, AbelianCount, Envelope	numerics identities, U(1) count, H-081 (B-NT); not needed by the targets

Files without a VERIFY file: AbelianCount, DetIdentities, Divergence, Gronwall, LemmaDDoublePrime, LemmaDE, LemmaH, OishiRump, Schur, StripBound, Telescoping. `lean/LEAN_CATALOGUE.md` lists about 90 Lean-able statements. It estimates that about 40% of the project's own proved steps can be formalized now, but only 10–15% of the load-bearing content of the ledger, because most of the load sits on Bałaban's walk expansions, polymer activities, and inductions, which stay in prose.

A Lean formalization

The shooting lemmas of Section 3.4, the arithmetic of L-009 (Theorem K) and L-011, and the bridge lemmas of the certifier are formalized in Lean 4 with Mathlib (25 files in `lean/YmLean/`); seven files formalize lemmas of the sources block (Schur test, block-valued Schur test, Combes–Thomas, interpolation and rescaling, resolvent and weighted Neumann bounds, the gauge condition of the constrained minimizer, the slab counterexample to CMP 96 (2.127)). The inventory is as

follows. Round 9: the directory has 45 files (waves 2–3 added Concatenation, ComplexGaussian, NeutralSpectrum, QConstants, TheoremL, ScaleChain, ParametrixCore, WGAlgebra, MagneticShift, and H1Majorant); the Lean support by ledger row is the table at the end of Section 7.7, and `lean/README.md` lists every file.

[LEAN] (no sorry; axioms propext, Classical.choice, Quot.sound)

- L-004 (Lemma D′) (a)–(d) (`YmLean.lemma_DPrime`), Step 8 (`lemma_DPrime_step8`), L-006 (Lemma D), L-007 (Lemma E) (Cg^2 form), L-006 (Lemma D)+E (`lean/YmLean/LemmaDPrime.lean`, `LemmaDE.lean`). One-argument β only.

- L-005 (Lemma D″) (`YmLean.lemma_DDoublePrime`, `lean/YmLean/LemmaDDoublePrime.lean`, 146 lines): (a″) bracket $(3b/4)(K-k) - 2R \le T(k,K) \le (5b/4)(K-k) + 2R$; (b) shooting for $g \in]0, g_1]$ with the clamp inactive; (c′) with $\beta = 3b/4$, $B' = 5b/4 + 2R$, $D' = 2R$, for every trajectory with couplings in $]0, \gamma]$; (d) for $n - m \ge 8R/b$, lower constant $b/2$. Hypotheses weaker than the note (continuity, any admissible γ, $R \ge 0$ derived). One-argument β only; L5 not formalized.

- Full inventory (`lean/README.md`; file, main theorems, source note):

  1. `LemmaDPrime`: `lemma_DPrime`; `LEMMA_DPRIME.md` §1–§2.
  2. `LemmaDE`: `lemma_D`, `lemma_E`, `lemma_DE`, `lemma_DPrime_step8`; `LEMMAS.md`.
  3. `LemmaH`: `gronwall_telescope`, `lemma_H`; `SOURCES_HBETA.md` L-182 (Lemma H).
  4. `DetIdentities`: `det_bordered`, `det_restrict_add_gauge`; `numerics/NUMERICS.md` §1.
  5. `Telescoping`: `telescoping_det`, `telescoping_log`; `R2_TELESCOPING.md` Prop. 3.1.
  6. `AbelianCount`: `su2_count`, `abelian_share`; `U1_INFINITE_LATTICE.md` Prop. 9.1 (used here only as a cross-check, Section 6.2).
  7. `LemmaDDoublePrime`: `lemma_DDoublePrime`; `KAPPA_BOUNDED.md` §5.3.
  8. `TheoremK`: `telescoping_bound`, `window_of_partial` (the window bound (5.1) from partial sums); `KAPPA_BOUNDED.md` §5.
  9. `Gronwall`: `discrete_gronwall`, `contraction_bound` ($N_{j+1} \le qN_j + Z$); `KAPPA_BOUNDED.md` Prop. 4.1.
  10. `CubeCount`: `cube_count_sharp`, `cube_count_lower` (L-236 arithmetic); `SOURCES_FX_01.md`.
  11. `BlockNorm`: `rho2_le`, the exact factorization $\rho_2 \le \rho_1(j) \max_l \rho_1(j-1, k_l)$; `H2_CERTIFY.md`.
  12. `OishiRump`: `oishi_rump`, `oishi_rump_eta`, `frobenius_power_bound`; `H2_CERTIFY.md` §5.
  13. `Envelope`: `envelope_nontrivial`, `envelope_not_scale_invariant` (H-081 (B-NT) in log form); `ASSEMBLY.md` §2.9.
  14. `Divergence`: `profile_divergence`, `P4_divergence` (identity (D) of premise P4); `H2_CERTIFY_P4.md`.
  15. `TheoremC1`: `c1_two_step`, `c1_kappa_bound`, `c1_window`; `B2_TWOSTEP.md` §2.
  16. `StripBound`: `strip_bound_linear`, `strip_bound_quadratic` (the $\frac{3}{2}\tau$ bootstrap); `SOURCES_HBETA_N3.md` §4.
  17. `CertBridge`: `cell_invertible`, `cell_sup_bound`, `vertex_bound` (interval-to-norm bridge of the cell certifier); `numerics/h1cert_cell.py`, `h1cert_cell2.py`.
  18. `Tower`: `tower_bounds`, `I1_compress`, `majorant_core`, `lemmaT_core` (algebraic core of the one-step majorant, Theorem 6.1); `H1_ANALYTIC.md`.
  19. `Schur`: `schur_test_weighted`, `schur_test` (weighted Schur test for the spectral norm); `SOURCES_LOC.md` §3.3, `SOURCES_S2_FORM.md` §2, `SOURCES_A3_SIGMA.md` 2.6.

20. `CombesThomas`: `norm_entry_le`, `coercive_inv`, `combes_thomas`, `combes_thomas_exp`, `combes_thomas_decay`, `combes_thomas_schur` (the proof structure of L-125 (Lemma CT), the kernel-decay reduction of ledger R-001); `SOURCES_A1_KERNEL.md` §7. The identification with the block operators, the kernel-to-row-sum step, (K)/H-057 (C), and the rescaling are not formalized (`lean/VERIFY_CombesThomas.md`); the block-valued Schur test is in `BlockSchur`.
21. `Interp`: `multilinear_convex`, `monomial_affine` (`SOURCES_A3_SIGMA.md` 3.2); `window_perturb`, `window_perturb_partial` (`SOURCES_T2_KUNIF.md` 5.2, chained to L-009 (Theorem K)); `rescale_exp`, `rescale_rate`, `K1_iff`, `K1_numeric` (`SOURCES_T2_KAPPA2.md`); `xexp_le`, `xexp_absorb` (`SOURCES_GGAIN_LF.md` §2).
22. `Resolvent`: `resolvent_identity`, `resolvent_bound`, `lower_transfer`, `neumann_perturb`, `resolvent_chain` (`SOURCES_A3_SIGMA.md` §4); `wnorm_mul_le`, `wnorm_neumann`, `pwnorm_mul_le`, `pwnorm_neumann` (weighted Neumann series, row half; `SOURCES_A2_R2.md` §2).
23. `SlabCounterexample`: `slab_fails_10`–`slab_fails_13`, `slab_fails`, `lambda1_lt` (the proof step CMP 96 (2.127) fails for every $L \geq 10$); `SOURCES_A3_GAMMA0_FULL.md` §3.2–§3.3.
24. `BlockSchur`: `block_row_le`, `block_schur_test` (Schur test with block norms $\|K_{xy}\|_2$ in place of $|K|$); `SOURCES_A1_KERNEL.md` §7, `SOURCES_S2_FORM.md` §2 Π3.
25. `GaugeMin`: `variational`, `constrMin_existsUnique`, `lemma5`, `lemma5_full` ($RD^*A_* = 0$ for the constrained minimizer, from (F1), (F3), (F4) as hypotheses); `SOURCES_A3_GAMMA0_FULL.md` §5.

Not formalized: interval arithmetic itself, the enclosure premises P1′–P4, the identity $\rho_1^2 L^2 = \lambda_{\max}(G)$, and the history form of the shooting lemmas.

B Index of results by status

This appendix indexes the results of the paper by status tag; the tags match the verifier files. Here v1–v4 denote the successive versions `THEOREM2_PROOF.md`, `THEOREM2_PROOF_v2.md`, `THEOREM2_PROOF_v3.md`, and `THEOREM2_PROOF_v4.md` of the Theorem 2 notes. The statements that the main theorems take on trust are listed by item in the trust ledger `TRUST_LEDGER.md`, condensed in Section 7.7; ledger IDs are given below where an entry corresponds to a ledger item.

[PROVED+VERIFIED]

- L-004 (Lemma D′) (a)–(d), K_0, Remark 4 (`LEMMA_DPRIME_VERIFY` grades 1, 2, 5).
- L-006 (Lemma D) (corrected interval), L-007 (Lemma E) (Cg^2 form), nested-interval variant of L-006 (Lemma D) (`LEMMAS_DE_VERIFY`). L-007 (Lemma E) C^1 form: trivial variant, same proof.
- L-005 (Lemma D″) (tail-sum form, no J) (`KAPPA_BOUNDED_VERIFY` §6).
- Telescoping $S_k = \sum \beta_j(0)$ for the $p \to 0$ coefficient (`BETA_FP_VERIFY` item 1, corrected scope).
- Prop. 2.1 anchor-free identity; exact at the constant field (`BETA_FP_LEMMA43_VERIFY` item 1; `BETA_FP_H_VERIFY` item 1).
- $c = 11N^2/(12\pi^2)$, normalization chain; R-027 (F) source (Savvidy eq. (1.4); ledger X-005) (`BETA_FP_LEMMA43_VERIFY` item 2; `BETA_FP_H_VERIFY` §4).
- KAPPA Prop. 1.1 as an identity given its fixed-level inputs; Lemma 2.1 (integration by parts); Prop. 2.2; level-independence of the degree −4 part; L-009 (Theorem K) and L-001 (Theorem 2w) as conditional reductions (`KAPPA_BOUNDED_VERIFY` (i)–(v)).

- $\sup_j |\sigma_j^{(2)}(0)| < \infty$; Prop. 4.1 algebra (`KAPPA_BOUNDED_VERIFY` (vi), reason for R-005 failure corrected: obstruction = $\partial\sigma^{(1)}$ in Q_j and in H-178 (W-2), §7 there).

- `B2_TWOSTEP` two-step recursion; L-011 (R-025 (H2) $\Rightarrow$ ($C1\sigma$), factor ρ_2^2/L^2) and L-012 (R-025 (H2) $\Rightarrow$ (B2), factor ρ_2^2/L^4), conditional on R-025 (H2); threshold $\rho_2 < L$ (no strip loss); R-025 (H2) weaker than (H″) at $n = 2$ (`B2_TWOSTEP_VERIFY` grades 1, 3, 4). Form of Q_j by read-off (grade 8; `U3_LEMMA51` (3.3) not re-derived).

- `h2_scan` code structure, one-colour reduction, H and Δ_{out} blocks, Gram identity, separable Prop. A form (`H2_LARGE_L_VERIFY` grade 1); tail-bound derivation (4.1)–(4.3) (grade 6, inputs corrected).

- Logic of L-088 (Theorem B) and Lemma 1.3 of `BETA_FP_PW`, as conditional implications (`BETA_FP_PW_VERIFY` §6 items 1–4 CONFIRMED: doubling identity and Lemma 1.3; Prop. 2.2 / Lemma 2.3; Theorems A, B; Lemma 4.1, conditional). Its inputs (U), (DF), (T1) are listed under [PROVED, UNVERIFIED] and [ARGUED].

- Prop. E evenness, L-309 (Theorem W0)/W, L-071, L-072, A0, Prop. R-003 (`U3_AW_VERIFY` grades 1–4).

- U3 Theorem 4.1 + O5, Prop. A, Prop. B, O6, Theorem 6.1 (given V1; Taylor-data norm), Lemma 5.1 neutral value bound, single-block $m = 2$ failure in $d = 4$, one-step recursion, R2 midpoint identity (`U3_VERIFY`, `U3_REPAIRS_VERIFY`, `U3_LEMMA51_VERIFY`, `U3_LEMMA51_R23_VERIFY`).

- Exact one-step factorization $\rho_2 \le \rho_1(j) \max_l \rho_1(j-1, k_l)$ (`H2_CERTIFY_VERIFY`; Lean `rho2_le`).

- Sources block: L-174 (Lemma G3) given (Q) (`SOURCES_G3_VERIFY`); LOC (`SOURCES_LOC_VERIFY`); H-143 (M-A) given (H−) and H-144 (M-A′) (`SOURCES_MA_VERIFY`, `SOURCES_MA2_VERIFY`); H-145 (M-B) with $p_0 \ge 17/(\gamma_0 A_1^2)$ (`SOURCES_MB_VERIFY`); condition (a) of L-232 (Lemma N3) (`SOURCES_N3A_VERIFY`); the sourced large-field draft B-S.D3, with corrections (`SOURCES_D3_VERIFY`, `SOURCES_D3R_VERIFY`, `SOURCES_D3W8_VERIFY`); R-045 (B-N1) given R-045 (S) (`SOURCES_HHIST_N1_VERIFY`).

- Cesàro: given convergence, $\kappa_k = o(k) \Leftrightarrow \beta_\infty = c \log L$ (equivalence, not a weaker form) (`THEOREM2_PROOF_VERIFY` Point 4).

- Item 7′: the running-maximum lift of C_β to non-monotone pasts (`SOURCES_GGAIN_NONMONO_VERIFY`).

- (KE') by citation and H-123 (H_Q) (`SOURCES_QE_KE_HQ_VERIFY`); S2 for G and G_1 (`SOURCES_S2_FORM_VERIFY`, `SOURCES_S2_G1_V7_VERIFY`); fixed-j H-244 (S1), H-247 (S3), H-245 (S2)$|_{t=0}$, H-246 (S2c) given (I3) (`SOURCES_T2_S123_VERIFY`).

- V1 at fixed level, history-free, conditional on R-030 (G2), Bałaban's inductive bounds, and the averaging (0.4) (`SOURCES_T2_V1_VERIFY`); k-uniform decay and moments of E_0, ledger R-050 (`SOURCES_T2_EUMOM_VERIFY`).

- Round 5, proved relative to listed inputs and verified: R-008 (Theorem 5.4), R-009 in corrected form (Theorem 5.5), R-010 in the corrected form R-011 (Theorem 5.6), L-174 (Lemma G3) = N2 with (Q) from R-008 + R-009 and G_Γ bounded uniformly in M (Theorem 5.9),

Prop. A15-N given H-141 (Loc), (Bd), (H), (Cnt) (Proposition 5.8), and (I3) at every step (`SOURCES_T2_I3_VERIFY`); R-012 closed as a verified list (Proposition 5.7).

- Round 6, proved relative to listed inputs and verified: L-026 (Theorem LOC) (Theorem 5.10), L-104 (Proposition BDRY) with the replaced R-032 (Proposition 5.11), (G'_A) on its stated f-range (Theorem 5.12); the canonical order of constants (`SOURCES_ORDER_CANON_VERIFY`).

- Round 7, proved relative to listed inputs and verified: (M_j) at fixed j (Theorem 5.14), L5 by nested intervals (Theorem 5.15), R-016 in tree form (Theorem 5.20), R-017 with $r = 1$ (Proposition 5.16); verified reductions: R-018 (Proposition 5.18), R-019 (Proposition 5.17), R-020 (Proposition 5.19); (C1) of the (H−) chain given R-019, R-018; ledger R-034 cleared (the sweep with the CMP 122 additions, `BETA_FP_M` Prop. 4.3 for $j \geq j_1$, the Prop. 3.6 assembly, L-192 (Lemma H″), N3″) (`SOURCES_B10_SWEEP_VERIFY`).

- Round 8, verified: R-039 a verified reduction with M2′, M4 derived and M5 not needed (`SOURCES_B15_M245_VERIFY`); H-241 (R1′) a verified reduction (Proposition 5.21); R-044 (W4)-R proved relative to listed inputs (Proposition 5.22); the Λ_j sub-item of R-044 proved relative to listed inputs (`SOURCES_B20_LAMBDAJ_VERIFY`); the H-098 (E14-c) clause reduced to R-045 (S) (`SOURCES_B21_E14C_VERIFY`); the page readings R-047, R-048, R-049, R-051 (`SOURCES_BOOKKEEPING_VERIFY`); R-027 a verified reduction (`SOURCES_T2_F_STEPB_VERIFY`).

- Round 9, verified: H-137 (L-LL) proved (Theorem 5.25; `SOURCES_T2_LLL_VERIFY`); L-039 (Lemma R) and L-284 (Lemma T), and the condition H-057 (C) as a verified reduction (Proposition 5.23; `SOURCES_C_CONDITION_VERIFY`); H-168 (S′) proved (Proposition 5.27); the s-rule convention K-046 admissible (`SOURCES_SRULE_CONVENTION_VERIFY`); the p. 254 transfer proved relative to listed inputs (Proposition 5.28); CMP 99 Thm 3.9 written out, paper-grade relative to tagged inputs (Theorem 5.29); H-105 (FX-D) = Q-060, closed (`SOURCES_FX_D_VERIFY`).

- Closures of the H-146 (M-C) inputs: H-133 (I-V0) (`SOURCES_MC_IV0_VERIFY`); (I-ε) and H-131 (I-ord) for CMP 109/116 (`SOURCES_T2_IEPS_VERIFY`) and for the large-field layer of CMP 119/122 (`SOURCES_T2_IEPS_LF_VERIFY`; residue [I], every branch closes); K-025 (M1) as convention K-025 and the bound "order $\leq$ 2" on coupling and threshold derivatives (`SOURCES_HIGHER_M1_VERIFY`).

[PROVED, UNVERIFIED]

- Round 9, verifier running: R-027 (F) by proper time (Theorem 5.26; `SOURCES_T2_MPT_STEPC`); H-035 (M1) closed as a list (`SOURCES_M1_SWEEP`); the write-out of CMP 99 Thm 3.10 (`THM310_FULL`); the local fixes of Theorem 5.30 (fix pass T-0128).

- `BETA_FP_U`, no verifier file: Theorem 2.4 ((U) $\Leftarrow$ (T1) + R-025 (H2)), Prop. 3.1 ($\Gamma^{ex} = \gamma_{rec} + \mathfrak{d}_F$), Prop. 3.2 ($\gamma_{rec}$ bounded), Corollary 4.1, L-001 (Theorem 2w) by the plane-wave route given (T1), R-025 (H2), (DF), LW (5.7), R-007, H-202 (H0); the argued step "R-025 (H2) closes ($C1\sigma$)" of §3.5 there is superseded by `B2_TWOSTEP` L-011 [PROVED+VERIFIED].

- `LEMMA_DPRIME_SWEEP`: 19 cross-scale uses covered by (b) + (c′) + K-018 (S) (4 of 19 table rows, 11, 12, 16, 18, verified on page images by `LEMMA_DPRIME_VERIFY` comparisons A–E).

- Step "L-001 (Theorem 2w) $\Rightarrow$ Balaban's Theorem 1" (depends on the sweep and on L5 in the sharpened form of Section 3.4.5; round 7: both verified, Section 5.11).

- Prop. 3.6 assembly with $C(1+j)^2 L^{-2j}$; neutral-sector K-derivative paragraph; $\sigma^{(1)}$ convergence sketch jL^{-2j}.
- Necessity of individual positivity for literal (0.31) at every K (elementary).
- `BETA_FP_M` Prop. 4.3 ($\log|\det|$ bound in the N–O window, given H-244 (S1)–H-247 (S3)); round 7: confirmed for $j \geq j_1$ in `SOURCES_B10_SWEEP_VERIFY`.
- Premise P4 of the one-step certificate (`H2_CERTIFY_P4`; identity (D) in Lean `P4_divergence`); the $L = 13$ certificate sizing after fix G1 (`H2_CERTIFY_L13_G1`).

[NUMERICAL]

- $\rho_1(\infty) = 1.760, \ldots, 2.0895$ at $L = 3, \ldots, 13$; at $L = 13$ the measured $\sup \rho_1^2 = 4.37$ (`H2_CERTIFY_L13`); the certified bound is below.

[CERTIFIED (INTERVAL ARITHMETIC, OUTWARD ROUNDING; VERIFIED)]

- R-025 (H2) at $L = 13$ (Theorem 6.1; ledger R-025): $\rho_1(j,K)^2 \leq 10 < 13$ for all $j \geq 1$ (including ∞) and all K, hence $\rho_2 \leq 10 < 13$ by `rho2_le` (`H1_ANALYTIC`; `H1_ANALYTIC_VERIFY` V0–V9, `H1_ANALYTIC_VERIFY2`, `H2_CERTIFY_P5_VERIFY`). Residue as in Section 6.7.

[VERIFIED REDUCTION]

- (Q) to CMP 99 (3.132), (E_U), and LOC (`SOURCES_Q_VERIFY`, `SOURCES_CDC_VERIFY`); (E_U) to kernel-moment bounds Bałaban does not print (`SOURCES_EU_VERIFY`; the bound $K \leq K_0(1 + \ldots)$ is a [DERIVED SKETCH]).
- N3 to R-040/R-041 (H-hol) + real R-037 (H-hist) + H-120 (H2-hist) (`SOURCES_HBETA_N3_VERIFY`).
- R-040 (H-hol), small field, to Bałaban's conditions with modified constants (`SOURCES_HHOL_VERIFY`); large field to H-143 (M-A), H-145 (M-B), H-146 (M-C), R-038 (H-hist2) (`SOURCES_HHOL_LF_VERIFY`); H-147 (M-D) to R-037 (H-hist) + R-039 (H-thr).
- (H−) to CMP 102 (189), an existing trust item (`SOURCES_MA2_VERIFY`).
- R-038 (H-hist2) to the real uniform R-037 (H-hist), no new input (`SOURCES_HHIST2_VERIFY`).
- Round 3 (ledger R-001–R-005, R-037, R-041, R-045, R-046): R-001 kernel-decay form via L-125 (Lemma CT) to CMP 99 Thms 3.1, 3.3, 3.12 + H-136 (KE') + H-123 (H_Q) + S2 (`SOURCES_QE_KE_HQ_VERIFY`, `SOURCES_S2_FORM_VERIFY`); R-002 to (W_G) + H-057 (C) + CMP 96 Lemma 2.1 (`SOURCES_A1_WALK_VERIFY`), now written out as Theorem 5.1, with (W_G) written out as Theorem 5.2 (`A1_WALK_FULL_VERIFY`, `WG_FULL_VERIFY`, paper-grade relative to their tagged inputs); R-003 with the V7 repair to (W^c) + R-008 + two [I] (`SOURCES_A2_R2_VERIFY`, `SOURCES_S2_G1_V7_VERIFY`), with (W^c) written out as Theorem 5.3 (`WC_FULL_VERIFY`, paper-grade relative to its tagged inputs); R-004, full proof with $\gamma_0 \geq m_L/(48d^2L^d)$, and its σ-localized clause (`SOURCES_A3_GAMMA0_FULL_VERIFY`, `SOURCES_A3_SIGMA_VERIFY`); $R - 028(M_j)$ to (I3), (G'_A), Prop. 4.4 region 1 (`SOURCES_T2_S123_VERIFY`); R-005 to R-001 + CMP 102 (`SOURCES_A1_A4_VERIFY`); large-field R-037 (H-hist) (`SOURCES_GGAIN_LF_VERIFY`); H-146 (M-C) to the printed CMP 116 §2 conditions (`SOURCES_MC_VERIFY`); the R-045 (S) clauses of R-045 (`SOURCES_B21_S_VERIFY`); the $\kappa \to \kappa/2$ recovery R-046 (`SOURCES_T2_KAPPA2_VERIFY`).

- R-030 (G2) at fixed j (item 6, was [FORMAL]): torus densities and coefficient extraction commute with $N \to \infty$, modulo $\tilde{Q}$ onto, (Gap$_f$), and the averaging-analyticity reading (`SOURCES_T2_G2_VERIFY`).
- The use of R-006 in L-001 (Theorem 2w) (item 7′): L-183 (Proposition H0-red), residue N1–N9 (`SOURCES_T2_H0_VERIFY`); N2 proved (Theorem 5.9).
- Item 9 (K-independence in the shooting) closed: β_j is the whole-lattice coefficient by definition, Lemmas D′/D″ run at fixed K (`SOURCES_T2_KUNIF_VERIFY`).

[INTERVAL ENCLOSURE, VERIFIED]

- Point enclosure $\rho_2(\infty, K^*) \in [1.9993, 2.1273] < 3$ at $L = 3$; P1 (loop-family coefficients exact rationals); Oishi–Rump bound on $\|\mathfrak{B}^{-1}\|$ (`H2_CERTIFY_VERIFY`).

[ARGUED]

- R-026 (PI); (M_j) (Prop. 4.4 of `BETA_FP_M`); (T1) (plane-wave form of R-004, `BETA_FP_U` §2.3); (DF) (`BETA_FP_U` §3.4); R-007 (v4 label, not ledger R-007) = plane-wave vs constrained-minimizer expansion point (uniform in k); $\mathfrak{d}_Y$, $\mathfrak{d}_{gf}$ bounds; (H″) ⇒ (A) in full ⇒ (B2) (`U3_AW` §4.7 sketch); C^2, C^3 of $\sigma^{(1)}$ under R-025 (H2) (not written); j_* size; the rate $O(\Lambda^{-2} \log \Lambda)$.

[OPEN]

- R-025 (H2) at $L \notin \{13, 15, 17, 21\}$ (numerical at $L = 3, 5, 9$; certified at $L = 13$ and, round 7, at $L = 15, 17, 21$, Section 6.8); R-027 (F), a verified reduction since round 8, with H-137 (L-LL) proved and verified (round 9), a full proof pending verification (Theorem 5.26), and the open tension of the single-step β_1 against $c \log L$ (Section 6.9); (B2) is reduced to it [VERIFIED REDUCTION]. The residue of R-002, R-003: the inputs of Theorems 5.1, 5.2, 5.3, namely the printed inputs of H-057 (C) (R-021), the printed sketches of CMP 99 Thms 3.7, 3.9, 3.10 (round 9: 3.7, 3.9 written out), R-022, R-023, WG-1 in the M_1 form, and [I] geometry and convention items (O3, O4 closed). (T1), (DF), R-007 (v4 label) for Route B (ledger R-033). The inputs of V1 and R-030 (G2): the averaging declaration (0.4), $\tilde{Q}$ onto, (Gap$_f$), and a j-uniform form of R-030 (G2). The residue of $R - 028(M_j)$: (I3), (G′$_A$), Prop. 4.4 region 1. L5 (domain of β; nested-interval D″ [NOT WRITTEN]). The finite-volume identification of the step, R-term part (ledger R-032; an input of Theorem 1). (A) in full. See Section 7.

[CORRECTED BY VERIFIERS SINCE V3]

- `B2_TWOSTEP` G1: the certification of R-025 (H2) is a computer-assisted proof project, not a check (`B2_TWOSTEP_VERIFY` grade 6).
- `H2_LARGE_L`: L-trend rate claim (grade 5); j_0 inputs, δ_m 45–90× larger, $a_*(L = 5) = 0.0107$ (grade 6), giving $j_0 \approx 8$ ($L = 3$), ≈ 6 ($L = 5$).
- `BETA_FP_PW`: comparison and numerics (`BETA_FP_PW_VERIFY` grades 5, 6).

[REFUTED]

- (HS) $\sup \|H_j\| < L\sqrt{L-1}$ at $L = 3$.
- Not a refuted statement, but a failed method: the one-step sup-norm C^1 recursion at $L = 3$ (factor 1.005–1.033 for $j \geq 2$; Section 3.7.2).
- `U3_LEMMA51` Theorem 4.1 for $m = 2$ with a single block-M constraint in $d = 4$.
- v1 claims withdrawn: "every early step must be certified for Theorem 1" (false: L-004 (Lemma D′)/D″); "(H′) is weaker than $\beta_\infty = c \log L$" (it is equivalent); "the whole $\log \Lambda$ sits in the fine piece" as a fact (it is equivalent to the open statement); "one-loop parts hold for SU(N)" (Lemma 5.1's charged/neutral split is the SU(2) Cartan split); (3.2) with Cj^2L^{-2j} (0 at $j = 0$); the (M) resolvent bound $\|\mathcal{K}_t^{-1}\| \leq C/|f|$ (Nielsen–Olesen zero); (H) with IR weights as stated (UNSUPPORTED).

[BALABAN]

- H-202 (H0) smoothness of β_{j+1} in g (CMP 109 (1.22), pp. 263–264), at odd $L > 11$ only; uniformity in j is our reading and uniformity in the history is not printed (ledger R-006); Theorem 1 and the inductive bounds of CMP 109, 116, 119, 122 and the propagator papers CMP 95, 96, 98, 99, 102. Uniform level bounds partly cited from CMP 95/96/99 (O6 replaced the main citation). Each Bałaban input that is printed without proof, or used without being printed, is a Group A row of the trust ledger.

[VERIFIED QUOTE]

- Theorem 2 and Theorem 1 wording, "separate paper", p. 298 history sentence, (0.1) with $1/g_k^2$ and χ_k, (1.3)–(1.5) (print misprint $E^{(j+k)}$ read as $E^{(j+1)}$), (2.6)–(2.9) of CMP 119, items A–E of CMP 122 (`BALABAN_EXCERPTS_CHECK`; `LEMMA_DPRIME_VERIFY`; `LEMMA_DPRIME_SWEEP` page images). `BALABAN_EXCERPTS_CHECK` item 10 (CMP 116 has no C^m-in-g statement) is unchanged since v3; the item 11 readings are verified, with corrections, in `SOURCES_BOOKKEEPING.md` (former ledger R-051).

[NUMERICAL] see Section 6.

C Label map

The numbered environments of this paper carry the names used in the project files. The first table maps them. The second table maps the section numbers of `THEOREM2_PROOF_v4.md` ("v4") and `ASSEMBLY.md` to the sections of this paper; section references inside the project files use the old numbers.

this paper	name in the project files	source
Theorem A	Route K summary: L-001 (Theorem 2w) ⇒ Theorem 1	`THEOREM2_PROOF_v4` §6, §10
Theorem B	Theorem 1.3 (summary)	`ASSEMBLY` §1.3
Theorem 3.1	L-001 (Theorem 2w)	`THEOREM2_PROOF_v4` §1.3
Lemma 3.2	L-004 (Lemma D′)	`LEMMA_DPRIME`; v4 §2.1
Remark 3.3	Remark 4	`LEMMA_DPRIME`; v4 §2.1

this paper	name in the project files	source
Lemma 3.4	L-005 (Lemma D″)	KAPPA_BOUNDED §5.3; v4 §2.2
Lemma 3.5	L-006 (Lemma D)	LEMMAS; v4 §2.4
Lemma 3.6	L-007 (Lemma E)	LEMMAS; v4 §2.4
Proposition 3.7	Prop. 2.1 (anchor-free identity)	BETA_FP_LEMMA43; v4 §3.2
Theorem 3.8	L-009 (Theorem K)	KAPPA_BOUNDED; v4 §4.1
Theorem 3.9 (Theorem B_{PW})	L-088 (Theorem B) = Theorem 3.2	BETA_FP_PW; v4 §4.2
Theorem 3.10	L-011	B2_TWOSTEP; v4 §5.3
Theorem 3.11	L-012	B2_TWOSTEP; v4 §5.3
Definition 4.1	H-116 (H-U)	ASSEMBLY §1.2
Theorem 4.2	Theorem 1.3	ASSEMBLY §1.3
Theorem 4.3	Theorem 1.4	ASSEMBLY §1.4
Proposition 4.4	Proposition of H-074 (B-C)	ASSEMBLY §2.4
Lemma 4.5	finite-form lemma of H-083 (B-RP)	ASSEMBLY §2.7
H-200 (GC1)–H-201 (GC3)	(G1)–(G3) of the dependency table	ASSEMBLY §3
(S_{src})	R-045 (S), final sourced bound	SOURCES_D4D5
items H, 1–13, B1–B8	items of the open list	THEOREM2_PROOF_v4 §10
Theorem 5.4	R-008 (round 5)	SOURCES_A7_CDC
Theorem 5.5	R-009, corrected form (round 5)	SOURCES_A8_POS
Theorem 5.6	R-011 (round 5)	SOURCES_A10_LOWER
Proposition 5.7	R-012 list (round 5)	SOURCES_A11_CONST
Proposition 5.8	Prop. A15-N (round 5)	SOURCES_A15_WHOLE
Theorem 5.9	L-174 (Lemma G3) = N2 (round 5)	SOURCES_G3_N2
Theorem 5.10	L-026 (Theorem LOC) (round 6)	SOURCES_A15_LOC §7.5
Proposition 5.11	L-104 (Proposition BDRY) (round 6)	SOURCES_A15_BDRY
Theorem 5.12	(G'_A), Thm 5.1 + Cor. 5.2 (round 6)	SOURCES_T2_GAPRIME2
Remark 5.13	range remark for (G'_A) (round 6)	SOURCES_T2_GAPRIME2, SOURCES_T2_FRANGE
Theorem 5.14	(M_j), Theorem 4.2 and §5 (round 7)	SOURCES_T2_MJ_REST
Theorem 5.15	Lemmas D″-nest, D′-nest, Route 2 (round 7)	SOURCES_T2_B7_CLAMP
Proposition 5.16	R-017 identity (round 7)	SOURCES_A17_A19
Proposition 5.17	R-019 reduction (round 7)	SOURCES_A17_A19 §4–§5
Proposition 5.18	R-018 reduction (round 7)	SOURCES_A18_REGULAR
Proposition 5.19	R-020 reduction (round 7)	SOURCES_A23_TSTEP
Theorem 5.20	R-016 in tree form (round 7)	SOURCES_A16_REPLACE2
Proposition 5.21	H-241 (R1′), frozen-threshold bounds (round 8)	SOURCES_HHOL_LF_SPRIME
Proposition 5.22	R-044 (W4)-R (round 8)	SOURCES_B20_W4R
Proposition 5.23	the H-057 (C) condition, §8 verdict (round 8; verified round 9)	SOURCES_C_CONDITION
Lemma 5.24	L-039 (Lemma R) (round 8; verified round 9)	SOURCES_C_CONDITION §3.2
Theorem 5.25	H-137 (L-LL), Theorem 5.1 (round 8; verified round 9)	SOURCES_T2_LLL
Theorem 5.26	H-148 (MPT) L-060 (Theorem A), L-112 (Lemma C), L-090 (Theorem B) (round 9, pending)	SOURCES_T2_MPT_STEPC

this paper	name in the project files	source
Proposition 5.27	H-168 (S′), L-127 (Lemma C′) (round 9)	`SOURCES_B21_S_PRIME`
Proposition 5.28	p. 254 transfer (round 9)	`SOURCES_P254_TRANSFER`
Theorem 5.29	L-044, L-210 (Lemma LB) (round 9)	`THM39_FULL`
Theorem 5.30	L-045, Cor. 3.8 forms (round 9)	`THM37_FULL`

old section	this paper
v4 §1.1	Section 1.2
v4 §1.2	Section 3.1
v4 §1.3	Section 3.2
v4 §1.4	Section 3.3
v4 §2 (notation)	Section 2.3
v4 §2.1	Section 3.4.1
v4 §2.2	Section 3.4.2
v4 §2.3	Section 3.4.3
v4 §2.4	Section 3.4.4
v4 §2.4 (L5)	Section 3.4.5
v4 §3.1	Section 3.5.1
v4 §3.2	Section 3.5.2
v4 §3.3	Section 3.5.3
v4 §3.4	Section 3.5.4
v4 §3.5	Section 3.5.5
v4 §4	Section 3.6
v4 §4.1	Section 3.6.1
v4 §4.2	Section 3.6.2
v4 §4.2a	Section 3.6.3
v4 §4.3	Section 3.6.4
v4 §5	Section 3.7
v4 §5.1	Section 3.7.1
v4 §5.2	Section 3.7.2
v4 §5.3	Section 3.7.3
v4 §5.4	Section 6.4
v4 §5.5 (arithmetic)	Section 3.7.4
v4 §5.5 (values)	Section 6.5
v4 §5.6	Section 6.11
v4 §5.7	Section 3.7.5
v4 §6	Section 3.8
v4 §7	Section 3.9
v4 §8.1	Section 6.1
v4 §8.2	Section 6.2
v4 §8.3	Section 6.3
v4 §9 (Lean)	Appendix A
v4 §9 (other tags)	Appendix B
v4 §10 (open list)	Section 7
v4 §10 (hypothesis table)	Section 4.6
v4 §11	Section 1.5
v4 preamble (tag legend)	Section 1.4
v4 preamble (range of L)	Section 2.2
v4 change logs	Appendix D

old section	this paper
ASSEMBLY §0 (notation)	Section 2.1
ASSEMBLY §0 (observables)	Section 2.5
ASSEMBLY §1.1	Section 4.1
ASSEMBLY §1.2	Section 4.2
ASSEMBLY §1.3	Section 4.3
ASSEMBLY §1.4	Section 4.4
ASSEMBLY §2	Section 4.5
ASSEMBLY §2.1	Section 4.5.1
ASSEMBLY §2.2	Section 4.5.2
ASSEMBLY §2.3	Section 4.5.3
ASSEMBLY §2.4	Section 4.5.4
ASSEMBLY §2.5	Section 4.5.5
ASSEMBLY §2.6	Section 4.5.6
ASSEMBLY §2.7	Section 4.5.7
ASSEMBLY §2.8	Section 4.5.8
ASSEMBLY §2.9	Section 4.5.9
ASSEMBLY §3	Section 4.6
ASSEMBLY §4	Section 4.7
ASSEMBLY corrections	Section 4.8
NUMERICS §7 (rewritten)	Section 6.9

D Revision history

- *Theorem 2 notes, v3.* Built on THEOREM2_PROOF_v2.md with the corrections of THEOREM2_PROOF_v2_VERIFY (cited there as v2_VERIFY): the range of L (odd, $L > 11$, or H-202 (H0) at this L as a hypothesis); L5 sharpened (the clamped shooting leaves Bałaban's domain; the nested-interval variant of L-005 (Lemma D$''$) is [NOT WRITTEN]); (H$''$) $\Rightarrow$ (B2) tagged [ARGUED]; "independent" routes replaced by "formally distinct", with the points $\theta_* < 1$, k-uniform β identification, and the first-order vertex smoothness of (U); the independence record completed (ONLINE_SWEEP §2 as the first channel; the O(3) paper cited); open list extended (item 9, R-029, item 10, hypothesis table); sweep count "4 of 19 table rows"; γ_* depends on R through B'. At that time BETA_FP_PW_VERIFY.md had no grade, and Route B stayed [PROVED, UNVERIFIED]. The grades of LEMMA_DPRIME_VERIFY and KAPPA_BOUNDED_VERIFY §9(iv) were cited explicitly.

- *Theorem 2 notes, v4.* Added the material verified after v3: B2_TWOSTEP.md with B2_TWOSTEP_VERIFY.md, H2_LARGE_L.md with H2_LARGE_L_VERIFY.md, lean/README.md (L-005 (Lemma D$''$) in Lean), BETA_FP_U.md (Route B inputs), and numerics/NUMERICS.md §7. The section on (B2) was rewritten around R-025 (H2) (Theorems L-011, L-012; threshold $\rho_2 < L$; one-step sufficiency at $L \geq 5$; certification plan; corrected L-trend and j_0 inputs). The ledger and open list were updated, with item H as the single uniform hypothesis of both routes. KAPPA_BOUNDED_VERIFY states the (B2) obstruction in its §7 and §9(vi). BALABAN_EXCERPTS_CHECK had no change since v3.

- *Tag change in the LaTeX version of v4.* The logic of Theorem $\mathrm{B_{PW}}$ is tagged [PROVED+VERIFIED, CONDITIONAL], citing BETA_FP_PW_VERIFY §6 items 1–4, per the tag decision in PLAN.md (2026-10-28). The Markdown v4 kept [PROVED, UNVERIFIED]. The inputs (U), (DF), (T1) from BETA_FP_U stay [PROVED, UNVERIFIED].

- *Assembly.* `ASSEMBLY.md` with the corrections of `ASSEMBLY_VERIFY.md` (Section 4.8). The verifier also found the dependency table stale in five rows (grade 7; no overclaim): B-T2.2, B-T2.3, B-T2.4, B-S.D2C, B-S.D3 lagged `PLAN.md`. The table was rebuilt with a tag for every block and the sub-blocks added later (Route K, Route B, R-025 (H2), R-039 (H-thr), H-111 (G-gain), R-043 (FX)). Minor items: `SOURCES.md` §1.3 gives $c_j = 1 + O(g_j^4)$, while this paper uses $O(g^3)$ per `SOURCES_VERIFY` item 8 (the corrected value); the variance estimate drops a harmless g_0^4 (logarithmic) factor.

- *This version.* The Theorem 2 chapter and the assembly chapter are merged into one article: one notation section, one status legend, numbered environments (Appendix C), one numerical section, one list of open items, and the hypothesis table of v4 §10 merged into the dependency table. Three names were changed to avoid collisions (Section 2.6). The dependency table and the open list were brought to the current state of `PLAN.md`: R-043 (FX) is reduced to N3 (Q-060 and R2 [PROVED+VERIFIED], Q-061 [PROVED+VERIFIED, CONDITIONAL ON N3], Q-062 avoided by the main form; previously Q-060, Q-061 [PROVED, UNVERIFIED] and R2, Q-062 [OPEN]), and the point enclosure of R-025 (H2) at $L = 3$ is verified (Section 6.6). No theorem, lemma, or proof changed.

- *Round 3.* R-001–R-005 regraded to verified reductions (R-004 a full proof with a corrected constant), R-002 added and reduced, item 9 closed (R-032 moved to the input set of Theorem 1), item 7 restated as 7′, G4 off-chain, R-037, R-041, R-045, R-046 regraded, Lean files 21 → 22 (`TRUST_LEDGER.md` round 3; `PLAN.md`).

- *Round 4 (integration of the write-outs).* Theorems 5.1, 5.2, 5.3 (Section 5.8) replace "reduced to printed sketches" for R-002, the final-propagator expansions, and (W^c); all three are paper-grade relative to their tagged inputs, and the tag stays [VERIFIED REDUCTION]. Corrected constants: WG-1 in the M_1 form; $K_b \le 64d^3L^{2d+1}$; $c_M \le 4\sqrt{N}$; the Schur-sum factor $1 + 4/(e\delta)$; the $\frac{1}{3}M_1$ band by one-layer enlargement. H-133 (I-V0), (I-ε), H-131 (I-ord) closed; K-025 (M1) became convention K-025; V1 proved and verified at fixed level; R-030 (G2) moved from [FORMAL] to [R]; R-050 proved and verified.

- *Round 5.* Statements with hypotheses and residues added in Section 5.9: R-008 proved relative to listed inputs (Theorem 5.4; C is the constraint map and $C^*\Delta_k C$ the precision; the reduction to R-001 + R-018 is superseded); R-009 proved in corrected form (Theorem 5.5; Bałaban's symmetry claim is not proved and not needed, the majorant uses $H_0 = \operatorname{sym} C_0^{-1}$; Lemma D with $e^{-\varepsilon'' M/8}$); R-010 proved in the corrected form R-011 (Theorem 5.6; the raw Taylor terms do not obey (1.24), the transformed terms do after the exact marginal cancellation); R-012 closed as a verified list (Proposition 5.7); R-015 narrowed (Proposition 5.8; exact equality cannot be derived; R-015 leaves the set of L-001 (Theorem 2w), and Bałaban's assertion that the $\mathbb{Z}^d$ limit exists moves into R-006); L-174 (Lemma G3) proved relative to listed inputs (Theorem 5.9; G_Γ bounded uniformly in M, verified; (Q) in the joint form, Λ corrected), so the use of R-006 in L-001 (Theorem 2w) is a verified reduction; (I3) proved at every step; β_H a free parameter, first in the order of constants. Trust table: rows R-006, R-008, R-009, R-010, R-012, R-015, R-050, R-028, R-030, R-037, K-005, K-025 rewritten; rows K-026–K-031 added. Binding sets: L-001 (Theorem 2w) = {R-006} + R-025–R-031 + X-005 (was R-006, R-015); L-001 (Theorem 2w) ⇒ Theorem 1 = 4 A + 9 B + X-005; Theorem B unchanged in size; the only Group A row without a verified proof or reduction is R-015. The headline stays "verified reduction".

- *Round 6.* Section 5.10 added: L-026 (Theorem LOC) (Theorem 5.10; H-141 (Loc) exact for non-wrapping polymers, relative to K-033–K-037); L-104 (Proposition BDRY) (Proposition 5.11; the $\partial\Lambda_{k+1}$ reading exact relative to K-038; threshold $N_k > 22M$, $\varepsilon(M) = C_j(22M)^6 e^{-11m_j M}$); R-032 found ill-posed and replaced by (a)–(c), proved; the volume hypothesis K-032 (Vol), $\ell > 22M$, added to the setting (Section 2.1) with the even form $\ell = 2L^{m_T}$, $m_T \geq m_M + 1$; (G'_A) proved for $0 < |f| \leq ((1-c)/100)^2/2$ with gap $2c|f|$ (Theorem 5.12), a narrow range, with the f-range check still running; the canonical acyclic order of constants, with two nominal cycles resolved by the split, 32 lower bounds on M, and none upper. Trust table: rows R-006, R-015, R-026, R-028, R-032, K-005, K-025, K-026 rewritten; rows K-032–K-038 added; counts A 19, B 22, C 37, X 5. Binding sets unchanged in size: L-001 (Theorem 2w) = {R-006} + R-025–R-031 + X-005, and R-006 has no bare assertion left; L-001 (Theorem 2w) ⇒ Theorem 1 = 4 A + 9 B + X-005, Group C adds K-029–K-038; Theorem B unchanged in size. No cited-but-unproved Group A row lacks a verified proof or reduction; the implicit rows R-016–R-019, R-020 (and R-014 inside R-013) have neither. The headline stays "verified reduction".

- *Round 7.* Section 5.11 added: (M_j) proved at fixed j (Theorem 5.14; region 1 as an asymptotic identity via the shifted-log split, neutral sector (N2); only the $o(f^2)$ form), with the proved (G'_A) range sufficient (`SOURCES_T2_FRANGE.md`); L5 proved by nested intervals on true trajectories, no clamp (Theorem 5.15); R-016 proved in tree form (Theorem 5.20; the first proof's locality step was false); R-017 proved by the exact identity $B' = g_k B$ on cut bonds (Proposition 5.16); R-018, R-019, R-020 verified reductions (Propositions 5.18, 5.17, 5.19); H-215 (K1) replaced by H-216 (K1′) with 13δ and $\delta = 0.025$; R-041 chain item 5; R-034 cleared; R-025 (H2) certified at $L = 15, 17, 21$; canon section F (28 entries, acyclic, four two-sided constants with existence conditions); R-027 reduced [R], unverified, with an open numerical tension on β_1 against $c \log L$. Trust table: rows R-016–R-019, R-020, R-025, R-027, R-028, R-031, R-034, R-041, R-046 rewritten; rows K-039–K-043 added; counts A 19, B 22, C 42, X 5. Binding sets unchanged in size. No Group A row remains without a verified proof or verified reduction; the only Group B row of Theorem B without one is R-027. The headline stays "verified reduction".

- *Round 8.* Section 5.12 added. R-039: M2′ and M4 derived, M5 not needed, a verified reduction with the new input H-241 (R1′), itself a verified reduction with three [I] (Proposition 5.21); R-038 is no longer an input. R-044: R-044 (W4)-R proved relative to inputs (Proposition 5.22; the R operation divides by nothing) and the Λ_j sub-item proved relative to inputs, both verified with corrections; residue W4-G [R] and the Ursell bound [I]. R-045: the H-098 (E14-c) clause reduced to R-045 (S), verified. The readings R-047, R-048, R-049, R-051 confirmed on page images, so the S1 downgrade is reversed and the round-7 binding sets stand. R-027: the reduction verified; H-137 (L-LL) proved by a proper-time route, verifier running (Theorem 5.25). H-057 (C): stated exactly in row R-001; a draft proof with L-039 (Lemma R) and a repair of a seam defect on the periodic lattice, verifier running (Proposition 5.23). K-009 unused. Section 6.9 added: `NUMERICS` §7 rewritten (the old claim withdrawn), $\beta_1(13)$ certified for rounding at $N_g = 8$, the converged-grid test running on the Windows GPU, the partial cross-check of the cell certificate (200 FAILED cells, not violations), the H-137 (L-LL) and H-057 (C) numerics. Trust table: rows R-001, R-027, R-039, R-044, R-045, K-009 rewritten; the Lean support table added; counts A 19, B 22, C 42, X 5 unchanged; binding sets unchanged. No Group A row remains without a verified proof or verified reduction; the only Group B row of Theorem B without one is R-027, now with H-137 (L-LL) proved, pending verification. The

headline stays "verified reduction".

- *Round 9.* The outlook sections were removed; the paper states results and scope only. Section 5.12 updated: H-057 (C) a verified reduction with L-039 (Lemma R) and L-284 (Lemma T) verified, the seam defect confirmed, and the torus-side hypothesis discharged by K-032 (Vol); H-137 (L-LL) proved and verified. Section 5.13 added: a proof of R-027 (F) by proper time, pending verification (Theorem 5.26); H-168 (S′) (Proposition 5.27); H-105 (FX-D) = Q-060; the s-rule convention K-046; H-035 (M1) closed as a list, verifier running; the p. 254 transfer (Proposition 5.28) with the gap of `WC_FULL.md` §3.1(2); CMP 99 Theorems 3.7 and 3.9 written out with corrections of printed errors, including H-180 (X-PU) (Theorems 5.30, 5.29); the canonical order stays acyclic; condition K-047. The (E_U) section is now Step 11. Section 6.10 added: the seam test of H-057 (C), H-137 (L-LL) to $\varphi = 0.00125$, and the restart of the β_1 run on the jax port (results pending). Trust table: rows R-001, R-002, R-003, R-027, R-043, R-045 rewritten; new rows R-021–R-023 and K-044–K-047; counts A 22, B 22, C 46, X 5; Theorem B rests on 20 Group A items; Group A rows without a verified proof or reduction: R-021, R-022, R-023; Group B rows without one: none. For L-001 (Theorem 2w) no genuine hypothesis remains. The Lean table lists 45 files. A remark on scope records that a volume-uniform unit-scale cluster expansion is expected to be impossible (Section 4.7). The headline stays "verified reduction".

References

[1] Tadeusz Bałaban. Propagators and renormalization transformations for lattice gauge theories. I. *Comm. Math. Phys.*, 95:17–40, 1984.

[2] Tadeusz Bałaban. Propagators and renormalization transformations for lattice gauge theories. II. *Comm. Math. Phys.*, 96:223–250, 1984.

[3] Tadeusz Bałaban. Averaging operations for lattice gauge theories. *Comm. Math. Phys.*, 98:17–51, 1985.

[4] Tadeusz Bałaban. Propagators for lattice gauge theories in a background field. *Comm. Math. Phys.*, 99:389–434, 1985.

[5] Tadeusz Bałaban. Spaces of regular gauge field configurations on a lattice and gauge fixing conditions. *Comm. Math. Phys.*, 99:75–102, 1985.

[6] Tadeusz Bałaban. Ultraviolet stability of three-dimensional lattice pure gauge field theories. *Comm. Math. Phys.*, 102:255–275, 1985.

[7] Tadeusz Bałaban. The variational problem and background fields in renormalization group method for lattice gauge theories. *Comm. Math. Phys.*, 102:277–309, 1985.

[8] Tadeusz Bałaban. Renormalization group approach to lattice gauge field theories. I. generation of effective actions in a small field approximation and a coupling constant renormalization in four dimensions. *Comm. Math. Phys.*, 109:249–301, 1987.

[9] Tadeusz Bałaban. Convergent renormalization expansions for lattice gauge theories. *Comm. Math. Phys.*, 119:243–285, 1988.

[10] Tadeusz Bałaban. Renormalization group approach to lattice gauge field theories. II. cluster expansions. *Comm. Math. Phys.*, 116:1–22, 1988.

[11] Tadeusz Bałaban. Large field renormalization. I. the basic step of the $\mathcal{R}$ operation. *Comm. Math. Phys.*, 122:175–202, 1989.

[12] Tadeusz Bałaban. Large field renormalization. II. localization, exponentiation, and bounds for the $\mathcal{R}$ operation. *Comm. Math. Phys.*, 122:355–392, 1989.

[13] Roger Dashen and David J. Gross. The relationship between lattice and continuum definitions of the gauge theory coupling. *Phys. Rev. D*, 23:2340–2348, 1981.

[14] Elvin. yang-mills: project repository (notes/ directory holds the project files cited as `FILE.md`), 2026. Local repository; no public remote.

[15] Arthur Jaffe and Edward Witten. Quantum Yang–Mills theory. In *The Millennium Prize Problems*, pages 129–152. Clay Mathematics Institute, American Mathematical Society, 2006. Official problem statement, Clay Mathematics Institute.

[16] Martin Lüscher and Peter Weisz. Background field technique and renormalization in lattice gauge theory. *Nucl. Phys. B*, 452:213–233, 1995.

[17] Paolo Menotti and Andrea Pelissetto. General proof of Osterwalder–Schrader positivity for the Wilson action. *Comm. Math. Phys.*, 113:369–373, 1987.

[18] OpenAI. The canonical massive continuum limit of the two-dimensional O(3) model. `https://github.com/openai/math`, family 215, preprints folder, 2026. October 4, 2026.

[19] Konrad Osterwalder and Robert Schrader. Axioms for Euclidean Green's functions. *Comm. Math. Phys.*, 31:83–112, 1973.

[20] Konrad Osterwalder and Robert Schrader. Axioms for Euclidean Green's functions. II. *Comm. Math. Phys.*, 42:281–305, 1975.

[21] Konrad Osterwalder and Erhard Seiler. Gauge field theories on a lattice. *Ann. Phys.*, 110:440–471, 1978.

[22] G. K. Savvidy. Infrared instability of the vacuum state of gauge theories and asymptotic freedom. *Phys. Lett. B*, 71:133–134, 1977.

[23] George Savvidy. From Heisenberg–Euler Lagrangian to the discovery of chromomagnetic gluon condensation. *Eur. Phys. J. C*, 80:165, 2020. arXiv:1910.00654.

[24] Hao Shen, Rongchan Zhu, and Xiangchan Zhu. A stochastic analysis approach to lattice Yang–Mills at strong coupling. *Comm. Math. Phys.*, 400(2):805–851, 2023.

[25] Ethan Sussman and Levent Alpöge. SU(2) Yang–Mills theory in four dimensions via Bałaban's renormalisation group: construction, axioms and non-triviality. ym4_manuscript_zeroth_draft.pdf, repository `https://github.com/ethanwsussman/Yang--Mills-LLM-experiment`, commit 1b92bd0347e3b8763cd8ecc1f96dc3c2a92dd310, 2026. Draft dated 6 October 2026. Accessed 2026-10-27.

[26] Ethan Sussman and Levent Alpöge. The Yang–Mills continuum limit along Bałaban's renormalization group: a complete account of Theorem 0. LaTeX files theorem_0_complete_account_raw_{master,I,II,III}.tex, repository `https://github.com/ethanwsussman/Yang--Mills-LLM-experiment`, commit 1b92bd0347e3b8763cd8ecc1f96dc3c2a92dd310 (2026-10-27), 2026. Written largely by LLM, with Claude; not vetted by the repository authors. Accessed 2026-10-27.